

MOROK TOWER

promin

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Appendices

These tables are a complete generated projection of canonical owners. Every entry shows an owner path and an executable validator, schema reference, predicate, or mutation-probe identifier.

Owner: *core/promin.manifest.json* | Validator: *bundle_digest + core_components*

Canonical owner trace

Owner	Sole function	Validator	SHA256
core/promin.manifest.json	Own Core bundle identity and exact digest binding for the five other Core artifacts.	urn:promin:contracts:1#/\$defs/StandardManifest	644f9461fca920d937417211c69e3b58ac78c95d9122e51b8c743d23ddbfb3cb3
core/semantic-model.json	Own the minimal persistent vocabulary, relation grammar, derived-projection boundary, and provider capability contracts.	urn:promin:contracts:1#/\$defs/SemanticModel	1f5af8d8ce12e99aa9964f03396d945c6781fba234afd4fb01ad3d9044aa6fe a
core/authority-model.json	Own action capabilities, runtime Grant semantics, trust modes, configured signature-adapter verification, root ceilings, and separation-of-duty constraints.	urn:promin:contracts:1#/\$defs/AuthorityModel	24c3993062893ace123fa5e8e5d85a76e2ed8e622f4422f8f2fbbdb00248c5ca1
core/policy-set.json	Own executable cross-record invariants through stable validator identifiers and no duplicated budget values.	urn:promin:contracts:1#/\$defs/PolicySet	88146d4e87aaee238ec497e155b8549d9c62a2b3d591da9d75069112c4c7902e
core/conformance.json	Own acceptance predicates, mutation families, structural hard ceilings, scale contracts, and production-claim rules.	urn:promin:contracts:1#/\$defs/Conformance	afcefec3a1316f8f7af570ac49ad32a6c36d94120f06b07173fa22f7676fc27d
core/contracts.schema.json	Own all structural, format, kind-specific payload, and strict critical-field contracts in one bundle.	Draft 2020-12 oneOf	bfbcb28d653b2a7356a6db0038d9cb0fbceb83f08300de0ff0f7eb72b9aa55e0a

Core bundle digest: 4ae2f4b86b408af59118160a00d0bae071778df66965867ec94def625910fbac

Selected preset outside Core identity: presets/semantic-morok-tower.json | SHA256: bc991f9c8bd8b7a3f52ea8305639203914164200ad48f17aac97a30480551184

Owner: *core/promin.manifest.json* | Validator: *canonical_name_only + core_components + bundle_digest*

Schema catalog - 127 definitions

Schema ID: urn:promin:contracts:1 | Dialect: <https://json-schema.org/draft/2020-12/schema> | Definitions: 127

1. ActionCapabilityDef

Type summary: type="object"; additionalProperties=false

Required: id, purpose

Property	Required	Structural rule
id	yes	enum=[16 items]
purpose	yes	type="string"; minLength=1
<pre>{ "additionalProperties": false, "properties": { "id": { "enum": ["standard.activate", "authority.manage", "task.plan", "task.execute", "lease.manage", "evidence.publish", "validation.evaluate", "finding.record", "finding.resolve",</pre>		


```

        "finding.waive",
        "candidate.promote",
        "release.decide",
        "export.create",
        "migration.manage",
        "projection.rebuild",
        "projection.read"
    ]
},
"purpose": {
    "minLength": 1,
    "type": "string"
}
},
"required": [
    "id",
    "purpose"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ActionCapabilityDef | Validator: urn:promin:contracts:1#/\$defs/ActionCapabilityDef

2. Activation

Type summary: type="object"; additionalProperties=false

Required: record_type, canonical_name, core_bundle_digest, preset_digest, init_file_digests, operating_profile, trust_mode, activation_digest, proofs, implementation_closure_digest

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
canonical_name	yes	const="promin"
core_bundle_digest	yes	\$ref="#/\$defs/Digest"
implementation_closure_digest	yes	\$ref="#/\$defs/Digest"
init_file_digests	yes	\$ref="#/\$defs/InitFileDigests"
operating_profile	yes	\$ref="#/\$defs/Id"
preset_digest	yes	\$ref="#/\$defs/Digest"
proofs	yes	type="array"; minItems=1; maxItems=32
record_type	yes	const="Activation"
trust_mode	yes	enum=["local-owner", "team-signed"]

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "canonical_name": {
      "const": "promin"
    },
    "core_bundle_digest": {
      "$ref": "#/$defs/Digest"
    },
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "init_file_digests": {
      "$ref": "#/$defs/InitFileDigests"
    },
    "operating_profile": {
      "$ref": "#/$defs/Id"
    },
    "preset_digest": {
      "$ref": "#/$defs/Digest"
    },
    "proofs": {
      "items": {
        "$ref": "#/$defs/ActivationProof"
      },
      "maxItems": 32,
      "minItems": 1,

```

```

    "type": "array"
  },
  "record_type": {
    "const": "Activation"
  },
  "trust_mode": {
    "enum": [
      "local-owner",
      "team-signed"
    ]
  }
},
"required": [
  "record_type",
  "canonical_name",
  "core_bundle_digest",
  "preset_digest",
  "init_file_digests",
  "operating_profile",
  "trust_mode",
  "activation_digest",
  "proofs",
  "implementation_closure_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Activation | Validator: urn:promin:contracts:1#/\$defs/Activation

3. ActivationProof

Type summary: oneOf[2]=#/\$defs/LocalActivationProof | #/\$defs/SignatureActivationProof

Required: -

```

{
  "oneOf": [
    {
      "$ref": "#/$defs/LocalActivationProof"
    },
    {
      "$ref": "#/$defs/SignatureActivationProof"
    }
  ]
}

```

Owner: core/contracts.schema.json#/\$defs/ActivationProof | Validator: urn:promin:contracts:1#/\$defs/ActivationProof

4. Artifact

Type summary: type="object"; additionalProperties=false; allOf[3]=keys:else,if,then,title | keys:if,then,title | keys:else,if,then,title

Required: record_type, artifact_id, artifact_kind, digest, media_type, size_bytes, retention_class, created_at

Property	Required	Structural rule
artifact_id	yes	\$ref="#/\$defs/Id"
artifact_kind	yes	enum=["product", "evidence", "log", "report", "diff"]
candidate_delta	no	\$ref="#/\$defs/CandidateDelta"
created_at	yes	\$ref="#/\$defs/Timestamp"
digest	yes	\$ref="#/\$defs/Digest"
evidence_binding	no	\$ref="#/\$defs/EvidenceBinding"
evidence_class	no	enum=["product-execution", "validator", "harness-generated", "migration"]
evidence_purpose	no	enum=["diagnostic", "gate", "product"]
media_type	yes	type="string"; minLength=1
outcome	no	enum=["pass", "fail", "blocked", "skipped", "error"]
product_credit_eligible	no	type="boolean"
record_type	yes	const="Artifact"
retention_class	yes	enum=["ephemeral", "project", "release", "audit"]
size_bytes	yes	type="integer"; minimum=0
stale	no	type="boolean"

Property	Required	Structural rule
unresolved	no	type="boolean"
<pre> { "additionalProperties": false, "allOf": [{ "else": { "not": { "anyOf": [{ "required": ["evidence_binding"] }, { "required": ["evidence_purpose"] }, { "required": ["outcome"] }, { "required": ["stale"] }], "required": ["unresolved"] }, "required": ["evidence_class"] }, "required": ["product_credit_eligible"] }], "if": { "properties": { "artifact_kind": { "const": "evidence" } }, "required": ["artifact_kind"] }, "then": { "required": ["evidence_binding", "evidence_purpose", "outcome", "stale", "unresolved", "evidence_class", "product_credit_eligible"], "title": "require replay-complete immutable evidence metadata" }, { "if": { "properties": { "product_credit_eligible": { "const": true } } } } } </pre>		

```

    },
    "required": [
      "product_credit_eligible"
    ]
  },
  "then": {
    "properties": {
      "artifact_kind": {
        "const": "evidence"
      },
      "evidence_binding": {
        "required": [
          "provider_invocations"
        ]
      },
      "evidence_class": {
        "const": "product-execution"
      },
      "evidence_purpose": {
        "const": "product"
      },
      "outcome": {
        "const": "pass"
      },
      "stale": {
        "const": false
      },
      "unresolved": {
        "const": false
      }
    },
    "required": [
      "evidence_binding",
      "evidence_purpose",
      "outcome",
      "stale",
      "unresolved",
      "evidence_class",
      "product_credit_eligible"
    ]
  },
  "title": "product credit requires fresh resolved passing product execution evidence"
},
{
  "else": {
    "not": {
      "required": [
        "candidate_delta"
      ]
    },
    "if": {
      "properties": {
        "artifact_kind": {
          "const": "diff"
        },
        "required": [
          "artifact_kind"
        ]
      },
      "then": {
        "required": [
          "candidate_delta"
        ]
      }
    }
  },
  "title": "diff Artifact is the canonical Candidate delta change manifest"
},
{
  "artifact_id": {
    "$ref": "#/$defs/Id"
  },
  "artifact_kind": {
    "enum": [
      "product",
      "evidence",

```

```

    "log",
    "report",
    "diff"
  ]
},
"candidate_delta": {
  "$ref": "#/$defs/CandidateDelta"
},
"created_at": {
  "$ref": "#/$defs/Timestamp"
},
"digest": {
  "$ref": "#/$defs/Digest"
},
"evidence_binding": {
  "$ref": "#/$defs/EvidenceBinding"
},

```

```

"evidence_class": {
  "enum": [
    "product-execution",
    "validator",
    "harness-generated",
    "migration"
  ]
},
"evidence_purpose": {
  "enum": [
    "diagnostic",
    "gate",
    "product"
  ]
},
"media_type": {
  "minLength": 1,
  "type": "string"
},
"outcome": {
  "enum": [
    "pass",
    "fail",
    "blocked",
    "skipped",
    "error"
  ]
},

```

```

"product_credit_eligible": {
  "type": "boolean"
},
"record_type": {
  "const": "Artifact"
},
"retention_class": {
  "enum": [
    "ephemeral",
    "project",
    "release",
    "audit"
  ]
},
"size_bytes": {
  "minimum": 0,
  "type": "integer"
},
"stale": {
  "type": "boolean"
},
"unresolved": {
  "type": "boolean"
}
},
"required": [
  "record_type",
  "artifact_id",

```

```

"artifact_kind",
"digest",
"media_type",
"size_bytes",

```

```

    "retention_class",
    "created_at"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Artifact | Validator: urn:promin:contracts:1#/\$defs/Artifact

5. AuthorityInit

Type summary: type="object"; additionalProperties=false; allOf[1]=keys;else;if,then

Required: record_type, trust_mode, subjects, roots

Property	Required	Structural rule
keys	no	type="array"; minItems=1; maxItems=64
record_type	yes	const="AuthorityInit"
roots	yes	type="array"; minItems=1; maxItems=16
subjects	yes	type="array"; minItems=1; maxItems=64
team_policy	no	\$ref="#/\$defs/TeamPolicy"
trust_mode	yes	enum=["local-owner", "team-signed"]

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "allOf": [
          {
            "not": {
              "required": [
                "team_policy"
              ]
            }
          },
          {
            "not": {
              "required": [
                "keys"
              ]
            }
          }
        ]
      },
      "if": {
        "properties": {
          "trust_mode": {
            "const": "team-signed"
          }
        }
      }
    },
    {
      "then": {
        "required": [
          "team_policy",
          "keys"
        ]
      }
    }
  ],
  "properties": {
    "keys": {
      "items": {
        "$ref": "#/$defs/TrustKey"
      },
      "maxItems": 64,
      "minItems": 1,
      "type": "array"
    },
    "record_type": {
      "const": "AuthorityInit"
    },
    "roots": {
      "items": {
        "$ref": "#/$defs/RootAuthority"
      },
      "maxItems": 16,

```

```

    "minItems": 1,
    "type": "array"
  },
  "subjects": {
    "items": {
      "$ref": "#/$defs/SubjectInit"
    },
    "maxItems": 64,
    "minItems": 1,
    "type": "array"
  },
  "team_policy": {
    "$ref": "#/$defs/TeamPolicy"
  },
  "trust_mode": {
    "enum": [
      "local-owner",
      "team-signed"
    ]
  }
},
"required": [
  "record_type",
  "trust_mode",
  "subjects",
  "roots"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/AuthorityInit | Validator: urn:promin:contracts:1#/\$defs/AuthorityInit

6. AuthorityModel

Type summary: type="object"; additionalProperties=false

Required: record_type, schema_ref, document, version, function, default, capabilities, command_capability_rules, command_effect_scope_rules, bootstrap_rule, trust_modes, role_presets_non_authoritative, separation_of_duties, model_tier_rule, command_mutation_claim_rule, continuation_access_rule, standard_release_decision_rule, release_evidence_authority_rule, canonical_timestamp_contract, delegation_depth_max, derived_event_store_policy_contract, derived_runtime_policy_contract, event_contract, grant_contract, scope_contract, separation_of_duties_contract

Property	Required	Structural rule
bootstrap_rule	yes	const="authority.json configures root ceilings and Activation only. Root command proofs bind one command intent and may issue only the first authority.manage Grant. Every Grant proof binds claim_digest; later Grants cite an active authority.manage issuer Grant or team signatures verified by the exact signature adapter selected in technologies.json."
canonical_timestamp_contract	yes	const=schema(keys:calendar,format,formatter_api,fractional_seconds,normalization,offset,parser_api,pattern,real_calendar_validation,resolution_seconds,year_max,year_min)
capabilities	yes	const=[16 items]
command_capability_rules	yes	const=[11 items]
command_effect_scope_rules	yes	const=[11 items]
command_mutation_claim_rule	yes	type="object"; additionalProperties=false
continuation_access_rule	yes	const=schema(keys:authority_effect,authorization_rule,capability_id,default_budget_source,default_depth_source,default_ttl_seconds,depth_max_source,depth_min,hard_budget_source,max_continuation_state_bytes_source,max_query_bytes_source,max_resume_binding_bytes,max_resume_binding_fields,max_resume_binding_key_chars,max_resume_binding_value_bytes,min_result_bytes,persistent_entity_types_source,ranking_algorithm_id,required_claims,required_resume_binding_fields,revocation_rule,secret_lifecycle,selected_profile_digest_rule,selected_profile_id_source,state_digest_rule,state_persistence,token_envelope,token_max_bytes_source,token_version,traversal_algorithm_id,ttl_min_seconds,ttl_seconds_max)
default	yes	const="deny"
delegation_depth_max	yes	const=8
derived_event_store_policy_contract	yes	const=schema(keys:allowed_state_binding_leaf_types_source,command_conditional_fields_source,command_mutation_fields_source,event_contract_source,lease_bound_command_kinds_source,lease_bound_task_transition_states_source,persistence,policy_digest_rule,required_bindings,scope_contract_source,timestamp_contract_source)

Property	Required	Structural rule
derived_runtime_policy_contract	yes	const=schema(keys:capability_ids_source,delegation_depth_source,grant_contract_source,persistence,policy_digest_rule,required_bindings,scope_contract_source,separation_of_duties_capability_ids_source,separation_of_duties_contract_source,separation_of_duties_source,timestamp_contract_source)
document	yes	const="promin.authority-model"
event_contract	yes	const=schema(keys:auxiliary_event_kinds,command_bytes_max,command_conditional_fields,command_required_fields,command_to_primary_event,derived_tail_batch_threshold,derived_tail_byte_threshold,-envelope_bytes_max,event_batch_commitment_rule,event_batch_required_fields,event_required_fields,events_per_batch_max,genesis_event_semantic_digest,genesis_previous_authority_commitment,lease_bound_command_kinds_source,lease_bound_task_transition_states_source,primary_event_count_per_command,requested_scope_items_max_source,runtime_overlay_compaction_depth,state_binding_algorithm,state_binding_algorithm_contract,state_binding_bytes_max,state_binding_identity_rule,state_binding_identity_rules,state_binding_leaf_type_source,state_binding_updates_per_batch_max,state_binding_value_rules,timestamp_contract_source)
function	yes	const="Own action capabilities, runtime Grant semantics, trust modes, configured signature-adapter verification, root ceilings, and separation-of-duty constraints."
grant_contract	yes	const=schema(keys:claim_digest_rule,nonce,required_fields,scope_items_max_source,trust_proof_kinds,trust_proofs_max,trust_proofs_min,trust_proofs_unique,unknown_proof_kind)
model_tier_rule	yes	const="Model strength changes decomposition, context, parallelism, and analysis depth; it never changes authority."
record_type	yes	const="AuthorityModel"
release_evidence_authority_rule	yes	const=schema(keys:attestation_algorithm,attestation_domain,decision_capability,evidence_capability,evidence_key_scope,required_claim_bindings,required_evidence_role_count,role_subject_key_uniqueness,separation_rule,unsigned_or_replayed)
role_presets_non_authoritative	yes	const=schema(keys:auditor,root-owner,validator,worker)
schema_ref	yes	const="urn:promin:contracts:1#/\$defs/AuthorityModel"
scope_contract	yes	const=schema(keys:all_selector,conjunctive_effect_scope,duplicate_selector,grant_scope_items_max,kinds,path_selector,project_selector,requested_scope_items_max,selector_fields,unique_items,unknown_kind,value_max_length,value_min_length,wildcard_outside_all)
separation_of_duties	yes	const=[5 items]
separation_of_duties_contract	yes	const=schema(keys:base_fields,capability_values_max,external_capability_ids,rules_max,supported_rule_fields,unknown_rule_field)
standard_release_decision_rule	yes	const=schema(keys:chain,chronology,decision_outcomes,derived_truths,evidence_binding,history_and_current_state,independent_trust_binding,location,product_acceptance_pass,product_public_approval,publication_rule,rule,signature_algorithm,signature_provider_id,standard_name,version_source)
trust_modes	yes	const=schema(keys:local-owner,team-signed)
version	yes	const="1.0.0-alpha.3"
<pre> { "additionalProperties": false, "properties": { "bootstrap_rule": { "const": "authority.json configures root ceilings and Activation only. Root command proofs bind one command intent and may issue only the first authority.manage Grant. Every Grant proof binds claim_digest; later Grants cite an active authority.manage issuer Grant or team signatures verified by the exact signature adapter selected in technologies.json." }, "canonical_timestamp_contract": { "const": { "calendar": "proleptic-gregorian", "format": "utc-second-z", "formatter_api": "promin.canonical.format_utc_second(value: datetime) -> str", "fractional_seconds": "reject", "normalization": "identity-round-trip", "offset": "Z-only", "parser_api": "promin.canonical.parse_utc_second(value: str) -> timezone-aware UTC datetime", "pattern": "^[0-9]{4}-[0-9]{2}-[0-9]{2}T[0-9]{2}: [0-9]{2}: [0-9]{2}Z\$", "real_calendar_validation": true, "resolution_seconds": 1, "year_max": 9999, "year_min": 1 } } }, }</pre>		


```

"capabilities": {
  "const": [
    {
      "id": "standard.activate",
      "purpose": "Activate one digest-bound repository configuration."
    },
    {
      "id": "authority.manage",
      "purpose": "Issue or revoke runtime Grants within a configured root ceiling."
    },
    {
      "id": "task.plan",
      "purpose": "Create, split, order, or block Tasks."
    },
    {
      "id": "task.execute",
      "purpose": "Execute one Task inside its WorkCard and valid Lease."
    },
    {
      "id": "lease.manage",
      "purpose": "Acquire, heartbeat, close, expire, or revoke Leases."
    },
    {
      "id": "evidence.publish",
      "purpose": "Publish candidate-bound evidence Artifacts."
    },
    {
      "id": "validation.evaluate",
      "purpose": "Evaluate gates without promotion or release authority."
    },
    {
      "id": "finding.record",
      "purpose": "Record Findings."
    },
    {
      "id": "finding.resolve",
      "purpose": "Resolve one Finding only with evidence bound to that Finding's canonical digest."
    },
    {
      "id": "finding.waive",
      "purpose": "Waive one Finding only through an expiring exception-policy Decision, Finding-bound
evidence, and separation of duties."
    },
    {
      "id": "candidate.promote",
      "purpose": "Promote a validated Candidate."
    },
    {
      "id": "release.decide",
      "purpose": "Record an immutable historical release judgement bound to an exact
acceptance-closure and provider-binding digest; current eligibility remains derived."
    },
    {
      "id": "export.create",
      "purpose": "Create a sanitized external package."
    },
    {
      "id": "migration.manage",
      "purpose": "Execute cutover, rollback, and deterministic recutover."
    },
    {
      "id": "projection.rebuild",
      "purpose": "Rebuild or verify non-authoritative projections."
    },
    {
      "id": "projection.read",
      "purpose": "Read one bounded current projection or resume its continuation under the exact
current Grant claim; it grants no mutation, validation, promotion, product acceptance, or standard
distribution authority."
    }
  ]
},
"command_capability_rules": {
  "const": [
    {
      "capability_id": "task.plan",
      "command_kind": "task.record"
    }
  ]
}

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```

    },
    {
      "capability_id": "task.plan",
      "command_kind": "task.transition"
    },
    {
      "capability_id": "task.execute",
      "command_kind": "candidate.record"
    },
    {
      "capability_by_value": {
        "diff": "task.execute",
        "evidence": "evidence.publish",
        "log": "task.execute",
        "product": "task.execute",
        "report": "projection.rebuild"
      }
    },
    {
      "command_kind": "artifact.record",
      "payload_field": "artifact_kind"
    },
    {
      "capability_by_value": {
        "execution": "task.execute",
        "validation": "validation.evaluate"
      },
      "command_kind": "run.record",
      "payload_field": "run_kind"
    },
    {
      "capability_id": "validation.evaluate",
      "command_kind": "gate.record"
    },
    {
      "capability_id": "finding.record",
      "command_kind": "finding.record"
    },
    {
      "capability_by_value": {
        "accept": "validation.evaluate",
        "promote": "candidate.promote",
        "reject": "validation.evaluate",
        "release": "release.decide",
        "resolve": "finding.resolve",
        "revoke": "authority.manage",
        "waive": "finding.waive"
      }
    },
    {
      "command_kind": "decision.record",
      "payload_field": "decision_kind"
    },
    {
      "capability_id": "authority.manage",
      "command_kind": "grant.issue"
    },
    {
      "capability_id": "authority.manage",
      "command_kind": "grant.revoke"
    },
    {
      "capability_id": "lease.manage",
      "command_kind": "lease.record"
    }
  ]
},
"command_effect_scope_rules": {
  "const": [
    {
      "command_kind": "task.record",
      "id_field": "task_id",
      "kind": "task",
      "mode": "payload-id"
    },
    {
      "command_kind": "task.transition",
      "id_field": "task_id",
      "kind": "task",
      "mode": "payload-id"
    }
  ],

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{
  "command_kind": "candidate.record",
  "id_field": "candidate_id",
  "kind": "candidate",
  "mode": "payload-id"
},
{
  "command_kind": "artifact.record",
  "id_field": "artifact_id",
  "kind": "artifact",
  "mode": "payload-id"
},
{
  "command_kind": "run.record",
  "mode": "payload-multi-id",
  "targets": [
    {
      "id_field": "task_id",
      "kind": "task"
    },
    {
      "id_field": "candidate_digest",
      "kind": "candidate"
    }
  ]
},
{
  "command_kind": "gate.record",
  "id_field": "candidate_digest",
  "kind": "candidate",
  "mode": "payload-id"
},
{
  "command_kind": "finding.record",
  "id_field": "finding_id",
  "kind": "finding",
  "mode": "payload-id"
},
{
  "command_kind": "decision.record",
  "grant_target_type": "Grant",
  "id_field": "target_id",
  "kind_field": "target_type",
  "kind_map": {
    "Candidate": "candidate",
    "Finding": "finding",
    "Task": "task"
  },
  "mode": "payload-dynamic-id-or-referenced-grant-scope"
},
{
  "command_kind": "grant.issue",
  "mode": "payload-scope",
  "scope_field": "scope"
},
{
  "command_kind": "grant.revoke",
  "grant_id_field": "grant_id",
  "mode": "referenced-grant-scope"
},
{
  "command_kind": "lease.record",
  "id_field": "task_id",
  "kind": "task",
  "mode": "payload-id"
}
]
},
"command_mutation_claim_rule": {
  "additionalProperties": false,
  "properties": {
    "binding_equalities": {
      "items": {
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 16,
      "minItems": 8,

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    "type": "array",
    "uniqueItems": true
  },
  "claim_rule": {
    "minLength": 1,
    "type": "string"
  },
  "command_authorization_source": {
    "const": "CommandRequest.authorization.grant_id + grant_claim_digest"
  },
  "grant_separation_rule": {
    "minLength": 1,
    "type": "string"
  },
  "holder_grant_capability": {
    "const": "task.execute"
  },
  "holder_grant_field": {
    "const": "holder_authorization.grant_id"
  },
  "lease_bound_command_kinds": {
    "items": {
      "enum": [
        "task.record",
        "task.transition",
        "candidate.record",
        "artifact.record",
        "run.record",
        "gate.record",
        "finding.record",
        "decision.record",
        "grant.issue",
        "grant.revoke",
        "lease.record"
      ]
    },
    "maxItems": 11,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "lease_bound_task_transition_states": {
    "items": {
      "enum": [
        "BLOCKED",
        "CANCELLED",
        "COMPLETED",
        "LEASED",
        "PLANNED",
        "READY",
        "RUNNING"
      ]
    },
    "maxItems": 7,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "lease_required_state": {
    "const": "ACTIVE"
  },
  "required_command_fields": {
    "const": [
      "workcard_task_id",
      "holder_authorization",
      "lease_id",
      "lease_generation",
      "fencing_token",
      "workcard_digest",
      "context_digest"
    ],
    "type": "array"
  },
  "required_workcard_fields": {
    "const": [
      "holder_grant_id",
      "lease_id",

```

<pre> "lease_generation", "fencing_token", "context_digest"], "type": "array" }, "stale_or_unresolved": { "const": "reject" } } </pre>	
<pre> }, "required": ["binding_equalities", "claim_rule", "command_authorization_source", "grant_separation_rule", "holder_grant_capability", "holder_grant_field", "lease_bound_command_kinds", "lease_bound_task_transition_states", "lease_required_state", "required_command_fields", "required_workcard_fields", "stale_or_unresolved"], "type": "object" }, "continuation_access_rule": { "const": { "authority_effect": false, "authorization_rule": "Initial retrieval and every continuation resume require a current projection.read Grant whose subject, grant_id, claim_digest, capability, Activation, conjunctive requested scope, interval, and revocation state match the token claims exactly.", "capability_id": "projection.read", "default_budget_source": "selected Preset profile max_context_bytes, max_entities, max_relations, max_fanout_per_entity, and top_k", "default_depth_source": "selected Preset profile default_dependency_depth", "default_ttl_seconds": 900, "depth_max_source": "Conformance.dependency_depth_hard_max", "depth_min": 1, "hard_budget_source": "Conformance.workcard_hard_ceiling", </pre>	
<pre> "max_continuation_state_bytes_source": "Conformance.scale_contracts.workcard.continuation_state_byt es_max", "max_query_bytes_source": "PolicySet.derived_result_contracts.RetrievalPage.query_bytes_max", "max_resume_binding_bytes": 4096, "max_resume_binding_fields": 8, "max_resume_binding_key_chars": 64, "max_resume_binding_value_bytes": 512, "min_result_bytes": 1024, "persistent_entity_types_source": "SemanticModel.persistent_entities.kind-in-owner-order", "ranking_algorithm_id": "bm25-v1", "required_claims": ["activation_digest", "subject_id", "grant_id", "grant_claim_digest", "capability_id", "requested_scope", "normalized_query_digest", "cursor", "head_event_id", "head_sequence", "head_digest", "projection_digest", "ranking_id", "traversal_id", "dependency_depth", "budgets", "issued_at", "expires_at", </pre>	
<pre> "ttl_seconds", "revocation_epoch", "implementation_closure_digest", "state_id"], "required_resume_binding_fields": ["activation_digest", "capability_id", </pre>	

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    "grant_claim_digest",
    "grant_id",
    "implementation_closure_digest",
    "requested_scope_digest",
    "revocation_epoch",
    "subject_id"
  ],
  "revocation_rule": "Expired, revoked, substituted, stale, mismatched, or unresolved Grant claims
reject before projection continuation.",
  "secret_lifecycle": "Generate one random 32-byte secret per Activation, store it only under .
promin/state/secrets with restrictive permissions, never derive it from public package or control digests,
never journal or export it, and rotate it with Activation.",
  "selected_profile_digest_rule": "sha256(canonical exact selected Preset profile object)",
  "selected_profile_id_source": "Activation.operating_profile",
  "state_digest_rule": "sha256(canonical full ContinuationTokenClaims); the compact token MAC binds
token version, opaque state_id, state digest, Activation, subject, Grant claim, and expiry",
  "state_persistence": "derived-live-only",
  "token_envelope": "promin-v2.<base64url opaque 128-bit state handle>.<lowercase state
digest>.<base64url HMAC-SHA256>",
  "token_max_bytes_source": "Conformance.scale_contracts.workcard.continuation_token_bytes_max",
  "token_version": 2,
  "traversal_algorithm_id": "typed-bfs-v2",
  "ttl_min_seconds": 1,
  "ttl_seconds_max": 3600
}

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},
"default": {
  "const": "deny"
},
"delegation_depth_max": {
  "const": 8
},
"derived_event_store_policy_contract": {
  "const": {
    "allowed_state_binding_leaf_types_source": "Activation plus SemanticModel.persistent_entities.kind
plus Relation",
    "command_conditional_fields_source": "event_contract.command_conditional_fields",
    "command_mutation_fields_source": "command_mutation_claim_rule.required_command_fields",
    "event_contract_source": "event_contract",
    "lease_bound_command_kinds_source": "command_mutation_claim_rule.lease_bound_command_kinds",
    "lease_bound_task_transition_states_source": "command_mutation_claim_rule.lease_bound_task_transiti
on_states",
    "persistence": "derived-live-only",
    "policy_digest_rule": "sha256(canonical EventStorePolicy without policy_digest)",
    "required_bindings": [
      "authority_model_digest",
      "max_command_bytes",
      "max_envelope_bytes",
      "max_state_binding_bytes",
      "max_state_binding_updates_per_batch",
      "max_events_per_batch",
      "max_requested_scope_items",
      "derived_tail_batch_threshold",
      "derived_tail_byte_threshold",
      "runtime_overlay_compaction_depth",

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      "command_required_fields",
      "command_conditional_fields",
      "command_mutation_fields",
      "lease_bound_command_kinds",
      "lease_bound_task_transition_states",
      "command_to_primary_event",
      "allowed_state_binding_leaf_types",
      "state_binding_identity_rules",
      "state_binding_algorithm_contract",
      "state_binding_value_rules",
      "canonical_timestamp_contract",
      "genesis_previous_authority_commitment",
      "genesis_event_semantic_digest",
      "policy_digest"
    ],
    "scope_contract_source": "scope_contract",
    "timestamp_contract_source": "canonical_timestamp_contract"
  }
},
"derived_runtime_policy_contract": {
  "const": {
    "capability_ids_source": "capabilities[].id-in-owner-order",
    "delegation_depth_source": "delegation_depth_max",

```

```

"grant_contract_source": "grant_contract",
"persistence": "derived-live-only",
"policy_digest_rule": "sha256(canonical AuthorityRuntimePolicy without policy_digest)",
"required_bindings": [
  "authority_model_digest",

  "capability_ids",
  "separation_of_duties",
  "separation_of_duties_capability_ids",
  "separation_of_duties_contract",
  "delegation_depth_max",
  "scope_contract",
  "grant_contract",
  "canonical_timestamp_contract",
  "policy_digest"
],
"scope_contract_source": "scope_contract",
"separation_of_duties_capability_ids_source": "capabilities[].id plus
separation_of_duties_contract.external_capability_ids",
"separation_of_duties_contract_source": "separation_of_duties_contract",
"separation_of_duties_source": "separation_of_duties-in-owner-order",
"timestamp_contract_source": "canonical_timestamp_contract"
}
},
"document": {
  "const": "promin.authority-model"
},
"event_contract": {
  "const": {
    "auxiliary_event_kinds": [
      "relation.recorded"
    ],
    "command_bytes_max": 1048576,
    "command_conditional_fields": [
      "definition_digest"
    ]
  },
  "command_required_fields": [
    "record_type",
    "command_id",
    "command_kind",
    "subject_id",
    "activation_digest",
    "idempotency_key",
    "requested_scope",
    "expected_head_digest",
    "issued_at",
    "payload",
    "intent_digest",
    "authorization"
  ],
  "command_to_primary_event": {
    "artifact.record": "artifact.recorded",
    "candidate.record": "candidate.recorded",
    "decision.record": "decision.recorded",
    "finding.record": "finding.recorded",
    "gate.record": "gate.recorded",
    "grant.issue": "grant.issued",
    "grant.revoke": "grant.revoked",
    "lease.record": "lease.recorded",
    "run.record": "run.recorded",
    "task.record": "task.recorded",
    "task.transition": "task.transitioned"
  }
},

"derived_tail_batch_threshold": 128,
"derived_tail_byte_threshold": 8388608,
"envelope_bytes_max": 2097152,
"event_batch_commitment_rule": {
  "authority_commitment": "sha256(canonical object with exactly previous_authority_commitment,
sequence, previous_batch_digest, event_count, event_semantic_digest, state_binding_update_count,
state_binding_digest)",
  "cumulative_event_count": "previous cumulative_event_count plus current events length",
  "cumulative_state_binding_update_count": "previous cumulative state-binding update count plus
current state_binding_delta length",
  "event_semantic_digest": "ordered append-only digest over every canonical Event",
  "previous_authority_commitment": "exact previous EventBatch authority_commitment; sequence one
uses genesis_previous_authority_commitment",
  "state_binding_delta": "nonempty canonical ordered unique set/delete updates over changed typed
authority leaves only",

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    "state_binding_digest": "exact typed-sparse-merkle-v1 post-batch authority root using runtime
domain separators for leaf and internal nodes; never a full serialized state hash"
  },
  "event_batch_required_fields": [
    "record_type",
    "batch_id",
    "sequence",
    "previous_digest",
    "previous_authority_commitment",
    "authority_commitment",
    "cumulative_event_count",
    "event_semantic_digest",
    "state_binding_delta",
    "cumulative_state_binding_update_count",
    "state_binding_digest",
    "created_at",
    "command_id",
    "idempotency_key",
    "activation_record_digest",
    "events",
    "subject_id",
    "command_intent_digest",
    "command_digest",
    "authorization_digest"
  ],
  "event_required_fields": [
    "record_type",
    "event_id",
    "event_kind",
    "activation_digest",
    "payload"
  ],
  "events_per_batch_max": 20,
  "genesis_event_semantic_digest": "e3b0c44298fclc149afb4c8996fb92427ae41e4649b934ca495991b7852b855",
  "genesis_previous_authority_commitment": "e3b0c44298fclc149afb4c8996fb92427ae41e4649b934ca495991b7
852b855",
  "lease_bound_command_kinds_source": "command_mutation_claim_rule.lease_bound_command_kinds",
  "lease_bound_task_transition_states_source": "command_mutation_claim_rule.lease_bound_task_transiti
on_states",
  "primary_event_count_per_command": 1,
  "requested_scope_items_max_source": "scope_contract.requested_scope_items_max",
  "runtime_overlay_compaction_depth": 64,
  "state_binding_algorithm": "typed-sparse-merkle-v1",
  "state_binding_algorithm_contract": {
    "delete_rule": "delete maps the keyed leaf to empty_leaf",
    "delta_order": [
      "leaf_type-ascending",
      "leaf_id-ascending"
    ]
  },
  "depth": 256,
  "depth_encoding": "unsigned-16-bit-big-endian",
  "empty_leaf_domain_utf8": "promin:typed-sparse-merkle-v1:empty-leaf",
  "internal_node_domain_utf8": "promin:typed-sparse-merkle-v1:node\u0000",
  "internal_node_rule": "sha256(domain + depth_uint16_be + left_32_bytes + right_32_bytes)",
  "leaf_domain_utf8": "promin:typed-sparse-merkle-v1:leaf\u0000",
  "leaf_key_rule": "sha256(canonical bytes of exact object leaf_type and leaf_id)",
  "set_leaf_rule": "sha256(domain + leaf_key_32_bytes + value_digest_32_bytes)"
},
"state_binding_bytes_max": 1048576,
"state_binding_identity_rule": "direct uses the exact id field; typed_digest_id uses prefix plus
first 48 lowercase hex characters of sha256(canonical ordered id_fields array)",
"state_binding_identity_rules": {
  "Activation": {
    "id_fields": [
      "activation_digest"
    ],
    "identity_mode": "direct"
  },
  "Artifact": {
    "id_fields": [
      "artifact_id"
    ],
    "identity_mode": "direct"
  },
  "Candidate": {
    "id_fields": [

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	<pre> "candidate_id"], </pre>
	<pre> "identity_mode": "direct" }, "Decision": { "id_fields": ["decision_id"], "identity_mode": "direct" }, "Finding": { "id_fields": ["finding_id"], "identity_mode": "direct" }, "GateResult": { "digest_algorithm": "sha256", "digest_hex_chars": 48, "digest_input": "canonical-ordered-id-field-values", "id_fields": ["gate_id", "run_id"], "identity_mode": "typed_digest_id", "prefix": "gate-result:" }, "Grant": { "id_fields": ["grant_id" </pre>
	<pre>], "identity_mode": "direct" }, "Lease": { "id_fields": ["lease_id"], "identity_mode": "direct" }, "Relation": { "id_fields": ["relation_id"], "identity_mode": "direct" }, "Run": { "id_fields": ["run_id"], "identity_mode": "direct" }, "Task": { "id_fields": ["task_id"], "identity_mode": "direct" } }, </pre>
<pre> Relation", </pre>	<pre> "state_binding_leaf_type_source": "Activation plus SemanticModel.persistent_entities.kind plus "state_binding_updates_per_batch_max": 20, "state_binding_value_rules": { "Activation": { "event_kinds": [], "value_definition": "Activation", "value_digest_rule": "sha256(canonical installed Activation record); installed before sequence one and immutable for the journal" }, "Artifact": { "event_kinds": ["artifact.recorded"], "value_definition": "Artifact", "value_digest_rule": "sha256(canonical exact Event.payload Artifact)" }, "Candidate": { "event_kinds": [</pre>

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        "candidate.recorded"
    ],
    "value_definition": "Candidate",
    "value_digest_rule": "sha256(canonical exact Event.payload Candidate)"
},
"Decision": {
    "event_kinds": [
        "decision.recorded"
    ],
    "value_definition": "Decision",
    "value_digest_rule": "sha256(canonical exact Event.payload Decision)"
},
"Finding": {
    "event_kinds": [
        "finding.recorded"
    ],
    "value_definition": "Finding",
    "value_digest_rule": "sha256(canonical exact Event.payload Finding)"
},
"GateResult": {
    "event_kinds": [
        "gate.recorded"
    ],
    "value_definition": "GateResult",
    "value_digest_rule": "sha256(canonical exact Event.payload GateResult)"
},
"Grant": {
    "event_kinds": [
        "grant.issued",
        "grant.revoked"
    ],
    "value_definition": "GrantAuthorityState",
    "value_digest_rule": "sha256(canonical exact GrantAuthorityState containing the current Grant
and null or exact current GrantRevocation)"
},
"Lease": {
    "event_kinds": [
        "lease.recorded"
    ],
    "value_definition": "Lease",
    "value_digest_rule": "sha256(canonical exact Event.payload post-state Lease)"
},
"Relation": {
    "event_kinds": [
        "relation.recorded"
    ],
    "value_definition": "Relation",
    "value_digest_rule": "sha256(canonical exact Event.payload Relation)"
},
"Run": {
    "event_kinds": [
        "run.recorded"
    ],
    "value_definition": "Run",
    "value_digest_rule": "sha256(canonical exact Event.payload Run)"
},
"Task": {
    "event_kinds": [
        "task.recorded",
        "task.transitioned"
    ],
    "value_definition": "Task",
    "value_digest_rule": "sha256(canonical exact Event.payload post-state Task)"
}
},
"timestamp_contract_source": "canonical_timestamp_contract"
}
},
"function": {
    "const": "Own action capabilities, runtime Grant semantics, trust modes, configured
signature-adaptor verification, root ceilings, and separation-of-duty constraints."
},
"grant_contract": {
    "const": {
        "claim_digest_rule": "sha256(canonical Grant without claim_digest and trust_proofs)",
        "nonce": {
            "max_length": 128,

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    "min_length": 16,
    "pattern": "^[A-Za-z0-9._:-]{16,128}$"
  },
  "required_fields": [
    "record_type",
    "grant_id",
    "subject_id",
    "capability_id",
    "scope",
    "activation_digest",
    "issued_at",
    "expires_at",
    "nonce",
    "claim_digest",
    "trust_proofs"
  ],
  "scope_items_max_source": "scope_contract.grant_scope_items_max",
  "trust_proof_kinds": {
    "issuer-grant": {
      "required_fields": [
        "kind",
        "issuer_grant_id",
        "issuer_signed_claim_digest",
        "signed_claim_digest"
      ]
    },
    "local-root": {
      "required_fields": [
        "kind",
        "root_subject_id",
        "authority_init_digest",
        "signed_claim_digest"
      ]
    },
    "signature": {
      "required_fields": [
        "kind",
        "key_id",
        "algorithm",
        "signed_digest",
        "signature"
      ],
      "signature_max_length": 8192,
      "signature_min_length": 1
    }
  },
  "trust_proofs_max": 32,
  "trust_proofs_min": 1,

  "trust_proofs_unique": true,
  "unknown_proof_kind": "reject"
},
"model_tier_rule": {
  "const": "Model strength changes decomposition, context, parallelism, and analysis depth; it never changes authority."
},
"record_type": {
  "const": "AuthorityModel"
},
"release_evidence_authority_rule": {
  "const": {
    "attestation_algorithm": "Ed25519",
    "attestation_domain": "promin.release-evidence.producer-attestation.v1",
    "decision_capability": "standard.distribute",
    "evidence_capability": "evidence.produce",
    "evidence_key_scope": "Every evidence producer key is independently pinned to an exact subject, evidence role, platform, public key, validity interval, and revocation state.",
    "required_claim_bindings": [
      "candidate_binding_digest",
      "record_type",
      "evidence_role",
      "platform",
      "invocation_id",
      "payload_digest",
      "raw_artifact_manifest_digest"
    ],
    "required_evidence_role_count": 7,
    "role_subject_key_uniqueness": "one-role-one-subject-one-key",

```

```

    "separation_rule": "For one candidate, an evidence-producing subject or key cannot hold
standard.distribute, and a distribution decider subject or key cannot hold evidence.produce.",
    "unsigned_or_replayed": "reject"
  }
},
"role_presets_non_authoritative": {
  "const": {
    "auditor": [
      "projection.read",
      "finding.record",
      "finding.resolve",
      "export.create"
    ],
    "root-owner": [
      "standard.activate",
      "authority.manage",
      "task.plan",
      "lease.manage",
      "finding.resolve",
      "finding.waive",
      "candidate.promote",
      "release.decide",
      "export.create",
      "migration.manage",
      "projection.rebuild",
      "projection.read"
    ],
    "validator": [
      "projection.read",
      "validation.evaluate",
      "evidence.publish",
      "finding.record"
    ],
    "worker": [
      "projection.read",
      "task.execute",
      "evidence.publish"
    ]
  }
},
"schema_ref": {
  "const": "urn:promin:contracts:1#/$defs/AuthorityModel"
},
"scope_contract": {
  "const": {
    "all_selector": {
      "exclusive": true,
      "kind": "all",
      "value": "*"
    },
    "conjunctive_effect_scope": true,
    "duplicate_selector": "reject",
    "grant_scope_items_max": 32,
    "kinds": [
      "project",
      "path",
      "task",
      "candidate",
      "artifact",
      "finding",
      "all"
    ],
    "path_selector": {
      "absolute": "reject",
      "dot_segments": "reject",
      "kind": "path",
      "normalization": "NFC POSIX relative path"
    },
    "project_selector": {
      "items_max": 1,
      "kind": "project"
    },
    "requested_scope_items_max": 32,
    "selector_fields": [
      "kind",
      "value"
    ],
    "unique_items": true,

```

```

    "unknown_kind": "reject",
    "value_max_length": 4096,
    "value_min_length": 1,
    "wildcard_outside_all": "reject"
  }
},
"separation_of_duties": {

```

```

  "const": [
    {
      "forbid_lease_derived_capabilities": [
        "validation.evaluate",
        "finding.resolve",
        "finding.waive",
        "candidate.promote",
        "release.decide"
      ],
      "id": "SOD-001",
      "rule": "lease-is-not-acceptance",
      "scope_kind": "lease"
    },
    {
      "forbid_same_subject_for_same_candidate": [
        "task.execute",
        "release.decide"
      ],
      "id": "SOD-002",
      "rule": "executor-not-release-decider",
      "scope_kind": "candidate"
    },
    {
      "forbid_same_subject_for_same_candidate": [
        "validation.evaluate",
        "candidate.promote"
      ],
      "id": "SOD-003",

```

```

      "rule": "validator-not-self-promoter",
      "scope_kind": "candidate"
    },
    {
      "forbid_same_subject_for_same_finding": [
        "finding.record",
        "finding.waive"
      ],
      "id": "SOD-004",
      "rule": "finder-not-sole-waiver",
      "scope_kind": "finding"
    },
    {
      "forbid_same_key_capabilities": [
        "evidence.produce",
        "standard.distribute"
      ],
      "forbid_same_subject_for_same_candidate": [
        "evidence.produce",
        "standard.distribute"
      ],
      "id": "SOD-005",
      "rule": "evidence-producer-not-distribution-decider",
      "scope_kind": "candidate-and-key"
    }
  ]
},
"separation_of_duties_contract": {

```

```

  "const": {
    "base_fields": [
      "id",
      "rule",
      "scope_kind"
    ],
    "capability_values_max": 32,
    "external_capability_ids": [
      "evidence.produce",
      "standard.distribute"
    ],
    "rules_max": 64,
    "supported_rule_fields": {
      "forbid_lease_derived_capabilities": {

```

```

        "cardinality": "unique-list",
        "items_min": 1
    },
    "forbid_same_key_capabilities": {
        "cardinality": "unique-pair",
        "items_exact": 2
    },
    "forbid_same_subject_for_same_candidate": {
        "cardinality": "unique-pair",
        "items_exact": 2
    },
    "forbid_same_subject_for_same_finding": {
        "cardinality": "unique-pair",
        "items_exact": 2
    }
},
"unknown_rule_field": "reject"
}
},
"standard_release_decision_rule": {
    "const": {
        "chain": "exact candidate bytes -> StandardReleaseCandidateBinding -> seven unique stable-read
authority-role records plus one non-authoritative exact eight-lane matrix and four CP314 supplemental
records -> physically resolved StandardReleaseEvidenceManifest -> signed approve or reject
StandardReleaseDecision -> independently pinned trust root -> derived distribution status",
        "chronology": "Every evidence invocation starts no later than it completes; its attestation time
equals completion and stays within bounded verifier skew. Nested physical runs complete no later than
their outer saturation audit, human review occurs inside its signed invocation, and
max_evidence_completed_at is derived from all seven authority records, all nested physical records, and
all four CP314 supplemental records. decided_at follows that maximum, is no more than 300 seconds beyond
verifier time, and every producer/decision key is valid at its event time.",
        "decision_outcomes": [
            "approve",
            "reject"
        ],
        "derived_truths": "ExactPackageVerification is recomputed by the verifier and multi-platform
implementation closure is derived only from fully authenticated Windows and Linux records; neither is an
independently supplied required role or public command.",
        "evidence_binding": "Every authority, supplemental, matrix, and nested physical record binds exact
path, SHA-256, and byte count and is read through a bounded handle-relative no-follow component walk. The
manifest rejects symlink/junction/reparse/root substitution and NFC/casefold collisions, validates exact
Draft 2020-12 schemas, verifies domain-separated producer attestations, binds producer/tool, role,
platform, invocation, raw artifacts, and exact candidate, and runs role-specific semantic recomputation.
The aggregate matrix remains non-authoritative with pass_credit=false and all acceptance/public flags
false.",
        "history_and_current_state": "The signed StandardReleaseDecision and its digest remain immutable
history. Current distribution eligibility is a derived view and becomes false when evidence, trust pin,
candidate identity, chronology, key validity, or any required gate no longer validates.",
        "independent_trust_binding": "Manifest construction and closure verification require an
independently operator-pinned SHA-256 for the exact trust configuration bytes. Exactly seven evidence
roles have one distinct key, subject, and public key each; CP313 and CP314 observations for one role reuse
only that role's configured producer. A mismatched, missing, duplicate, over-broad, or invalid scope
cannot yield evidence credit or approval.",
        "location": "external-to-canonical-standard-folder",
        "product_acceptance_pass": false,
        "product_public_approval": "not_approved",
        "publication_rule": "Only outcome approve signed by an active configured Ed25519 key with
standard.distribute capability and verified against an independently pinned trust-root SHA-256 yields
approved distribution status. Missing, rejected, stale, unsigned, mismatched, unpinned, fabricated, or
unresolved records fail closed.",
        "rule": "StandardReleaseDecision is external to the canonical standard bytes, binds candidate and
evidence manifest digests, includes explicit outcome and anti-replay nonce, proves decider authority
through a configured trust root and signature provider, and cannot elevate a trust root delivered by the
same untrusted source without an independent digest pin.",
        "signature_algorithm": "Ed25519",
        "signature_provider_id": "cryptography-ed25519-v1",
        "standard_name": "promin",
        "version_source": "StandardReleaseCandidateBinding.version"
    }
},
},
"trust_modes": {
    "const": {
        "local-owner": {
            "boundary": "single-machine owner-controlled root authority",
            "signer_required": false
        },
        "team-signed": {
            "boundary": "multiple actors or remote transport",

```

```

        "minimum_threshold": 1,
        "provider_capability": "signature",
        "provider_resolution": "Resolve the exact technologies.json signature binding for init and every
ActivationGuard, service, CLI, replay, and mutation path. A process-local injected adapter is explicitly
non-reconstructable and valid only while bound to the same project, Activation, provider, and identity;
cross-process CLI workflows require a reconstructable configured adapter.",
        "signer_required": true,
        "unsupported_binding": "reject-before-workflow"
    }
},
"version": {
    "const": "1.0.0-alpha.3"
}
},
"required": [
    "record_type",
    "schema_ref",
    "document",
    "version",
    "function",
    "default",
    "capabilities",
    "command_capability_rules",
    "command_effect_scope_rules",
    "bootstrap_rule",
    "trust_modes",
    "role_presets_non_authoritative",
    "separation_of_duties",
    "model_tier_rule",
    "command_mutation_claim_rule",
    "continuation_access_rule",
    "standard_release_decision_rule",
    "release_evidence_authority_rule",
    "canonical_timestamp_contract",
    "delegation_depth_max",
    "derived_event_store_policy_contract",
    "derived_runtime_policy_contract",
    "event_contract",
    "grant_contract",
    "scope_contract",
    "separation_of_duties_contract"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/AuthorityModel | Validator: urn:promin:contracts:1#/\$defs/AuthorityModel

7. AuthorityRuntimePolicy

Type summary: type="object"; additionalProperties=false

Required: record_type, authority_model_digest, capability_ids, separation_of_duties, separation_of_duties_capability_ids, separation_of_duties_contract, delegation_depth_max, scope_contract, grant_contract, canonical_timestamp_contract, policy_digest

Property	Required	Structural rule
authority_model_digest	yes	\$ref="#/\$defs/Digest"
canonical_timestamp_contract	yes	const=schema(keys:calendar,format,formatter_api,fractional_seconds,normalization,offset,parser_api,pattern,real_calendar_validation,resolution_seconds,year_max,year_min)
capability_ids	yes	const=[16 items]
delegation_depth_max	yes	type="integer"; const=8
grant_contract	yes	const=schema(keys:claim_digest_rule,nonce,required_fields,scope_items_max_source,trust_proof_kinds,trust_proofs_max,trust_proofs_min,trust_proofs_unique,unknown_proof_kind)
policy_digest	yes	\$ref="#/\$defs/Digest"
record_type	yes	const="AuthorityRuntimePolicy"
scope_contract	yes	const=schema(keys:all_selector,conjunctive_effect_scope,duplicate_selector,grant_scope_items_max,kinds,path_selector,project_selector,requested_scope_items_max,selector_fields,unique_items,unknown_kind,value_max_length,value_min_length,wildcard_outside_all)
separation_of_duties	yes	const=[5 items]

Property	Required	Structural rule
separation_of_duties_capability_ids	yes	const=[18 items]
separation_of_duties_contract	yes	const=schema(keys:base_fields,capability_values_max,external_capability_ids,rules_max,supported_rule_fields,unknown_rule_field)
<pre> { "additionalProperties": false, "properties": { "authority_model_digest": { "\$ref": "#/\$defs/Digest" }, "canonical_timestamp_contract": { "const": { "calendar": "proleptic-gregorian", "format": "utc-second-z", "formatter_api": "promin.canonical.format_utc_second(value: datetime) -> str", "fractional_seconds": "reject", "normalization": "identity-round-trip", "offset": "Z-only", "parser_api": "promin.canonical.parse_utc_second(value: str) -> timezone-aware UTC datetime", "pattern": "^[0-9]{4}-[0-9]{2}-[0-9]{2}T[0-9]{2}:[0-9]{2}:[0-9]{2}Z\$", "real_calendar_validation": true, "resolution_seconds": 1, "year_max": 9999, "year_min": 1 } }, "capability_ids": { "const": ["standard.activate", "authority.manage", "task.plan", "task.execute", "lease.manage", "evidence.publish", "validation.evaluate", "finding.record", "finding.resolve", "finding.waive", "candidate.promote", "release.decide", "export.create", "migration.manage", "projection.rebuild", "projection.read"] }, "delegation_depth_max": { "const": 8, "type": "integer" }, "grant_contract": { "const": { "claim_digest_rule": "sha256(canonical Grant without claim_digest and trust_proofs)", "nonce": { "max_length": 128, "min_length": 16, "pattern": "^[A-Za-z0-9._-]{16,128}\$" } }, "required_fields": ["record_type", "grant_id", "subject_id", "capability_id", "scope", "activation_digest", "issued_at", "expires_at", "nonce", "claim_digest", "trust_proofs"], "scope_items_max_source": "scope_contract.grant_scope_items_max", "trust_proof_kinds": { "issuer-grant": { "required_fields": [</pre>		


```

        "kind",
        "issuer_grant_id",
        "issuer_signed_claim_digest",
        "signed_claim_digest"
    ]
},
"local-root": {
    "required_fields": [
        "kind",
        "root_subject_id",
        "authority_init_digest",
        "signed_claim_digest"
    ]
}

```

```

    },
    "signature": {
        "required_fields": [
            "kind",
            "key_id",
            "algorithm",
            "signed_digest",
            "signature"
        ],
        "signature_max_length": 8192,
        "signature_min_length": 1
    }
},
"trust_proofs_max": 32,
"trust_proofs_min": 1,
"trust_proofs_unique": true,
"unknown_proof_kind": "reject"
}
},
"policy_digest": {
    "$ref": "#/$defs/Digest"
},
"record_type": {
    "const": "AuthorityRuntimePolicy"
},
"scope_contract": {
    "const": {
        "all_selector": {

```

```

            "exclusive": true,
            "kind": "all",
            "value": "*"
        },
    },
    "conjunctive_effect_scope": true,
    "duplicate_selector": "reject",
    "grant_scope_items_max": 32,
    "kinds": [
        "project",
        "path",
        "task",
        "candidate",
        "artifact",
        "finding",
        "all"
    ],
    "path_selector": {
        "absolute": "reject",
        "dot_segments": "reject",
        "kind": "path",
        "normalization": "NFC POSIX relative path"
    },
    "project_selector": {
        "items_max": 1,
        "kind": "project"
    },
    "requested_scope_items_max": 32,
    "selector_fields": [

```

```

        "kind",
        "value"
    ],
    "unique_items": true,
    "unknown_kind": "reject",
    "value_max_length": 4096,
    "value_min_length": 1,
    "wildcard_outside_all": "reject"
}

```

```

    }
  },
  "separation_of_duties": {
    "const": [
      {
        "forbid_lease_derived_capabilities": [
          "validation.evaluate",
          "finding.resolve",
          "finding.waive",
          "candidate.promote",
          "release.decide"
        ],
        "id": "SOD-001",
        "rule": "lease-is-not-acceptance",
        "scope_kind": "lease"
      },
      {
        "forbid_same_subject_for_same_candidate": [
          "task.execute",
          "release.decide"
        ],
        "id": "SOD-002",
        "rule": "executor-not-release-decider",
        "scope_kind": "candidate"
      },
      {
        "forbid_same_subject_for_same_candidate": [
          "validation.evaluate",
          "candidate.promote"
        ],
        "id": "SOD-003",
        "rule": "validator-not-self-promoter",
        "scope_kind": "candidate"
      },
      {
        "forbid_same_subject_for_same_finding": [
          "finding.record",
          "finding.waive"
        ],
        "id": "SOD-004",
        "rule": "finder-not-sole-waiver",
        "scope_kind": "finding"
      },
      {
        "forbid_same_key_capabilities": [
          "evidence.produce",
          "standard.distribute"
        ],
        "id": "SOD-005",
        "rule": "evidence-producer-not-distribution-decider",
        "scope_kind": "candidate-and-key"
      }
    ]
  },
  "separation_of_duties_capability_ids": {
    "const": [
      "standard.activate",
      "authority.manage",
      "task.plan",
      "task.execute",
      "lease.manage",
      "evidence.publish",
      "validation.evaluate",
      "finding.record",
      "finding.resolve",
      "finding.waive",
      "candidate.promote",
      "release.decide",
      "export.create",
      "migration.manage",
      "projection.rebuild",
      "projection.read",
    ]
  }
}

```

```

        "evidence.produce",
        "standard.distribute"
    ]
},
"separation_of_duties_contract": {
    "const": {
        "base_fields": [
            "id",
            "rule",
            "scope_kind"
        ],
        "capability_values_max": 32,
        "external_capability_ids": [
            "evidence.produce",
            "standard.distribute"
        ],
        "rules_max": 64,
        "supported_rule_fields": {
            "forbid_lease_derived_capabilities": {
                "cardinality": "unique-list",
                "items_min": 1
            },
            "forbid_same_key_capabilities": {
                "cardinality": "unique-pair",
                "items_exact": 2
            },
            "forbid_same_subject_for_same_candidate": {
                "cardinality": "unique-pair",
                "items_exact": 2
            },
            "forbid_same_subject_for_same_finding": {
                "cardinality": "unique-pair",
                "items_exact": 2
            }
        },
        "unknown_rule_field": "reject"
    }
},
"required": [
    "record_type",
    "authority_model_digest",
    "capability_ids",
    "separation_of_duties",
    "separation_of_duties_capability_ids",
    "separation_of_duties_contract",
    "delegation_depth_max",
    "scope_contract",
    "grant_contract",
    "canonical_timestamp_contract",
    "policy_digest"
],
"type": "object",
"x-promin-persistence": "derived-live-only"
}

```

Owner: *core/contracts.schema.json#/\$defs/AuthorityRuntimePolicy* | Validator: *urn:promin:contracts:1#/\$defs/AuthorityRuntimePolicy*

8. AuthorizationRef

Type summary: *oneOf[2]=#/\$defs/GrantAuthorization | #/\$defs/RootAuthorization*

Required: -

```

{
  "oneOf": [
    {
      "$ref": "#/$defs/GrantAuthorization"
    },
    {
      "$ref": "#/$defs/RootAuthorization"
    }
  ]
}

```

Owner: *core/contracts.schema.json#/\$defs/AuthorizationRef* | Validator: *urn:promin:contracts:1#/\$defs/AuthorizationRef*

9. Budget

Type summary: type="object"; additionalProperties=false

Required: max_bytes, max_entities, max_relations, max_fanout_per_entity, top_k

Property	Required	Structural rule
max_bytes	yes	type="integer"; minimum=1024
max_entities	yes	type="integer"; minimum=1
max_fanout_per_entity	yes	type="integer"; minimum=1
max_relations	yes	type="integer"; minimum=0
top_k	yes	type="integer"; minimum=1
<pre>{ "additionalProperties": false, "properties": { "max_bytes": { "minimum": 1024, "type": "integer" }, "max_entities": { "minimum": 1, "type": "integer" }, "max_fanout_per_entity": { "minimum": 1, "type": "integer" }, "max_relations": { "minimum": 0, "type": "integer" }, "top_k": { "minimum": 1, "type": "integer" } }, "required": ["max_bytes", "max_entities", "max_relations", "max_fanout_per_entity", "top_k"], "type": "object" }</pre>		

Owner: core/contracts.schema.json#/\$defs/Budget | Validator: urn:promin:contracts:1#/\$defs/Budget

10. Candidate

Type summary: type="object"; additionalProperties=false; allOf[2]=keys:else,if,then,title | keys:else,if,then,title

Required: record_type, candidate_id, candidate_digest, inventory_digest, product_root_digest, control_excluded, candidate_recipe_digest, consistency_mode, creditable

Property	Required	Structural rule
baseline_kind	no	enum=["specification", "product", "hybrid"]
candidate_digest	yes	\$ref="#/\$defs/Digest"
candidate_id	yes	\$ref="#/\$defs/Id"
candidate_recipe_digest	yes	\$ref="#/\$defs/Digest"
consistency_mode	yes	enum=["observational-best-effort", "immutable-vcs-tree"]
control_excluded	yes	const=true
creditable	yes	type="boolean"
inventory_digest	yes	\$ref="#/\$defs/Digest"
product_root_digest	yes	\$ref="#/\$defs/Digest"
record_type	yes	const="Candidate"
snapshot_digest	no	\$ref="#/\$defs/Digest"
snapshot_provider_id	no	\$ref="#/\$defs/Id"

Property	Required	Structural rule
specification_digest	no	\$ref="#/\$defs/Digest"
<pre> { "additionalProperties": false, "allOf": [{ "else": { "not": { "required": ["specification_digest"] } }, "if": { "properties": { "baseline_kind": { "enum": ["specification", "hybrid"] } }, "required": ["baseline_kind"] }, "then": { "required": ["specification_digest"] } }], "title": "Candidate specification baseline binding is exact" }, { "else": { "not": { "anyOf": [{ "required": ["snapshot_provider_id"] }, { "required": ["snapshot_digest"] }] }, "properties": { "creditable": { "const": false } } }, "if": { "properties": { "consistency_mode": { </pre>		
<pre> "enum": ["immutable-vcs-tree"] }, "required": ["consistency_mode"] }, "then": { "properties": { "creditable": { "const": true } }, "required": ["snapshot_provider_id", "snapshot_digest" </pre>		

```

    ],
    "title": "creditable Candidate requires provider-proven immutable snapshot"
  },
  "properties": {
    "baseline_kind": {
      "enum": [
        "specification",
        "product",
        "hybrid"
      ]
    },
    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_id": {
      "$ref": "#/$defs/Id"
    },
    "candidate_recipe_digest": {
      "$ref": "#/$defs/Digest"
    },
    "consistency_mode": {
      "enum": [
        "observational-best-effort",
        "immutable-vcs-tree"
      ]
    },
    "control_excluded": {
      "const": true
    },
    "creditable": {
      "type": "boolean"
    },
    "inventory_digest": {
      "$ref": "#/$defs/Digest"
    },
    "product_root_digest": {
      "$ref": "#/$defs/Digest"
    },
    "record_type": {
      "const": "Candidate"
    },
    "snapshot_digest": {
      "$ref": "#/$defs/Digest"
    },
    "snapshot_provider_id": {
      "$ref": "#/$defs/Id"
    },
    "specification_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "record_type",
    "candidate_id",
    "candidate_digest",
    "inventory_digest",
    "product_root_digest",
    "control_excluded",
    "candidate_recipe_digest",
    "consistency_mode",
    "creditable"
  ],
  "type": "object"
}

```

Owner: `core/contracts.schema.json#/$defs/Candidate` | Validator: `urn:promin:contracts:1#/$defs/Candidate`

11. CandidateDelta

Type summary: `type="object"`; `additionalProperties=false`

Required: `base_candidate_digest`, `new_candidate_digest`, `workcard_digest`, `changed_paths`

Property	Required	Structural rule
<code>base_candidate_digest</code>	yes	<code>\$ref="#/\$defs/Digest"</code>

Property	Required	Structural rule
changed_paths	yes	type="array"; minItems=1; maxItems=100000; uniqueItems=true
new_candidate_digest	yes	\$ref="#/\$defs/Digest"
workcard_digest	yes	\$ref="#/\$defs/Digest"
<pre>{ "additionalProperties": false, "properties": { "base_candidate_digest": { "\$ref": "#/\$defs/Digest" }, "changed_paths": { "items": { "\$ref": "#/\$defs/RelativePath" }, "maxItems": 100000, "minItems": 1, "type": "array", "uniqueItems": true }, "new_candidate_digest": { "\$ref": "#/\$defs/Digest" }, "workcard_digest": { "\$ref": "#/\$defs/Digest" } }, "required": ["base_candidate_digest", "new_candidate_digest", "workcard_digest", "changed_paths"], "type": "object" }</pre>		
Owner: core/contracts.schema.json#/\$defs/CandidateDelta Validator: urn:promin:contracts:1#/\$defs/CandidateDelta		

12. CandidateRecipe

Type summary: type="object"; additionalProperties=false; allOf[1]=keys:else,if,then,title

Required: inventory_mode, include, exclude, symlink_policy, path_identity, collision_policy, product_identity_excludes_control_state

Property	Required	Structural rule
collision_policy	yes	const="reject-nfc-and-casefold-collisions"
exclude	yes	type="array"; minItems=1; maxItems=128; uniqueItems=true
include	yes	type="array"; minItems=1; maxItems=64; uniqueItems=true
inventory_mode	yes	const="explicit"
path_identity	yes	const="nfc-posix-relative"
product_identity_excludes_control_state	yes	const=true
snapshot_consistency	no	enum=["observational-best-effort", "immutable-vcs-tree"]
snapshot_provider_id	no	\$ref="#/\$defs/Id"
symlink_policy	yes	enum=["reject", "hash-link-metadata"]
<pre>{ "additionalProperties": false, "allOf": [{ "else": { "not": { "required": ["snapshot_provider_id"] } }, "if": { "properties": { "snapshot_consistency": { "enum": [</pre>		

```

        "immutable-vcs-tree"
      ]
    }
  },
  "required": [
    "snapshot_consistency"
  ]
},
"then": {
  "required": [
    "snapshot_provider_id"
  ]
}
},

```

```

    "title": "immutable snapshot recipe requires exact provider"
  }
],
"properties": {
  "collision_policy": {
    "const": "reject-nfc-and-casefold-collisions"
  },
  "exclude": {
    "items": {
      "$ref": "#/$defs/RelativePath"
    },
    "maxItems": 128,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "include": {
    "items": {
      "$ref": "#/$defs/RelativePath"
    },
    "maxItems": 64,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "inventory_mode": {
    "const": "explicit"
  },
}

```

```

"path_identity": {
  "const": "nfc-posix-relative"
},
"product_identity_excludes_control_state": {
  "const": true
},
"snapshot_consistency": {
  "enum": [
    "observational-best-effort",
    "immutable-vcs-tree"
  ]
},
"snapshot_provider_id": {
  "$ref": "#/$defs/Id"
},
"symlink_policy": {
  "enum": [
    "reject",
    "hash-link-metadata"
  ]
}
},
"required": [
  "inventory_mode",
  "include",
  "exclude",
  "symlink_policy",
  "path_identity",

```

```

    "collision_policy",
    "product_identity_excludes_control_state"
  ],
  "type": "object"
}

```

Owner: [core/contracts.schema.json#/\\$defs/CandidateRecipe](#) | Validator: [urn:promin:contracts:1#/\\$defs/CandidateRecipe](#)

13. CapacityReconciliation

Type summary: type="object"; additionalProperties=false

Required: reconciled_by, grant_id, grant_claim_digest, reconciled_at, generation, fencing_token

Property	Required	Structural rule
fencing_token	yes	type="integer"; minimum=1
generation	yes	type="integer"; minimum=1
grant_claim_digest	yes	\$ref="#/\$defs/Digest"
grant_id	yes	\$ref="#/\$defs/Id"
reconciled_at	yes	\$ref="#/\$defs/Timestamp"
reconciled_by	yes	\$ref="#/\$defs/Id"

```
{
  "additionalProperties": false,
  "properties": {
    "fencing_token": {
      "minimum": 1,
      "type": "integer"
    },
    "generation": {
      "minimum": 1,
      "type": "integer"
    },
    "grant_claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "grant_id": {
      "$ref": "#/$defs/Id"
    },
    "reconciled_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "reconciled_by": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "reconciled_by",
    "grant_id",
    "grant_claim_digest",
    "reconciled_at",
    "generation",
    "fencing_token"
  ],
  "type": "object"
}
```

Owner: core/contracts.schema.json#/\$defs/CapacityReconciliation | Validator: urn:promin:contracts:1#/\$defs/CapacityReconciliation

14. CloseAck

Type summary: type="object"; additionalProperties=false

Required: acknowledged_by, grant_id, grant_claim_digest, acknowledged_at

Property	Required	Structural rule
acknowledged_at	yes	\$ref="#/\$defs/Timestamp"
acknowledged_by	yes	\$ref="#/\$defs/Id"
grant_claim_digest	yes	\$ref="#/\$defs/Digest"
grant_id	yes	\$ref="#/\$defs/Id"

```
{
  "additionalProperties": false,
  "properties": {
    "acknowledged_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "acknowledged_by": {
      "$ref": "#/$defs/Id"
    },
    "grant_claim_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "acknowledged_by",
    "grant_id",
    "grant_claim_digest",
    "acknowledged_at"
  ],
  "type": "object"
}
```

```

    },
    "grant_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "acknowledged_by",
    "grant_id",
    "grant_claim_digest",
    "acknowledged_at"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/CloseAck | Validator: urn:promin:contracts:1#/\$defs/CloseAck

15. CommandCapabilityRule

Type summary: type="object"; additionalProperties=false; oneOf[2]=keys:not,required | keys:not,required

Required: command_kind

Property	Required	Structural rule
capability_by_value	no	type="object"; additionalProperties=schema(keys:\$ref)
capability_id	no	\$ref="#/\$defs/Id"
command_kind	yes	\$ref="#/\$defs/Id"
payload_field	no	\$ref="#/\$defs/Id"

```

{
  "additionalProperties": false,
  "oneOf": [
    {
      "not": {
        "anyOf": [
          {
            "required": [
              "payload_field"
            ]
          },
          {
            "required": [
              "capability_by_value"
            ]
          }
        ]
      },
      "required": [
        "capability_id"
      ]
    },
    {
      "not": {
        "required": [
          "capability_id"
        ]
      }
    ]
  },
  "required": [
    "payload_field",
    "capability_by_value"
  ]
},
{
  "not": {
    "required": [
      "capability_id"
    ]
  }
},
{
  "required": [
    "payload_field",
    "capability_by_value"
  ]
},
{
  "properties": {
    "capability_by_value": {
      "additionalProperties": {
        "$ref": "#/$defs/Id"
      },
      "minProperties": 1,
      "type": "object"
    },
    "capability_id": {
      "$ref": "#/$defs/Id"
    },
    "command_kind": {
      "$ref": "#/$defs/Id"
    },
    "payload_field": {

```

```

    "$ref": "#/$defs/Id"
  },
  "required": [
    "command_kind"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/CommandCapabilityRule | Validator: urn:promin:contracts:1#/\$defs/CommandCapabilityRule

16. CommandEffectScopeRule

Type summary: type="object"; additionalProperties=false; allOf[6]=keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then

Required: command_kind, mode

Property	Required	Structural rule
command_kind	yes	\$ref="#/\$defs/Id"
grant_id_field	no	\$ref="#/\$defs/Id"
grant_target_type	no	const="Grant"
id_field	no	\$ref="#/\$defs/Id"
kind	no	\$ref="#/\$defs/Id"
kind_field	no	\$ref="#/\$defs/Id"
kind_map	no	type="object"; additionalProperties=schema(keys:\$ref)
mode	yes	enum=["payload-id", "payload-multi-id", "payload-dynamic-id", "payload-scope", "referenced-grant-scope", "payload-dynamic-id-or-referenced-grant-scope"]
scope_field	no	\$ref="#/\$defs/Id"
targets	no	type="array"; minItems=1; maxItems=8; uniqueItems=true

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "if": {
        "properties": {
          "mode": {
            "const": "payload-id"
          }
        }
      },
      "then": {
        "required": [
          "kind",
          "id_field"
        ]
      }
    },
    {
      "if": {
        "properties": {
          "mode": {
            "const": "payload-multi-id"
          }
        }
      },
      "then": {
        "required": [
          "targets"
        ]
      }
    },
    {
      "if": {
        "properties": {
          "mode": {
            "const": "payload-dynamic-id"
          }
        }
      },
      "then": {
        "required": [
          "targets"
        ]
      }
    }
  ]
}

```

```

    "then": {
      "required": [
        "kind_field",
        "kind_map",
        "id_field"
      ]
    },
  },
  {
    "if": {
      "properties": {
        "mode": {
          "const": "payload-scope"
        }
      }
    }
  },
},

```

```

    "then": {
      "required": [
        "scope_field"
      ]
    },
  },
  {
    "if": {
      "properties": {
        "mode": {
          "const": "referenced-grant-scope"
        }
      }
    },
    "then": {
      "required": [
        "grant_id_field"
      ]
    }
  },
},
  {
    "if": {
      "properties": {
        "mode": {
          "const": "payload-dynamic-id-or-referenced-grant-scope"
        }
      }
    }
  },
},

```

```

    "then": {
      "required": [
        "kind_field",
        "kind_map",
        "id_field",
        "grant_target_type"
      ]
    },
  },
],
"properties": {
  "command_kind": {
    "$ref": "#/$defs/Id"
  },
  "grant_id_field": {
    "$ref": "#/$defs/Id"
  },
  "grant_target_type": {
    "const": "Grant"
  },
  "id_field": {
    "$ref": "#/$defs/Id"
  },
  "kind": {
    "$ref": "#/$defs/Id"
  },
  "kind_field": {
    "$ref": "#/$defs/Id"
  },

```

```

},
"kind_map": {
  "additionalProperties": {
    "$ref": "#/$defs/Id"
  },
},

```

```

    "minProperties": 1,
    "type": "object"
  },
  "mode": {
    "enum": [
      "payload-id",
      "payload-multi-id",
      "payload-dynamic-id",
      "payload-scope",
      "referenced-grant-scope",
      "payload-dynamic-id-or-referenced-grant-scope"
    ]
  },
  "scope_field": {
    "$ref": "#/$defs/Id"
  },
  "targets": {
    "items": {
      "$ref": "#/$defs/EffectTarget"
    },
    "maxItems": 8,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  }
},
"required": [
  "command_kind",
  "mode"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/CommandEffectScopeRule | Validator: urn:promin:contracts:1#/\$defs/CommandEffectScopeRule

17. CommandRequest

Type summary: type="object"; additionalProperties=false; allOf[16]=keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then,title | keys:if,then,title | keys:if,then,title | keys:if,then,title | keys:if,then,title

Required: record_type, command_id, command_kind, subject_id, activation_digest, idempotency_key, requested_scope, expected_head_digest, issued_at, payload, intent_digest, authorization

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
authorization	yes	\$ref="#/\$defs/AuthorizationRef"
command_id	yes	\$ref="#/\$defs/Id"
command_kind	yes	enum=["task.record", "task.transition", "candidate.record", "artifact.record", "run.record", "gate.record", "finding.record", "decision.record", "grant.issue", "grant.revoke", "lease.record"]
context_digest	no	\$ref="#/\$defs/Digest"
definition_digest	no	\$ref="#/\$defs/Digest"
expected_head_digest	yes	oneOf[2]=#/\$defs/Digest type:null
fencing_token	no	type="integer"; minimum=1
holder_authorization	no	\$ref="#/\$defs/GrantAuthorization"
idempotency_key	yes	type="string"; pattern="^[A-Za-z0-9._-]{16,128}\$"
intent_digest	yes	\$ref="#/\$defs/Digest"
issued_at	yes	\$ref="#/\$defs/Timestamp"
lease_generation	no	type="integer"; minimum=1
lease_id	no	\$ref="#/\$defs/Id"
payload	yes	composed structural definition
record_type	yes	const="CommandRequest"
requested_scope	yes	type="array"; minItems=1; maxItems=32
subject_id	yes	\$ref="#/\$defs/Id"
workcard_digest	no	\$ref="#/\$defs/Digest"

Property	Required	Structural rule
workcard_task_id	no	\$ref="#/\$defs/Id"
<pre> { "additionalProperties": false, "allOf": [{ "if": { "properties": { "command_kind": { "const": "task.record" } } }, "then": { "properties": { "payload": { "\$ref": "#/\$defs/Task" } } } }, { "if": { "properties": { "command_kind": { "const": "task.transition" } } }, "then": { "properties": { "payload": { "\$ref": "#/\$defs/TaskTransition" } } } }, { "if": { "properties": { "command_kind": { "const": "candidate.record" } } }, "then": { "properties": { "payload": { "\$ref": "#/\$defs/Candidate" } } } }, { "if": { "properties": { "command_kind": { "const": "artifact.record" } } }, "then": { "properties": { "payload": { "\$ref": "#/\$defs/Artifact" } } } }, { "if": { "properties": { "command_kind": { "const": "run.record" } } } }] } </pre>		

```

    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/Run"
        }
      }
    }
  },
  {

```

```

    "if": {
      "properties": {
        "command_kind": {
          "const": "gate.record"
        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/GateResult"
        }
      }
    }
  },
  {
    "if": {
      "properties": {
        "command_kind": {
          "const": "finding.record"
        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/Finding"
        }
      }
    }

```

```

  }
},
{
  "if": {
    "properties": {
      "command_kind": {
        "const": "decision.record"
      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/Decision"
      }
    }
  }
},
{
  "if": {
    "properties": {
      "command_kind": {
        "const": "grant.issue"
      }
    }
  },
  "then": {

```

```

    "properties": {
      "payload": {
        "$ref": "#/$defs/Grant"
      }
    }
  }
},
{
  "if": {
    "properties": {
      "command_kind": {

```

```
{
  "const": "grant.revoke"
}
},
{
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/GrantRevocation"
      }
    }
  },
  "if": {
    "properties": {
      "command_kind": {
        "const": "lease.record"
      }
    }
  }
}
```

```

    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/Lease"
      }
    }
  }
},
{
  "if": {
    "properties": {
      "command_kind": {
        "enum": [
          "artifact.record",
          "run.record",
          "gate.record",
          "finding.record"
        ]
      }
    },
    "required": [
      "command_kind"
    ]
  },
  "then": {

```

```

    "required": [
      "workcard_task_id",
      "holder_authorization",
      "lease_id",
      "lease_generation",
      "fencing_token",
      "workcard_digest",
      "context_digest"
    ]
  },
  "title": "require current Lease and bounded WorkCard for direct mutation"
},
{
  "if": {
    "properties": {
      "command_kind": {
        "const": "task.transition"
      },
      "payload": {
        "properties": {
          "to_state": {
            "enum": [
              "LEASED",
              "RUNNING",
              "COMPLETED"
            ]
          }
        }
      }
    },
    "required": [
      "to_state"
    ]
  }
}

```



```

    },
    "required": [
        "command_kind",
        "payload"
    ]
},
"then": {
    "required": [
        "workcard_task_id",
        "holder_authorization",
        "lease_id",
        "lease_generation",
        "fencing_token",
        "workcard_digest",
        "context_digest"
    ]
},
"title": "require current Lease and bounded WorkCard for execution transition"
},
{
    "if": {
        "not": {
            "anyOf": [
                {

```

```

                    "properties": {
                        "command_kind": {
                            "enum": [
                                "artifact.record",
                                "run.record",
                                "gate.record",
                                "finding.record"
                            ]
                        }
                    },
                    "required": [
                        "command_kind"
                    ]
                },
                {
                    "properties": {
                        "command_kind": {
                            "const": "task.transition"
                        },
                        "payload": {
                            "properties": {
                                "to_state": {
                                    "enum": [
                                        "LEASED",
                                        "RUNNING",
                                        "COMPLETED"
                                    ]
                                }
                            }
                        }
                    }
                }
            ]
        }
    }

```

```

    },
    "required": [
        "to_state"
    ]
},
"required": [
    "command_kind",
    "payload"
]
}
}
},
"then": {
    "not": {
        "anyOf": [
            {
                "required": [
                    "workcard_task_id"
                ]
            },
            {
                "required": [
                    "holder_authorization"
                ]
            }
        ]
    }
}

```

```

    },
    {
      "required": [
        "lease_id"
      ]
    },
    {
      "required": [
        "lease_generation"
      ]
    },
    {
      "required": [
        "fencing_token"
      ]
    },
    {
      "required": [
        "workcard_digest"
      ]
    },
    {
      "required": [
        "context_digest"
      ]
    }
  ]
},
"title": "forbid mutation binding on non-lease command"

```

```

},
{
  "if": {
    "properties": {
      "command_kind": {
        "enum": [
          "run.record",
          "gate.record"
        ]
      }
    }
  },
  "then": {
    "required": [
      "definition_digest"
    ]
  },
  "title": "run and gate commands bind an immutable definition digest"
},

```

```

{
  "if": {
    "properties": {
      "command_kind": {
        "enum": [
          "task.record",
          "task.transition",
          "candidate.record",
          "artifact.record",
          "finding.record",
          "decision.record",
          "grant.issue",
          "grant.revoke",
          "lease.record"
        ]
      }
    }
  },
  "then": {
    "not": {
      "required": [
        "definition_digest"
      ]
    }
  },
  "title": "other commands cannot claim a gate run definition"
}
],

```

```

"properties": {
  "activation_digest": {
    "$ref": "#/$defs/Digest"
  },
  "authorization": {
    "$ref": "#/$defs/AuthorizationRef"
  },
  "command_id": {
    "$ref": "#/$defs/Id"
  },
  "command_kind": {
    "enum": [
      "task.record",
      "task.transition",
      "candidate.record",
      "artifact.record",
      "run.record",
      "gate.record",
      "finding.record",
      "decision.record",
      "grant.issue",
      "grant.revoke",
      "lease.record"
    ]
  },
  "context_digest": {
    "$ref": "#/$defs/Digest"
  },
  "definition_digest": {
    "$ref": "#/$defs/Digest"
  },
  "expected_head_digest": {
    "oneOf": [
      {
        "$ref": "#/$defs/Digest"
      },
      {
        "type": "null"
      }
    ]
  },
  "fencing_token": {
    "minimum": 1,
    "type": "integer"
  },
  "holder_authorization": {
    "$ref": "#/$defs/GrantAuthorization"
  },
  "idempotency_key": {
    "pattern": "^[A-Za-z0-9._:-]{16,128}$",
    "type": "string"
  },
  "intent_digest": {
    "$ref": "#/$defs/Digest"
  },
  "issued_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "lease_generation": {
    "minimum": 1,
    "type": "integer"
  },
  "lease_id": {
    "$ref": "#/$defs/Id"
  },
  "payload": {},
  "record_type": {
    "const": "CommandRequest"
  },
  "requested_scope": {
    "items": {
      "$ref": "#/$defs/ScopeSelector"
    },
    "maxItems": 32,
    "minItems": 1,
    "type": "array"
  },

```

```

    "subject_id": {
      "$ref": "#/$defs/Id"
    },
    "workcard_digest": {
      "$ref": "#/$defs/Digest"
    },
    "workcard_task_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "record_type",
    "command_id",
    "command_kind",
    "subject_id",
    "activation_digest",

    "idempotency_key",
    "requested_scope",
    "expected_head_digest",
    "issued_at",
    "payload",
    "intent_digest",
    "authorization"
  ],
  "type": "object",
  "x-promin-max-canonical-bytes": 1048576
}

```

Owner: core/contracts.schema.json#/\$defs/CommandRequest | Validator: urn:promin:contracts:1#/\$defs/CommandRequest

18. CommandResult

Type summary: type="object"; additionalProperties=false; allOf[1]=keys:else,if,then,title

Required: record_type, command_id, outcome, batch_digest, reason_codes, facts, created_at

Property	Required	Structural rule
batch_digest	yes	oneOf[2]=#/\$defs/Digest type:null
command_id	yes	\$ref="#/\$defs/Id"
created_at	yes	\$ref="#/\$defs/Timestamp"
facts	yes	type="object"
operation_metrics	no	\$ref="#/\$defs/OperationMetrics"
outcome	yes	enum=["committed", "rejected", "idempotent-replay", "conflict"]
reason_codes	yes	type="array"; minItems=0; maxItems=32; uniqueItems=true
record_type	yes	const="CommandResult"

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "required": [
            "operation_metrics"
          ]
        }
      },
      "if": {
        "properties": {
          "outcome": {
            "enum": [
              "committed",
              "idempotent-replay"
            ]
          }
        }
      },
      "required": [
        "outcome"
      ]
    },
    {
      "then": {
        "required": [
          "operation_metrics"
        ]
      }
    }
  ]
}

```

```

    },
    "title": "committed command result carries exact operation metrics"
  }
],
"properties": {
  "batch_digest": {
    "oneOf": [
      {
        "$ref": "#/$defs/Digest"
      },
      {
        "type": "null"
      }
    ]
  },
  "command_id": {
    "$ref": "#/$defs/Id"
  },
  "created_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "facts": {
    "type": "object"
  },
  "operation_metrics": {
    "$ref": "#/$defs/OperationMetrics"
  },
  "outcome": {
    "enum": [
      "committed",
      "rejected",
      "idempotent-replay",
      "conflict"
    ]
  },
  "reason_codes": {
    "items": {
      "$ref": "#/$defs/Id"
    },
    "maxItems": 32,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
  },
  "record_type": {
    "const": "CommandResult"
  }
},
"required": [
  "record_type",
  "command_id",
  "outcome",
  "batch_digest",
  "reason_codes",
  "facts",
  "created_at"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/CommandResult | Validator: urn:promin:contracts:1#/\$defs/CommandResult

19. Conformance

Type summary: type="object"; additionalProperties=false

Required: record_type, schema_ref, document, version, function, structural_budgets, workcard_hard_ceiling, scale_contracts, reference_benchmarks, required_acceptance, mutation_families, production_claim_rule, dependency_depth_hard_max, acceptance_proof_class_counts, acceptance_proof_classes, command_latency_contracts

Property	Required	Structural rule
acceptance_proof_class_counts	yes	const=schema(keys:direct-semantic,human-decision,local-runtime-reexecute,migration-external-no-credit,package-static,saturation-signed,signed-role-platfor m-evidence,standard-release-signed-manifest)

Property	Required	Structural rule
acceptance_proof_classes	yes	const=schema(keys:activation-digest-binding,artifact-kind-resolves-correct-capability,authenticated-role-platform-evidence-attestations,authorization-binds-exact-grant-claim,bounded-incremental-commit-and-compaction,bounded-workcard-through-depth-twelve,broad-query-bounded-seed-refinement,candidate-delta-within-task-and-grant-scope,candidate-evidence-decision-acyclic-chain,candidate-evidence-provider-binding,candidate-recipe-applied-once,canonical-exact-zip-distribution,canonical-name-only,command-capability-and-scope-resolution,command-effect-target-in-requested-scope,configuration-exact-idempotency,conjunctive-scope-containment,continuation-secret-subject-grant-binding,continuation-token-bytes-at-most-256,continuation-v2-complete-page-union,creditable-candidate-immutable-vcs-tree,current-interpreter-and-clean-install-modes,current-release-closure-revalidation,decision-after-evidence-with-bounded-skew,decision-grant-matches-command-authorization,decision-kind-target-scope-binding,dependency-graph-acyclic-and-current-ready-frontier,depth-aware-seed-budget,deterministic-activation-identity,deterministic-replay,evidence-manifest-exact-eight-lane-matrix-resolution,evidence-manifest-four-cp314-supplemental-records,evidence-manifest-nested-and-supplemental-chronology,evidence-manifest-path-digest-size-closure,evidence-manifest-physical-resolution,exact-command-primary-event-correspondence,exact-five-init-records,exact-one-selected-preset,exact-six-base-command-surface,exact-six-core-artifacts,explicit-scale-marker-selection,external-standard-release-decision-exact-binding,finding-bound-resolution-and-waiver-evidence,fresh-semantic-control-state-per-saturation-iteration,grant-issuance-chain-closure,grant-scope-expiry-nonce-proof,handle-relative-evidence-root-safety,historical-decision-current-eligibility-separation,idempotent-state-commands,implementation-closure-bound-and-current,installed-platform-observation-closure,inventory-stream-memory-amplification-at-most-32,lease-generation-fence-and-close-ack,lease-holder-has-task-execute-grant,lossless-continuation-in-reference-100k-scenario,mutation-workcard-requires-current-lease,no-false-success,non-skipped-credit-result-has-evidence,normalization-collision-rejection,offline-installed-no-degradation,offline-release-online-compatibility,one-command-one-atomic-batch,physical-100k-one-artifact-proxy-per-file,policy-owned-task-and-lease-state-machines,preset-outside-core-digest,product-public-approval-separation,profile-bound-physical-100k-performance,profile-default-with-core-depth-twelve,projection-disposable-and-current,provider-substitution-invalidates-derived-state,raw-scale-summary-recomputation,read-workcard-without-lease,required-no-degradation-tests-fail-closed,required-provider-preflight-before-commit,rollback-and-recut-over-equivalence,root-bootstrap-command-limited,root-ceiling-without-bootstrap-grant-cycle,safe-recursive-export,selected-provider-adapter-evidence,semver-from-candidate-binding,seven-distinct-role-producer-scopes,signed-standard-decision-authority-proof,single-derived-platform-closure-owner,single-pass-inventory,standard-distribution-separate-from-product-acceptance,strict-critical-fields,team-signatures-cryptographically-verified,team-signed-client-to-end,three-zero-new-saturation-iterations,truthful-operation-reporting,truthful-process-invocation-status,typed-human-document-verification,typed-reference-domain-range-closure,verified-head-checkpoint-delta-replay,verified-inventory-result-provenance,windows-linux-exact-zip-matrix,zero-compiled-cache-files,zero-core-provider-lock-in,zero-dead-evidence-budget-fields,zero-forbidden-name-occurrences,zero-product-scan-during-init,zero-unowned-normative-facts)
command_latency_contracts	yes	const=[9 items]
dependency_depth_hard_max	yes	type="integer"; const=12
document	yes	const="promin.conformance"
function	yes	const="Own acceptance predicates, mutation families, structural hard ceilings, scale contracts, and production-claim rules."
mutation_families	yes	const=[77 items]; minItems=77; maxItems=77
production_claim_rule	yes	const="Standard distribution success never grants product acceptance. Live product credit separately requires a creditable immutable-vcs-tree Candidate, all P0/P1 findings closed with exact target-bound evidence, current project release closure eligibility, rollback and recut-over equivalence, and an authorized human project Decision. product_acceptance_pass remains false for promin v1 distribution evidence."
record_type	yes	const="Conformance"
reference_benchmarks	yes	const=schema(keys:core_valid_relation_count,depths_tested,file_count,forced_continuation,inventory_passes,performance_profiles,physical_relation_corpus,query_count,raw_file_proxy_ratio,rebuild_product_passes,reference_targets_are_environment_specific,semantic_corpus,silent_truncations_max,synthetic_task_ratio)
required_acceptance	yes	const=[102 items]; minItems=102; maxItems=102
scale_contracts	yes	const=schema(keys:event_commit,init,inventory,normal_one_shot_replay,projection,replay,saturation,workcard)
schema_ref	yes	const="urn:promin:contracts:1#/\$defs/Conformance"

Property	Required	Structural rule
structural_budgets	yes	const=schema(keys:authority_bearing_init_records,base_user_commands_max,core_normative_artifacts,persistent_entity_kinds_max,relation_kinds_max,resolved_plan_bytes_max,resolved_plan_source_samples_max,selected_presets)
version	yes	const="1.0.0-alpha.3"
workcard_hard_ceiling	yes	const=schema(keys:max_bytes,max_entities,max_fanout_per_entity,max_relations,top_k)
<pre> { "additionalProperties": false, "properties": { "acceptance_proof_class_counts": { "const": { "direct-semantic": 35, "human-decision": 9, "local-runtime-reexecute": 30, "migration-external-no-credit": 1, "package-static": 4, "saturation-signed": 1, "signed-role-platform-evidence": 17, "standard-release-signed-manifest": 5 } }, "acceptance_proof_classes": { "const": { "activation-digest-binding": "direct-semantic", "artifact-kind-resolves-correct-capability": "local-runtime-reexecute", "authenticated-role-platform-evidence-attestations": "signed-role-platform-evidence", "authorization-binds-exact-grant-claim": "local-runtime-reexecute", "bounded-incremental-commit-and-compaction": "signed-role-platform-evidence", "bounded-workcard-through-depth-twelve": "direct-semantic", "broad-query-bounded-seed-refinement": "direct-semantic", "candidate-delta-within-task-and-grant-scope": "direct-semantic", "candidate-evidence-decision-acyclic-chain": "human-decision", "candidate-evidence-provider-binding": "local-runtime-reexecute", "candidate-recipe-applied-once": "direct-semantic", "canonical-exact-zip-distribution": "direct-semantic", "canonical-name-only": "direct-semantic", "command-capability-and-scope-resolution": "local-runtime-reexecute", "command-effect-target-in-requested-scope": "local-runtime-reexecute", "configuration-exact-idempotency": "local-runtime-reexecute", "conjunctive-scope-containment": "local-runtime-reexecute", "continuation-secret-subject-grant-binding": "direct-semantic", "continuation-token-bytes-at-most-256": "direct-semantic", "continuation-v2-complete-page-union": "direct-semantic", "creditable-candidate-immutable-vcs-tree": "direct-semantic", "current-interpretter-and-clean-install-modes": "signed-role-platform-evidence", "current-release-closure-revalidation": "human-decision", "decision-after-evidence-with-bounded-skew": "human-decision", "decision-grant-matches-command-authorization": "local-runtime-reexecute", "decision-kind-target-scope-binding": "local-runtime-reexecute", "dependency-graph-acyclic-and-current-ready-frontier": "direct-semantic", "depth-aware-seed-budget": "local-runtime-reexecute", "deterministic-activation-identity": "local-runtime-reexecute", "deterministic-replay": "local-runtime-reexecute", "evidence-manifest-exact-eight-lane-matrix-resolution": "standard-release-signed-manifest", "evidence-manifest-four-cp314-supplemental-records": "standard-release-signed-manifest", "evidence-manifest-nested-and-supplemental-chronology": "standard-release-signed-manifest", "evidence-manifest-path-digest-size-closure": "standard-release-signed-manifest", "evidence-manifest-physical-resolution": "signed-role-platform-evidence", "exact-command-primary-event-correspondence": "local-runtime-reexecute", "exact-five-init-records": "direct-semantic", "exact-one-selected-preset": "direct-semantic", "exact-six-base-command-surface": "direct-semantic", "exact-six-core-artifacts": "direct-semantic", "explicit-scale-marker-selection": "direct-semantic", "external-standard-release-decision-exact-binding": "human-decision", "finding-bound-resolution-and-waiver-evidence": "direct-semantic", "fresh-semantic-control-state-per-saturation-iteration": "saturation-signed", "grant-issuance-chain-closure": "local-runtime-reexecute", "grant-scope-expiry-nonce-proof": "local-runtime-reexecute", "handle-relative-evidence-root-safety": "signed-role-platform-evidence", "historical-decision-current-eligibility-separation": "human-decision", "idempotent-state-commands": "local-runtime-reexecute", "implementation-closure-bound-and-current": "direct-semantic", "installed-platform-observation-closure": "signed-role-platform-evidence", </pre>		

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"inventory-stream-memory-amplification-at-most-32": "direct-semantic",
"lease-generation-fence-and-close-ack": "local-runtime-reexecute",
"lease-holder-has-task-execute-grant": "local-runtime-reexecute",
"lossless-continuation-in-reference-100k-scenario": "direct-semantic",
"mutation-workcard-requires-current-lease": "direct-semantic",
"no-false-success": "direct-semantic",
"non-skipped-credit-result-has-evidence": "local-runtime-reexecute",
"normalization-collision-rejection": "direct-semantic",
"offline-installed-no-degradation": "signed-role-platform-evidence",
"offline-release-online-compatibility": "signed-role-platform-evidence",
"one-command-one-atomic-batch": "local-runtime-reexecute",
"physical-100k-one-artifact-proxy-per-file": "signed-role-platform-evidence",
"policy-owned-task-and-lease-state-machines": "local-runtime-reexecute",
"preset-outside-core-digest": "direct-semantic",
"product-public-approval-separation": "human-decision",
"profile-bound-physical-100k-performance": "signed-role-platform-evidence",

"profile-default-with-core-depth-twelve": "direct-semantic",
"projection-disposable-and-current": "local-runtime-reexecute",
"provider-substitution-invalidates-derived-state": "local-runtime-reexecute",
"raw-scale-summary-recomputation": "signed-role-platform-evidence",
"read-workcard-without-lease": "direct-semantic",
"required-no-degradation-tests-fail-closed": "signed-role-platform-evidence",
"required-provider-preflight-before-commit": "local-runtime-reexecute",
"rollback-and-recutover-equivalence": "migration-external-no-credit",
"root-bootstrap-command-limited": "local-runtime-reexecute",
"root-ceiling-without-bootstrap-grant-cycle": "local-runtime-reexecute",
"safe-recursive-export": "direct-semantic",
"selected-provider-adapter-evidence": "direct-semantic",
"semver-from-candidate-binding": "human-decision",
"seven-distinct-role-producer-scopes": "standard-release-signed-manifest",
"signed-standard-decision-authority-proof": "human-decision",
"single-derived-platform-closure-owner": "signed-role-platform-evidence",
"single-pass-inventory": "local-runtime-reexecute",
"standard-distribution-separate-from-product-acceptance": "human-decision",
"strict-critical-fields": "direct-semantic",
"team-signatures-cryptographically-verified": "local-runtime-reexecute",
"team-signed-cli-end-to-end": "direct-semantic",
"three-zero-new-saturation-iterations": "signed-role-platform-evidence",
"truthful-operation-reporting": "local-runtime-reexecute",
"truthful-process-invocation-status": "signed-role-platform-evidence",
"typed-human-document-verification": "signed-role-platform-evidence",
"typed-reference-domain-range-closure": "direct-semantic",
"verified-head-checkpoint-delta-replay": "direct-semantic",
"verified-inventory-result-provenance": "direct-semantic",

"windows-linux-exact-zip-matrix": "signed-role-platform-evidence",
"zero-compiled-cache-files": "package-static",
"zero-core-provider-lock-in": "package-static",
"zero-dead-evidence-budget-fields": "direct-semantic",
"zero-forbidden-name-occurrences": "package-static",
"zero-product-scan-during-init": "local-runtime-reexecute",
"zero-unowned-normative-facts": "package-static"
}
},
"command_latency_contracts": {
  "const": [
    {
      "budget_ms": 5000,
      "command": "audit",
      "max_files": 32,
      "max_total_bytes": 65536,
      "measured_runs": 3,
      "measurement_mode": "cli_cold",
      "percentile": "p95",
      "projection_state": "not-required",
      "provider_state": "no-install",
      "status": "provisional",
      "warmup_runs": 1,
      "workload_id": "bounded-audit-32-v1"
    },
    {
      "budget_ms": 2000,
      "command": "context",
      "max_files": 32,
      "max_total_bytes": 65536,
      "measured_runs": 3,
      "measurement_mode": "runtime_warm",
      "percentile": "p95",

```



```

"projection_state": "current",
"provider_state": "installed-current",
"status": "provisional",
"warmup_runs": 1,
"workload_id": "current-projection-32-v1"
},
{
  "budget_ms": 30000,
  "command": "init apply small",
  "max_files": 32,
  "max_total_bytes": 65536,
  "measured_runs": 3,
  "measurement_mode": "cli_cold",
  "percentile": "p95",
  "projection_state": "empty",
  "provider_state": "no-install",
  "status": "provisional",
  "warmup_runs": 1,
  "workload_id": "bounded-first-apply-32-v1"
},
{
  "budget_ms": 5000,
  "command": "init review",

```

```

"max_files": 32,
"max_total_bytes": 65536,
"measured_runs": 3,
"measurement_mode": "cli_cold",
"percentile": "p95",
"projection_state": "not-required",
"provider_state": "no-install",
"status": "provisional",
"warmup_runs": 1,
"workload_id": "bounded-review-32-v1"
},
{
  "budget_ms": 2000,
  "command": "next",
  "max_files": 32,
  "max_total_bytes": 65536,
  "measured_runs": 3,
  "measurement_mode": "runtime_warm",
  "percentile": "p95",
  "projection_state": "current",
  "provider_state": "installed-current",
  "status": "provisional",
  "warmup_runs": 1,
  "workload_id": "current-projection-32-v1"
},
{
  "budget_ms": 2000,
  "command": "status",

```

```

"max_files": 32,
"max_total_bytes": 65536,
"measured_runs": 3,
"measurement_mode": "operation_incremental",
"percentile": "p95",
"projection_state": "current",
"provider_state": "installed-current",
"status": "provisional",
"warmup_runs": 1,
"workload_id": "current-projection-32-v1"
},
{
  "budget_ms": 2000,
  "command": "status",
  "max_files": 32,
  "max_total_bytes": 65536,
  "measured_runs": 3,
  "measurement_mode": "runtime_warm",
  "percentile": "p95",
  "projection_state": "current",
  "provider_state": "installed-current",
  "status": "provisional",
  "warmup_runs": 1,
  "workload_id": "current-projection-32-v1"
},
{

```

```

    "budget_ms": 10000,
    "command": "validate",

    "max_files": 32,
    "max_total_bytes": 65536,
    "measured_runs": 3,
    "measurement_mode": "cli_cold",
    "percentile": "p95",
    "projection_state": "current",
    "provider_state": "installed-current",
    "status": "provisional",
    "warmup_runs": 1,
    "workload_id": "current-project-no-replay-v1"
  },
  {
    "budget_ms": 3000,
    "command": "validate",
    "max_files": 32,
    "max_total_bytes": 65536,
    "measured_runs": 3,
    "measurement_mode": "runtime_warm",
    "percentile": "p95",
    "projection_state": "current",
    "provider_state": "installed-current",
    "status": "provisional",
    "warmup_runs": 1,
    "workload_id": "current-project-no-replay-v1"
  }
],
"dependency_depth_hard_max": {
  "const": 12,
  "type": "integer"
},
"document": {
  "const": "promin.conformance"
},
"function": {
  "const": "Own acceptance predicates, mutation families, structural hard ceilings, scale contracts,
and production-claim rules."
},
"mutation_families": {
  "const": [
    "status-credit-contradiction",
    "missing-critical-field",
    "unknown-critical-field",
    "activation-drift",
    "grant-replay",
    "scope-escalation",
    "expired-grant",
    "revoked-grant",
    "stale-fence",
    "candidate-drift",
    "provider-substitution",
    "event-fork",
    "batch-command-mismatch",
    "duplicate-idempotency-key",
    "projection-corruption",
    "projection-staleness",
    "reference-break",
    "relation-domain-break",
    "context-flood",
    "symlink-core-input",
    "shadow-core-file",
    "symlink-product-entry",
    "secret-export",
    "archive-expansion",
    "preset-authority-escalation",
    "normalization-key-collision",
    "path-casefold-collision",
    "forbidden-name-reintroduction",
    "reinit-configuration-conflict",
    "clock-order-violation",
    "capability-confusion",
    "command-scope-escalation",
    "grant-issuer-chain-break",
    "unverified-signature-acceptance",
    "provider-declared-but-unavailable",

```

```

"illegal-task-transition",
"illegal-lease-transition",
"decision-target-confusion",
"credit-without-evidence",
"command-primary-event-mismatch",
"duplicate-event-id",
"scope-selector-drop-broadening",
"authorization-grant-claim-substitution",
"command-effect-outside-requested-scope",
"lease-holder-grant-mismatch",

```

```

"decision-grant-provenance-mismatch",
"artifact-kind-capability-confusion",
"continuation-frontier-loss",
"continuation-duplicate-emission",
"candidate-recipe-bypass",
"observational-candidate-pass-credit",
"snapshot-provider-substitution",
"unsupported-provider-adapter",
"stale-release-closure-credit",
"finding-evidence-target-mismatch",
"candidate-delta-scope-escape",
"checkpoint-head-mismatch",
"checkpoint-corruption-credit",
"normal-command-full-replay",
"profile-default-depth-bypass",
"continuation-public-material-mismatch",
"inventory-provenance-override",
"implementation-closure-drift",
"noncanonical-archive-layout",
"inventory-graph-inflation",
"workcard-budget-bypass",
"fabricated-evidence-digest",
"unsigned-standard-decision",
"unconfigured-standard-decision-key",
"standard-decision-version-replay",
"candidate-decision-cycle",
"broad-query-corpus-pagination",
"continuation-token-context-inflation",

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"inventory-stream-materialization",
"pdf-header-only-credit",
"scale-environment-skip-credit",
"content-search-omission"
],

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"items": {
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    "expired-grant",
    "revoked-grant",
    "stale-fence",
    "candidate-drift",
    "provider-substitution",
    "event-fork",
    "batch-command-mismatch",
    "duplicate-idempotency-key",
    "projection-corruption",
    "projection-staleness",
    "reference-break",
    "relation-domain-break",
    "context-flood",
    "symlink-core-input",
    "shadow-core-file",

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"symlink-product-entry",
"secret-export",
"archive-expansion",
"preset-authority-escalation",
"normalization-key-collision",
"path-casefold-collision",
"forbidden-name-reintroduction",
"reinit-configuration-conflict",
"clock-order-violation",
"capability-confusion",
"command-scope-escalation",

```

```

"grant-issuer-chain-break",
"unverified-signature-acceptance",
"provider-declared-but-unavailable",
"illegal-task-transition",
"illegal-lease-transition",
"decision-target-confusion",
"credit-without-evidence",
"command-primary-event-mismatch",
"duplicate-event-id",
"scope-selector-drop-broadening",
"authorization-grant-claim-substitution",
"command-effect-outside-requested-scope",
"lease-holder-grant-mismatch",
"decision-grant-provenance-mismatch",
"artifact-kind-capability-confusion",
"continuation-frontier-loss",
"continuation-duplicate-emission",

```

```

"candidate-recipe-bypass",
"observational-candidate-pass-credit",
"snapshot-provider-substitution",
"unsupported-provider-adapter",
"stale-release-closure-credit",
"finding-evidence-target-mismatch",
"candidate-delta-scope-escape",
"checkpoint-head-mismatch",
"checkpoint-corruption-credit",
"normal-command-full-replay",
"profile-default-depth-bypass",
"continuation-public-material-mismatch",
"inventory-provenance-override",
"implementation-closure-drift",
"noncanonical-archive-layout",
"inventory-graph-inflation",
"workcard-budget-bypass",
"fabricated-evidence-digest",
"unsigned-standard-decision",
"unconfigured-standard-decision-key",
"standard-decision-version-replay",
"candidate-decision-cycle",
"broad-query-corpus-pagination",
"continuation-token-context-inflation",
"inventory-stream-materialization",
"pdf-header-only-credit",
"scale-environment-skip-credit",
"content-search-omission"

```

```

]
},
"maxItems": 77,
"minItems": 77
},
"production_claim_rule": {
  "const": "Standard distribution success never grants product acceptance. Live product credit
separately requires a creditable immutable-vcs-tree Candidate, all P0/P1 findings closed with exact
target-bound evidence, current project release closure eligibility, rollback and recutover equivalence,
and an authorized human project Decision. product_acceptance_pass remains false for promin v1 distribution
evidence."
},
"record_type": {
  "const": "Conformance"
},
"reference_benchmarks": {
  "const": {
    "core_valid_relation_count": 198999,
    "depths_tested": [
      1,
      2,
      3,
      4,
      5,
      6,
      7,
      8,
      9,
      10,
      11,
      12
    ]
  }
},
],

```

```

"file_count": 100000,
"forced_continuation": true,
"inventory_passes": 1,
"performance_profiles": {
  "portable-local-v1": {
    "commit_bytes_per_changed_record_max": 32768,
    "commit_p95_ms_max": 500,
    "commit_p99_ms_max": 1000,
    "compatible_platforms": [
      "linux",
      "windows"
    ],
    "database_bytes_max": 402653184,
    "p50_ms_max": 100,
    "p95_ms_max": 150,
    "p99_ms_max": 500,
    "peak_rss_bytes_max": 805306368,
    "projection_amplification_max": 20,
    "requires_same_runner_no_degradation": true,
    "runtime_checkpoint_count_max": 24,
    "semantic_inflation_max": 3.25,
    "semantic_ingestion_seconds_max": 600
  }
},
"physical_relation_corpus": {
  "artifact_target_coverage": 1,
  "expected_counts_source": "fixture-manifest",
  "generation": "explicit-authorized-command-events",

  "relation_count": 198971,
  "relation_kind": "READS",
  "separate_from_raw_inventory": true,
  "task_count": 1567,
  "total_core_valid_relations": 198999
},
"query_count": 600,
"raw_file_proxy_ratio": 1,
"rebuild_product_passes": 0,
"reference_targets_are_environment_specific": true,
"semantic_corpus": {
  "depths": [
    1,
    2,
    3,
    4,
    5,
    6,
    7,
    8,
    9,
    10,
    11,
    12
  ],
  "expected_counts_source": "fixture-manifest",
  "relation_count": 28,
  "relation_kind": "DEPENDS_ON",

  "separate_from_raw_inventory": true,
  "task_count": 32
},
"silent_truncations_max": 0,
"synthetic_task_ratio": 0
},
"required_acceptance": {
  "const": [
    "canonical-name-only",
    "exact-six-core-artifacts",
    "preset-outside-core-digest",
    "exact-one-selected-preset",
    "exact-five-init-records",
    "zero-product-scan-during-init",
    "single-pass-inventory",
    "no-false-success",
    "strict-critical-fields",
    "activation-digest-binding",
    "root-ceiling-without-bootstrap-grant-cycle",
    "grant-scope-expiry-nonce-proof",
    "lease-generation-fence-and-close-ack",

```

"candidate-evidence-provider-binding",
"one-command-one-atomic-batch",
"deterministic-replay",
"projection-disposable-and-current",
"bounded-workcard-through-depth-twelve",
"typed-reference-domain-range-closure",

"dependency-graph-acyclic-and-current-ready-frontier",
"safe-recursive-export",
"idempotent-state-commands",
"normalization-collision-rejection",
"provider-substitution-invalidates-derived-state",
"rollback-and-recutover-equivalence",
"zero-forbidden-name-occurrences",
"zero-unowned-normative-facts",
"zero-core-provider-lock-in",
"zero-compiled-cache-files",
"physical-100k-one-artifact-proxy-per-file",
"external-standard-release-decision-exact-binding",
"root-bootstrap-command-limited",
"deterministic-activation-identity",
"configuration-exact-idempotency",
"truthful-operation-reporting",
"depth-aware-seed-budget",
"lossless-continuation-in-reference-100k-scenario",
"read-workcard-without-lease",
"mutation-workcard-requires-current-lease",
"command-capability-and-scope-resolution",
"grant-issuance-chain-closure",
"team-signatures-cryptographically-verified",
"required-provider-preflight-before-commit",
"policy-owned-task-and-lease-state-machines",
"decision-kind-target-scope-binding",
"non-skipped-credit-result-has-evidence",
"exact-command-primary-event-correspondence",

"conjunctive-scope-containment",
"authorization-binds-exact-grant-claim",
"command-effect-target-in-requested-scope",
"lease-holder-has-task-execute-grant",
"decision-grant-matches-command-authorization",
"artifact-kind-resolves-correct-capability",
"continuation-v2-complete-page-union",
"candidate-recipe-applied-once",
"creditable-candidate-immutable-vcs-tree",
"selected-provider-adapter-evidence",
"team-signed-cli-end-to-end",
"current-release-closure-revalidation",
"finding-bound-resolution-and-waiver-evidence",
"candidate-delta-within-task-and-grant-scope",
"verified-head-checkpoint-delta-replay",
"profile-default-with-core-depth-twelve",
"canonical-exact-zip-distribution",
"current-interpreter-and-clean-install-modes",
"required-no-degradation-tests-fail-closed",
"continuation-secret-subject-grant-binding",
"implementation-closure-bound-and-current",
"verified-inventory-result-provenance",
"exact-six-base-command-surface",
"zero-dead-evidence-budget-fields",
"profile-bound-physical-100k-performance",
"windows-linux-exact-zip-matrix",
"fresh-semantic-control-state-per-saturation-iteration",
"three-zero-new-saturation-iterations",

"standard-distribution-separate-from-product-acceptance",
"candidate-evidence-decision-acyclic-chain",
"evidence-manifest-physical-resolution",
"evidence-manifest-exact-eight-lane-matrix-resolution",
"evidence-manifest-four-cp314-supplemental-records",
"evidence-manifest-path-digest-size-closure",
"evidence-manifest-nested-and-supplemental-chronology",
"seven-distinct-role-producer-scopes",
"signed-standard-decision-authority-proof",
"semver-from-candidate-binding",
"typed-human-document-verification",
"broad-query-bounded-seed-refinement",
"continuation-token-bytes-at-most-256",
"inventory-stream-memory-amplification-at-most-32",
"explicit-scale-marker-selection",

```

"authenticated-role-platform-evidence-attestations",
"installed-platform-observation-closure",
"handle-relative-evidence-root-safety",
"raw-scale-summary-recomputation",
"bounded-incremental-commit-and-compaction",
"offline-installed-no-degradation",
"offline-release-online-compatibility",
"decision-after-evidence-with-bounded-skew",
"single-derived-platform-closure-owner",
"truthful-process-invocation-status",
"product-public-approval-separation",
"historical-decision-current-eligibility-separation"
],

```

```

"items": {
  "enum": [
    "canonical-name-only",
    "exact-six-core-artifacts",
    "preset-outside-core-digest",
    "exact-one-selected-preset",
    "exact-five-init-records",
    "zero-product-scan-during-init",
    "single-pass-inventory",
    "no-false-success",
    "strict-critical-fields",
    "activation-digest-binding",
    "root-ceiling-without-bootstrap-grant-cycle",
    "grant-scope-expiry-nonce-proof",
    "lease-generation-fence-and-close-ack",
    "candidate-evidence-provider-binding",
    "one-command-one-atomic-batch",
    "deterministic-replay",
    "projection-disposable-and-current",
    "bounded-workcard-through-depth-twelve",
    "typed-reference-domain-range-closure",
    "dependency-graph-acyclic-and-current-ready-frontier",
    "safe-recursive-export",
    "idempotent-state-commands",
    "normalization-collision-rejection",
    "provider-substitution-invalidates-derived-state",
    "rollback-and-recutover-equivalence",
    "zero-forbidden-name-occurrences",

```

```

"zero-unowned-normative-facts",
"zero-core-provider-lock-in",
"zero-compiled-cache-files",
"physical-100k-one-artifact-proxy-per-file",
"external-standard-release-decision-exact-binding",
"root-bootstrap-command-limited",
"deterministic-activation-identity",
"configuration-exact-idempotency",
"truthful-operation-reporting",
"depth-aware-seed-budget",
"lossless-continuation-in-reference-100k-scenario",
"read-workcard-without-lease",
"mutation-workcard-requires-current-lease",
"command-capability-and-scope-resolution",
"grant-issuance-chain-closure",
"team-signatures-cryptographically-verified",
"required-provider-preflight-before-commit",
"policy-owned-task-and-lease-state-machines",
"decision-kind-target-scope-binding",
"non-skipped-credit-result-has-evidence",
"exact-command-primary-event-correspondence",
"conjunctive-scope-containment",
"authorization-binds-exact-grant-claim",
"command-effect-target-in-requested-scope",
"lease-holder-has-task-execute-grant",
"decision-grant-matches-command-authorization",
"artifact-kind-resolves-correct-capability",
"continuation-v2-complete-page-union",

```

```

"candidate-recipe-applied-once",
"creditable-candidate-immutable-vcs-tree",
"selected-provider-adapter-evidence",
"team-signed-cli-end-to-end",
"current-release-closure-revalidation",
"finding-bound-resolution-and-waiver-evidence",
"candidate-delta-within-task-and-grant-scope",
"verified-head-checkpoint-delta-replay",

```

```

"profile-default-with-core-depth-twelve",
"canonical-exact-zip-distribution",
"current-interpreter-and-clean-install-modes",
"required-no-degradation-tests-fail-closed",
"continuation-secret-subject-grant-binding",
"implementation-closure-bound-and-current",
"verified-inventory-result-provenance",
"exact-six-base-command-surface",
"zero-dead-evidence-budget-fields",
"profile-bound-physical-100k-performance",
"windows-linux-exact-zip-matrix",
"fresh-semantic-control-state-per-saturation-iteration",
"three-zero-new-saturation-iterations",
"standard-distribution-separate-from-product-acceptance",
"candidate-evidence-decision-acyclic-chain",
"evidence-manifest-physical-resolution",
"evidence-manifest-exact-eight-lane-matrix-resolution",
"evidence-manifest-four-cp314-supplemental-records",
"evidence-manifest-path-digest-size-closure",
"evidence-manifest-nested-and-supplemental-chronology",

"seven-distinct-role-producer-scopes",
"signed-standard-decision-authority-proof",
"semver-from-candidate-binding",
"typed-human-document-verification",
"broad-query-bounded-seed-refinement",
"continuation-token-bytes-at-most-256",
"inventory-stream-memory-amplification-at-most-32",
"explicit-scale-marker-selection",
"authenticated-role-platform-evidence-attestations",
"installed-platform-observation-closure",
"handle-relative-evidence-root-safety",
"raw-scale-summary-recomputation",
"bounded-incremental-commit-and-compaction",
"offline-installed-no-degradation",
"offline-release-online-compatibility",
"decision-after-evidence-with-bounded-skew",
"single-derived-platform-closure-owner",
"truthful-process-invocation-status",
"product-public-approval-separation",
"historical-decision-current-eligibility-separation"
]
},
"maxItems": 102,
"minItems": 102
},
"scale_contracts": {
  "const": {
    "event_commit": {
      "complexity": "0(batch)",
      "head_lookup": "0(1)"
    },
    "init": {
      "complexity": "0(core+plan)",
      "project_tree_passes": 0
    },
    "inventory": {
      "complexity": "0(files+bytes)",
      "creditable_consistency_modes": [
        "immutable-vcs-tree"
      ],
      "derived_artifact_proxies_per_raw_file": 1,
      "manifest_verified_jsonl": true,
      "memory_amplification_max": 32,
      "observational_mode_pass_credit": false,
      "project_tree_passes_max": 1,
      "search_text_bytes_max": 4096,
      "synthetic_relations_per_raw_file": 0,
      "synthetic_tasks_per_raw_file": 0
    },
    "normal_one_shot_replay": {
      "checkpoint_authoritative": false,
      "complexity": "0(delta-since-verified-checkpoint)",
      "full_replay_allowed": "doctor, validate, recovery, or verified fallback only"
    },
    "projection": {
      "public_rebuild_ingress": "verified InventoryResult or persisted verified manifest only",

```



```

    "rebuild_product_tree_passes": 0,
    "reserved_provenance_rejected": true
  },
  "replay": {
    "complexity": "0(events)"
  },
  "saturation": {
    "fresh_control_state_per_iteration": true,
    "init_product_tree_scans": 0,
    "semantic_state_reused": false
  },
  "workcard": {
    "broad_query_behavior": "deterministic bounded top-k seeds, refinement_required=true, no corpus pagination",
    "complexity": "0(top_k+capped_typed_closure)",
    "continuation_state_bytes_max": 16384,
    "continuation_token_bytes_max": 256,
    "corpus_independent_output_bound": true,
    "seed_budget": "effective_top_k=min(profile.top_k,max(2,floor(max_entities/(1+observed_branching*requested_depth))))",
    "selected_closure_union_completeness": 1
  }
},
"schema_ref": {
  "const": "urn:promin:contracts:1#/$defs/Conformance"
},
"structural_budgets": {
  "const": {
    "authority_bearing_init_records": 5,

    "base_user_commands_max": 10,
    "core_normative_artifacts": 6,
    "persistent_entity_kinds_max": 9,
    "relation_kinds_max": 6,
    "resolved_plan_bytes_max": 8192,
    "resolved_plan_source_samples_max": 64,
    "selected_presets": 1
  }
},
"version": {
  "const": "1.0.0-alpha.3"
},
"workcard_hard_ceiling": {
  "const": {
    "max_bytes": 16384,
    "max_entities": 32,
    "max_fanout_per_entity": 8,
    "max_relations": 48,
    "top_k": 12
  }
},
"required": [
  "record_type",
  "schema_ref",
  "document",
  "version",
  "function",

  "structural_budgets",
  "workcard_hard_ceiling",
  "scale_contracts",
  "reference_benchmarks",
  "required_acceptance",
  "mutation_families",
  "production_claim_rule",
  "dependency_depth_hard_max",
  "acceptance_proof_class_counts",
  "acceptance_proof_classes",
  "command_latency_contracts"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Conformance | Validator: urn:promin:contracts:1#/\$defs/Conformance

20. ContinuationTokenClaims

Type summary: type="object"; additionalProperties=false

Required: record_type, token_version, activation_digest, subject_id, grant_id, grant_claim_digest, capability_id, requested_scope, normalized_query_digest, cursor, head_event_id, head_sequence, head_digest, projection_digest, ranking_id, traversal_id, dependency_depth, budgets, issued_at, expires_at, ttl_seconds, revocation_epoch, implementation_closure_digest, state_id

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
budgets	yes	\$ref="#/\$defs/Budget"
capability_id	yes	const="projection.read"
cursor	yes	type="integer"; minimum=0
dependency_depth	yes	type="integer"; minimum=1; maximum=12
expires_at	yes	\$ref="#/\$defs/Timestamp"
grant_claim_digest	yes	\$ref="#/\$defs/Digest"
grant_id	yes	\$ref="#/\$defs/Id"
head_digest	yes	oneOf[2]=#/\$defs/Digest type:null
head_event_id	yes	oneOf[2]=#/\$defs/Id type:null
head_sequence	yes	type="integer"; minimum=0
implementation_closure_digest	yes	\$ref="#/\$defs/Digest"
issued_at	yes	\$ref="#/\$defs/Timestamp"
normalized_query_digest	yes	\$ref="#/\$defs/Digest"
projection_digest	yes	\$ref="#/\$defs/Digest"
ranking_id	yes	const="bm25-v1"
record_type	yes	const="ContinuationTokenClaims"
requested_scope	yes	type="array"; minItems=1; maxItems=32
revocation_epoch	yes	\$ref="#/\$defs/Digest"
state_id	yes	\$ref="#/\$defs/Id"
subject_id	yes	\$ref="#/\$defs/Id"
token_version	yes	type="integer"; const=2
traversal_id	yes	const="typed-bfs-v2"
ttl_seconds	yes	type="integer"; minimum=1; maximum=3600

```
{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "budgets": {
      "$ref": "#/$defs/Budget"
    },
    "capability_id": {
      "const": "projection.read"
    },
    "cursor": {
      "minimum": 0,
      "type": "integer"
    },
    "dependency_depth": {
      "maximum": 12,
      "minimum": 1,
      "type": "integer"
    },
    "expires_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "grant_claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "grant_id": {
```

```

    "$ref": "#/$defs/Id"
  },
  "head_digest": {
    "oneOf": [
      {
        "$ref": "#/$defs/Digest"
      },
      {
        "type": "null"
      }
    ]
  },
  "head_event_id": {
    "oneOf": [
      {
        "$ref": "#/$defs/Id"
      },
      {
        "type": "null"
      }
    ]
  },
  "head_sequence": {
    "minimum": 0,
    "type": "integer"
  },
  "implementation_closure_digest": {
    "$ref": "#/$defs/Digest"
  }

```

```

  },
  "issued_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "normalized_query_digest": {
    "$ref": "#/$defs/Digest"
  },
  "projection_digest": {
    "$ref": "#/$defs/Digest"
  },
  "ranking_id": {
    "const": "bm25-v1"
  },
  "record_type": {
    "const": "ContinuationTokenClaims"
  },
  "requested_scope": {
    "items": {
      "$ref": "#/$defs/ScopeSelector"
    },
    "maxItems": 32,
    "minItems": 1,
    "type": "array"
  },
  "revocation_epoch": {
    "$ref": "#/$defs/Digest"
  },
  "state_id": {

```

```

    "$ref": "#/$defs/Id"
  },
  "subject_id": {
    "$ref": "#/$defs/Id"
  },
  "token_version": {
    "const": 2,
    "type": "integer"
  },
  "traversal_id": {
    "const": "typed-bfs-v2"
  },
  "ttl_seconds": {
    "maximum": 3600,
    "minimum": 1,
    "type": "integer"
  }
},
"required": [
  "record_type",
  "token_version",
  "activation_digest",

```

```

"subject_id",
"grant_id",
"grant_claim_digest",
"capability_id",
"requested_scope",
"normalized_query_digest",

"cursor",
"head_event_id",
"head_sequence",
"head_digest",
"projection_digest",
"ranking_id",
"traversal_id",
"dependency_depth",
"budgets",
"issued_at",
"expires_at",
"ttl_seconds",
"revocation_epoch",
"implementation_closure_digest",
"state_id"
],
"type": "object",
"x-promin-max-state-bytes": 16384,
"x-promin-max-token-bytes": 256,
"x-promin-persistence": "derived-live-only"
}

```

Owner: `core/contracts.schema.json#/$defs/ContinuationTokenClaims` | Validator: `urn:promin:contracts:1#/$defs/ContinuationTokenClaims`

21. ContinueResult

Type summary: `type="object"`; `additionalProperties=false`; `allOf[2]=keys:else,if,then` | `keys:if,then`

Required: `record_type`, `status`, `subject_id`, `activation_digest`, `work_card`, `continuation`

Property	Required	Structural rule
<code>activation_digest</code>	yes	<code>\$ref="#/\$defs/Digest"</code>
<code>continuation</code>	yes	<code>oneOf[2]=type:null type:string</code>
<code>record_type</code>	yes	<code>const="ContinueResult"</code>
<code>status</code>	yes	<code>enum=["ready", "empty"]</code>
<code>subject_id</code>	yes	<code>\$ref="#/\$defs/Id"</code>
<code>work_card</code>	yes	<code>oneOf[2]=#/\$defs/WorkCard type:null</code>

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "properties": {
          "continuation": {
            "type": "null"
          },
          "work_card": {
            "type": "null"
          }
        }
      },
      "if": {
        "properties": {
          "status": {
            "const": "ready"
          }
        },
        "required": [
          "status"
        ]
      },
      "then": {
        "properties": {
          "work_card": {
            "$ref": "#/$defs/WorkCard"
          }
        }
      }
    }
  ]
}

```

```

    }
  },
  {
    "if": {
      "properties": {
        "continuation": {
          "type": "string"
        }
      },
      "required": [
        "continuation"
      ]
    },
    "then": {
      "properties": {
        "work_card": {
          "properties": {
            "truncated": {
              "const": true
            }
          },
          "required": [
            "truncated"
          ]
        }
      }
    }
  }
}

```

```

    }
  },
  {
    "properties": {
      "activation_digest": {
        "$ref": "#/$defs/Digest"
      },
      "continuation": {
        "oneOf": [
          {
            "type": "null"
          },
          {
            "maxLength": 256,
            "minLength": 1,
            "pattern": "^promin-v2\\. [A-Za-z0-9_-]{22}\\.[0-9a-f]{64}\\.[A-Za-z0-9_-]{43}$",
            "type": "string"
          }
        ]
      },
      "record_type": {
        "const": "ContinueResult"
      },
      "status": {
        "enum": [
          "ready",
          "empty"
        ]
      }
    }
  }
}

```

```

  },
  "subject_id": {
    "$ref": "#/$defs/Id"
  },
  "work_card": {
    "oneOf": [
      {
        "$ref": "#/$defs/WorkCard"
      },
      {
        "type": "null"
      }
    ]
  }
},
"required": [
  "record_type",
  "status",
  "subject_id",
  "activation_digest",
  "work_card",
  "continuation"
],
"type": "object",

```

```
"x-promin-persistence": "derived-live-only"
}
```

Owner: core/contracts.schema.json#/\$defs/ContinueResult | Validator: urn:promin:contracts:1#/\$defs/ContinueResult

22. CoreComponent

Type summary: type="object"; additionalProperties=false

Required: path, sha256

Property	Required	Structural rule
path	yes	enum=["semantic-model.json", "authority-model.json", "policy-set.json", "contracts.schema.json", "conformance.json"]
sha256	yes	\$ref="#/\$defs/Digest"
<pre>{ "additionalProperties": false, "properties": { "path": { "enum": ["semantic-model.json", "authority-model.json", "policy-set.json", "contracts.schema.json", "conformance.json"] }, "sha256": { "\$ref": "#/\$defs/Digest" } }, "required": ["path", "sha256"], "type": "object" }</pre>		

Owner: core/contracts.schema.json#/\$defs/CoreComponent | Validator: urn:promin:contracts:1#/\$defs/CoreComponent

23. CurrentInventory

Type summary: type="object"; additionalProperties=false

Required: record_type, candidate_digest, manifest_digest, inventory_digest, stream_digest, stream_bytes, entry_count, projection_row_count, project_tree_passes, inventory_passes, rebuild_product_passes, search_text_rule, projection_authoritative

Property	Required	Structural rule
candidate_digest	yes	\$ref="#/\$defs/Digest"
entry_count	yes	type="integer"; minimum=0
inventory_digest	yes	\$ref="#/\$defs/Digest"
inventory_passes	yes	type="integer"; const=1
manifest_digest	yes	\$ref="#/\$defs/Digest"
project_tree_passes	yes	type="integer"; minimum=0; maximum=1
projection_authoritative	yes	const=false
projection_row_count	yes	type="integer"; minimum=0
rebuild_product_passes	yes	type="integer"; const=0
record_type	yes	const="CurrentInventory"
search_text_rule	yes	const="exact NFC-normalized canonical path, UTF-8 bounded by Conformance scale_contracts.inventory.search_text_bytes_max"
stream_bytes	yes	type="integer"; minimum=1
stream_digest	yes	\$ref="#/\$defs/Digest"
<pre>{ "additionalProperties": false, "properties": { "candidate_digest": { "\$ref": "#/\$defs/Digest" }, "entry_count": {</pre>		

```

    "minimum": 0,
    "type": "integer"
  },
  "inventory_digest": {
    "$ref": "#/$defs/Digest"
  },
  "inventory_passes": {
    "const": 1,
    "type": "integer"
  },
  "manifest_digest": {
    "$ref": "#/$defs/Digest"
  },
  "project_tree_passes": {
    "maximum": 1,
    "minimum": 0,
    "type": "integer"
  },
  "projection_authoritative": {
    "const": false
  },
  "projection_row_count": {
    "minimum": 0,
    "type": "integer"
  },
  "rebuild_product_passes": {
    "const": 0,
    "type": "integer"
  },
  "record_type": {
    "const": "CurrentInventory"
  },
  "search_text_rule": {
    "const": "exact NFC-normalized canonical path, UTF-8 bounded by Conformance
scale_contracts.inventory.search_text_bytes_max"
  },
  "stream_bytes": {
    "minimum": 1,
    "type": "integer"
  },
  "stream_digest": {
    "$ref": "#/$defs/Digest"
  }
},
"required": [
  "record_type",
  "candidate_digest",
  "manifest_digest",
  "inventory_digest",
  "stream_digest",
  "stream_bytes",
  "entry_count",
  "projection_row_count",
  "project_tree_passes",
  "inventory_passes",
  "rebuild_product_passes",
  "search_text_rule",
  "projection_authoritative"
],
"type": "object",
"x-promin-ingress-operations": [
  "rebuild"
],
"x-promin-persistence": "derived-rebuild-only"
}

```

Owner: `core/contracts.schema.json#/$defs/CurrentInventory` | Validator: `urn:promin:contracts:1#/$defs/CurrentInventory`

24. Decision

Type summary: `type="object"; additionalProperties=false; allOf[7]=keys:if,then | keys:if,then | keys:if,then | keys:else,if,then,title | keys:else,if,then,title | keys:else,if,then,title | keys:else,if,then,title`

Required: `record_type`, `decision_id`, `decision_kind`, `subject_id`, `grant_id`, `grant_claim_digest`, `activation_digest`, `target_type`, `target_id`, `target_digest`, `rationale`, `evidence_artifacts`, `created_at`

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
candidate_digest	no	\$ref="#/\$defs/Digest"
created_at	yes	\$ref="#/\$defs/Timestamp"
decision_id	yes	\$ref="#/\$defs/Id"
decision_kind	yes	enum=["accept", "reject", "resolve", "waive", "promote", "release", "revoke"]
evidence_artifacts	yes	type="array"; minItems=1; maxItems=64; uniqueItems=true
exception_policy_digest	no	\$ref="#/\$defs/Digest"
expires_at	no	\$ref="#/\$defs/Timestamp"
finding_digest	no	\$ref="#/\$defs/Digest"
grant_claim_digest	yes	\$ref="#/\$defs/Digest"
grant_id	yes	\$ref="#/\$defs/Id"
provider_binding_digest	no	\$ref="#/\$defs/Digest"
rationale	yes	type="string"; minLength=1
record_type	yes	const="Decision"
release_closure_digest	no	\$ref="#/\$defs/Digest"
subject_id	yes	\$ref="#/\$defs/Id"
target_digest	yes	\$ref="#/\$defs/Digest"
target_id	yes	\$ref="#/\$defs/Id"
target_type	yes	enum=["Task", "Candidate", "Finding", "Grant"]

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "if": {
        "properties": {
          "decision_kind": {
            "enum": [
              "resolve",
              "waive"
            ]
          }
        }
      },
      "then": {
        "properties": {
          "target_type": {
            "const": "Finding"
          }
        }
      }
    },
    {
      "if": {
        "properties": {
          "decision_kind": {
            "enum": [
              "promote",
              "release"
            ]
          }
        }
      },
      "then": {
        "properties": {
          "target_type": {
            "const": "Candidate"
          }
        }
      }
    },
    {
      "if": {
        "properties": {
          "decision_kind": {
            "enum": [

```



```

        "accept",
        "reject"
    ]
    }
  },
  "then": {
    "properties": {
      "target_type": {
        "enum": [

```

```

          "Task",
          "Candidate"
        ]
      }
    }
  },
  {
    "else": {
      "not": {
        "anyOf": [
          {
            "required": [
              "release_closure_digest"
            ]
          },
          {
            "required": [
              "provider_binding_digest"
            ]
          }
        ]
      }
    },
    "if": {
      "properties": {
        "decision_kind": {
          "const": "release"

```

```

        }
      },
      "required": [
        "decision_kind"
      ]
    },
    "then": {
      "required": [
        "release_closure_digest",
        "provider_binding_digest"
      ]
    },
    "title": "release Decision binds exact historical closure and providers"
  },
  {
    "else": {
      "not": {
        "required": [
          "finding_digest"
        ]
      }
    },
    "if": {
      "properties": {
        "decision_kind": {
          "enum": [
            "resolve",
            "waive"

```

```

        ]
      }
    },
    "required": [
      "decision_kind"
    ]
  },
  "then": {
    "required": [
      "finding_digest"
    ]

```

```

    },
    "title": "Finding disposition binds exact canonical Finding digest"
  },
  {
    "else": {
      "not": {
        "anyOf": [
          {
            "required": [
              "exception_policy_digest"
            ]
          },
          {
            "required": [
              "expires_at"
            ]
          }
        ]
      }
    }
  }

```

```

  ]
}
},
"if": {
  "properties": {
    "decision_kind": {
      "const": "waive"
    }
  },
  "required": [
    "decision_kind"
  ]
},
"then": {
  "required": [
    "exception_policy_digest",
    "expires_at"
  ]
},
"title": "waiver binds exception policy and expiry"
},
{
  "else": {
    "required": [
      "candidate_digest"
    ]
  },
  "if": {

```

```

    "properties": {
      "decision_kind": {
        "const": "revoke"
      }
    },
    "required": [
      "decision_kind"
    ]
  },
  "then": {
    "not": {
      "required": [
        "candidate_digest"
      ]
    },
    "properties": {
      "target_type": {
        "const": "Grant"
      }
    }
  }
},
"title": "Grant revocation has no Candidate binding"
}
],
"properties": {
  "activation_digest": {
    "$ref": "#/$defs/Digest"
  },

```

```

  "candidate_digest": {
    "$ref": "#/$defs/Digest"
  },
  "created_at": {

```

```

    "$ref": "#/$defs/Timestamp"
  },
  "decision_id": {
    "$ref": "#/$defs/Id"
  },
  "decision_kind": {
    "enum": [
      "accept",
      "reject",
      "resolve",
      "waive",
      "promote",
      "release",
      "revoke"
    ]
  },
  "evidence_artifacts": {
    "items": {
      "additionalProperties": false,
      "properties": {
        "artifact_id": {
          "$ref": "#/$defs/Id"
        },
        "artifact_record_digest": {

```

```

          "$ref": "#/$defs/Digest"
        }
      },
      "required": [
        "artifact_id",
        "artifact_record_digest"
      ],
      "type": "object"
    },
    "maxItems": 64,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "exception_policy_digest": {
    "$ref": "#/$defs/Digest"
  },
  "expires_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "finding_digest": {
    "$ref": "#/$defs/Digest"
  },
  "grant_claim_digest": {
    "$ref": "#/$defs/Digest"
  },
  "grant_id": {
    "$ref": "#/$defs/Id"

```

```

  },
  "provider_binding_digest": {
    "$ref": "#/$defs/Digest"
  },
  "rationale": {
    "minLength": 1,
    "type": "string"
  },
  "record_type": {
    "const": "Decision"
  },
  "release_closure_digest": {
    "$ref": "#/$defs/Digest"
  },
  "subject_id": {
    "$ref": "#/$defs/Id"
  },
  "target_digest": {
    "$ref": "#/$defs/Digest"
  },
  "target_id": {
    "$ref": "#/$defs/Id"
  },
  "target_type": {
    "enum": [
      "Task",

```

```

        "Candidate",
        "Finding",

        "Grant"
    ]
},
"required": [
    "record_type",
    "decision_id",
    "decision_kind",
    "subject_id",
    "grant_id",
    "grant_claim_digest",
    "activation_digest",
    "target_type",
    "target_id",
    "target_digest",
    "rationale",
    "evidence_artifacts",
    "created_at"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Decision | Validator: urn:promin:contracts:1#/\$defs/Decision

25. Digest

Type summary: type="string"; pattern="^[0-9a-f]{64}\$"

Required: -

```

{
  "pattern": "^[0-9a-f]{64}$",
  "type": "string"
}

```

Owner: core/contracts.schema.json#/\$defs/Digest | Validator: urn:promin:contracts:1#/\$defs/Digest

26. DoctorResult

Type summary: type="object"; additionalProperties=false

Required: record_type, status, claim_scope, source_scope, report_authoritative, pass_credit, product_acceptance_pass, product_public_approval, activation_digest, implementation_closure_digest, head, recovery, replay, projection, provider_adapters, metrics, components, archival_diagnostics, operation_metrics, product_tree_scans

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
archival_diagnostics	yes	type="object"; additionalProperties=false
claim_scope	yes	const="runtime-health-only"
components	yes	type="object"; additionalProperties=false
head	yes	type="object"; additionalProperties=false
implementation_closure_digest	yes	\$ref="#/\$defs/Digest"
metrics	yes	type="object"; additionalProperties=false
operation_metrics	yes	oneOf[2]=#/\$defs/OperationMetrics type:null
pass_credit	yes	const=false
product_acceptance_pass	yes	const=false
product_public_approval	yes	const="not_approved"
product_tree_scans	yes	const=0
projection	yes	oneOf[2]=type:object type:object
provider_adapters	yes	type="array"; minItems=1; maxItems=32
record_type	yes	const="DoctorResult"
recovery	yes	type="object"; additionalProperties=false
replay	yes	oneOf[2]=type:null type:object
report_authoritative	yes	const=false

Property	Required	Structural rule
source_scope	yes	const="installed-core-and-live-operational-state-only"
status	yes	enum=["healthy", "degraded", "failed", "incomplete"]
<pre> { "additionalProperties": false, "properties": { "activation_digest": { "\$ref": "#/\$defs/Digest" }, "archival_diagnostics": { "additionalProperties": false, "properties": { "affects_current_status": { "const": false }, "entries": { "items": { "additionalProperties": false, "properties": { "detail_digest": { "\$ref": "#/\$defs/Digest" }, "diagnostic_id": { "\$ref": "#/\$defs/Id" }, "recorded_at": { "\$ref": "#/\$defs/Timestamp" }, "status": { "enum": ["healthy", "degraded", "failed", "incomplete"] } }, "required": ["diagnostic_id", "status", "recorded_at", "detail_digest"], "type": "object" }, "maxItems": 256, "minItems": 0, "type": "array" }, "pass_credit": { "const": false } }, "required": ["affects_current_status", "pass_credit", "entries"], "type": "object" }, "claim_scope": { "const": "runtime-health-only" }, "components": { "additionalProperties": false, "properties": { "core": { "additionalProperties": false, "properties": { "observed_digest": { "oneOf": [{ "\$ref": "#/\$defs/Digest" }, { "type": "null" }] } } } } } } } </pre>		

```

    }
  ]
},
"reason_codes": {
  "items": {
    "$ref": "#/$defs/Id"
  },
  "maxItems": 32,
  "minItems": 0,
  "type": "array",
  "uniqueItems": true
}

```

```

},
"status": {
  "enum": [
    "healthy",
    "degraded",
    "failed",
    "incomplete"
  ]
}
},
"required": [
  "status",
  "reason_codes",
  "observed_digest"
],
"type": "object"
},
"init": {
  "additionalProperties": false,
  "properties": {
    "observed_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    }
  }
}

```

```

]
},
"reason_codes": {
  "items": {
    "$ref": "#/$defs/Id"
  },
  "maxItems": 32,
  "minItems": 0,
  "type": "array",
  "uniqueItems": true
},
"status": {
  "enum": [
    "healthy",
    "degraded",
    "failed",
    "incomplete"
  ]
}
},
"required": [
  "status",
  "reason_codes",
  "observed_digest"
],
"type": "object"
},
"projection": {

```

```

  "additionalProperties": false,
  "properties": {
    "observed_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    }
  }
}

```

```

    ]
  },
  "reason_codes": {
    "items": {
      "$ref": "#/$defs/Id"
    },
    "maxItems": 32,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
  },
  "status": {
    "enum": [
      "healthy",
      "degraded",
      "failed",
      "incomplete"
    ]
  }
]

```

```

    }
  },
  "required": [
    "status",
    "reason_codes",
    "observed_digest"
  ],
  "type": "object"
},
"providers": {
  "additionalProperties": false,
  "properties": {
    "observed_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    }
  },
  "reason_codes": {
    "items": {
      "$ref": "#/$defs/Id"
    },
    "maxItems": 32,
    "minItems": 0,

```

```

    "type": "array",
    "uniqueItems": true
  },
  "status": {
    "enum": [
      "healthy",
      "degraded",
      "failed",
      "incomplete"
    ]
  }
},
"required": [
  "status",
  "reason_codes",
  "observed_digest"
],
"type": "object"
},
"recovery": {
  "additionalProperties": false,
  "properties": {
    "observed_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    }
  }
}
]

```

```

    },
    "reason_codes": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 32,
      "minItems": 0,
      "type": "array",
      "uniqueItems": true
    },
    "status": {
      "enum": [
        "healthy",
        "degraded",
        "failed",
        "incomplete"
      ]
    }
  },
  "required": [
    "status",
    "reason_codes",
    "observed_digest"
  ],
  "type": "object"

```

```

},
"replay": {
  "additionalProperties": false,
  "properties": {
    "observed_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    }
  },
  "reason_codes": {
    "items": {
      "$ref": "#/$defs/Id"
    },
    "maxItems": 32,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
  },
  "status": {
    "enum": [
      "healthy",
      "degraded",
      "failed",

```

```

      "incomplete"
    ]
  },
  "required": [
    "status",
    "reason_codes",
    "observed_digest"
  ],
  "type": "object"
}
},
"required": [
  "core",
  "init",
  "providers",
  "recovery",
  "replay",
  "projection"
],
"type": "object"
},
"head": {
  "additionalProperties": false,
  "properties": {

```



```

"batch_digest": {
  "oneOf": [
    {
      "$ref": "#/$defs/Digest"
    },
    {
      "type": "null"
    }
  ]
},
"batch_id": {
  "oneOf": [
    {
      "$ref": "#/$defs/Id"
    },
    {
      "type": "null"
    }
  ]
},
"sequence": {
  "minimum": 0,
  "type": "integer"
}
],
"required": [
  "sequence",
  "batch_id",
  "batch_digest"
],
"type": "object"
},
"implementation_closure_digest": {
  "$ref": "#/$defs/Digest"
},
"metrics": {
  "additionalProperties": false,
  "properties": {
    "core_artifacts_verified": {
      "minimum": 0,
      "type": "integer"
    },
    "event_batches": {
      "minimum": 0,
      "type": "integer"
    },
    "init_records_verified": {
      "minimum": 0,
      "type": "integer"
    },
    "projection_entities": {
      "oneOf": [
        {
          "minimum": 0,
          "type": "integer"
        },
        {
          "type": "null"
        }
      ]
    }
  ]
},
"projection_relations": {
  "oneOf": [
    {
      "minimum": 0,
      "type": "integer"
    },
    {
      "type": "null"
    }
  ]
},
"provider_healthchecks_executed": {
  "minimum": 0,
  "type": "integer"
}
},

```

```

    "required": [
      "core_artifacts_verified",
      "event_batches",
      "init_records_verified",
      "projection_entities",
      "projection_relations",
      "provider_healthchecks_executed"
    ],
    "type": "object"
  },

```

```

    "operation_metrics": {
      "oneOf": [
        {
          "$ref": "#/$defs/OperationMetrics"
        },
        {
          "type": "null"
        }
      ]
    },
    "pass_credit": {
      "const": false
    },
    "product_acceptance_pass": {
      "const": false
    },
    "product_public_approval": {
      "const": "not_approved"
    },
    "product_tree_scans": {
      "const": 0
    },
    "projection": {
      "oneOf": [
        {
          "additionalProperties": false,
          "properties": {
            "projection_authoritative": {

```

```

              "const": false
            },
            "status": {
              "const": "missing"
            }
          },
          "required": [
            "status",
            "projection_authoritative"
          ],
          "type": "object"
        },
        {
          "additionalProperties": false,
          "properties": {
            "activation_digest": {
              "$ref": "#/$defs/Digest"
            },
            "built_at": {
              "$ref": "#/$defs/Timestamp"
            },
            "entity_count": {
              "minimum": 0,
              "type": "integer"
            },
            "event_count": {
              "minimum": 0,
              "type": "integer"

```

```

            },
            "head_digest": {
              "oneOf": [
                {
                  "$ref": "#/$defs/Digest"
                },
                {
                  "type": "null"
                }
              ]
            }
          ],
        },

```

```

"head_sequence": {
  "minimum": 0,
  "type": "integer"
},
"implementation_closure_digest": {
  "$ref": "#/$defs/Digest"
},
"incremental_changed_records": {
  "minimum": 0,
  "type": "integer"
},
"incremental_commit_count": {
  "minimum": 0,
  "type": "integer"
},
"inventory_entries": {
  "minimum": 0,

```

```

    "type": "integer"
  },
  "inventory_passes": {
    "minimum": 0,
    "type": "integer"
  },
  "inventory_proxies": {
    "minimum": 0,
    "type": "integer"
  },
  "inventory_relations": {
    "minimum": 0,
    "type": "integer"
  },
  "product_passes": {
    "minimum": 0,
    "type": "integer"
  },
  "projection_authoritative": {
    "const": false
  },
  "projection_compaction_count": {
    "minimum": 0,
    "type": "integer"
  },
  "projection_db_bytes": {
    "minimum": 0,
    "type": "integer"

```

```

  },
  "raw_file_proxy_ratio": {
    "minimum": 0,
    "type": "number"
  },
  "relation_count": {
    "minimum": 0,
    "type": "integer"
  },
  "semantic_digest": {
    "$ref": "#/$defs/Digest"
  },
  "semantic_digest_algorithm": {
    "maxLength": 128,
    "minLength": 1,
    "type": "string"
  },
  "storage_layout": {
    "maxLength": 128,
    "minLength": 1,
    "type": "string"
  },
  "synthetic_task_count": {
    "const": 0,
    "type": "integer"
  },
  "synthetic_task_ratio": {
    "const": 0.0,

```

```

    "type": "number"
  }
},
"required": [

```

```

"activation_digest",
"head_digest",
"head_sequence",
"semantic_digest",
"entity_count",
"relation_count",
"inventory_entries",
"inventory_proxies",
"inventory_relations",
"inventory_passes",
"product_passes",
"event_count",
"built_at",
"projection_authoritative",
"implementation_closure_digest",
"semantic_digest_algorithm",
"storage_layout",
"incremental_commit_count",
"incremental_changed_records",
"projection_compaction_count",
"raw_file_proxy_ratio",
"synthetic_task_count",
"synthetic_task_ratio",
"projection_db_bytes"

```

```

],
"type": "object"
}
]
},
"provider_adapters": {
  "items": {
    "additionalProperties": false,
    "oneOf": [
      {
        "properties": {
          "adapter_id": {
            "const": "python-runtime-v1"
          },
          "capability_id": {
            "const": "control-runtime"
          },
          "identity_kind": {
            "const": "file-digest"
          },
          "invocation_kind": {
            "const": "python-runtime"
          },
          "operation": {
            "const": "healthcheck"
          },
          "operation_contract_digest": {
            "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"

```

```

},
"protocol_id": {
  "const": "promin.control-runtime.v1"
}
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "shape-validation"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
    },

```

```

    "protocol_id": {
      "const": "promin.shape-validation.v1"
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "content-identity"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "healthcheck"
      },
      "operation_contract_digest": {
        "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
      },
      "protocol_id": {
        "const": "promin.content-identity.v1"
      }
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "local-serialization"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "healthcheck"
      },
      "operation_contract_digest": {
        "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
      },
      "protocol_id": {
        "const": "promin.local-serialization.v1"
      }
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "query-projection"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "healthcheck"
      },
      "operation_contract_digest": {
        "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
      },
      "protocol_id": {
        "const": "promin.query-projection.v1"
      }
    }
  }
}

```

```

    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "git-executable-v1"
      },
      "capability_id": {
        "const": "filesystem-inventory"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "executable"
      },
      "operation": {
        "const": "healthcheck"
      },
      "operation_contract_digest": {
        "const": "053c4181d6d35bd4ca5a13b8a1f18c19571c21b3426644bfa6bda99066dce5b3"
      },
      "protocol_id": {
        "const": "promin.filesystem-inventory.v1"
      }
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "executable-export-scan-v1"
      },
      "capability_id": {
        "const": "export-scan"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "executable"
      },
      "operation": {
        "const": "healthcheck"
      },
      "operation_contract_digest": {
        "const": "116e5e8a519592f63585f0fae7b90f86863ba3369f0b006913a0e4f7d51eaf2e"
      },
      "protocol_id": {
        "const": "promin.export-scan.v1"
      }
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "executable-signature-v1"
      },
      "capability_id": {
        "const": "signature"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "executable"
      },
      "operation": {
        "const": "healthcheck"
      },
      "operation_contract_digest": {
        "const": "0e2de1aec47aec67eachbec7838d7f7f19f457bcff9c3fae57c9e88a72b6bb6be"
      },
      "protocol_id": {
        "const": "promin.signature.v1"
      }
    }
  }
}

```

```

},
{
  "properties": {
    "adapter_id": {
      "const": "python-module-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "module-file-digest"
    },
    "invocation_kind": {
      "const": "python-module"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "0e2delaec47aec67eachbec7838d7f7f19f457bcff9c3fae57c9e88a72b6bb6be"
    },
    "protocol_id": {
      "const": "promin.signature.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-build-dependency-v1"
    },
    "capability_id": {
      "const": "build-dependency"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "116e5e8a519592f63585f0fae7b90f86863ba3369f0b006913a0e4f7d51eaf2e"
    },
    "protocol_id": {
      "const": "promin.build-dependency.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "$ref": "#/$defs/Id"
    },
    "authoritative": {
      "const": false
    },
    "capability_id": {
      "enum": [
        "control-runtime",
        "shape-validation",
        "content-identity",
        "local-serialization",
        "query-projection",
        "filesystem-inventory",
        "export-scan",
        "signature",
        "build-dependency"
      ]
    }
  }
},
{
  "completed_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "dependency_receipt_digest": {
    "$ref": "#/$defs/Digest"
  }
}

```

```

    },
    "exit_code": {
      "oneOf": [
        {
          "type": "integer"
        },
        {
          "type": "null"
        }
      ]
    },
    "identity_digest": {
      "$ref": "#/$defs/Digest"
    },
    "identity_kind": {
      "enum": [
        "file-digest",
        "module-file-digest",
        "distribution-tree-digest",
        "module-closure-digest",
        "platform-attestation"
      ]
    }
  },

```

```

    "invocation_kind": {
      "enum": [
        "python-runtime",
        "python-module",
        "executable",
        "platform-service"
      ]
    },
    "invocation_receipt_digest": {
      "$ref": "#/$defs/Digest"
    },
    "invoked": {
      "const": true
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "$ref": "#/$defs/Digest"
    },
    "outcome": {
      "enum": [
        "healthy",
        "degraded",
        "failed"
      ]
    },
    "pass_credit": {

```

```

      "const": false
    },
    "persistence_scope": {
      "const": "content-addressed-receipt"
    },
    "protocol_id": {
      "$ref": "#/$defs/Id"
    },
    "provider_id": {
      "$ref": "#/$defs/Id"
    },
    "reconstructable": {
      "const": true
    },
    "started_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "stderr_digest": {
      "$ref": "#/$defs/Digest"
    },
    "stderr_size_bytes": {
      "maximum": 1048576,
      "minimum": 0,
      "type": "integer"
    },
    "stdout_digest": {

```



```
    "$ref": "#/$defs/Digest"
  },
```

```
    "stdout_size_bytes": {
      "maximum": 1048576,
      "minimum": 0,
      "type": "integer"
    }
  },
  "required": [
    "capability_id",
    "provider_id",
    "invocation_kind",
    "identity_kind",
    "identity_digest",
    "adapter_id",
    "protocol_id",
    "operation",
    "operation_contract_digest",
    "invocation_receipt_digest",
    "dependency_receipt_digest",
    "persistence_scope",
    "reconstructable",
    "invoked",
    "started_at",
    "completed_at",
    "outcome",
    "exit_code",
    "stdout_digest",
    "stdout_size_bytes",
    "stderr_digest",
```

```
    "stderr_size_bytes",
    "authoritative",
    "pass_credit"
  ],
  "type": "object"
},
"maxItems": 32,
"minItems": 1,
"type": "array"
},
"record_type": {
  "const": "DoctorResult"
},
"recovery": {
  "additionalProperties": false,
  "properties": {
    "batch_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    }
  }
},
"batch_id": {
  "oneOf": [
```

```
    {
      "$ref": "#/$defs/Id"
    },
    {
      "type": "null"
    }
  ]
},
"sequence": {
  "minimum": 0,
  "type": "integer"
}
},
"required": [
  "sequence",
  "batch_id",
  "batch_digest"
],
"type": "object"
```

```

},
"replay": {
  "oneOf": [
    {
      "type": "null"
    },
    {
      "additionalProperties": false,
      "properties": {

```

```

        "event_batches": {
          "minimum": 0,
          "type": "integer"
        },
        "head": {
          "additionalProperties": false,
          "properties": {
            "batch_digest": {
              "oneOf": [
                {
                  "$ref": "#/$defs/Digest"
                },
                {
                  "type": "null"
                }
              ]
            },
            "batch_id": {
              "oneOf": [
                {
                  "$ref": "#/$defs/Id"
                },
                {
                  "type": "null"
                }
              ]
            }
          },
          "sequence": {

```

```

            "minimum": 0,
            "type": "integer"
          }
        },
        "required": [
          "sequence",
          "batch_id",
          "batch_digest"
        ],
        "type": "object"
      },
      "record_type": {
        "const": "ReplayResult"
      },
      "runtime_checkpoint": {
        "oneOf": [
          {
            "additionalProperties": false,
            "properties": {
              "authoritative": {
                "const": false
              },
              "batch_threshold": {
                "minimum": 1,
                "type": "integer"
              },
              "byte_threshold": {
                "minimum": 1,

```

```

                "type": "integer"
              },
              "checkpoint_bytes": {
                "minimum": 0,
                "type": "integer"
              },
              "checkpoint_count": {
                "minimum": 0,
                "type": "integer"
              },
              "tail_batches": {
                "minimum": 0,

```

```

        "type": "integer"
    },
    "tail_bytes": {
        "minimum": 0,
        "type": "integer"
    },
    "written": {
        "const": false
    }
},
"required": [
    "authoritative",
    "batch_threshold",
    "byte_threshold",
    "checkpoint_bytes",
    "checkpoint_count",
    "tail_batches",
    "tail_bytes",
    "written"
],
"type": "object"
},
{
    "additionalProperties": false,
    "properties": {
        "authoritative": {
            "const": false
        },
        "batch_threshold": {
            "minimum": 1,
            "type": "integer"
        },
        "byte_threshold": {
            "minimum": 1,
            "type": "integer"
        },
        "checkpoint_bytes": {
            "minimum": 0,
            "type": "integer"
        },
        "checkpoint_count": {
            "minimum": 0,
            "type": "integer"
        },
        "compacted_tail_batches": {
            "minimum": 0,
            "type": "integer"
        },
        "compacted_tail_bytes": {
            "minimum": 0,
            "type": "integer"
        },
        "tail_batches": {
            "minimum": 0,
            "type": "integer"
        },
        "tail_bytes": {
            "minimum": 0,
            "type": "integer"
        },
        "written": {
            "const": true
        }
    },
    "required": [
        "authoritative",
        "batch_threshold",
        "byte_threshold",
        "checkpoint_bytes",
        "checkpoint_count",
        "tail_batches",
        "tail_bytes",
        "written",
        "compacted_tail_batches",
        "compacted_tail_bytes"
    ],
    "type": "object"
}

```

```

    }
  ],
  "runtime_checkpoint_authoritative": {
    "const": false
  },
  "runtime_checkpoint_written": {
    "type": "boolean"
  },
  "status": {
    "const": "pass"
  }
},
"required": [
  "record_type",
  "status",
  "event_batches",
  "head",
  "runtime_checkpoint_written",
  "runtime_checkpoint",
  "runtime_checkpoint_authoritative"
],
"type": "object"

```

```

    }
  ],
  "report_authoritative": {
    "const": false
  },
  "source_scope": {
    "const": "installed-core-and-live-operational-state-only"
  },
  "status": {
    "enum": [
      "healthy",
      "degraded",
      "failed",
      "incomplete"
    ]
  }
},
"required": [
  "record_type",
  "status",
  "claim_scope",
  "source_scope",
  "report_authoritative",
  "pass_credit",
  "product_acceptance_pass",
  "product_public_approval",
  "activation_digest",

```

```

  "implementation_closure_digest",
  "head",
  "recovery",
  "replay",
  "projection",
  "provider_adapters",
  "metrics",
  "components",
  "archival_diagnostics",
  "operation_metrics",
  "product_tree_scans"
],
"type": "object",
"x-promin-persistence": "derived-live-only"
}

```

Owner: `core/contracts.schema.json#/$defs/DoctorResult` | Validator: `urn:promin:contracts:1#/$defs/DoctorResult`

27. EffectTarget

Type summary: type="object"; additionalProperties=false

Required: kind, id_field

Property	Required	Structural rule
id_field	yes	<code>\$ref="#/\$defs/Id"</code>

Property	Required	Structural rule
kind	yes	\$ref="#/\$defs/Id"
<pre>{ "additionalProperties": false, "properties": { "id_field": { "\$ref": "#/\$defs/Id" }, "kind": { "\$ref": "#/\$defs/Id" } }, "required": ["kind", "id_field"], "type": "object" }</pre>		
Owner: core/contracts.schema.json#/\$defs/EffectTarget Validator: urn:promin:contracts:1#/\$defs/EffectTarget		

28. Event

Type summary: type="object"; additionalProperties=false; allOf[12]=keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then

Required: record_type, event_id, event_kind, activation_digest, payload

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
event_id	yes	\$ref="#/\$defs/Id"
event_kind	yes	enum=["artifact.recorded", "candidate.recorded", "decision.recorded", "finding.recorded", "gate.recorded", "grant.issued", "grant.revoked", "lease.recorded", "run.recorded", "task.recorded", "task.transitioned", "relation.recorded"]
payload	yes	composed structural definition
record_type	yes	const="Event"
<pre>{ "additionalProperties": false, "allOf": [{ "if": { "properties": { "event_kind": { "const": "task.recorded" } } }, "then": { "properties": { "payload": { "\$ref": "#/\$defs/Task" } } } }, { "if": { "properties": { "event_kind": { "const": "task.transitioned" } } }, "then": { "properties": { "payload": { "\$ref": "#/\$defs/TaskTransition" } } } }], { </pre>		

```

    "if": {
      "properties": {
        "event_kind": {
          "const": "candidate.recorded"
        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/Candidate"
        }
      }
    }
  },
  {
    "if": {
      "properties": {
        "event_kind": {
          "const": "artifact.recorded"

```

```

        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/Artifact"
        }
      }
    }
  },
  {
    "if": {
      "properties": {
        "event_kind": {
          "const": "run.recorded"
        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/Run"
        }
      }
    }
  },
  {

```

```

    "if": {
      "properties": {
        "event_kind": {
          "const": "gate.recorded"
        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/GateResult"
        }
      }
    }
  },
  {
    "if": {
      "properties": {
        "event_kind": {
          "const": "finding.recorded"
        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/Finding"
        }
      }
    }
  }
}

```

```

    }
  },
  {
    "if": {
      "properties": {
        "event_kind": {
          "const": "decision.recorded"
        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/Decision"
        }
      }
    }
  },
  {
    "if": {
      "properties": {
        "event_kind": {
          "const": "grant.issued"
        }
      }
    },
    "then": {

```

```

      "properties": {
        "payload": {
          "$ref": "#/$defs/Grant"
        }
      }
    },
    {
      "if": {
        "properties": {
          "event_kind": {
            "const": "grant.revoked"
          }
        }
      },
      "then": {
        "properties": {
          "payload": {
            "$ref": "#/$defs/GrantRevocation"
          }
        }
      }
    },
    {
      "if": {
        "properties": {
          "event_kind": {
            "const": "lease.recorded"

```

```

      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/Lease"
        }
      }
    }
  },
  {
    "if": {
      "properties": {
        "event_kind": {
          "const": "relation.recorded"
        }
      }
    },
    "then": {
      "properties": {
        "payload": {

```

```

        "$ref": "#/$defs/Relation"
      }
    }
  }
},
],
"properties": {
  "activation_digest": {
    "$ref": "#/$defs/Digest"
  },
  "event_id": {
    "$ref": "#/$defs/Id"
  },
  "event_kind": {
    "enum": [
      "artifact.recorded",
      "candidate.recorded",
      "decision.recorded",
      "finding.recorded",
      "gate.recorded",
      "grant.issued",
      "grant.revoked",
      "lease.recorded",
      "run.recorded",
      "task.recorded",
      "task.transitioned",
      "relation.recorded"
    ]
  },
  "payload": {},
  "record_type": {
    "const": "Event"
  }
},
"required": [
  "record_type",
  "event_id",
  "event_kind",
  "activation_digest",
  "payload"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Event | Validator: urn:promin:contracts:1#/\$defs/Event

29. EventBatch

Type summary: type="object"; additionalProperties=false

Required: record_type, batch_id, sequence, previous_digest, previous_authority_commitment, authority_commitment, cumulative_event_count, event_semantic_digest, state_binding_delta, cumulative_state_binding_update_count, state_binding_digest, created_at, command_id, idempotency_key, activation_record_digest, events, subject_id, command_intent_digest, command_digest, authorization_digest

Property	Required	Structural rule
activation_record_digest	yes	\$ref="#/\$defs/Digest"
authority_commitment	yes	\$ref="#/\$defs/Digest"
authorization_digest	yes	\$ref="#/\$defs/Digest"
batch_id	yes	\$ref="#/\$defs/Id"
command_digest	yes	\$ref="#/\$defs/Digest"
command_id	yes	\$ref="#/\$defs/Id"
command_intent_digest	yes	\$ref="#/\$defs/Digest"
created_at	yes	\$ref="#/\$defs/Timestamp"
cumulative_event_count	yes	type="integer"; minimum=1
cumulative_state_binding_update_count	yes	type="integer"; minimum=1
event_semantic_digest	yes	\$ref="#/\$defs/Digest"
events	yes	type="array"; minItems=1; maxItems=20
idempotency_key	yes	type="string"; pattern="^[A-Za-z0-9._-]{16,128}\$"

Property	Required	Structural rule
previous_authority_commitment	yes	\$ref="#/\$defs/Digest"
previous_digest	yes	oneOf[2]=#/\$defs/Digest type:null
record_type	yes	const="EventBatch"
sequence	yes	type="integer"; minimum=1
state_binding_delta	yes	type="array"; minItems=1; maxItems=20; uniqueItems=true
state_binding_digest	yes	\$ref="#/\$defs/Digest"
subject_id	yes	\$ref="#/\$defs/Id"
<pre> { "additionalProperties": false, "properties": { "activation_record_digest": { "\$ref": "#/\$defs/Digest" }, "authority_commitment": { "\$ref": "#/\$defs/Digest" }, "authorization_digest": { "\$ref": "#/\$defs/Digest" }, "batch_id": { "\$ref": "#/\$defs/Id" }, "command_digest": { "\$ref": "#/\$defs/Digest" }, "command_id": { "\$ref": "#/\$defs/Id" }, "command_intent_digest": { "\$ref": "#/\$defs/Digest" }, "created_at": { "\$ref": "#/\$defs/Timestamp" }, "cumulative_event_count": { "minimum": 1, "type": "integer" }, "cumulative_state_binding_update_count": { "minimum": 1, "type": "integer" }, "event_semantic_digest": { "\$ref": "#/\$defs/Digest" }, "events": { "items": { "\$ref": "#/\$defs/Event" }, "maxItems": 20, "minItems": 1, "type": "array" }, "idempotency_key": { "pattern": "^[A-Za-z0-9._:-]{16,128}\$", "type": "string" }, "previous_authority_commitment": { "\$ref": "#/\$defs/Digest" }, "previous_digest": { "oneOf": [{ "\$ref": "#/\$defs/Digest" }, { "type": "null" }] }, "record_type": { "const": "EventBatch" } } } </pre>		

```

    },
    "sequence": {
      "minimum": 1,
      "type": "integer"
    },
    "state_binding_delta": {
      "items": {
        "$ref": "#/$defs/StateBindingUpdate"
      },
      "maxItems": 20,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "state_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "record_type",
    "batch_id",
    "sequence",
    "previous_digest",
    "previous_authority_commitment",
    "authority_commitment",
    "cumulative_event_count",
    "event_semantic_digest",
    "state_binding_delta",
    "cumulative_state_binding_update_count",
    "state_binding_digest",
    "created_at",
    "command_id",
    "idempotency_key",
    "activation_record_digest",
    "events",
    "subject_id",
    "command_intent_digest",
    "command_digest",
    "authorization_digest"
  ],
  "type": "object",
  "x-promin-max-canonical-bytes": 2097152,
  "x-promin-max-state-binding-bytes": 1048576,
  "x-promin-primary-event-count-per-command": 1
}

```

Owner: `core/contracts.schema.json#/$defs/EventBatch` | Validator: `urn:promin:contracts:1#/$defs/EventBatch`

30. EventStorePolicy

Type summary: `type="object"; additionalProperties=false`

Required: `record_type`, `authority_model_digest`, `max_command_bytes`, `max_envelope_bytes`, `max_state_binding_bytes`, `max_state_binding_updates_per_batch`, `max_events_per_batch`, `max_requested_scope_items`, `derived_tail_batch_threshold`, `derived_tail_byte_threshold`, `runtime_overlay_compaction_depth`, `command_required_fields`, `command_conditional_fields`, `command_mutation_fields`, `lease_bound_command_kinds`, `lease_bound_task_transition_states`, `command_to_primary_event`, `allowed_state_binding_leaf_types`, `state_binding_identity_rules`, `state_binding_algorithm_contract`, `state_binding_value_rules`, `canonical_timestamp_contract`, `genesis_previous_authority_commitment`, `genesis_event_semantic_digest`, `policy_digest`

Property	Required	Structural rule
<code>allowed_state_binding_leaf_types</code>	yes	<code>const=["Activation", "Task", "Candidate", "Artifact", "Run", "Finding", "Decision", "GateResult", "Grant", "Lease", "Relation"]</code>
<code>authority_model_digest</code>	yes	<code>\$ref="#/\$defs/Digest"</code>
<code>canonical_timestamp_contract</code>	yes	<code>const=schema(keys:calendar,format,formatter_api,fractional_seconds,normalization,offset,parser_api,pattern,real_calendar_validation,resolution_seconds,year_max,year_min)</code>
<code>command_conditional_fields</code>	yes	<code>const=["definition_digest"]</code>
<code>command_mutation_fields</code>	yes	<code>const=["workcard_task_id", "holder_authorization", "lease_id", "lease_generation", "fencing_token", "workcard_digest", "context_digest"]</code>

Property	Required	Structural rule
command_required_fields	yes	const=["record_type", "command_id", "command_kind", "subject_id", "activation_digest", "idempotency_key", "requested_scope", "expected_head_digest", "issued_at", "payload", "intent_digest", "authorization"]
command_to_primary_event	yes	const=[11 items]
derived_tail_batch_threshold	yes	type="integer"; const=128
derived_tail_byte_threshold	yes	type="integer"; const=8388608
genesis_event_semantic_digest	yes	const="e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855"
genesis_previous_authority_commitment	yes	const="e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855"
lease_bound_command_kinds	yes	const=["artifact.record", "run.record", "gate.record", "finding.record"]
lease_bound_task_transition_states	yes	const=["LEASED", "RUNNING", "COMPLETED"]
max_command_bytes	yes	type="integer"; const=1048576
max_envelope_bytes	yes	type="integer"; const=2097152
max_events_per_batch	yes	type="integer"; const=20
max_requested_scope_items	yes	type="integer"; const=32
max_state_binding_bytes	yes	type="integer"; const=1048576
max_state_binding_updates_per_batch	yes	type="integer"; const=20
policy_digest	yes	\$ref="#/\$defs/Digest"
record_type	yes	const="EventStorePolicy"
runtime_overlay_compaction_depth	yes	type="integer"; const=64
state_binding_algorithm_contract	yes	const=schema(keys:delete_rule,delta_order,depth,depth_encoding,empty_leaf_domain_utf8,internal_node_domain_utf8,internal_node_rule,leaf_domain_utf8,leaf_key_rule,set_leaf_rule)
state_binding_identity_rules	yes	const=schema(keys:Activation,Artifact,Candidate,Decision,Finding,GateResult,Grant,Lease,Relation,Run,Task)
state_binding_value_rules	yes	const=schema(keys:Activation,Artifact,Candidate,Decision,Finding,GateResult,Grant,Lease,Relation,Run,Task)

```

{
  "additionalProperties": false,
  "properties": {
    "allowed_state_binding_leaf_types": {
      "const": [
        "Activation",
        "Task",
        "Candidate",
        "Artifact",
        "Run",
        "Finding",
        "Decision",
        "GateResult",
        "Grant",
        "Lease",
        "Relation"
      ]
    },
    "authority_model_digest": {
      "$ref": "#/$defs/Digest"
    },
    "canonical_timestamp_contract": {
      "const": {
        "calendar": "proleptic-gregorian",
        "format": "utc-second-z",
        "formatter_api": "promin.canonical.format_utc_second(value: datetime) -> str",
        "fractional_seconds": "reject",
        "normalization": "identity-round-trip",
        "offset": "Z-only",
        "parser_api": "promin.canonical.parse_utc_second(value: str) -> timezone-aware UTC datetime",
        "pattern": "^[0-9]{4}-[0-9]{2}-[0-9]{2}T[0-9]{2}:[0-9]{2}:[0-9]{2}Z$",
        "real_calendar_validation": true,
        "resolution_seconds": 1,
        "year_max": 9999,

```

```

        "year_min": 1
    }
},
"command_conditional_fields": {
    "const": [
        "definition_digest"
    ]
},
"command_mutation_fields": {
    "const": [
        "workcard_task_id",
        "holder_authorization",
        "lease_id",
        "lease_generation",
        "fencing_token",
        "workcard_digest",
        "context_digest"
    ]
},
"command_required_fields": {
    "const": [
        "record_type",

```

```

        "command_id",
        "command_kind",
        "subject_id",
        "activation_digest",
        "idempotency_key",
        "requested_scope",
        "expected_head_digest",
        "issued_at",
        "payload",
        "intent_digest",
        "authorization"
    ]
},
"command_to_primary_event": {
    "const": [
        [
            "artifact.record",
            "artifact.recorded"
        ],
        [
            "candidate.record",
            "candidate.recorded"
        ],
        [
            "decision.record",
            "decision.recorded"
        ],
        [

```

```

            "finding.record",
            "finding.recorded"
        ],
        [
            "gate.record",
            "gate.recorded"
        ],
        [
            "grant.issue",
            "grant.issued"
        ],
        [
            "grant.revoke",
            "grant.revoked"
        ],
        [
            "lease.record",
            "lease.recorded"
        ],
        [
            "run.record",
            "run.recorded"
        ],
        [
            "task.record",
            "task.recorded"
        ],
        [

```

```

        "task.transition",
        "task.transitioned"
    ]
},
"derived_tail_batch_threshold": {
    "const": 128,
    "type": "integer"
},
"derived_tail_byte_threshold": {
    "const": 8388608,
    "type": "integer"
},
"genesis_event_semantic_digest": {
    "const": "e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855"
},
"genesis_previous_authority_commitment": {
    "const": "e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855"
},
"lease_bound_command_kinds": {
    "const": [
        "artifact.record",
        "run.record",
        "gate.record",
        "finding.record"
    ]
},
"lease_bound_task_transition_states": {

```

```

    "const": [
        "LEASED",
        "RUNNING",
        "COMPLETED"
    ]
},
"max_command_bytes": {
    "const": 1048576,
    "type": "integer"
},
"max_envelope_bytes": {
    "const": 2097152,
    "type": "integer"
},
"max_events_per_batch": {
    "const": 20,
    "type": "integer"
},
"max_requested_scope_items": {
    "const": 32,
    "type": "integer"
},
"max_state_binding_bytes": {
    "const": 1048576,
    "type": "integer"
},
"max_state_binding_updates_per_batch": {
    "const": 20,

```

```

    "type": "integer"
},
"policy_digest": {
    "$ref": "#/$defs/Digest"
},
"record_type": {
    "const": "EventStorePolicy"
},
"runtime_overlay_compaction_depth": {
    "const": 64,
    "type": "integer"
},
"state_binding_algorithm_contract": {
    "const": {
        "delete_rule": "delete maps the keyed leaf to empty_leaf",
        "delta_order": [
            "leaf_type-ascending",
            "leaf_id-ascending"
        ],
        "depth": 256,
        "depth_encoding": "unsigned-16-bit-big-endian",
        "empty_leaf_domain_utf8": "promin:typed-sparse-merkle-v1:empty-leaf",

```

```

    "internal_node_domain_utf8": "promin:typed-sparse-merkle-v1:node\u0000",
    "internal_node_rule": "sha256(domain + depth_uint16_be + left_32_bytes + right_32_bytes)",
    "leaf_domain_utf8": "promin:typed-sparse-merkle-v1:leaf\u0000",
    "leaf_key_rule": "sha256(canonical bytes of exact object leaf_type and leaf_id)",
    "set_leaf_rule": "sha256(domain + leaf_key_32_bytes + value_digest_32_bytes)"
}

```

```

},
"state_binding_identity_rules": {
  "const": {
    "Activation": {
      "id_fields": [
        "activation_digest"
      ],
      "identity_mode": "direct"
    },
    "Artifact": {
      "id_fields": [
        "artifact_id"
      ],
      "identity_mode": "direct"
    },
    "Candidate": {
      "id_fields": [
        "candidate_id"
      ],
      "identity_mode": "direct"
    },
    "Decision": {
      "id_fields": [
        "decision_id"
      ],
      "identity_mode": "direct"
    },
    "Finding": {

```

```

      "id_fields": [
        "finding_id"
      ],
      "identity_mode": "direct"
    },
    "GateResult": {
      "digest_algorithm": "sha256",
      "digest_hex_chars": 48,
      "digest_input": "canonical-ordered-id-field-values",
      "id_fields": [
        "gate_id",
        "run_id"
      ],
      "identity_mode": "typed_digest_id",
      "prefix": "gate-result:"
    },
    "Grant": {
      "id_fields": [
        "grant_id"
      ],
      "identity_mode": "direct"
    },
    "Lease": {
      "id_fields": [
        "lease_id"
      ],
      "identity_mode": "direct"
    },
  },

```

```

  "Relation": {
    "id_fields": [
      "relation_id"
    ],
    "identity_mode": "direct"
  },
  "Run": {
    "id_fields": [
      "run_id"
    ],
    "identity_mode": "direct"
  },
  "Task": {
    "id_fields": [
      "task_id"
    ],

```

```

    ],
    "identity_mode": "direct"
  }
},
"state_binding_value_rules": {
  "const": {
    "Activation": {
      "event_kinds": [],
      "value_definition": "Activation",
      "value_digest_rule": "sha256(canonical installed Activation record); installed before sequence
one and immutable for the journal"
    },
    "Artifact": {
      "event_kinds": [
        "artifact.recorded"
      ],
      "value_definition": "Artifact",
      "value_digest_rule": "sha256(canonical exact Event.payload Artifact)"
    },
    "Candidate": {
      "event_kinds": [
        "candidate.recorded"
      ],
      "value_definition": "Candidate",
      "value_digest_rule": "sha256(canonical exact Event.payload Candidate)"
    },
    "Decision": {
      "event_kinds": [
        "decision.recorded"
      ],
      "value_definition": "Decision",
      "value_digest_rule": "sha256(canonical exact Event.payload Decision)"
    },
    "Finding": {
      "event_kinds": [
        "finding.recorded"
      ],
      "value_definition": "Finding",
      "value_digest_rule": "sha256(canonical exact Event.payload Finding)"
    },
    "GateResult": {
      "event_kinds": [
        "gate.recorded"
      ],
      "value_definition": "GateResult",
      "value_digest_rule": "sha256(canonical exact Event.payload GateResult)"
    },
    "Grant": {
      "event_kinds": [
        "grant.issued",
        "grant.revoked"
      ],
      "value_definition": "GrantAuthorityState",
      "value_digest_rule": "sha256(canonical exact GrantAuthorityState containing the current Grant
and null or exact current GrantRevocation)"
    },
    "Lease": {
      "event_kinds": [
        "lease.recorded"
      ],
      "value_definition": "Lease",
      "value_digest_rule": "sha256(canonical exact Event.payload post-state Lease)"
    },
    "Relation": {
      "event_kinds": [
        "relation.recorded"
      ],
      "value_definition": "Relation",
      "value_digest_rule": "sha256(canonical exact Event.payload Relation)"
    },
    "Run": {
      "event_kinds": [
        "run.recorded"
      ],
      "value_definition": "Run",
      "value_digest_rule": "sha256(canonical exact Event.payload Run)"
    }
  }
}

```

```

    },
    "Task": {
      "event_kinds": [
        "task.recorded",
        "task.transitioned"
      ],
      "value_definition": "Task",
      "value_digest_rule": "sha256(canonical exact Event.payload post-state Task)"
    }
  }
},
"required": [
  "record_type",
  "authority_model_digest",
  "max_command_bytes",
  "max_envelope_bytes",
  "max_state_binding_bytes",
  "max_state_binding_updates_per_batch",
  "max_events_per_batch",
  "max_requested_scope_items",
  "derived_tail_batch_threshold",
  "derived_tail_byte_threshold",
  "runtime_overlay_compaction_depth",
  "command_required_fields",
  "command_conditional_fields",
  "command_mutation_fields",
  "lease_bound_command_kinds",
  "lease_bound_task_transition_states",
  "command_to_primary_event",
  "allowed_state_binding_leaf_types",
  "state_binding_identity_rules",
  "state_binding_algorithm_contract",
  "state_binding_value_rules",
  "canonical_timestamp_contract",
  "genesis_previous_authority_commitment",
  "genesis_event_semantic_digest",
  "policy_digest"
],
"type": "object",
"x-promin-persistence": "derived-live-only"
}

```

Owner: core/contracts.schema.json#/\$defs/EventStorePolicy | Validator: urn:promin:contracts:1#/\$defs/EventStorePolicy

31. EvidenceBinding

Type summary: type="object"; additionalProperties=false

Required: activation_digest, implementation_closure_digest, candidate_digest, policy_digest, tool_digest, input_digests

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
candidate_digest	yes	\$ref="#/\$defs/Digest"
finding_digest	no	\$ref="#/\$defs/Digest"
implementation_closure_digest	yes	\$ref="#/\$defs/Digest"
input_digests	yes	type="array"; minItems=0; maxItems=128; uniqueItems=true
policy_digest	yes	\$ref="#/\$defs/Digest"
provider_invocations	no	type="array"; minItems=1; maxItems=32; uniqueItems=true
tool_digest	yes	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "finding_digest": {
      "$ref": "#/$defs/Digest"
    }
  }
}

```



```

"implementation_closure_digest": {
  "$ref": "#/$defs/Digest"
},
"input_digests": {
  "items": {
    "$ref": "#/$defs/Digest"
  },
  "maxItems": 128,
  "minItems": 0,
  "type": "array",
  "uniqueItems": true
},
"policy_digest": {
  "$ref": "#/$defs/Digest"
},
"provider_invocations": {
  "items": {
    "$ref": "#/$defs/ProviderInvocationEvidence"
  },
  "maxItems": 32,
  "minItems": 1,
  "type": "array",
  "uniqueItems": true
},
"tool_digest": {
  "$ref": "#/$defs/Digest"
}
},
"required": [
  "activation_digest",
  "implementation_closure_digest",
  "candidate_digest",
  "policy_digest",
  "tool_digest",
  "input_digests"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/EvidenceBinding | Validator: urn:promin:contracts:1#/\$defs/EvidenceBinding

32. EvidenceProducerAttestation

Type summary: type="object"; additionalProperties=false

Required: record_type, claim_domain, candidate_binding_digest, evidence_role, evidence_record_type, platform, invocation_id, payload_digest, raw_artifact_manifest_digest, producer_id, trust_root_id, signature_provider_id, key_id, public_key, nonce, signed_at, signed_claim_digest, signature

Property	Required	Structural rule
candidate_binding_digest	yes	\$ref="#/\$defs/Digest"
claim_domain	yes	const="promin.release-evidence.producer-attestation.v1"
evidence_record_type	yes	enum=["HumanDocumentVerification", "NoDegradationResult", "PlatformVerificationResult", "SaturationAudit", "SaturationEvidence"]
evidence_role	yes	enum=["human-documents", "linux", "linux-no-degradation", "physical-scale", "saturation-audit", "windows", "windows-no-degradation"]
invocation_id	yes	\$ref="#/\$defs/Id"
key_id	yes	\$ref="#/\$defs/Id"
nonce	yes	type="string"; minLength=24; maxLength=128
payload_digest	yes	\$ref="#/\$defs/Digest"
platform	yes	enum=["cross-platform", "darwin", "linux", "windows"]
producer_id	yes	\$ref="#/\$defs/Id"
public_key	yes	type="string"; pattern="^[A-Za-z0-9+/]{43}=\$"; minLength=44; maxLength=44
raw_artifact_manifest_digest	yes	\$ref="#/\$defs/Digest"
record_type	yes	const="EvidenceProducerAttestation"
signature	yes	type="string"; minLength=80; maxLength=128
signature_provider_id	yes	const="cryptography-ed25519-v1"
signed_at	yes	\$ref="#/\$defs/Timestamp"

Property	Required	Structural rule
signed_claim_digest	yes	\$ref="#/\$defs/Digest"
trust_root_id	yes	\$ref="#/\$defs/Id"
<pre> { "additionalProperties": false, "properties": { "candidate_binding_digest": { "\$ref": "#/\$defs/Digest" }, "claim_domain": { "const": "promin.release-evidence.producer-attestation.v1" }, "evidence_record_type": { "enum": ["HumanDocumentVerification", "NoDegradationResult", "PlatformVerificationResult", "SaturationAudit", "SaturationEvidence"] }, "evidence_role": { "enum": ["human-documents", "linux", "linux-no-degradation", "physical-scale", "saturation-audit", "windows", "windows-no-degradation"] }] }, "invocation_id": { "\$ref": "#/\$defs/Id" }, "key_id": { "\$ref": "#/\$defs/Id" }, "nonce": { "maxLength": 128, "minLength": 24, "type": "string" }, "payload_digest": { "\$ref": "#/\$defs/Digest" }, "platform": { "enum": ["cross-platform", "darwin", "linux", "windows"] }, "producer_id": { "\$ref": "#/\$defs/Id" }, "public_key": { "maxLength": 44, "minLength": 44, "pattern": "^[A-Za-z0-9+/{43}]=\$", "type": "string" }, "raw_artifact_manifest_digest": { "\$ref": "#/\$defs/Digest" }, "record_type": { "const": "EvidenceProducerAttestation" }, "signature": { "maxLength": 128, "minLength": 80, "type": "string" }, "signature_provider_id": { "const": "cryptography-ed25519-v1" } } </pre>		

```

    },
    "signed_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "signed_claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "trust_root_id": {
      "$ref": "#/$defs/Id"
    }
  }
},

```

```

"required": [
  "record_type",
  "claim_domain",
  "candidate_binding_digest",
  "evidence_role",
  "evidence_record_type",
  "platform",
  "invocation_id",
  "payload_digest",
  "raw_artifact_manifest_digest",
  "producer_id",
  "trust_root_id",
  "signature_provider_id",
  "key_id",
  "public_key",
  "nonce",
  "signed_at",
  "signed_claim_digest",
  "signature"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/EvidenceProducerAttestation | Validator: urn:promin:contracts:1#/\$defs/EvidenceProducerAttestation

33. Finding

Type summary: type="object"; additionalProperties=false; allOf[1]=keys:else,if,then

Required: record_type, finding_id, status, severity, blocking, statement, activation_digest, candidate_digest, evidence_artifacts, created_at

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
blocking	yes	type="boolean"
candidate_digest	yes	\$ref="#/\$defs/Digest"
created_at	yes	\$ref="#/\$defs/Timestamp"
disposition_decision_id	no	\$ref="#/\$defs/Id"
evidence_artifacts	yes	type="array"; minItems=1; maxItems=64; uniqueItems=true
finding_id	yes	\$ref="#/\$defs/Id"
record_type	yes	const="Finding"
severity	yes	enum=["P0", "P1", "P2", "P3"]
statement	yes	type="string"; minLength=1
status	yes	enum=["OPEN", "RESOLVED", "WAIVED"]

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "required": [
            "disposition_decision_id"
          ]
        }
      },
      "if": {
        "properties": {
          "status": {
            "enum": [
              "RESOLVED",

```

```

        "WAIVED"
      ]
    }
  },
  "then": {
    "properties": {
      "blocking": {
        "const": false
      }
    }
  },
  "required": [
    "disposition_decision_id"
  ]
},
],
"properties": {
  "activation_digest": {
    "$ref": "#/$defs/Digest"
  },
  "blocking": {
    "type": "boolean"
  },
  "candidate_digest": {
    "$ref": "#/$defs/Digest"
  },
  "created_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "disposition_decision_id": {
    "$ref": "#/$defs/Id"
  },
  "evidence_artifacts": {
    "items": {
      "additionalProperties": false,
      "properties": {
        "artifact_id": {
          "$ref": "#/$defs/Id"
        }
      },
      "artifact_record_digest": {
        "$ref": "#/$defs/Digest"
      }
    },
    "required": [
      "artifact_id",
      "artifact_record_digest"
    ],
    "type": "object"
  },
  "maxItems": 64,
  "minItems": 1,
  "type": "array",
  "uniqueItems": true
},
  "finding_id": {
    "$ref": "#/$defs/Id"
  },
  "record_type": {
    "const": "Finding"
  },
  "severity": {
    "enum": [
      "P0",
      "P1",
      "P2",
      "P3"
    ]
  },
  "statement": {
    "minLength": 1,
    "type": "string"
  },
  "status": {
    "enum": [
      "OPEN",
      "RESOLVED",

```

```

        "WAIVED"
    ]
}
},
"required": [
    "record_type",
    "finding_id",
    "status",
    "severity",
    "blocking",
    "statement",
    "activation_digest",
    "candidate_digest",
    "evidence_artifacts",
    "created_at"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Finding | Validator: urn:promin:contracts:1#/\$defs/Finding

34. GateEvidenceArtifactBinding

Type summary: type="object"; additionalProperties=false

Required: artifact_id, artifact_record_digest, run_id, run_digest

Property	Required	Structural rule
artifact_id	yes	\$ref="#/\$defs/Id"
artifact_record_digest	yes	\$ref="#/\$defs/Digest"
run_digest	yes	\$ref="#/\$defs/Digest"
run_id	yes	\$ref="#/\$defs/Id"

```

{
  "additionalProperties": false,
  "properties": {
    "artifact_id": {
      "$ref": "#/$defs/Id"
    },
    "artifact_record_digest": {
      "$ref": "#/$defs/Digest"
    },
    "run_digest": {
      "$ref": "#/$defs/Digest"
    },
    "run_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "artifact_id",
    "artifact_record_digest",
    "run_id",
    "run_digest"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/GateEvidenceArtifactBinding | Validator: urn:promin:contracts:1#/\$defs/GateEvidenceArtifactBinding

35. GateResult

Type summary: type="object"; additionalProperties=false; oneOf[4]=keys:properties,required | keys:properties,required | keys:properties,required | keys:properties,required

Required: record_type, task_id, gate_id, run_id, run_digest, definition_digest, status, outcome, pass_credit, activation_digest, candidate_digest, policy_digest, tool_digest, evidence_class, evidence_artifacts

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
candidate_digest	yes	\$ref="#/\$defs/Digest"
definition_digest	yes	\$ref="#/\$defs/Digest"
evidence_artifacts	yes	type="array"; minItems=0; maxItems=64; uniqueItems=true
evidence_class	yes	enum=["product-execution", "validator", "harness-generated", "migration"]

Property	Required	Structural rule
gate_id	yes	\$ref="#/\$defs/Id"
outcome	yes	enum=["pass", "fail", "blocked", "skipped"]
pass_credit	yes	type="boolean"
policy_digest	yes	\$ref="#/\$defs/Digest"
reason	no	type="string"; minLength=1; maxLength=2048
record_type	yes	const="GateResult"
run_digest	yes	\$ref="#/\$defs/Digest"
run_id	yes	\$ref="#/\$defs/Id"
status	yes	enum=["pass", "fail", "blocked", "skipped"]
task_id	yes	\$ref="#/\$defs/Id"
tool_digest	yes	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "oneOf": [
    {
      "properties": {
        "evidence_artifacts": {
          "minItems": 1
        },
        "outcome": {
          "const": "pass"
        },
        "status": {
          "const": "pass"
        }
      },
      "required": [
        "status",
        "outcome",
        "pass_credit"
      ]
    },
    {
      "properties": {
        "evidence_artifacts": {
          "minItems": 1
        },
        "outcome": {
          "const": "fail"
        },
        "pass_credit": {
          "const": false
        },
        "status": {
          "const": "fail"
        }
      },
      "required": [
        "status",
        "outcome",
        "pass_credit"
      ]
    },
    {
      "properties": {
        "evidence_artifacts": {
          "minItems": 1
        },
        "outcome": {
          "const": "blocked"
        },
        "pass_credit": {
          "const": false
        },
        "status": {
          "const": "blocked"
        }
      },
      "required": [

```

```

        "status",
        "outcome",
        "pass_credit",
        "reason"
    ]
},
{
    "properties": {
        "evidence_artifacts": {
            "maxItems": 0
        },
        "outcome": {
            "const": "skipped"
        },
        "pass_credit": {
            "const": false
        },
        "status": {
            "const": "skipped"
        }
    },
    "required": [
        "status",
        "outcome",
        "pass_credit",
        "reason"
    ]
}

```

```

    ]
}
],
"properties": {
    "activation_digest": {
        "$ref": "#/$defs/Digest"
    },
    "candidate_digest": {
        "$ref": "#/$defs/Digest"
    },
    "definition_digest": {
        "$ref": "#/$defs/Digest"
    },
    "evidence_artifacts": {
        "items": {
            "$ref": "#/$defs/GateEvidenceArtifactBinding"
        },
        "maxItems": 64,
        "minItems": 0,
        "type": "array",
        "uniqueItems": true
    },
    "evidence_class": {
        "enum": [
            "product-execution",
            "validator",
            "harness-generated",
            "migration"
        ]
    }
}

```

```

    ]
},
"gate_id": {
    "$ref": "#/$defs/Id"
},
"outcome": {
    "enum": [
        "pass",
        "fail",
        "blocked",
        "skipped"
    ]
},
"pass_credit": {
    "type": "boolean"
},
"policy_digest": {
    "$ref": "#/$defs/Digest"
},
"reason": {
    "maxLength": 2048,
    "minLength": 1,
    "type": "string"
},
}

```

```

"record_type": {
  "const": "GateResult"
},
"run_digest": {
  "$ref": "#/$defs/Digest"
},
"run_id": {
  "$ref": "#/$defs/Id"
},
"status": {
  "enum": [
    "pass",
    "fail",
    "blocked",
    "skipped"
  ]
},
"task_id": {
  "$ref": "#/$defs/Id"
},
"tool_digest": {
  "$ref": "#/$defs/Digest"
}
},
"required": [
  "record_type",
  "task_id",
  "gate_id",
  "run_id",
  "run_digest",
  "definition_digest",
  "status",
  "outcome",
  "pass_credit",
  "activation_digest",
  "candidate_digest",
  "policy_digest",
  "tool_digest",
  "evidence_class",
  "evidence_artifacts"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/GateResult | Validator: urn:promin:contracts:1#/\$defs/GateResult

36. GateRunDefinition

Type summary: type="object"; additionalProperties=false; oneOf[4]=keys:properties,required | keys:properties,required | keys:properties,required | keys:properties,required; allOf[1]=keys:if,then

Required: definition_kind, definition_id, owner_kind, owner_digest, defined_at_head_digest, gate_id, run_kind, expected_evidence_class, expected_evidence_purpose, product_credit_required, target_kind, target_digest, target_scope, candidate_digest, policy_digest, tool_digest, implementation_closure_digest, provider_binding_digest, input_digests, activation_digest

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
candidate_digest	yes	\$ref="#/\$defs/Digest"
defined_at_head_digest	yes	oneOf[2]=#/\$defs/Digest type:null
definition_id	yes	\$ref="#/\$defs/Id"
definition_kind	yes	const="GateRunDefinition"
expected_evidence_class	yes	enum=["product-execution", "validator", "harness-generated", "migration"]
expected_evidence_purpose	yes	enum=["diagnostic", "gate", "product"]
gate_id	yes	\$ref="#/\$defs/Id"
implementation_closure_digest	yes	\$ref="#/\$defs/Digest"
input_digests	yes	type="array"; minItems=0; maxItems=128; uniqueItems=true
owner_digest	yes	\$ref="#/\$defs/Digest"
owner_kind	yes	const="Task"
policy_digest	yes	\$ref="#/\$defs/Digest"

Property	Required	Structural rule
product_credit_required	yes	type="boolean"
provider_binding_digest	yes	\$ref="#/\$defs/Digest"
run_kind	yes	enum=["execution", "validation"]
target_digest	yes	\$ref="#/\$defs/Digest"
target_kind	yes	enum=["candidate", "finding"]
target_scope	yes	type="array"; minItems=1; maxItems=32; uniqueItems=true
tool_digest	yes	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "if": {
        "properties": {
          "product_credit_required": {
            "const": true
          }
        },
        "required": [
          "product_credit_required"
        ]
      },
      "then": {
        "properties": {
          "expected_evidence_class": {
            "const": "product-execution"
          },
          "expected_evidence_purpose": {
            "const": "product"
          }
        }
      }
    }
  ],
  "oneOf": [
    {
      "properties": {
        "expected_evidence_class": {
          "const": "validator"
        },
        "expected_evidence_purpose": {
          "const": "diagnostic"
        }
      },
      "required": [
        "expected_evidence_class",
        "expected_evidence_purpose"
      ]
    },
    {
      "properties": {
        "expected_evidence_class": {
          "const": "validator"
        },
        "expected_evidence_purpose": {
          "const": "gate"
        }
      },
      "required": [
        "expected_evidence_class",
        "expected_evidence_purpose"
      ]
    }
  ],
  {
    "properties": {
      "expected_evidence_class": {
        "const": "product-execution"
      },
      "expected_evidence_purpose": {
        "const": "gate"
      }
    },
    "required": [

```

```

        "expected_evidence_class",
        "expected_evidence_purpose"
    ]
},
{
    "properties": {
        "expected_evidence_class": {
            "const": "product-execution"
        },
        "expected_evidence_purpose": {
            "const": "product"
        }
    },
    "required": [
        "expected_evidence_class",
        "expected_evidence_purpose"
    ]
}
],

```

```

"properties": {
    "activation_digest": {
        "$ref": "#/$defs/Digest"
    },
    "candidate_digest": {
        "$ref": "#/$defs/Digest"
    },
    "defined_at_head_digest": {
        "oneOf": [
            {
                "$ref": "#/$defs/Digest"
            },
            {
                "type": "null"
            }
        ]
    },
    "definition_id": {
        "$ref": "#/$defs/Id"
    },
    "definition_kind": {
        "const": "GateRunDefinition"
    },
    "expected_evidence_class": {
        "enum": [
            "product-execution",
            "validator",
            "harness-generated",

```

```

            "migration"
        ]
    },
    "expected_evidence_purpose": {
        "enum": [
            "diagnostic",
            "gate",
            "product"
        ]
    },
    "gate_id": {
        "$ref": "#/$defs/Id"
    },
    "implementation_closure_digest": {
        "$ref": "#/$defs/Digest"
    },
    "input_digests": {
        "items": {
            "$ref": "#/$defs/Digest"
        },
        "maxItems": 128,
        "minItems": 0,
        "type": "array",
        "uniqueItems": true
    },
    "owner_digest": {
        "$ref": "#/$defs/Digest"
    },

```

```

"owner_kind": {
    "const": "Task"
}

```

```

    },
    "policy_digest": {
      "$ref": "#/$defs/Digest"
    },
    "product_credit_required": {
      "type": "boolean"
    },
    "provider_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "run_kind": {
      "enum": [
        "execution",
        "validation"
      ]
    },
    "target_digest": {
      "$ref": "#/$defs/Digest"
    },
    "target_kind": {
      "enum": [
        "candidate",
        "finding"
      ]
    },
    "target_scope": {
      "items": {
        "$ref": "#/$defs/ScopeSelector"
      },
      "maxItems": 32,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "tool_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "definition_kind",
    "definition_id",
    "owner_kind",
    "owner_digest",
    "defined_at_head_digest",
    "gate_id",
    "run_kind",
    "expected_evidence_class",
    "expected_evidence_purpose",
    "product_credit_required",
    "target_kind",
    "target_digest",
    "target_scope",
    "candidate_digest",
    "policy_digest",
    "tool_digest",
    "implementation_closure_digest",
    "provider_binding_digest",
    "input_digests",
    "activation_digest"
  ],
  "type": "object"
}

```

Owner: *core/contracts.schema.json#/\$defs/GateRunDefinition* | Validator: *urn:promin:contracts:1#/\$defs/GateRunDefinition*

37. Grant

Type summary: type="object"; additionalProperties=false

Required: record_type, grant_id, subject_id, capability_id, scope, activation_digest, issued_at, expires_at, nonce, claim_digest, trust_proofs

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
capability_id	yes	enum=[16 items]
claim_digest	yes	\$ref="#/\$defs/Digest"

Property	Required	Structural rule
expires_at	yes	\$ref="#/\$defs/Timestamp"
grant_id	yes	\$ref="#/\$defs/Id"
issued_at	yes	\$ref="#/\$defs/Timestamp"
nonce	yes	type="string"; pattern="^[A-Za-z0-9._-]{16,128}\$"; minLength=16; maxLength=128
record_type	yes	const="Grant"
scope	yes	type="array"; minItems=1; maxItems=32; uniqueItems=true
subject_id	yes	\$ref="#/\$defs/Id"
trust_proofs	yes	type="array"; minItems=1; maxItems=32; uniqueItems=true

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "capability_id": {
      "enum": [
        "standard.activate",
        "authority.manage",
        "task.plan",
        "task.execute",
        "lease.manage",
        "evidence.publish",
        "validation.evaluate",
        "finding.record",
        "finding.resolve",
        "finding.waive",
        "candidate.promote",
        "release.decide",
        "export.create",
        "migration.manage",
        "projection.rebuild",
        "projection.read"
      ]
    },
    "claim_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "expires_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "grant_id": {
    "$ref": "#/$defs/Id"
  },
  "issued_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "nonce": {
    "maxLength": 128,
    "minLength": 16,
    "pattern": "^[A-Za-z0-9._-]{16,128}$",
    "type": "string"
  },
  "record_type": {
    "const": "Grant"
  },
  "scope": {
    "items": {
      "$ref": "#/$defs/ScopeSelector"
    },
    "maxItems": 32,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "subject_id": {
    "$ref": "#/$defs/Id"
  },
  "trust_proofs": {
    "items": {
      "$ref": "#/$defs/GrantTrustProof"
    }
  }
}

```

```

    },
    "maxItems": 32,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  }
},
"required": [
  "record_type",
  "grant_id",
  "subject_id",
  "capability_id",
  "scope",
  "activation_digest",
  "issued_at",
  "expires_at",
  "nonce",
  "claim_digest",
  "trust_proofs"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Grant | Validator: urn:promin:contracts:1#/\$defs/Grant

38. GrantAuthorityState

Type summary: type="object"; additionalProperties=false

Required: record_type, grant, revocation

Property	Required	Structural rule
grant	yes	\$ref="#/\$defs/Grant"
record_type	yes	const="GrantAuthorityState"
revocation	yes	oneOf[2]=#/\$defs/GrantRevocation type:null

```

{
  "additionalProperties": false,
  "properties": {
    "grant": {
      "$ref": "#/$defs/Grant"
    },
    "record_type": {
      "const": "GrantAuthorityState"
    },
    "revocation": {
      "oneOf": [
        {
          "$ref": "#/$defs/GrantRevocation"
        },
        {
          "type": "null"
        }
      ]
    }
  },
  "required": [
    "record_type",
    "grant",
    "revocation"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/GrantAuthorityState | Validator: urn:promin:contracts:1#/\$defs/GrantAuthorityState

39. GrantAuthorization

Type summary: type="object"; additionalProperties=false

Required: kind, grant_id, grant_claim_digest

Property	Required	Structural rule
grant_claim_digest	yes	\$ref="#/\$defs/Digest"
grant_id	yes	\$ref="#/\$defs/Id"
kind	yes	const="grant"

```

{
  "additionalProperties": false,
  "properties": {
    "grant_claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "grant_id": {
      "$ref": "#/$defs/Id"
    },
    "kind": {
      "const": "grant"
    }
  },
  "required": [
    "kind",
    "grant_id",
    "grant_claim_digest"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/GrantAuthorization | Validator: urn:promin:contracts:1#/\$defs/GrantAuthorization

40. GrantRevocation

Type summary: type="object"; additionalProperties=false

Required: grant_id, decision_id, reason, revoked_at

Property	Required	Structural rule
decision_id	yes	\$ref="#/\$defs/Id"
grant_id	yes	\$ref="#/\$defs/Id"
reason	yes	type="string"; minLength=1
revoked_at	yes	\$ref="#/\$defs/Timestamp"

```

{
  "additionalProperties": false,
  "properties": {
    "decision_id": {
      "$ref": "#/$defs/Id"
    },
    "grant_id": {
      "$ref": "#/$defs/Id"
    },
    "reason": {
      "minLength": 1,
      "type": "string"
    },
    "revoked_at": {
      "$ref": "#/$defs/Timestamp"
    }
  },
  "required": [
    "grant_id",
    "decision_id",
    "reason",
    "revoked_at"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/GrantRevocation | Validator: urn:promin:contracts:1#/\$defs/GrantRevocation

41. GrantTrustProof

Type summary: oneOf[3]=#/\$defs/IssuerGrantTrustProof | #/\$defs/LocalTrustProof | #/\$defs/SignatureTrustProof

Required: -

```

{
  "oneOf": [
    {
      "$ref": "#/$defs/IssuerGrantTrustProof"
    },
    {
      "$ref": "#/$defs/LocalTrustProof"
    },
    {

```

```

    "$ref": "#/$defs/SignatureTrustProof"
  }
]
}

```

Owner: core/contracts.schema.json#/\$defs/GrantTrustProof | Validator: urn:promin:contracts:1#/\$defs/GrantTrustProof

42. Healthcheck

Type summary: type="object"; additionalProperties=false

Required: argv, timeout_ms, expected_exit

Property	Required	Structural rule
argv	yes	type="array"; minItems=1; maxItems=16
expected_exit	yes	type="integer"; minimum=0; maximum=255
timeout_ms	yes	type="integer"; minimum=100; maximum=60000

```

{
  "additionalProperties": false,
  "properties": {
    "argv": {
      "items": {
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 16,
      "minItems": 1,
      "type": "array"
    },
    "expected_exit": {
      "maximum": 255,
      "minimum": 0,
      "type": "integer"
    },
    "timeout_ms": {
      "maximum": 60000,
      "minimum": 100,
      "type": "integer"
    }
  },
  "required": [
    "argv",
    "timeout_ms",
    "expected_exit"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Healthcheck | Validator: urn:promin:contracts:1#/\$defs/Healthcheck

43. HumanDocumentVerification

Type summary: type="object"; additionalProperties=false

Required: record_type, status, candidate_binding_digest, documents, deterministic_rebuild, rebuild_digest, page_count, extraction_diagnostics, visual_review_scope, font_bindings, generator_result_digest, render_environment, render_manifest, render_manifest_digest, product_acceptance_pass, producer, invocation, raw_artifact_manifest_digest, producer_attestation, result_digest

Property	Required	Structural rule
candidate_binding_digest	yes	\$ref="#/\$defs/Digest"
deterministic_rebuild	yes	const=true
documents	yes	type="array"; minItems=4; maxItems=4; uniqueItems=true
extraction_diagnostics	yes	type="object"; additionalProperties=false
font_bindings	yes	type="object"; additionalProperties=false
generator_result_digest	yes	\$ref="#/\$defs/Digest"
invocation	yes	\$ref="#/\$defs/ReleaseEvidenceInvocation"
page_count	yes	type="integer"; minimum=1
producer	yes	\$ref="#/\$defs/ReleaseEvidenceProducer"
producer_attestation	yes	\$ref="#/\$defs/EvidenceProducerAttestation"

Property	Required	Structural rule
product_acceptance_pass	yes	const=false
raw_artifact_manifest_digest	yes	\$ref="#/\$defs/Digest"
rebuild_digest	yes	\$ref="#/\$defs/Digest"
record_type	yes	const="HumanDocumentVerification"
render_environment	yes	type="object"; additionalProperties=false
render_manifest	yes	type="array"; minItems=4; maxItems=4096; uniqueItems=true
render_manifest_digest	yes	\$ref="#/\$defs/Digest"
result_digest	yes	\$ref="#/\$defs/Digest"
status	yes	const="pass"
visual_review_scope	yes	type="object"; additionalProperties=false

```

{
  "additionalProperties": false,
  "properties": {
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "deterministic_rebuild": {
      "const": true
    },
    "documents": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "blank_pages": {
            "maxItems": 0,
            "type": "array"
          },
          "extracted_characters": {
            "minimum": 1,
            "type": "integer"
          },
          "extraction_errors": {
            "maxItems": 0,
            "type": "array"
          },
          "page_count": {
            "minimum": 1,
            "type": "integer"
          }
        }
      },
      "path": {
        "enum": [
          "human/promin_appendices_en.pdf",
          "human/promin_appendices_ua.pdf",
          "human/promin_main_en.pdf",
          "human/promin_main_ua.pdf"
        ]
      },
      "sha256": {
        "$ref": "#/$defs/Digest"
      },
      "size_bytes": {
        "minimum": 1,
        "type": "integer"
      }
    },
    "required": [
      "path",
      "sha256",
      "size_bytes",
      "page_count",
      "extracted_characters",
      "blank_pages",
      "extraction_errors"
    ],
    "type": "object"
  },
  "maxItems": 4,
  "minItems": 4,
  "type": "array",
  "uniqueItems": true
}

```



```

},
"extraction_diagnostics": {
  "additionalProperties": false,
  "properties": {
    "blank_pages": {
      "maxItems": 0,
      "type": "array"
    },
    "documents_parsed": {
      "const": 4
    },
    "errors": {
      "maxItems": 0,
      "type": "array"
    },
    "total_extracted_characters": {
      "minimum": 1,
      "type": "integer"
    }
  }
},
"required": [
  "documents_parsed",
  "total_extracted_characters",
  "blank_pages",

```

```

    "errors"
  ],
  "type": "object"
},
"font_bindings": {
  "additionalProperties": false,
  "properties": {
    "bold": {
      "additionalProperties": false,
      "properties": {
        "bytes": {
          "minimum": 1,
          "type": "integer"
        },
        "pdf_font_name": {
          "minLength": 1,
          "type": "string"
        },
        "sha256": {
          "$ref": "#/$defs/Digest"
        }
      }
    },
    "required": [
      "bytes",
      "pdf_font_name",
      "sha256"
    ],
    "type": "object"

```

```

},
"italic": {
  "additionalProperties": false,
  "properties": {
    "bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "pdf_font_name": {
      "minLength": 1,
      "type": "string"
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "bytes",
    "pdf_font_name",
    "sha256"
  ],
  "type": "object"
},
"mono": {
  "additionalProperties": false,
  "properties": {

```

```

    "bytes": {
      "minimum": 1,

      "type": "integer"
    },
    "pdf_font_name": {
      "minLength": 1,
      "type": "string"
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "bytes",
    "pdf_font_name",
    "sha256"
  ],
  "type": "object"
},
"regular": {
  "additionalProperties": false,
  "properties": {
    "bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "pdf_font_name": {
      "minLength": 1,
      "type": "string"
    }
  },

```

```

    "sha256": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "bytes",
    "pdf_font_name",
    "sha256"
  ],
  "type": "object"
}
},
"required": [
  "regular",
  "bold",
  "italic",
  "mono"
],
"type": "object"
},
"generator_result_digest": {
  "$ref": "#/$defs/Digest"
},
"invocation": {
  "$ref": "#/$defs/ReleaseEvidenceInvocation"
},
"page_count": {
  "minimum": 1,

```

```

    "type": "integer"
  },
  "producer": {
    "$ref": "#/$defs/ReleaseEvidenceProducer"
  },
  "producer_attestation": {
    "$ref": "#/$defs/EvidenceProducerAttestation"
  },
  "product_acceptance_pass": {
    "const": false
  },
  "raw_artifact_manifest_digest": {
    "$ref": "#/$defs/Digest"
  },
  "rebuild_digest": {
    "$ref": "#/$defs/Digest"
  },
  "record_type": {
    "const": "HumanDocumentVerification"
  }
}

```

```

},
"render_environment": {
  "additionalProperties": false,
  "properties": {
    "font_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
  },
  "generator_sha256": {
    "$ref": "#/$defs/Digest"
  }
}

```

```

},
"platform": {
  "minLength": 1,
  "type": "string"
},
"python_version": {
  "minLength": 1,
  "type": "string"
},
"reportlab_version": {
  "minLength": 1,
  "type": "string"
}
},
"required": [
  "generator_sha256",
  "reportlab_version",
  "python_version",
  "platform",
  "font_binding_digest"
],
"type": "object"
},
"render_manifest": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "document_path": {

```

```

        "enum": [
          "human/promin_appendices_en.pdf",
          "human/promin_appendices_ua.pdf",
          "human/promin_main_en.pdf",
          "human/promin_main_ua.pdf"
        ]
      },
      "image_bytes": {
        "maximum": 33554432,
        "minimum": 1,
        "type": "integer"
      },
      "image_path": {
        "$ref": "#/$defs/RelativePath"
      },
      "image_sha256": {
        "$ref": "#/$defs/Digest"
      },
      "page": {
        "minimum": 1,
        "type": "integer"
      }
    },
    "required": [
      "document_path",
      "page",
      "image_path",
      "image_sha256",

```

```

      "image_bytes"
    ],
    "type": "object"
  },
  "maxItems": 4096,
  "minItems": 4,
  "type": "array",
  "uniqueItems": true
},
"render_manifest_digest": {
  "$ref": "#/$defs/Digest"
},

```

```

"result_digest": {
  "$ref": "#/$defs/Digest"
},
"status": {
  "const": "pass"
},
"visual_review_scope": {
  "additionalProperties": false,
  "properties": {
    "clipping_detected": {
      "const": false
    },
    "completed": {
      "const": true
    },
  },
  "pages_reviewed": {

```

```

    "minimum": 1,
    "type": "integer"
  },
  "render_manifest_digest": {
    "$ref": "#/$defs/Digest"
  },
  "rendered_page_count": {
    "minimum": 1,
    "type": "integer"
  },
  "reviewed_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "reviewer_id": {
    "$ref": "#/$defs/Id"
  },
  "unreadable_text_detected": {
    "const": false
  }
},
"required": [
  "completed",
  "reviewer_id",
  "reviewed_at",
  "rendered_page_count",
  "pages_reviewed",
  "clipping_detected",
  "unreadable_text_detected",

```

```

    "render_manifest_digest"
  ],
  "type": "object"
}
},
"required": [
  "record_type",
  "status",
  "candidate_binding_digest",
  "documents",
  "deterministic_rebuild",
  "rebuild_digest",
  "page_count",
  "extraction_diagnostics",
  "visual_review_scope",
  "font_bindings",
  "generator_result_digest",
  "render_environment",
  "render_manifest",
  "render_manifest_digest",
  "product_acceptance_pass",
  "producer",
  "invocation",
  "raw_artifact_manifest_digest",
  "producer_attestation",
  "result_digest"
],
"type": "object"

```

```

}

```

Owner: [core/contracts.schema.json#/\\$defs/HumanDocumentVerification](#) | Validator: [urn:promin:contracts:1#/\\$defs/HumanDocumentVerification](#)

44. Id

Type summary: type="string"; pattern="^[A-Za-z0-9][A-Za-z0-9._-]{0,127}\$"

Required: -

```
{
  "pattern": "^[A-Za-z0-9][A-Za-z0-9._-]{0,127}$",
  "type": "string"
}
```

Owner: core/contracts.schema.json#/\$defs/Id | Validator: urn:promin:contracts:1#/\$defs/Id

45. ImplementationClosure

Type summary: type="object"; additionalProperties=false

Required: runtime, interpreter, components, adapters, closure_digest

Property	Required	Structural rule
adapters	yes	type="array"; minItems=1; maxItems=1; uniqueItems=true
closure_digest	yes	\$ref="#/\$defs/Digest"
components	yes	type="array"; minItems=1; maxItems=4096; uniqueItems=true
interpreter	yes	type="object"; additionalProperties=false
runtime	yes	type="object"; additionalProperties=false

```
{
  "additionalProperties": false,
  "properties": {
    "adapters": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "adapter_id": {
            "$ref": "#/$defs/Id"
          },
          "capability_id": {
            "enum": [
              "control-runtime",
              "shape-validation",
              "content-identity",
              "local-serialization",
              "query-projection",
              "filesystem-inventory",
              "export-scan",
              "signature",
              "build-dependency"
            ]
          },
          "dependency_receipt_digest": {
            "$ref": "#/$defs/Digest"
          },
          "digest": {
            "$ref": "#/$defs/Digest"
          },
          "operation_contract_digest": {
            "$ref": "#/$defs/Digest"
          },
          "protocol_id": {
            "$ref": "#/$defs/Id"
          },
          "version": {
            "minLength": 1,
            "type": "string"
          }
        },
        "required": [
          "adapter_id",
          "capability_id",
          "version",
          "digest",
          "protocol_id",
          "operation_contract_digest",
          "dependency_receipt_digest"
        ],
        "type": "object"
      },
      "type": "array"
    }
  }
}
```

```

    "maxItems": 1,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },

```

```

    "closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "components": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "component_id": {
            "$ref": "#/$defs/Id"
          },
          "component_kind": {
            "enum": [
              "native-runtime",
              "provider-file",
              "provider-tree",
              "python-distribution"
            ]
          },
          "digest": {
            "$ref": "#/$defs/Digest"
          },
          "size_bytes": {
            "minimum": 1,
            "type": "integer"
          },
          "source": {
            "maxLength": 4096,
            "minLength": 1,

```

```

            "type": "string"
          },
          "version": {
            "maxLength": 256,
            "minLength": 1,
            "type": "string"
          }
        },
        "required": [
          "component_id",
          "component_kind",
          "source",
          "version",
          "digest",
          "size_bytes"
        ],
        "type": "object"
      },
      "maxItems": 4096,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "interpreter": {
      "additionalProperties": false,
      "properties": {
        "executable_digest": {
          "$ref": "#/$defs/Digest"

```

```

        },
        "name": {
          "minLength": 1,
          "type": "string"
        },
        "version": {
          "minLength": 1,
          "type": "string"
        }
      },
      "required": [
        "name",
        "version",
        "executable_digest"
      ],
      "type": "object"
    }
  },

```

```

    },
    "runtime": {
      "additionalProperties": false,
      "properties": {
        "digest": {
          "$ref": "#/$defs/Digest"
        },
        "name": {
          "const": "promin"
        },
        "version": {
          "minLength": 1,
            "type": "string"
          }
      },
      "required": [
        "name",
        "version",
        "digest"
      ],
      "type": "object"
    },
    "type": "object"
  },
  "required": [
    "runtime",
    "interpreter",
    "components",
    "adapters",
    "closure_digest"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ImplementationClosure | Validator: urn:promin:contracts:1#/\$defs/ImplementationClosure

46. InitFileDigests

Type summary: type="object"; additionalProperties=false

Required: project.json, standards.json, technologies.json, authority.json

Property	Required	Structural rule
authority.json	yes	\$ref="#/\$defs/Digest"
project.json	yes	\$ref="#/\$defs/Digest"
standards.json	yes	\$ref="#/\$defs/Digest"
technologies.json	yes	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "properties": {
    "authority.json": {
      "$ref": "#/$defs/Digest"
    },
    "project.json": {
      "$ref": "#/$defs/Digest"
    },
    "standards.json": {
      "$ref": "#/$defs/Digest"
    },
    "technologies.json": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "project.json",
    "standards.json",
    "technologies.json",
    "authority.json"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/InitFileDigests | Validator: urn:promin:contracts:1#/\$defs/InitFileDigests

47. InventoryInputManifest

Type summary: type="object"; additionalProperties=false

Required: record_type, candidate_digest, inventory_digest, stream_artifact_id, stream_artifact_record_digest, stream_digest, stream_bytes, entry_count, row_schema_digest, consistency_mode, snapshot_provider_id, snapshot_digest

Property	Required	Structural rule
candidate_digest	yes	\$ref="#/\$defs/Digest"
consistency_mode	yes	enum=["immutable-vcs-tree", "observational-best-effort"]
entry_count	yes	type="integer"; minimum=0
inventory_digest	yes	\$ref="#/\$defs/Digest"
record_type	yes	const="InventoryInputManifest"
row_schema_digest	yes	\$ref="#/\$defs/Digest"
snapshot_digest	yes	\$ref="#/\$defs/Digest"
snapshot_provider_id	yes	\$ref="#/\$defs/Id"
stream_artifact_id	yes	\$ref="#/\$defs/Id"
stream_artifact_record_digest	yes	\$ref="#/\$defs/Digest"
stream_bytes	yes	type="integer"; minimum=1
stream_digest	yes	\$ref="#/\$defs/Digest"

```
{
  "additionalProperties": false,
  "properties": {
    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "consistency_mode": {
      "enum": [
        "immutable-vcs-tree",
        "observational-best-effort"
      ]
    },
    "entry_count": {
      "minimum": 0,
      "type": "integer"
    },
    "inventory_digest": {
      "$ref": "#/$defs/Digest"
    },
    "record_type": {
      "const": "InventoryInputManifest"
    },
    "row_schema_digest": {
      "$ref": "#/$defs/Digest"
    },
    "snapshot_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
```

```
    "snapshot_provider_id": {
      "$ref": "#/$defs/Id"
    },
    "stream_artifact_id": {
      "$ref": "#/$defs/Id"
    },
    "stream_artifact_record_digest": {
      "$ref": "#/$defs/Digest"
    },
    "stream_bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "stream_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "record_type",
    "candidate_digest",
    "inventory_digest",
    "stream_artifact_id",
```



```

    "stream_artifact_record_digest",
    "stream_digest",
    "stream_bytes",
    "entry_count",
    "row_schema_digest",
    "consistency_mode",

    "snapshot_provider_id",
    "snapshot_digest"
  ],
  "type": "object",
  "x-promin-ingress-operations": [
    "rebuild"
  ],
  "x-promin-persistence": "transient-ingress-only"
}

```

Owner: core/contracts.schema.json#/\$defs/InventoryInputManifest | Validator: urn:promin:contracts:1#/\$defs/InventoryInputManifest

48. InventoryJsonlRow

Type summary: type="object"; additionalProperties=false

Required: path, digest, size, search_text

Property	Required	Structural rule
digest	yes	\$ref="#/\$defs/Digest"
path	yes	\$ref="#/\$defs/RelativePath"
search_text	yes	type="string"; minLength=1; maxLength=4096
size	yes	type="integer"; minimum=0

```

{
  "additionalProperties": false,
  "properties": {
    "digest": {
      "$ref": "#/$defs/Digest"
    },
    "path": {
      "$ref": "#/$defs/RelativePath"
    },
    "search_text": {
      "maxLength": 4096,
      "minLength": 1,
      "type": "string"
    },
    "size": {
      "minimum": 0,
      "type": "integer"
    }
  },
  "required": [
    "path",
    "digest",
    "size",
    "search_text"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/InventoryJsonlRow | Validator: urn:promin:contracts:1#/\$defs/InventoryJsonlRow

49. InventoryProjectionRow

Type summary: type="object"; additionalProperties=false

Required: record_type, path, digest, size, search_text, semantic_proxy

Property	Required	Structural rule
digest	yes	\$ref="#/\$defs/Digest"
path	yes	\$ref="#/\$defs/RelativePath"
record_type	yes	const="InventoryProjectionRow"
search_text	yes	type="string"; minLength=1; maxLength=4096
semantic_proxy	yes	type="object"; additionalProperties=false

Property	Required	Structural rule
size	yes	type="integer"; minimum=0
<pre> { "additionalProperties": false, "properties": { "digest": { "\$ref": "#/\$defs/Digest" }, "path": { "\$ref": "#/\$defs/RelativePath" }, "record_type": { "const": "InventoryProjectionRow" }, "search_text": { "maxLength": 4096, "minLength": 1, "type": "string" }, "semantic_proxy": { "additionalProperties": false, "properties": { "entity_type": { "const": "Artifact" }, "id": { "\$ref": "#/\$defs/Id" }, "payload": { "\$ref": "#/\$defs/Artifact" } } }, "required": ["id", "entity_type", "payload"], "type": "object" }, "size": { "minimum": 0, "type": "integer" } }, "required": ["record_type", "path", "digest", "size", "search_text", "semantic_proxy"], "type": "object", "x-promin-ingress-operations": ["rebuild"], "x-promin-persistence": "derived-rebuild-only" } </pre>		
Owner: core/contracts.schema.json#/\$defs/InventoryProjectionRow Validator: urn:promin:contracts:1#/\$defs/InventoryProjectionRow		

50. Invocation

Type summary: type="object"; additionalProperties=false

Required: kind, value

Property	Required	Structural rule
kind	yes	enum=["python-runtime", "python-module", "executable", "platform-service"]
value	yes	type="string"; minLength=1
<pre> { "additionalProperties": false, "properties": { "kind": { "enum": [</pre>		

```

        "python-runtime",
        "python-module",
        "executable",
        "platform-service"
    ]
},
"value": {
    "minLength": 1,
    "type": "string"
}
},
"required": [
    "kind",
    "value"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Invocation | Validator: urn:promin:contracts:1#/\$defs/Invocation

51. IssuerGrantTrustProof

Type summary: type="object"; additionalProperties=false

Required: kind, issuer_grant_id, issuer_signed_claim_digest, signed_claim_digest

Property	Required	Structural rule
issuer_grant_id	yes	\$ref="#/\$defs/Id"
issuer_signed_claim_digest	yes	\$ref="#/\$defs/Digest"
kind	yes	const="issuer-grant"
signed_claim_digest	yes	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "properties": {
    "issuer_grant_id": {
      "$ref": "#/$defs/Id"
    },
    "issuer_signed_claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "kind": {
      "const": "issuer-grant"
    },
    "signed_claim_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "kind",
    "issuer_grant_id",
    "issuer_signed_claim_digest",
    "signed_claim_digest"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/IssuerGrantTrustProof | Validator: urn:promin:contracts:1#/\$defs/IssuerGrantTrustProof

52. Lease

Type summary: type="object"; additionalProperties=false; allOf[3]=keys:else,if,then | keys:if,then,title | keys:else,if,then,title

Required: record_type, lease_id, task_id, holder_grant_claim_digest, manager_grant_claim_digest, manager_grant_id, manager_subject_id, holder_subject_id, holder_grant_id, generation, fencing_token, state, acquired_at, heartbeat_at, expires_at, activation_digest

Property	Required	Structural rule
acquired_at	yes	\$ref="#/\$defs/Timestamp"
activation_digest	yes	\$ref="#/\$defs/Digest"
capacity_reconciliation	no	\$ref="#/\$defs/CapacityReconciliation"
close_ack	no	\$ref="#/\$defs/CloseAck"
expires_at	yes	\$ref="#/\$defs/Timestamp"

Property	Required	Structural rule
fencing_token	yes	type="integer"; minimum=1
generation	yes	type="integer"; minimum=1
heartbeat_at	yes	\$ref="#/\$defs/Timestamp"
holder_grant_claim_digest	yes	\$ref="#/\$defs/Digest"
holder_grant_id	yes	\$ref="#/\$defs/Id"
holder_subject_id	yes	\$ref="#/\$defs/Id"
lease_id	yes	\$ref="#/\$defs/Id"
manager_grant_claim_digest	yes	\$ref="#/\$defs/Digest"
manager_grant_id	yes	\$ref="#/\$defs/Id"
manager_subject_id	yes	\$ref="#/\$defs/Id"
record_type	yes	const="Lease"
state	yes	\$ref="#/\$defs/LeaseState"
task_id	yes	\$ref="#/\$defs/Id"
termination	no	type="object"; additionalProperties=false

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "required": [
            "close_ack"
          ]
        }
      },
      "if": {
        "properties": {
          "state": {
            "const": "CLOSED"
          }
        }
      },
      "then": {
        "required": [
          "close_ack"
        ]
      }
    },
    {
      "if": {
        "properties": {
          "state": {
            "not": {
              "enum": [
                "EXPIRED",
                "REVOKED"
              ]
            }
          }
        }
      },
      "then": {
        "not": {
          "required": [
            "capacity_reconciliation"
          ]
        }
      },
      "title": "capacity reconciliation exists only on reconciled expired or revoked Lease"
    },
    {
      "else": {
        "not": {
          "required": [
            "termination"
          ]
        }
      }
    }
  ],
  "termination": false
}

```

```

    "if": {
      "properties": {
        "state": {
          "enum": [
            "EXPIRED",
            "REVOKED"
          ]
        },
        "required": [
          "state"
        ]
      },
      "then": {
        "required": [
          "termination"
        ]
      },
      "title": "terminal Lease carries exact termination provenance"
    }
  ],
  "properties": {
    "acquired_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "capacity_reconciliation": {
      "$ref": "#/$defs/CapacityReconciliation"
    },
    "close_ack": {
      "$ref": "#/$defs/CloseAck"
    },
    "expires_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "fencing_token": {
      "minimum": 1,
      "type": "integer"
    },
    "generation": {
      "minimum": 1,
      "type": "integer"
    },
    "heartbeat_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "holder_grant_claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "holder_grant_id": {
      "$ref": "#/$defs/Id"
    },
    "holder_subject_id": {
      "$ref": "#/$defs/Id"
    },
    "lease_id": {
      "$ref": "#/$defs/Id"
    },
    "manager_grant_claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "manager_grant_id": {
      "$ref": "#/$defs/Id"
    },
    "manager_subject_id": {
      "$ref": "#/$defs/Id"
    },
    "record_type": {
      "const": "Lease"
    },
    "state": {
      "$ref": "#/$defs/LeaseState"
    },
    "task_id": {
      "$ref": "#/$defs/Id"
    }
  }
}

```

```

    },
    "termination": {
      "additionalProperties": false,
      "properties": {
        "fencing_token": {
          "minimum": 1,
          "type": "integer"
        },
        "generation": {

```

```

          "minimum": 1,
          "type": "integer"
        },
        "grant_claim_digest": {
          "$ref": "#/$defs/Digest"
        },
        "grant_id": {
          "$ref": "#/$defs/Id"
        },
        "state": {
          "enum": [
            "EXPIRED",
            "REVOKED"
          ]
        },
        "terminated_at": {
          "$ref": "#/$defs/Timestamp"
        },
        "terminated_by": {
          "$ref": "#/$defs/Id"
        }
      }
    },
    "required": [
      "state",
      "terminated_by",
      "grant_id",
      "grant_claim_digest",
      "terminated_at",

```

```

      "generation",
      "fencing_token"
    ],
    "type": "object"
  }
},
"required": [
  "record_type",
  "lease_id",
  "task_id",
  "holder_grant_claim_digest",
  "manager_grant_claim_digest",
  "manager_grant_id",
  "manager_subject_id",
  "holder_subject_id",
  "holder_grant_id",
  "generation",
  "fencing_token",
  "state",
  "acquired_at",
  "heartbeat_at",
  "expires_at",
  "activation_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Lease | Validator: urn:promin:contracts:1#/\$defs/Lease

53. LeaseState

Type summary: enum=["ACTIVE", "CLOSED", "CLOSING", "EXPIRED", "REVOKED"]

Required: -

```

{
  "enum": [
    "ACTIVE",
    "CLOSED",
    "CLOSING",
    "EXPIRED",

```

```

    "REVOKED"
  ]
}

```

Owner: core/contracts.schema.json#/\$defs/LeaseState | Validator: urn:promin:contracts:1#/\$defs/LeaseState

54. LeaseStateMachine

Type summary: const=schema(keys:ACTIVE,CLOSED,CLOSING,EXPIRED,REVOKED)

Required: -

```

{
  "const": {
    "ACTIVE": [
      "CLOSING",
      "EXPIRED",
      "REVOKED"
    ],
    "CLOSED": [],
    "CLOSING": [
      "CLOSED",
      "EXPIRED",
      "REVOKED"
    ],
    "EXPIRED": [],
    "REVOKED": []
  }
}

```

Owner: core/contracts.schema.json#/\$defs/LeaseStateMachine | Validator: urn:promin:contracts:1#/\$defs/LeaseStateMachine

55. LicenseBinding

Type summary: type="object"; additionalProperties=false

Required: expression, source_uris, review_state

Property	Required	Structural rule
expression	yes	type="string"; minLength=1
review_state	yes	enum=["reviewed", "owner-review-required", "source-verified"]
source_uris	yes	type="array"; minItems=1; maxItems=8; uniqueItems=true

```

{
  "additionalProperties": false,
  "properties": {
    "expression": {
      "minLength": 1,
      "type": "string"
    },
    "review_state": {
      "enum": [
        "reviewed",
        "owner-review-required",
        "source-verified"
      ]
    },
    "source_uris": {
      "items": {
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 8,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    }
  },
  "required": [
    "expression",
    "source_uris",
    "review_state"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/LicenseBinding | Validator: urn:promin:contracts:1#/\$defs/LicenseBinding

56. LocalActivationProof

Type summary: type="object"; additionalProperties=false

Required: kind, subject_id, authority_init_digest

Property	Required	Structural rule
authority_init_digest	yes	\$ref="#/\$defs/Digest"
kind	yes	const="local-root"
subject_id	yes	\$ref="#/\$defs/Id"

```
{
  "additionalProperties": false,
  "properties": {
    "authority_init_digest": {
      "$ref": "#/$defs/Digest"
    },
    "kind": {
      "const": "local-root"
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "kind",
    "subject_id",
    "authority_init_digest"
  ],
  "type": "object"
}
```

Owner: core/contracts.schema.json#/\$defs/LocalActivationProof | Validator: urn:promin:contracts:1#/\$defs/LocalActivationProof

57. LocalRootCommandProof

Type summary: type="object"; additionalProperties=false

Required: kind, subject_id, authority_init_digest, signed_intent_digest

Property	Required	Structural rule
authority_init_digest	yes	\$ref="#/\$defs/Digest"
kind	yes	const="local-root-command"
signed_intent_digest	yes	\$ref="#/\$defs/Digest"
subject_id	yes	\$ref="#/\$defs/Id"

```
{
  "additionalProperties": false,
  "properties": {
    "authority_init_digest": {
      "$ref": "#/$defs/Digest"
    },
    "kind": {
      "const": "local-root-command"
    },
    "signed_intent_digest": {
      "$ref": "#/$defs/Digest"
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "kind",
    "subject_id",
    "authority_init_digest",
    "signed_intent_digest"
  ],
  "type": "object"
}
```

Owner: core/contracts.schema.json#/\$defs/LocalRootCommandProof | Validator: urn:promin:contracts:1#/\$defs/LocalRootCommandProof

58. LocalTrustProof

Type summary: type="object"; additionalProperties=false

Required: kind, root_subject_id, authority_init_digest, signed_claim_digest

Property	Required	Structural rule
authority_init_digest	yes	\$ref="#/\$defs/Digest"
kind	yes	const="local-root"
root_subject_id	yes	\$ref="#/\$defs/Id"
signed_claim_digest	yes	\$ref="#/\$defs/Digest"

```
{
  "additionalProperties": false,
  "properties": {
    "authority_init_digest": {
      "$ref": "#/$defs/Digest"
    },
    "kind": {
      "const": "local-root"
    },
    "root_subject_id": {
      "$ref": "#/$defs/Id"
    },
    "signed_claim_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "kind",
    "root_subject_id",
    "authority_init_digest",
    "signed_claim_digest"
  ],
  "type": "object"
}
```

Owner: core/contracts.schema.json#/\$defs/LocalTrustProof | Validator: urn:promin:contracts:1#/\$defs/LocalTrustProof

59. NextResult

Type summary: type="object"; additionalProperties=false; allOf[2]=keys:else,if,then | keys:if,then

Required: record_type, status, subject_id, activation_digest, work_card, continuation

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
continuation	yes	oneOf[2]=type:null type:string
record_type	yes	const="NextResult"
status	yes	enum=["ready", "empty"]
subject_id	yes	\$ref="#/\$defs/Id"
work_card	yes	oneOf[2]=#/\$defs/WorkCard type:null

```
{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "properties": {
          "continuation": {
            "type": "null"
          },
          "work_card": {
            "type": "null"
          }
        }
      },
      "if": {
        "properties": {
          "status": {
            "const": "ready"
          }
        }
      },
      "required": [
        "status"
      ]
    }
  ]
}
```

```

    ]
  },
  "then": {
    "properties": {
      "work_card": {
        "$ref": "#/$defs/WorkCard"

```

```

      }
    }
  },
  {
    "if": {
      "properties": {
        "continuation": {
          "type": "string"
        }
      },
      "required": [
        "continuation"
      ]
    },
    "then": {
      "properties": {
        "work_card": {
          "properties": {
            "truncated": {
              "const": true
            }
          },
          "required": [
            "truncated"
          ]
        }
      }
    }
  }
}

```

```

    }
  },
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "continuation": {
      "oneOf": [
        {
          "type": "null"
        },
        {
          "maxLength": 256,
          "minLength": 1,
          "pattern": "^promin-v2\\. [A-Za-z0-9_-]{22}\\.[0-9a-f]{64}\\.[A-Za-z0-9_-]{43}$",
          "type": "string"
        }
      ]
    },
    "record_type": {
      "const": "NextResult"
    },
    "status": {
      "enum": [
        "ready",
        "empty"
      ]
    }
  }
}

```

```

  },
  "subject_id": {
    "$ref": "#/$defs/Id"
  },
  "work_card": {
    "oneOf": [
      {
        "$ref": "#/$defs/WorkCard"
      },
      {
        "type": "null"
      }
    ]
  }
}

```

```

"required": [
  "record_type",
  "status",
  "subject_id",
  "activation_digest",
  "work_card",
  "continuation"
],
"type": "object",
"x-promin-persistence": "derived-live-only"
}

```

Owner: core/contracts.schema.json#/\$defs/NextResult | Validator: urn:promin:contracts:1#/\$defs/NextResult

60. NoDegradationResult

Type summary: type="object"; additionalProperties=false

Required: record_type, status, platform, no_degradation, source, root, artifact_binding, candidate_binding_digest, test_manifest, install_mode, required_predicates, validation, tests, passed, pass_credit, acceptance_pass, product_acceptance_pass, product_public_approval, scale_selection, artifact_binding_unchanged, standard_distribution_gate_pass, phases, execution_budget, producer, invocation, raw_artifact_manifest_digest, producer_attestation, result_digest

Property	Required	Structural rule
acceptance_pass	yes	const=false
artifact_binding	yes	type="object"
artifact_binding_unchanged	yes	const=true
candidate_binding_digest	yes	\$ref="#/\$defs/Digest"
execution_budget	yes	type="object"; additionalProperties=false
install_mode	yes	const="offline-wheelhouse"
invocation	yes	\$ref="#/\$defs/ReleaseEvidenceInvocation"
no_degradation	yes	const=true
pass_credit	yes	const=false
passed	yes	const=true
phases	yes	type="array"; minItems=2; maxItems=2
platform	yes	enum=["linux", "windows"]
producer	yes	\$ref="#/\$defs/ReleaseEvidenceProducer"
producer_attestation	yes	\$ref="#/\$defs/EvidenceProducerAttestation"
product_acceptance_pass	yes	const=false
product_public_approval	yes	const="not_approved"
raw_artifact_manifest_digest	yes	\$ref="#/\$defs/Digest"
record_type	yes	const="NoDegradationResult"
required_predicates	yes	type="object"; additionalProperties=false
result_digest	yes	\$ref="#/\$defs/Digest"
root	yes	type="string"; minLength=1
scale_selection	yes	type="object"; additionalProperties=false
source	yes	const="live_files_and_executable_tests"
standard_distribution_gate_pass	yes	const=false
status	yes	const="pass"
test_manifest	yes	type="object"; additionalProperties=false
tests	yes	type="object"; additionalProperties=false
validation	yes	type="object"; additionalProperties=false

```

{
  "additionalProperties": false,
  "properties": {
    "acceptance_pass": {
      "const": false
    },
    "artifact_binding": {
      "type": "object"
    },

```

```

"artifact_binding_unchanged": {
  "const": true
},
"candidate_binding_digest": {
  "$ref": "#/$defs/Digest"
},
"execution_budget": {
  "additionalProperties": false,
  "properties": {
    "completed_at": {},
    "elapsed_ms": {},
    "started_at": {},
    "test_timeout_seconds": {},
    "total_timeout_seconds": {},
    "validation_timeout_seconds": {}
  },
  "required": [
    "total_timeout_seconds",
    "validation_timeout_seconds",
    "test_timeout_seconds",
    "elapsed_ms",
    "started_at",
    "completed_at"
  ],
  "type": "object"
},
"install_mode": {
  "const": "offline-wheelhouse"
},
"invocation": {
  "$ref": "#/$defs/ReleaseEvidenceInvocation"
},
"no_degradation": {
  "const": true
},
"pass_credit": {
  "const": false
},
"passed": {
  "const": true
},
"phases": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "elapsed_ms": {},
      "phase": {},
      "process_group_cleanup": {},
      "returncode": {},
      "status": {},
      "timeout_seconds": {}
    },
    "required": [
      "phase",
      "status",
      "returncode",
      "elapsed_ms",
      "timeout_seconds",
      "process_group_cleanup"
    ],
    "type": "object"
  },
  "maxItems": 2,
  "minItems": 2,
  "type": "array"
},
"platform": {
  "enum": [
    "linux",
    "windows"
  ]
},
"producer": {
  "$ref": "#/$defs/ReleaseEvidenceProducer"
},
"producer_attestation": {
  "$ref": "#/$defs/EvidenceProducerAttestation"
}

```

```

    },
    "product_acceptance_pass": {
      "const": false
    },
    "product_public_approval": {
      "const": "not_approved"
    },
    "raw_artifact_manifest_digest": {
      "$ref": "#/$defs/Digest"
    },
    "record_type": {
      "const": "NoDegradationResult"
    },
    "required_predicates": {
      "additionalProperties": false,
      "properties": {
        "acceptance_predicates": {},
        "complete": {},
        "missing_acceptance_predicates": {},
        "missing_mutation_families": {},
        "missing_policies": {},
        "mutation_families": {},
        "policies": {}
      }
    },
    "required": [
      "policies",
      "acceptance_predicates",
      "mutation_families",
      "missing_policies",
      "missing_acceptance_predicates",
      "missing_mutation_families",
      "complete"
    ],
    "type": "object"
  },
  "result_digest": {
    "$ref": "#/$defs/Digest"
  },
  "root": {
    "minLength": 1,
    "type": "string"
  },
  "scale_selection": {
    "additionalProperties": false,
    "properties": {
      "command": {},
      "missing_workspace_is_failure": {},
      "pass_credit": {},
      "runs_separately": {}
    },
    "required": [
      "command",
      "runs_separately",
      "missing_workspace_is_failure",
      "pass_credit"
    ],
    "type": "object"
  },
  "source": {
    "const": "live_files_and_executable_tests"
  },
  "standard_distribution_gate_pass": {
    "const": false
  },
  "status": {
    "const": "pass"
  },
  "test_manifest": {
    "additionalProperties": false,
    "properties": {
      "digest": {},
      "file_count": {},
      "files": {}
    },
    "required": [
      "files",
      "file_count",
      "digest"
    ]
  }
}

```

```

    ],
    "type": "object"
  },
  "tests": {

```

```

    "additionalProperties": false,
    "properties": {
      "command": {},
      "elapsed_ms": {},
      "installed_environment": {
        "additionalProperties": false,
        "properties": {
          "build_dependency_source": {
            "const": "offline-wheelhouse"
          },
          "console_script": {
            "additionalProperties": false,
            "properties": {
              "declared": {
                "const": true
              },
              "installed_wrapper_checked": {
                "const": true
              },
              "path": {
                "minLength": 1,
                "type": "string"
              },
              "posix_execute_bits": {
                "type": [
                  "boolean",
                  "null"
                ]
              }
            }
          }
        }
      }
    }
  },
  "present": {
    "const": true
  },
  "sha256": {
    "$ref": "#/$defs/Digest"
  },
  "size_bytes": {
    "minimum": 1,
    "type": "integer"
  },
  "source_module_invoked": {
    "const": false
  }
},
"required": [
  "present",
  "declared",
  "installed_wrapper_checked",
  "source_module_invoked",
  "posix_execute_bits",
  "path",
  "sha256",
  "size_bytes"
],
"type": "object"
},
"declared_build_dependencies": {

```

```

  "items": {
    "type": "string"
  },
  "minItems": 1,
  "type": "array"
},
"declared_python_requirement": {
  "minLength": 1,
  "type": "string"
},
"declared_runtime_dependencies": {
  "items": {
    "type": "string"
  },
  "minItems": 1,
  "type": "array"
},

```

```

"dependency_closure_check": {
  "const": "pass"
},
"environment": {
  "const": "clean-venv"
},
"installation_performed": {
  "const": true
},
"installed_environment_observation": {
  "additionalProperties": false,

```

```

"properties": {
  "build_backend": {
    "additionalProperties": false,
    "properties": {
      "distribution": {
        "const": "setuptools"
      },
      "name": {
        "const": "setuptools.build_meta"
      },
    },
    "record_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "version": {
      "minLength": 1,
      "type": "string"
    }
  },
  "required": [
    "name",
    "distribution",
    "version",
    "record_sha256"
  ],
  "type": "object"
},
"console_wrapper": {
  "additionalProperties": false,

```

```

"properties": {
  "bytes": {
    "minimum": 1,
    "type": "integer"
  },
  "path": {
    "minLength": 1,
    "type": "string"
  },
  "sha256": {
    "$ref": "#/$defs/Digest"
  }
},
"required": [
  "path",
  "sha256",
  "bytes"
],
"type": "object"
},
"license_closure": {
  "additionalProperties": false,
  "properties": {
    "complete": {
      "const": true
    },
    "digest": {
      "$ref": "#/$defs/Digest"

```

```

},
"entries": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "license": {
        "maxLength": 2048,
        "minLength": 1,
        "not": {
          "const": "UNKNOWN"

```

```

    },
    "type": "string"
  },
  "name": {
    "minLength": 1,
    "type": "string"
  },
  "version": {
    "minLength": 1,
    "type": "string"
  }
},
"required": [
  "name",
  "version",
  "license"
],
"type": "object"

```

```

    },
    "maxItems": 256,
    "minItems": 1,
    "type": "array"
  }
},
"required": [
  "entries",
  "complete",
  "digest"
],
"type": "object"
},
"observation_digest": {
  "$ref": "#/$defs/Digest"
},
"platform": {
  "additionalProperties": false,
  "properties": {
    "machine": {
      "minLength": 1,
      "type": "string"
    },
    "release": {
      "minLength": 1,
      "type": "string"
    }
  },
  "sys_platform": {

```

```

    "minLength": 1,
    "type": "string"
  },
  "system": {
    "enum": [
      "linux",
      "windows"
    ]
  },
  "tags": {
    "items": {
      "minLength": 1,
      "type": "string"
    },
    "maxItems": 64,
    "minItems": 1,
    "type": "array"
  }
},
"required": [
  "system",
  "machine",
  "release",
  "sys_platform",
  "tags"
],
"type": "object"
},

```

```

"promin": {
  "additionalProperties": false,
  "properties": {

```



```

    "installed_file_count": {
      "maximum": 4096,
      "minimum": 1,
      "type": "integer"
    },
    "installed_files": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "bytes": {
            "minimum": 0,
            "type": "integer"
          },
          "path": {
            "minLength": 1,
            "type": "string"
          },
          "sha256": {
            "$ref": "#/$defs/Digest"
          }
        },
        "required": [
          "path",
          "sha256",
          "bytes"
        ]
      },
      "type": "object"
    },
    "maxItems": 4096,
    "minItems": 1,
    "type": "array"
  },
  "installed_files_digest": {
    "$ref": "#/$defs/Digest"
  },
  "module_file": {
    "minLength": 1,
    "type": "string"
  },
  "module_sha256": {
    "$ref": "#/$defs/Digest"
  },
  "record": {
    "additionalProperties": false,
    "properties": {
      "bytes": {
        "minimum": 1,
        "type": "integer"
      },
      "path": {
        "minLength": 1,
        "pattern": "\\\\.dist-info/RECORD$",
        "type": "string"
      },
      "sha256": {
        "$ref": "#/$defs/Digest"
      }
    },
    "required": [
      "path",
      "sha256",
      "bytes"
    ],
    "type": "object"
  },
  "version": {
    "$ref": "#/$defs/SemVer"
  }
},
"required": [
  "version",
  "module_file",
  "module_sha256",
  "record",
  "installed_files",
  "installed_files_digest",
  "installed_file_count"
],

```

```

    "type": "object"
  },
  "python": {
    "additionalProperties": false,
    "properties": {
      "abi_tag": {
        "minLength": 1,
        "type": "string"
      },
      "base_executable": {
        "minLength": 1,
        "type": "string"
      },
      "base_executable_sha256": {
        "$ref": "#/$defs/Digest"
      },
      "base_prefix": {
        "minLength": 1,
        "type": "string"
      },
      "executable": {
        "minLength": 1,
        "type": "string"
      },
      "executable_sha256": {
        "$ref": "#/$defs/Digest"
      },
      "implementation": {
        "minLength": 1,
        "type": "string"
      }
    },
    "prefix": {
      "minLength": 1,
      "type": "string"
    },
    "version": {
      "minLength": 1,
      "type": "string"
    }
  },
  "required": [
    "executable",
    "executable_sha256",
    "base_executable",
    "base_executable_sha256",
    "implementation",
    "version",
    "abi_tag",
    "prefix",
    "base_prefix"
  ],
  "type": "object"
},
"sbom": {
  "additionalProperties": false,
  "properties": {
    "components": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "name": {
            "minLength": 1,
            "type": "string"
          },
          "record_sha256": {
            "$ref": "#/$defs/Digest"
          },
          "version": {
            "minLength": 1,
            "type": "string"
          }
        },
        "required": [
          "name",
          "version",
          "record_sha256"
        ],

```

```

        "type": "object"
      },
      "maxItems": 256,
      "minItems": 1,
      "type": "array"
    },
    "digest": {
      "$ref": "#/$defs/Digest"
    },
    "format": {

```

```

      "const": "promin-spx-lite-v1"
    }
  },
  "required": [
    "format",
    "components",
    "digest"
  ],
  "type": "object"
},
"site_packages": {
  "minLength": 1,
  "type": "string"
},
"sqlite_version": {
  "minLength": 1,
  "type": "string"
},
"transitive_distribution_digest": {
  "$ref": "#/$defs/Digest"
},
"transitive_distributions": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "license": {
        "maxLength": 2048,
        "minLength": 1,

```

```

        "type": "string"
      },
      "name": {
        "minLength": 1,
        "type": "string"
      },
      "record": {
        "additionalProperties": false,
        "properties": {
          "bytes": {
            "minimum": 1,
            "type": "integer"
          },
          "path": {
            "minLength": 1,
            "pattern": "\\..dist-info/RECORD$",
            "type": "string"
          },
          "sha256": {
            "$ref": "#/$defs/Digest"
          }
        },
        "required": [
          "path",
          "sha256",
          "bytes"
        ],
        "type": "object"

```

```

    },
    "requires": {
      "items": {
        "minLength": 1,
        "type": "string"
      },
      "type": "array"
    },
    "version": {
      "minLength": 1,
      "type": "string"

```

```

    }
  },
  "required": [
    "name",
    "version",
    "record",
    "requires",
    "license"
  ],
  "type": "object"
},
"maxItems": 256,
"minItems": 1,
"type": "array"
}
},
"required": [

```

```

  "python",
  "platform",
  "sqlite_version",
  "site_packages",
  "promin",
  "console_wrapper",
  "transitive_distributions",
  "transitive_distribution_digest",
  "sbom",
  "license_closure",
  "build_backend",
  "observation_digest"
],
"type": "object"
},
"interpreter": {
  "additionalProperties": false,
  "properties": {
    "executable": {
      "minLength": 1,
      "type": "string"
    },
    "nested_venv_created": {
      "const": true
    }
  }
},
"required": [
  "executable",

```

```

    "nested_venv_created"
  ],
  "type": "object"
},
"invocations": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "argv_digest": {
        "$ref": "#/$defs/Digest"
      },
      "command": {
        "maxLength": 128,
        "minLength": 1,
        "type": "string"
      },
      "returncode": {
        "const": 0
      },
      "stderr_bytes": {
        "maximum": 1048576,
        "minimum": 0,
        "type": "integer"
      },
      "stderr_sha256": {
        "$ref": "#/$defs/Digest"
      },
      "stdout_bytes": {

```

```

        "maximum": 1048576,
        "minimum": 0,
        "type": "integer"
      },
    },

```

```

        "stdout_sha256": {
            "$ref": "#/$defs/Digest"
        }
    },
    "required": [
        "command",
        "argv_digest",
        "returncode",
        "stdout_sha256",
        "stdout_bytes",
        "stderr_sha256",
        "stderr_bytes"
    ],
    "type": "object"
},
"maxItems": 7,
"minItems": 7,
"type": "array"
},
"mode": {
    "const": "offline-wheelhouse"
},
"nested_venv_created": {
    "const": true

```

```

},
"network_disabled": {
    "const": true
},
"origin_assertion": {
    "additionalProperties": false,
    "properties": {
        "inside_site_packages": {
            "const": true
        },
        "isolated_mode": {
            "const": true
        },
        "module_file": {
            "minLength": 1,
            "type": "string"
        },
        "module_sha256": {
            "$ref": "#/$defs/Digest"
        },
        "outside_install_source": {
            "const": true
        },
        "safe_path": {
            "const": true
        },
        "site_packages": {
            "minLength": 1,

```

```

            "type": "string"
        },
        "source_absent_from_sys_path": {
            "const": true
        },
        "sys_path_digest": {
            "$ref": "#/$defs/Digest"
        }
    },
    "required": [
        "isolated_mode",
        "safe_path",
        "module_file",
        "module_sha256",
        "site_packages",
        "inside_site_packages",
        "outside_install_source",
        "source_absent_from_sys_path",
        "sys_path_digest"
    ],
    "type": "object"
},
"performed": {
    "const": true
},
"pip_report": {

```

```
"additionalProperties": false,  
"properties": {
```

```
  "artifact_digest": {  
    "$ref": "#/$defs/Digest"  
  },  
  "report_count": {  
    "const": 2  
  },  
  "reports": {  
    "items": {  
      "additionalProperties": false,  
      "properties": {  
        "artifact_digest": {  
          "$ref": "#/$defs/Digest"  
        },  
        "artifacts": {  
          "items": {  
            "additionalProperties": false,  
            "properties": {  
              "hashes": {  
                "items": {  
                  "minLength": 3,  
                  "type": "string"  
                },  
                "type": "array"  
              },  
              "is_direct": {  
                "type": "boolean"  
              },  
              "name": {
```

```
                "minLength": 1,  
                "type": "string"  
              },  
              "version": {  
                "minLength": 1,  
                "type": "string"  
              }  
            },  
            "required": [  
              "name",  
              "version",  
              "is_direct",  
              "hashes"  
            ],  
            "type": "object"  
          },  
          "maxItems": 256,  
          "minItems": 1,  
          "type": "array"  
        },  
        "bytes": {  
          "maximum": 16777216,  
          "minimum": 1,  
          "type": "integer"  
        },  
        "install_count": {  
          "maximum": 256,  
          "minimum": 1,
```

```
          "type": "integer"  
        },  
        "pip_version": {  
          "minLength": 1,  
          "type": "string"  
        },  
        "role": {  
          "enum": [  
            "build-requirements",  
            "candidate-install"  
          ]  
        },  
        "sha256": {  
          "$ref": "#/$defs/Digest"  
        }  
      },  
      "required": [  
        "role",  
        "sha256",
```

```

        "bytes",
        "pip_version",
        "install_count",
        "artifacts",
        "artifact_digest"
    ],
    "type": "object"
},
"maxItems": 2,

    "minItems": 2,
    "type": "array"
}
},
"required": [
    "reports",
    "report_count",
    "artifact_digest"
],
"type": "object"
},
"platform": {
    "additionalProperties": false,
    "properties": {
        "os_name": {
            "enum": [
                "nt",
                "posix"
            ]
        },
        "python": {
            "minLength": 1,
            "type": "string"
        },
        "sys_platform": {
            "minLength": 1,
            "type": "string"
        }
    }
}

```

```

},
"required": [
    "os_name",
    "sys_platform",
    "python"
],
"type": "object"
},
"product_acceptance_pass": {
    "const": false
},
"python_distribution_version": {
    "$ref": "#/$defs/SemVer"
},
"resolved_runtime_dependencies": {
    "additionalProperties": false,
    "properties": {
        "cryptography": {
            "additionalProperties": false,
            "properties": {
                "requirement": {
                    "minLength": 1,
                    "type": "string"
                },
                "version": {
                    "minLength": 1,
                    "type": "string"
                }
            }
        }
    }
}

```

```

},
"required": [
    "requirement",
    "version"
],
"type": "object"
},
"jsonschema": {
    "additionalProperties": false,
    "properties": {
        "requirement": {
            "minLength": 1,

```

```

        "type": "string"
      },
      "version": {
        "minLength": 1,
        "type": "string"
      }
    },
    "required": [
      "requirement",
      "version"
    ],
    "type": "object"
  },
  "pypdf": {
    "additionalProperties": false,
    "properties": {
      "requirement": {
        "minLength": 1,
        "type": "string"
      },
      "version": {
        "minLength": 1,
        "type": "string"
      }
    },
    "required": [
      "requirement",
      "version"
    ],
    "type": "object"
  },
  "required": [
    "cryptography",
    "jsonschema",
    "pypdf"
  ],
  "type": "object"
},
"runtime_dependency_source": {
  "const": "offline-wheelhouse"
},
"test_dependencies_installed": {
  "const": true
},
"verified": {
  "const": true
},
"version": {
  "$ref": "#/$defs/SemVer"
},
"wheelhouse_binding": {
  "additionalProperties": false,
  "properties": {
    "file_count": {
      "maximum": 256,
      "minimum": 1,
      "type": "integer"
    },
    "files": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "bytes": {
            "minimum": 1,
            "type": "integer"
          },
          "path": {
            "minLength": 1,
            "type": "string"
          },
          "sha256": {
            "$ref": "#/$defs/Digest"
          }
        },
        "required": [
          "path",

```



```

        "bytes",
        "sha256"
    ],
    "type": "object"
},
"maxItems": 256,
"minItems": 1,
"type": "array"
},
"manifest_sha256": {
    "$ref": "#/$defs/Digest"
}
},
"required": [
    "files",
    "file_count",
    "manifest_sha256"
],
"type": "object"
},
"wheelhouse_lock": {
    "additionalProperties": false,
    "properties": {

```

```

        "copy_verified": {
            "const": true
        },
        "locked_copy_binding": {
            "additionalProperties": false,
            "properties": {
                "file_count": {
                    "maximum": 256,
                    "minimum": 1,
                    "type": "integer"
                },
                "files": {
                    "items": {
                        "additionalProperties": false,
                        "properties": {
                            "bytes": {
                                "minimum": 1,
                                "type": "integer"
                            },
                            "path": {
                                "minLength": 1,
                                "type": "string"
                            },
                            "sha256": {
                                "$ref": "#/$defs/Digest"
                            }
                        }
                    },
                    "required": [

```

```

                        "path",
                        "bytes",
                        "sha256"
                    ],
                    "type": "object"
                },
                "maxItems": 256,
                "minItems": 1,
                "type": "array"
            },
            "manifest_sha256": {
                "$ref": "#/$defs/Digest"
            }
        },
        "required": [
            "files",
            "file_count",
            "manifest_sha256"
        ],
        "type": "object"
    },
    "source_binding": {
        "additionalProperties": false,
        "properties": {
            "file_count": {
                "maximum": 256,

```

```

        "minimum": 1,
        "type": "integer"
    },
    "files": {
        "items": {
            "additionalProperties": false,
            "properties": {
                "bytes": {
                    "minimum": 1,
                    "type": "integer"
                },
                "path": {
                    "minLength": 1,
                    "type": "string"
                },
                "sha256": {
                    "$ref": "#/$defs/Digest"
                }
            },
            "required": [
                "path",
                "bytes",
                "sha256"
            ],
            "type": "object"
        },
        "maxItems": 256,
        "minItems": 1,
        "type": "array"
    },

```

```

        "manifest_sha256": {
            "$ref": "#/$defs/Digest"
        }
    },
    "required": [
        "files",
        "file_count",
        "manifest_sha256"
    ],
    "type": "object"
},
"source_unchanged": {
    "const": true
}
},
"required": [
    "source_binding",
    "locked_copy_binding",
    "source_unchanged",
    "copy_verified"
],
"type": "object"
}
},
"required": [
    "performed",
    "verified",
    "mode",

```

```

"environment",
"runtime_dependency_source",
"build_dependency_source",
"network_disabled",
"installation_performed",
"nested_venv_created",
"wheelhouse_binding",
"declared_runtime_dependencies",
"declared_build_dependencies",
"declared_python_requirement",
"resolved_runtime_dependencies",
"dependency_closure_check",
"installed_environment_observation",
"pip_report",
"wheelhouse_lock",
"origin_assertion",
"test_dependencies_installed",
"version",
"python_distribution_version",

```

```

        "console_script",
        "invocations",
        "interpreter",
        "platform",
        "product_acceptance_pass"
    ],
    "type": "object"
},
"junit_sha256": {},

```

```

"marker_expression": {},
"optional_platform_tests_excluded": {},
"passed": {},
"process_group_cleanup": {},
"ran": {},
"raw_artifact_manifest_digest": {
    "$ref": "#/$defs/Digest"
},
"raw_artifacts": {
    "items": {
        "additionalProperties": false,
        "properties": {
            "bytes": {
                "maximum": 16777216,
                "minimum": 0,
                "type": "integer"
            },
            "data": {
                "type": "string"
            },
            "encoding": {
                "const": "base64"
            },
            "media_type": {
                "enum": [
                    "application/xml",
                    "text/plain; charset=utf-8"
                ]
            }
        }
    }
}

```

```

    },
    "role": {
        "enum": [
            "required-junit",
            "required-stdout",
            "required-stderr"
        ]
    },
    "sha256": {
        "$ref": "#/$defs/Digest"
    }
},
"required": [
    "role",
    "media_type",
    "bytes",
    "sha256",
    "encoding",
    "data"
],
"type": "object"
},
"maxItems": 3,
"minItems": 3,
"type": "array"
},
"required_counts": {},
"required_skips_fail_closed": {},

```

```

"returncode": {},
"scale_environment_configured": {},
"selection_expression": {},
"source_tree_shadowing_disabled": {
    "const": true
},
"stderr": {
    "additionalProperties": false,
    "properties": {
        "bytes": {
            "minimum": 0,
            "type": "integer"
        }
    }
}

```

```

    },
    "selection": {
      "enum": [
        "complete",
        "tail"
      ]
    },
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    },
    },
    "text": {
      "type": "string"
    },
    },
    "truncated": {
      "type": "boolean"
    }
  }

```

```

    },
    "required": [
      "text",
      "bytes",
      "sha256",
      "truncated",
      "selection"
    ],
    "type": "object"
  },
  "stdout": {
    "additionalProperties": false,
    "properties": {
      "bytes": {
        "minimum": 0,
        "type": "integer"
      },
      "selection": {
        "enum": [
          "complete",
          "tail"
        ]
      },
      },
      "sha256": {
        "$ref": "#/$defs/Digest"
      },
      },
      "text": {
        "type": "string"
      }
    }
  },
  "truncated": {
    "type": "boolean"
  }
},
"required": [
  "text",
  "bytes",
  "sha256",
  "truncated",
  "selection"
],
"type": "object"
},
"timed_out": {},
"timeout_seconds": {}
},
"required": [
  "ran",
  "command",
  "returncode",
  "stdout",
  "stderr",
  "timed_out",
  "timeout_seconds",
  "elapsed_ms",
  "process_group_cleanup",
  "required_counts",

```

```

    },
    "truncated": {
      "type": "boolean"
    }
  },
  "required": [
    "text",
    "bytes",
    "sha256",
    "truncated",
    "selection"
  ],
  "type": "object"
},
"timed_out": {},
"timeout_seconds": {}
},
"required": [
  "ran",
  "command",
  "returncode",
  "stdout",
  "stderr",
  "timed_out",
  "timeout_seconds",
  "elapsed_ms",
  "process_group_cleanup",
  "required_counts",

```

```

  "required_skips_fail_closed",
  "optional_platform_tests_excluded",
  "marker_expression",
  "selection_expression",
  "junit_sha256",

```

```

    "passed",
    "scale_environment_configured",
    "raw_artifacts",
    "raw_artifact_manifest_digest",
    "installed_environment",
    "source_tree_shadowing_disabled"
  ],
  "type": "object"
},
"validation": {
  "additionalProperties": false,
  "properties": {
    "checks": {},
    "errors": {},
    "record_type": {},
    "root": {},
    "valid": {}
  },
  "required": [
    "checks",
    "errors",
    "record_type",
    "root",

```

```

    "valid"
  ],
  "type": "object"
}
},
"required": [
  "record_type",
  "status",
  "platform",
  "no_degradation",
  "source",
  "root",
  "artifact_binding",
  "candidate_binding_digest",
  "test_manifest",
  "install_mode",
  "required_predicates",
  "validation",
  "tests",
  "passed",
  "pass_credit",
  "acceptance_pass",
  "product_acceptance_pass",
  "product_public_approval",
  "scale_selection",
  "artifact_binding_unchanged",
  "standard_distribution_gate_pass",
  "phases",

```

```

  "execution_budget",
  "producer",
  "invocation",
  "raw_artifact_manifest_digest",
  "producer_attestation",
  "result_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/NoDegradationResult | Validator: urn:promin:contracts:1#/\$defs/NoDegradationResult

61. OperationMetrics

Type summary: type="object"; additionalProperties=false

Required: record_type, claim_scope, authoritative, duration_ms, changed_records, event_write, projection_update, runtime_checkpoint, physical_payload_bytes, bytes_per_changed_record

Property	Required	Structural rule
authoritative	yes	const=false
bytes_per_changed_record	yes	type="number"; minimum=0
changed_records	yes	type="integer"; minimum=0
claim_scope	yes	const="single-command-observation-only"

Property	Required	Structural rule
duration_ms	yes	type="number"; minimum=0
event_write	yes	type="object"; additionalProperties=false
physical_payload_bytes	yes	type="integer"; minimum=0
projection_update	yes	oneOf[3]=type:object type:object type:object
record_type	yes	const="OperationMetrics"
runtime_checkpoint	yes	oneOf[2]=type:object type:object

```

{
  "additionalProperties": false,
  "properties": {
    "authoritative": {
      "const": false
    },
    "bytes_per_changed_record": {
      "minimum": 0,
      "type": "number"
    },
    "changed_records": {
      "minimum": 0,
      "type": "integer"
    },
    "claim_scope": {
      "const": "single-command-observation-only"
    },
    "duration_ms": {
      "minimum": 0,
      "type": "number"
    },
    "event_write": {
      "additionalProperties": false,
      "properties": {
        "changed_records": {
          "minimum": 0,
          "type": "integer"
        }
      }
    },
    "derived_index_bytes": {
      "minimum": 0,
      "type": "integer"
    },
    "head_bytes": {
      "minimum": 0,
      "type": "integer"
    },
    "journal_authority_bytes": {
      "minimum": 0,
      "type": "integer"
    },
    "journal_checkpoint_bytes": {
      "minimum": 0,
      "type": "integer"
    },
    "logical_final_bytes": {
      "minimum": 0,
      "type": "integer"
    },
    "physical_payload_bytes": {
      "minimum": 0,
      "type": "integer"
    },
    "temporary_staging_bytes": {
      "minimum": 0,
      "type": "integer"
    }
  },
  "required": [
    "changed_records",
    "journal_authority_bytes",
    "temporary_staging_bytes",
    "head_bytes",
    "derived_index_bytes",
    "journal_checkpoint_bytes",
    "logical_final_bytes",
    "physical_payload_bytes"
  ]
}

```

```

    ],
    "type": "object"
  },
  "physical_payload_bytes": {
    "minimum": 0,
    "type": "integer"
  },
  "projection_update": {
    "oneOf": [
      {
        "additionalProperties": false,
        "properties": {
          "bytes_per_changed_record": {
            "minimum": 0,
            "type": "number"
          },
          "changed_records": {
            "minimum": 0,

```

```

            "type": "integer"
          },
          "changed_shards": {
            "minimum": 0,
            "type": "integer"
          },
          "physical_payload_bytes": {
            "minimum": 0,
            "type": "integer"
          },
          "projection_authoritative": {
            "const": false
          },
          "status": {
            "enum": [
              "already-current",
              "missing",
              "stale-rebuild-required"
            ]
          }
        }
      },
      "required": [
        "status",
        "projection_authoritative",
        "changed_records",
        "changed_shards",
        "physical_payload_bytes",
        "bytes_per_changed_record"

```

```

    ],
    "type": "object"
  },
  {
    "additionalProperties": false,
    "properties": {
      "bytes_per_changed_record": {
        "minimum": 0,
        "type": "number"
      },
      "changed_records": {
        "minimum": 0,
        "type": "integer"
      },
      "changed_shards": {
        "minimum": 0,
        "type": "integer"
      },
      "error": {
        "maxLength": 4096,
        "minLength": 1,
        "type": "string"
      },
      "physical_payload_bytes": {
        "minimum": 0,
        "type": "integer"
      },
      "projection_authoritative": {

```

```

        "const": false
      },
      "status": {

```

```

        "const": "failed-rebuild-required"
    }
},
"required": [
    "status",
    "projection_authoritative",
    "changed_records",
    "changed_shards",
    "physical_payload_bytes",
    "bytes_per_changed_record",
    "error"
],
"type": "object"
},
{
    "additionalProperties": false,
    "properties": {
        "bytes_per_changed_record": {
            "minimum": 0,
            "type": "number"
        },
        "changed_records": {
            "minimum": 0,
            "type": "integer"
        },

```

```

        "changed_shards": {
            "minimum": 0,
            "type": "integer"
        },
        "head": {
            "additionalProperties": false,
            "properties": {
                "batch_digest": {
                    "oneOf": [
                        {
                            "$ref": "#/$defs/Digest"
                        },
                        {
                            "type": "null"
                        }
                    ]
                },
                "batch_id": {
                    "oneOf": [
                        {
                            "$ref": "#/$defs/Id"
                        },
                        {
                            "type": "null"
                        }
                    ]
                },
            },
        },
        "sequence": {

```

```

            "minimum": 0,
            "type": "integer"
        }
    },
    "required": [
        "sequence",
        "batch_id",
        "batch_digest"
    ],
    "type": "object"
},
"logical_payload_bytes": {
    "minimum": 0,
    "type": "integer"
},
"physical_payload_bytes": {
    "minimum": 0,
    "type": "integer"
},
"projection_authoritative": {
    "const": false
},
"projection_growth_bytes": {
    "minimum": 0,
    "type": "integer"
}

```



```

    },
    "semantic_digest": {
      "$ref": "#/$defs/Digest"
    },
    "status": {
      "const": "updated"
    }
  },
  "required": [
    "status",
    "projection_authoritative",
    "changed_records",
    "changed_shards",
    "physical_payload_bytes",
    "bytes_per_changed_record",
    "head",
    "semantic_digest",
    "logical_payload_bytes",
    "projection_growth_bytes"
  ],
  "type": "object"
}
]
},
"record_type": {
  "const": "OperationMetrics"
},
"runtime_checkpoint": {
  "oneOf": [
    {
      "additionalProperties": false,

      "properties": {
        "authoritative": {
          "const": false
        },
        "batch_threshold": {
          "minimum": 1,
          "type": "integer"
        },
        "byte_threshold": {
          "minimum": 1,
          "type": "integer"
        },
        "checkpoint_bytes": {
          "minimum": 0,
          "type": "integer"
        },
        "checkpoint_count": {
          "minimum": 0,
          "type": "integer"
        },
        "tail_batches": {
          "minimum": 0,
          "type": "integer"
        },
        "tail_bytes": {
          "minimum": 0,
          "type": "integer"
        }
      },

      "written": {
        "const": false
      }
    },
    {
      "required": [
        "authoritative",
        "batch_threshold",
        "byte_threshold",
        "checkpoint_bytes",
        "checkpoint_count",
        "tail_batches",
        "tail_bytes",
        "written"
      ],
      "type": "object"
    }
  ],
  {
    "additionalProperties": false,

```

```

    "properties": {
      "authoritative": {
        "const": false
      },
      "batch_threshold": {
        "minimum": 1,
        "type": "integer"
      },
      "byte_threshold": {
        "minimum": 1,
        "type": "integer"
      },
      "checkpoint_bytes": {
        "minimum": 0,
        "type": "integer"
      },
      "checkpoint_count": {
        "minimum": 0,
        "type": "integer"
      },
      "compacted_tail_batches": {
        "minimum": 0,
        "type": "integer"
      },
      "compacted_tail_bytes": {
        "minimum": 0,
        "type": "integer"
      },
      "tail_batches": {
        "minimum": 0,
        "type": "integer"
      },
      "tail_bytes": {
        "minimum": 0,
        "type": "integer"
      },
      "written": {
        "const": true
      }
    },
    "required": [
      "authoritative",
      "batch_threshold",
      "byte_threshold",
      "checkpoint_bytes",
      "checkpoint_count",
      "tail_batches",
      "tail_bytes",
      "written",
      "compacted_tail_batches",
      "compacted_tail_bytes"
    ],
    "type": "object"
  }
]
}
},
"required": [
  "record_type",
  "claim_scope",
  "authoritative",
  "duration_ms",
  "changed_records",
  "event_write",
  "projection_update",
  "runtime_checkpoint",
  "physical_payload_bytes",
  "bytes_per_changed_record"
],
"type": "object",
"x-promin-persistence": "derived-live-only"
}

```

Owner: core/contracts.schema.json#/\$defs/OperationMetrics | Validator: urn:promin:contracts:1#/\$defs/OperationMetrics

62. PersistentEntityDef

Type summary: type="object"; additionalProperties=false

Required: kind, purpose, admission_rule

Property	Required	Structural rule
admission_rule	yes	type="string"; minLength=1; maxLength=256
kind	yes	enum=["Task", "Candidate", "Artifact", "Run", "Finding", "Decision", "GateResult", "Grant", "Lease"]
purpose	yes	type="string"; minLength=1
<pre>{ "additionalProperties": false, "properties": { "admission_rule": { "maxLength": 256, "minLength": 1, "type": "string" }, "kind": { "enum": ["Task", "Candidate", "Artifact", "Run", "Finding", "Decision", "GateResult", "Grant", "Lease"] }, "purpose": { "minLength": 1, "type": "string" } }, "required": ["kind", "purpose", "admission_rule"], "type": "object" }</pre>		

Owner: core/contracts.schema.json#/\$defs/PersistentEntityDef | Validator: urn:promin:contracts:1#/\$defs/PersistentEntityDef

63. PlanProposal

Type summary: type="object"; additionalProperties=false

Required: record_type, proposal_id, project_id, project_mode, goal, source_plan_digest, created_at, tasks, authority, pass_credit, proposal_digest

Property	Required	Structural rule
authority	yes	const=false
created_at	yes	\$ref="#/\$defs/Timestamp"
goal	yes	type="string"; minLength=1; maxLength=8192
pass_credit	yes	const=false
project_id	yes	\$ref="#/\$defs/Id"
project_mode	yes	enum=["greenfield", "existing-code", "hybrid"]
proposal_digest	yes	\$ref="#/\$defs/Digest"
proposal_id	yes	\$ref="#/\$defs/Id"
record_type	yes	const="PlanProposal"
source_plan_digest	yes	\$ref="#/\$defs/Digest"
tasks	yes	type="array"; maxItems=1024
<pre>{ "additionalProperties": false, "properties": {</pre>		

```

"authority": {
  "const": false
},
"created_at": {
  "$ref": "#/$defs/Timestamp"
},
"goal": {
  "maxLength": 8192,
  "minLength": 1,
  "type": "string"
},
"pass_credit": {
  "const": false
},
"project_id": {
  "$ref": "#/$defs/Id"
},
"project_mode": {
  "enum": [
    "greenfield",
    "existing-code",
    "hybrid"
  ]
},
"proposal_digest": {
  "$ref": "#/$defs/Digest"
},
"proposal_id": {
  "$ref": "#/$defs/Id"
},
"record_type": {
  "const": "PlanProposal"
},
"source_plan_digest": {
  "$ref": "#/$defs/Digest"
},
"tasks": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "acceptance_predicate": {
        "maxLength": 8192,
        "minLength": 1,
        "type": "string"
      },
      "allowed_paths": {
        "items": {
          "maxLength": 4096,
          "minLength": 1,
          "type": "string"
        },
        "maxItems": 256,
        "minItems": 1,
        "type": "array",
        "uniqueItems": true
      },
      "authority": {
        "const": false
      },
      "depends_on": {
        "items": {
          "$ref": "#/$defs/Id"
        },
        "maxItems": 256,
        "type": "array",
        "uniqueItems": true
      },
      "operation": {
        "enum": [
          "read",
          "mutate"
        ]
      },
      "pass_credit": {
        "const": false
      },
      "source_bindings": {
        "items": {

```

```

    "maxLength": 4096,
    "minLength": 1,
    "type": "string"
  },
  "maxItems": 256,
  "minItems": 1,
  "type": "array",
  "uniqueItems": true
},
"task_id": {
  "$ref": "#/$defs/Id"
},
"title": {
  "maxLength": 4096,
  "minLength": 1,
  "type": "string"
}
},
"required": [
  "task_id",
  "title",
  "operation",
  "depends_on",
  "acceptance_predicate",
  "allowed_paths",
  "source_bindings",
  "authority"
],
"type": "object"
},
"maxItems": 1024,

```

```

  "type": "array"
}
},
"required": [
  "record_type",
  "proposal_id",
  "project_id",
  "project_mode",
  "goal",
  "source_plan_digest",
  "created_at",
  "tasks",
  "authority",
  "pass_credit",
  "proposal_digest"
],
"type": "object",
"x-promin-persistence": "transient-ingress-only"
}

```

Owner: `core/contracts.schema.json#/$defs/PlanProposal` | Validator: `urn:promin:contracts:1#/$defs/PlanProposal`

64. PlatformVerificationResult

Type summary: `type="object"; additionalProperties=false`

Required: `record_type`, `status`, `candidate_binding_digest`, `platform`, `platform_identity`, `exact_candidate_verified`, `archive_sha256`, `archive_bytes`, `portable_implementation_closure_digest`, `observed_environment_closure_digest`, `install_mode`, `installed_command_verified`, `installation`, `product_acceptance_pass`, `product_public_approval`, `producer`, `invocation`, `raw_artifact_manifest_digest`, `producer_attestation`, `result_digest`

Property	Required	Structural rule
<code>archive_bytes</code>	yes	<code>type="integer"; minimum=1</code>
<code>archive_sha256</code>	yes	<code>\$ref="#/\$defs/Digest"</code>
<code>candidate_binding_digest</code>	yes	<code>\$ref="#/\$defs/Digest"</code>
<code>exact_candidate_verified</code>	yes	<code>const=true</code>
<code>install_mode</code>	yes	<code>const="online-clean"</code>
<code>installation</code>	yes	<code>type="object"; additionalProperties=false</code>
<code>installed_command_verified</code>	yes	<code>const=true</code>
<code>invocation</code>	yes	<code>\$ref="#/\$defs/ReleaseEvidenceInvocation"</code>

Property	Required	Structural rule
observed_environment_closure_digest	yes	\$ref="#/\$defs/Digest"
platform	yes	enum=["linux", "windows"]
platform_identity	yes	type="object"; additionalProperties=false
portable_implementation_closure_digest	yes	\$ref="#/\$defs/Digest"
producer	yes	\$ref="#/\$defs/ReleaseEvidenceProducer"
producer_attestation	yes	\$ref="#/\$defs/EvidenceProducerAttestation"
product_acceptance_pass	yes	const=false
product_public_approval	yes	const="not_approved"
raw_artifact_manifest_digest	yes	\$ref="#/\$defs/Digest"
record_type	yes	const="PlatformVerificationResult"
result_digest	yes	\$ref="#/\$defs/Digest"
status	yes	const="pass"

```

{
  "additionalProperties": false,
  "properties": {
    "archive_bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "archive_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "exact_candidate_verified": {
      "const": true
    },
    "install_mode": {
      "const": "online-clean"
    },
    "installation": {
      "additionalProperties": false,
      "properties": {
        "build_backend": {
          "additionalProperties": false,
          "properties": {
            "distribution": {
              "const": "setuptools"
            }
          }
        },
        "name": {
          "const": "setuptools.build_meta"
        },
        "record_sha256": {
          "$ref": "#/$defs/Digest"
        },
        "version": {
          "minLength": 1,
          "type": "string"
        }
      }
    },
    "required": [
      "name",
      "distribution",
      "version",
      "record_sha256"
    ],
    "type": "object"
  },
  "build_dependency_source": {
    "const": "online-index-explicit"
  },
  "command_invocations": {
    "items": {
      "additionalProperties": false,
      "properties": {

```

```

    "argv_digest": {
      "$ref": "#/$defs/Digest"
    },

```

```

    },
    "command": {
      "maxLength": 128,
      "minLength": 1,
      "type": "string"
    },
    "returncode": {
      "const": 0
    },
    "stderr_bytes": {
      "maximum": 1048576,
      "minimum": 0,
      "type": "integer"
    },
    "stderr_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "stdout_bytes": {
      "maximum": 1048576,
      "minimum": 0,
      "type": "integer"
    },
    "stdout_sha256": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "command",

```

```

    "argv_digest",
    "returncode",
    "stdout_sha256",
    "stdout_bytes",
    "stderr_sha256",
    "stderr_bytes"
  ],
  "type": "object"
},
"maxItems": 7,
"minItems": 7,
"type": "array"
},
"console_script": {
  "additionalProperties": false,
  "properties": {
    "declared": {
      "const": true
    },
    "installed_wrapper_checked": {
      "const": true
    },
    "path": {
      "minLength": 1,
      "type": "string"
    },
    "posix_execute_bits": {
      "type": [

```

```

        "boolean",
        "null"
      ]
    },
    "present": {
      "const": true
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    },
    "size_bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "source_module_invoked": {
      "const": false
    }
  },
  "required": [

```

```

    "present",
    "declared",
    "installed_wrapper_checked",
    "source_module_invoked",
    "posix_execute_bits",
    "path",
    "sha256",
    "size_bytes"
  ],

```

```

    "type": "object"
  },
  "declared_python_requirement": {
    "minLength": 1,
    "type": "string"
  },
  "dependency_closure": {
    "additionalProperties": false,
    "properties": {
      "closure_check": {
        "const": "pass"
      },
    },
    "declared_build": {
      "items": {
        "type": "string"
      },
      "minItems": 1,
      "type": "array"
    },
    "declared_runtime": {
      "items": {
        "type": "string"
      },
      "minItems": 1,
      "type": "array"
    },
  },
  "dependency_artifact_digest": {
    "$ref": "#/$defs/Digest"
  }

```

```

},
"digest": {
  "$ref": "#/$defs/Digest"
},
"resolved_runtime": {
  "additionalProperties": false,
  "properties": {
    "cryptography": {
      "additionalProperties": false,
      "properties": {
        "requirement": {
          "minLength": 1,
          "type": "string"
        },
      },
      "version": {
        "minLength": 1,
        "type": "string"
      }
    },
    "required": [
      "requirement",
      "version"
    ],
    "type": "object"
  },
},
"jsonschema": {
  "additionalProperties": false,
  "properties": {

```

```

    "requirement": {
      "minLength": 1,
      "type": "string"
    },
    "version": {
      "minLength": 1,
      "type": "string"
    }
  },
  "required": [
    "requirement",
    "version"
  ]

```



```

    },
    "type": "object"
  },
  "pypdf": {
    "additionalProperties": false,
    "properties": {
      "requirement": {
        "minLength": 1,
        "type": "string"
      },
      "version": {
        "minLength": 1,
        "type": "string"
      }
    }
  },
  "required": [
    "requirement",
    "version"
  ],
  "type": "object"
},
"required": [
  "cryptography",
  "jsonschema",
  "pypdf"
],
"type": "object"
},
"transitive_distribution_digest": {
  "$ref": "#/$defs/Digest"
},
"transitive_distributions": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "license": {
        "maxLength": 2048,
        "minLength": 1,
        "type": "string"
      },
      "name": {
        "minLength": 1,
        "type": "string"
      }
    }
  },
  "record": {
    "additionalProperties": false,
    "properties": {
      "bytes": {
        "minimum": 1,
        "type": "integer"
      },
      "path": {
        "minLength": 1,
        "pattern": "\\\\.dist-info/RECORD$",
        "type": "string"
      },
      "sha256": {
        "$ref": "#/$defs/Digest"
      }
    }
  },
  "required": [
    "path",
    "sha256",
    "bytes"
  ],
  "type": "object"
},
"requires": {
  "items": {
    "minLength": 1,
    "type": "string"
  }
},
"type": "array"
},
"version": {
  "minLength": 1,

```

```

        "type": "string"
      }
    },
    "required": [
      "name",
      "version",
      "record",
      "requires",
      "license"
    ],
    "type": "object"
  },
  "maxItems": 256,
  "minItems": 1,
  "type": "array"
}

```

```

},
"required": [
  "declared_runtime",
  "declared_build",
  "resolved_runtime",
  "closure_check",
  "transitive_distributions",

```

```

    "transitive_distribution_digest",
    "dependency_artifact_digest",
    "digest"
  ],
  "type": "object"
},
"environment": {
  "const": "clean-venv"
},
"installation_performed": {
  "const": true
},
"installed_distribution": {
  "additionalProperties": false,
  "properties": {
    "name": {
      "const": "promin"
    },
    "runtime_version": {
      "$ref": "#/$defs/SemVer"
    },
    "version": {
      "$ref": "#/$defs/SemVer"
    }
  }
},
"required": [
  "name",
  "version",

```

```

    "runtime_version"
  ],
  "type": "object"
},
"installed_environment_observation": {
  "additionalProperties": false,
  "properties": {
    "build_backend": {
      "additionalProperties": false,
      "properties": {
        "distribution": {
          "const": "setuptools"
        },
        "name": {
          "const": "setuptools.build_meta"
        }
      },
      "record_sha256": {
        "$ref": "#/$defs/Digest"
      },
      "version": {
        "minLength": 1,
        "type": "string"
      }
    }
  },
  "required": [
    "name",

```

```

    "distribution",
    "version",

    "record_sha256"
  ],
  "type": "object"
},
"console_wrapper": {
  "additionalProperties": false,
  "properties": {
    "bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "path": {
      "minLength": 1,
      "type": "string"
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "path",
    "sha256",
    "bytes"
  ],
  "type": "object"
},
"license_closure": {
  "additionalProperties": false,

```

```

    "properties": {
      "complete": {
        "const": true
      },
      "digest": {
        "$ref": "#/$defs/Digest"
      },
      "entries": {
        "items": {
          "additionalProperties": false,
          "properties": {
            "license": {
              "maxLength": 2048,
              "minLength": 1,
              "not": {
                "const": "UNKNOWN"
              },
              "type": "string"
            },
            "name": {
              "minLength": 1,
              "type": "string"
            },
            "version": {
              "minLength": 1,
              "type": "string"
            }
          }
        },

```

```

      "required": [
        "name",
        "version",
        "license"
      ],
      "type": "object"
    },
    "maxItems": 256,
    "minItems": 1,
    "type": "array"
  }
},
"required": [
  "entries",
  "complete",
  "digest"
],
"type": "object"
},

```

```

"observation_digest": {
  "$ref": "#/$defs/Digest"
},
"platform": {
  "additionalProperties": false,
  "properties": {
    "machine": {
      "minLength": 1,
      "type": "string"

```

```

    },
    "release": {
      "minLength": 1,
      "type": "string"
    },
    "sys_platform": {
      "minLength": 1,
      "type": "string"
    },
    "system": {
      "enum": [
        "linux",
        "windows"
      ]
    },
    "tags": {
      "items": {
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 64,
      "minItems": 1,
      "type": "array"
    }
  },
  "required": [
    "system",
    "machine",

```

```

    "release",
    "sys_platform",
    "tags"
  ],
  "type": "object"
},
"promin": {
  "additionalProperties": false,
  "properties": {
    "installed_file_count": {
      "maximum": 4096,
      "minimum": 1,
      "type": "integer"
    },
    "installed_files": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "bytes": {
            "minimum": 0,
            "type": "integer"
          },
          "path": {
            "minLength": 1,
            "type": "string"
          },
          "sha256": {
            "$ref": "#/$defs/Digest"

```

```

        }
      },
      "required": [
        "path",
        "sha256",
        "bytes"
      ],
      "type": "object"
    },
    "maxItems": 4096,
    "minItems": 1,
    "type": "array"

```

```

    },
    "installed_files_digest": {
      "$ref": "#/$defs/Digest"
    },
    },
    "module_file": {
      "minLength": 1,
      "type": "string"
    },
    },
    "module_sha256": {
      "$ref": "#/$defs/Digest"
    },
    },
    "record": {
      "additionalProperties": false,
      "properties": {
        "bytes": {
          "minimum": 1,

          "type": "integer"
        },
        "path": {
          "minLength": 1,
          "pattern": "\\\\.dist-info/RECORD$",
          "type": "string"
        },
        "sha256": {
          "$ref": "#/$defs/Digest"
        }
      },
      "required": [
        "path",
        "sha256",
        "bytes"
      ],
      "type": "object"
    },
    },
    "version": {
      "$ref": "#/$defs/SemVer"
    }
  },
  "required": [
    "version",
    "module_file",
    "module_sha256",
    "record",
    "installed_files",
    "installed_files_digest",
    "installed_file_count"
  ],
  "type": "object"
},
"python": {
  "additionalProperties": false,
  "properties": {
    "abi_tag": {
      "minLength": 1,
      "type": "string"
    },
    "base_executable": {
      "minLength": 1,
      "type": "string"
    },
    "base_executable_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "base_prefix": {
      "minLength": 1,
      "type": "string"
    },
    "executable": {
      "minLength": 1,
      "type": "string"
    },
    "executable_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "implementation": {
      "minLength": 1,
      "type": "string"
    }
  }
}

```

```

    },
    "prefix": {
      "minLength": 1,
      "type": "string"
    },
    "version": {
      "minLength": 1,
      "type": "string"
    }
  },
  "required": [
    "executable",
    "executable_sha256",
    "base_executable",
    "base_executable_sha256",
    "implementation",
    "version",
    "abi_tag",
    "prefix",
    "base_prefix"
  ],
  "type": "object"
},

```

```

"sbom": {
  "additionalProperties": false,
  "properties": {
    "components": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "name": {
            "minLength": 1,
            "type": "string"
          },
          "record_sha256": {
            "$ref": "#/$defs/Digest"
          },
          "version": {
            "minLength": 1,
            "type": "string"
          }
        },
        "required": [
          "name",
          "version",
          "record_sha256"
        ],
        "type": "object"
      },
      "maxItems": 256,
      "minItems": 1,

```

```

      "type": "array"
    },
    "digest": {
      "$ref": "#/$defs/Digest"
    },
    "format": {
      "const": "promin-spx-lite-v1"
    }
  },
  "required": [
    "format",
    "components",
    "digest"
  ],
  "type": "object"
},
"site_packages": {
  "minLength": 1,
  "type": "string"
},
"sqlite_version": {
  "minLength": 1,
  "type": "string"
},
"transitive_distribution_digest": {
  "$ref": "#/$defs/Digest"

```

```

    },
    "transitive_distributions": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "license": {
            "maxLength": 2048,
            "minLength": 1,
            "type": "string"
          },
          "name": {
            "minLength": 1,
            "type": "string"
          },
          "record": {
            "additionalProperties": false,
            "properties": {
              "bytes": {
                "minimum": 1,
                "type": "integer"
              },
              "path": {
                "minLength": 1,
                "pattern": "\\\\.dist-info/RECORD$",
                "type": "string"
              },
              "sha256": {
                "$ref": "#/$defs/Digest"
              }
            }
          }
        },
        "required": [
          "path",
          "sha256",
          "bytes"
        ],
        "type": "object"
      },
      "requires": {
        "items": {
          "minLength": 1,
          "type": "string"
        },
        "type": "array"
      },
      "version": {
        "minLength": 1,
        "type": "string"
      }
    },
    "required": [
      "name",
      "version",
      "record",
      "requires",
      "license"
    ],
    "type": "object"
  },
  "maxItems": 256,
  "minItems": 1,
  "type": "array"
}
},
"required": [
  "python",
  "platform",
  "sqlite_version",
  "site_packages",
  "promin",
  "console_wrapper",
  "transitive_distributions",
  "transitive_distribution_digest",
  "sbom",
  "license_closure",
  "build_backend",
  "observation_digest"
],

```

```

    "type": "object"
  },
  "license_closure": {
    "additionalProperties": false,
    "properties": {
      "complete": {
        "const": true
      },
      "digest": {

```

```

        "$ref": "#/$defs/Digest"
      },
      "entries": {
        "items": {
          "additionalProperties": false,
          "properties": {
            "license": {
              "maxLength": 2048,
              "minLength": 1,
              "not": {
                "const": "UNKNOWN"
              },
              "type": "string"
            },
            "name": {
              "minLength": 1,
              "type": "string"
            },
            "version": {
              "minLength": 1,
              "type": "string"
            }
          },
          "required": [
            "name",
            "version",
            "license"
          ],

```

```

    "type": "object"
  },
  "maxItems": 256,
  "minItems": 1,
  "type": "array"
},
"required": [
  "entries",
  "complete",
  "digest"
],
"type": "object"
},
"nested_venv_created": {
  "const": true
},
"network_disabled": {
  "const": false
},
"pip_report": {
  "additionalProperties": false,
  "properties": {
    "artifact_digest": {
      "$ref": "#/$defs/Digest"
    },
    "report_count": {
      "const": 2

```

```

  },
  "reports": {
    "items": {
      "additionalProperties": false,
      "properties": {
        "artifact_digest": {
          "$ref": "#/$defs/Digest"
        },
        "artifacts": {
          "items": {
            "additionalProperties": false,
            "properties": {

```



```

        "hashes": {
            "items": {
                "minLength": 3,
                "type": "string"
            },
            "type": "array"
        },
        "is_direct": {
            "type": "boolean"
        },
        "name": {
            "minLength": 1,
            "type": "string"
        },
        "version": {
            "minLength": 1,

```

```

                "type": "string"
            }
        },
        "required": [
            "name",
            "version",
            "is_direct",
            "hashes"
        ],
        "type": "object"
    },
    "maxItems": 256,
    "minItems": 1,
    "type": "array"
},
"bytes": {
    "maximum": 16777216,
    "minimum": 1,
    "type": "integer"
},
"install_count": {
    "maximum": 256,
    "minimum": 1,
    "type": "integer"
},
"pip_version": {
    "minLength": 1,
    "type": "string"

```

```

},
"role": {
    "enum": [
        "build-requirements",
        "candidate-install"
    ]
},
"sha256": {
    "$ref": "#/$defs/Digest"
}
},
"required": [
    "role",
    "sha256",
    "bytes",
    "pip_version",
    "install_count",
    "artifacts",
    "artifact_digest"
],
"type": "object"
},
"maxItems": 2,
"minItems": 2,
"type": "array"
}
},
"required": [

```

```

    "reports",
    "report_count",
    "artifact_digest"
],
"type": "object"

```

```

},
"runtime_dependency_source": {
  "const": "online-index"
},
"s bom": {
  "additionalProperties": false,
  "properties": {
    "components": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "name": {
            "minLength": 1,
            "type": "string"
          },
          "record_sha256": {
            "$ref": "#/$defs/Digest"
          },
          "version": {
            "minLength": 1,
            "type": "string"
          }
        }
      },
    },
  },
},

```

```

    "required": [
      "name",
      "version",
      "record_sha256"
    ],
    "type": "object"
  },
  "maxItems": 256,
  "minItems": 1,
  "type": "array"
},
"digest": {
  "$ref": "#/$defs/Digest"
},
"format": {
  "const": "promin-spx-lite-v1"
}
},
"required": [
  "format",
  "components",
  "digest"
],
"type": "object"
}

```

```

},
"required": [
  "environment",
  "installation_performed",
  "nested_venv_created",
  "runtime_dependency_source",
  "build_dependency_source",
  "network_disabled",
  "declared_python_requirement",
  "installed_distribution",
  "console_script",
  "command_invocations",
  "dependency_closure",
  "installed_environment_observation",
  "pip_report",
  "s bom",
  "license_closure",
  "build_backend"
],
"type": "object"
},
"installed_command_verified": {
  "const": true
},
"invocation": {
  "$ref": "#/$defs/ReleaseEvidenceInvocation"
},
"observed_environment_closure_digest": {
  "$ref": "#/$defs/Digest"
}

```

```

},
"platform": {
  "enum": [
    "linux",
    "windows"
  ]
},
"platform_identity": {
  "additionalProperties": false,
  "properties": {
    "machine": {
      "minLength": 1,
      "type": "string"
    },
    "platform": {
      "enum": [
        "linux",
        "windows"
      ]
    },
    "platform_release": {
      "minLength": 1,
      "type": "string"
    },
    "platform_tags": {
      "items": {
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 64,
      "minItems": 1,
      "type": "array"
    },
    "python_abi_tag": {
      "minLength": 1,
      "type": "string"
    },
    "python_executable": {
      "minLength": 1,
      "type": "string"
    },
    "python_executable_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "python_implementation": {
      "minLength": 1,
      "type": "string"
    },
    "python_version": {
      "minLength": 1,
      "type": "string"
    },
    "sys_platform": {
      "minLength": 1,
      "type": "string"
    }
  }
},
"required": [
  "platform",
  "machine",
  "platform_release",
  "sys_platform",
  "platform_tags",
  "python_version",
  "python_implementation",
  "python_executable",
  "python_executable_sha256",
  "python_abi_tag"
],
"type": "object"
},
"portable_implementation_closure_digest": {
  "$ref": "#/$defs/Digest"
},
"producer": {
  "$ref": "#/$defs/ReleaseEvidenceProducer"
},

```

```

"producer_attestation": {
  "$ref": "#/$defs/EvidenceProducerAttestation"
},
"product_acceptance_pass": {
  "const": false
},
"product_public_approval": {
  "const": "not_approved"
},

```

```

"raw_artifact_manifest_digest": {
  "$ref": "#/$defs/Digest"
},
"record_type": {
  "const": "PlatformVerificationResult"
},
"result_digest": {
  "$ref": "#/$defs/Digest"
},
"status": {
  "const": "pass"
}

```

```

},
"required": [
  "record_type",
  "status",
  "candidate_binding_digest",
  "platform",
  "platform_identity",
  "exact_candidate_verified",
  "archive_sha256",
  "archive_bytes",
  "portable_implementation_closure_digest",
  "observed_environment_closure_digest",
  "install_mode",
  "installed_command_verified",
  "installation",
  "product_acceptance_pass",

```

```

"product_public_approval",
"producer",
"invocation",
"raw_artifact_manifest_digest",
"producer_attestation",
"result_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/PlatformVerificationResult | Validator: urn:promin:contracts:1#/\$defs/PlatformVerificationResult

65. PolicyDef

Type summary: type="object"; additionalProperties=false

Required: id, name, validator_id, severity, statement

Property	Required	Structural rule
id	yes	enum=[68 items]
name	yes	\$ref="#/\$defs/Id"
severity	yes	enum=["P0", "P1", "P2"]
statement	yes	type="string"; minLength=1
validator_id	yes	\$ref="#/\$defs/Id"

```

{
  "additionalProperties": false,
  "properties": {
    "id": {
      "enum": [
        "P-001",
        "P-002",
        "P-003",
        "P-004",
        "P-005",
        "P-006",
        "P-007",
        "P-008",

```

```
"P-009",
"P-010",
"P-011",
"P-012",
"P-013",
"P-014",
"P-015",
"P-016",
"P-017",
"P-018",
"P-019",
"P-020",
"P-021",
"P-022",
"P-023",
```

```
"P-024",
"P-025",
"P-026",
"P-027",
"P-028",
"P-029",
"P-030",
"P-031",
"P-032",
"P-033",
"P-034",
"P-035",
"P-036",
"P-037",
"P-038",
"P-039",
"P-040",
"P-041",
"P-042",
"P-043",
"P-044",
"P-045",
"P-046",
"P-047",
"P-048",
"P-049",
"P-050",
"P-051",
```

```
"P-052",
"P-053",
"P-054",
"P-055",
"P-056",
"P-057",
"P-058",
"P-059",
"P-060",
"P-061",
"P-062",
"P-063",
"P-064",
"P-065",
"P-066",
"P-067",
"P-068"
]
},
"name": {
  "$ref": "#/$defs/Id"
},
"severity": {
  "enum": [
    "P0",
    "P1",
    "P2"
  ]
}
```

```
},
"statement": {
  "minLength": 1,
  "type": "string"
},
"validator_id": {
```

```

    "$ref": "#/$defs/Id"
  },
  "required": [
    "id",
    "name",
    "validator_id",
    "severity",
    "statement"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/PolicyDef | Validator: urn:promin:contracts:1#/\$defs/PolicyDef

66. PolicySet

Type summary: type="object"; additionalProperties=false

Required: record_type, schema_ref, document, version, function, validator_registry_version, optimization_gate, state_machines, policies, derived_result_contracts, gate_run_definition_contract, research_draft_sanitation, semantic_export_contract

Property	Required	Structural rule
derived_result_contracts	yes	const=schema(keys:DoctorResult,OperationMetrics,ReadyFrontier,RetrievalPage,WorkCardProjection)
document	yes	const="promin.policy-set"
function	yes	const="Own executable cross-record invariants through stable validator identifiers and no duplicated budget values."
gate_run_definition_contract	yes	const=schema(keys:artifact_binding_fields,artifact_record_digest_rule,authorization_scope_contract,command_binding,definition_change,definition_digest_rule,definition_owner,definition_provenance_fields,evidence_class_purpose_pairs,expected_owner_fields,implementation_closure_binding_rule,inline_definition_submission,mixed_evidence_classes,payload_digest_as_artifact_record_binding,persistence,product_credit_required_rule,provider_binding_digest_rule,run_binding_rule,status_outcome_rules,submitted_artifacts_define_expectations,target_binding_rule,task_binding,task_owner_digest_rule)
optimization_gate	yes	const=schema(keys:base_predicate,formula,stress_predicate)
policies	yes	const=[68 items]
record_type	yes	const="PolicySet"
research_draft_sanitation	yes	const=schema(keys:allowed_license_disposition,canonical_bytes_max,citation_required_rule,claim_digest_rule,claim_kinds,evidence_reference_rule,forbidden_normative_claim_classes,forbidden_requested_flag_mutations,metadata_only_rule,migration_rule,normative_candidate_rule,persistence,result_flags,result_persistence,result_record_type,source_binding,source_license_rule,source_payload_bytes_max,source_payload_total_bytes_max,verification_statuses)
schema_ref	yes	const="urn:promin:contracts:1#/\$defs/PolicySet"
semantic_export_contract	yes	const=schema(keys:export_kind,input_digests_max,input_digests_rule,output_artifact_kind,output_artifact_record_digest_rule,output_bytes_max,output_bytes_rule,output_media_type,output_record_type,output_records_max,package_or_archive,persistence,product_flags,record_order,required_bindings)
state_machines	yes	const=schema(keys:capacity_release,lease,task)
validator_registry_version	yes	const=1
version	yes	const="1.0.0-alpha.3"

```

{
  "additionalProperties": false,
  "properties": {
    "derived_result_contracts": {
      "const": {
        "DoctorResult": {
          "archival_diagnostics_effect": "none",
          "claim_scope": "runtime-health-only",
          "component_status_values": [
            "healthy",
            "degraded",
            "failed",
            "incomplete"
          ],
          "current_components": [
            "core",
            "init",
            "providers",

```

```

    "recovery",
    "replay",
    "projection"
  ],
  "forbidden_allocation_scopes": [
    "batch",
    "goal",
    "per-run",
    "session",
    "workspace"
  ],
  "forbidden_metric_families": [
    "cost",
    "token",
    "usage"
  ],
  "live_only": true,
  "metrics": [
    "core_artifacts_verified",
    "event_batches",
    "init_records_verified",
    "projection_entities",
    "projection_relations",
    "provider_healthchecks_executed"
  ],
  "pass_credit": false,
  "product_acceptance_pass": false,
  "product_public_approval": "not_approved",
  "product_tree_scans": 0,
  "report_authoritative": false,
  "rollup_precedence": [
    "failed",
    "incomplete",
    "degraded",
    "healthy"
  ],
  "source_scope": "installed-core-and-live-operational-state-only"
},
"OperationMetrics": {
  "authoritative": false,
  "claim_scope": "single-command-observation-only",
  "forbidden_allocation_scopes": [
    "batch",
    "goal",
    "per-run",
    "session",
    "workspace"
  ],
  "forbidden_metric_families": [
    "cost",
    "token",
    "usage"
  ],
  "live_only": true,
  "metric_groups": [
    "duration_ms",
    "changed_records",
    "event_write",
    "projection_update",
    "runtime_checkpoint",
    "physical_payload_bytes",
    "bytes_per_changed_record"
  ]
},
"ReadyFrontier": {
  "blocking_finding_statuses": [
    "OPEN"
  ],
  "cycle": "reject",
  "dependency_relation_kind": "DEPENDS_ON",
  "eligible_task_state": "READY",
  "gate_currentness_bindings": [
    "activation_digest",
    "candidate_digest",
    "definition_digest"
  ],
  "gate_outcome": "pass",

```

```

    "gate_pass_credit": true,
    "live_only": true,
    "ordering": [
      "created_at-ascending",
      "task_id-ascending"
    ],
    "projection_authoritative": false,
    "required_dependency_state": "COMPLETED",
    "silent_truncation": false
  },
  "RetrievalPage": {
    "canonical_cursor_fields": [
      "stream_cursor",
      "next_stream_cursor"
    ],
    "forbidden_compatibility_fields": [
      "seed_cursor",
        "next_seed_cursor"
      ],
    "live_only": true,
    "projection_authoritative": false,
    "query_bytes_max": 4096,
    "refinement_hints_max": 8,
    "silent_truncation": false
  },
  "WorkCardProjection": {
    "continuation_exposed": false,
    "live_only": true,
    "projection_authoritative": false,
    "selected_task_only": true
  }
},
"document": {
  "const": "promin.policy-set"
},
"function": {
  "const": "Own executable cross-record invariants through stable validator identifiers and no
duplicated budget values."
},
"gate_run_definition_contract": {
  "const": {
    "artifact_binding_fields": [
      "artifact_id",
      "artifact_record_digest",
      "run_id",
        "run_digest"
      ],
    "artifact_record_digest_rule": "sha256(canonical finalized record with exact artifact and
authoritative commit_binding)",
    "authorization_scope_contract": {
      "all_selector": "reject",
      "allowed_extra_kinds": [
        "project",
        "task"
      ],
      "lease_bound_task_selector_required": true,
      "project_items_max": 1,
      "target_scope_relation": "subset",
      "task_items_max": 1,
      "task_value_rule": "GateResult.task_id",
      "unrelated_selector": "reject"
    },
    "command_binding": "CommandRequest.definition_digest equals the immutable GateRunDefinition digest
and the payload definition_digest",
    "definition_change": "reject after Task creation; define a new Task identity for changed
requirements",
    "definition_digest_rule": "sha256(canonical GateRunDefinition)",
    "definition_owner": "Task.gate_run_definitions",
    "definition_provenance_fields": [
      "owner_kind",
      "owner_digest",
      "defined_at_head_digest"
    ],
    "evidence_class_purpose_pairs": {
      "diagnostic": [
        "validator"
      ]
    }
  }
}

```



```

    ],
    "gate": [
        "validator",
        "product-execution"
    ],
    "product": [
        "product-execution"
    ]
},
"expected_owner_fields": [
    "expected_evidence_class",
    "expected_evidence_purpose",
    "product_credit_required",
    "target_kind",
    "target_digest",
    "target_scope",
    "candidate_digest",
    "policy_digest",
    "tool_digest",
    "implementation_closure_digest",
    "provider_binding_digest",
    "input_digests"
],
"implementation_closure_binding_rule": "GateRunDefinition, Run, and every exact bound Artifact
EvidenceBinding carry the same implementation_closure_digest",
"inline_definition_submission": "reject",
"mixed_evidence_classes": "reject",
"payload_digest_as_artifact_record_binding": "forbidden",
"persistence": "immutable-definition-input",

    "product_credit_required_rule": "true requires the product purpose and product-execution class;
false may use any declared evidence class-purpose pair",
    "provider_binding_digest_rule": "sha256(canonical installed provider-binding evidence array);
actual ProviderInvocationEvidence records remain exact immutable Artifact inputs and are independently
validated against the active provider bindings when present",
    "run_binding_rule": "GateResult.run_id and run_digest and every GateEvidenceArtifactBinding.run_id
and run_digest bind the exact canonical Run record",
    "status_outcome_rules": {
        "blocked": {
            "evidence_binding": "required",
            "outcome": "blocked",
            "pass_credit": false,
            "reason": "required"
        },
        "fail": {
            "evidence_binding": "required",
            "outcome": "fail",
            "pass_credit": false,
            "reason": "optional"
        },
        "pass": {
            "diagnostic_pass_with_no_credit": true,
            "evidence_binding": "required",
            "outcome": "pass",
            "pass_credit_rule": "true only when definition product_credit_required is true and every exact
bound Artifact is fresh, resolved, passing, product-execution evidence"
        },
        "skipped": {
            "evidence_binding": "forbidden",
            "outcome": "skipped",
            "pass_credit": false,
            "reason": "required"
        }
    }
},
"submitted_artifacts_define_expectations": false,
"target_binding_rule": "candidate target_digest equals candidate_digest; finding target_digest is
the immutable exact OPEN Finding record digest captured at definition time and is not recomputed from
later mutable Finding state",
"task_binding": "GateResult.task_id and definition_digest resolve one exact embedded Task
definition",
"task_owner_digest_rule": "sha256(canonical immutable Task fields excluding state and
gate_run_definitions)"
}
},
"optimization_gate": {
    "const": {
        "base_predicate": "NetValue>0",
        "formula": "NetValue=3*RiskReduction+2*AutomationGain+1.5*QueryabilityGain+ReuseGain-1.5*Implementa
tionCost-2*MaintenanceCost-2*CognitiveNoise-1.5*ReferenceCost-MigrationCost",

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    "stress_predicate": "NetValue remains >0 when benefits decrease 20 percent and costs increase 20
percent"
  }
},
"policies": {
  "const": [
    {
      "id": "P-001",
      "name": "no-false-success",
      "severity": "P0",
      "statement": "Gate pass credit is true exactly when normalized status is pass and every required
gate passes.",
      "validator_id": "no_false_success"
    },
    {
      "id": "P-002",
      "name": "activation-binding",
      "severity": "P0",
      "statement": "Every authoritative record binds the exact active Core, preset, init, and profile
configuration.",
      "validator_id": "activation_binding"
    },
    {
      "id": "P-003",
      "name": "grant-validity",
      "severity": "P0",
      "statement": "Subject, capability, conjunctive scope, interval, nonce, exact claim digest, proof
chain, revocation state, and Activation all match.",
      "validator_id": "grant_validity"
    },
    {
      "id": "P-004",
      "name": "lease-fence",
      "severity": "P0",
      "statement": "Every mutation uses the current active Lease generation and monotonic fencing
token.",
      "validator_id": "lease_fence"
    },
    {
      "id": "P-005",
      "name": "candidate-evidence-binding",
      "severity": "P0",
      "statement": "Every evidence Artifact requires an explicit diagnostic, gate, or product purpose
and binds Activation, candidate, policy, provider/tool, and input digests inside the immutable event
payload.",
      "validator_id": "candidate_evidence_binding"
    },
    {
      "id": "P-006",
      "name": "atomic-event-commit",
      "severity": "P0",
      "statement": "One command produces one immutable batch under lock, compare-head, sync, atomic
replace, and recovery.",
      "validator_id": "atomic_event_commit"
    },
    {
      "id": "P-007",
      "name": "projection-is-derived",
      "severity": "P0",
      "statement": "Projection deletion and deterministic replay cannot change normative meaning.",
      "validator_id": "projection_derived"
    },
    {
      "id": "P-008",
      "name": "minimal-graph",
      "severity": "P1",
      "statement": "Each verified raw file creates exactly one non-authoritative derived Artifact
proxy. Raw inventory creates no Task, READS, or PRODUCES record; Tasks and typed relations require
explicit semantic content.",
      "validator_id": "minimal_graph"
    },
    {
      "id": "P-009",
      "name": "bounded-workcard",
      "severity": "P0",
      "statement": "Every WorkCard stays within both Core hard ceilings and its selected profile byte,
entity, relation, fanout, and top-k budgets; every truncation is explicit and losslessly resumable.",
      "validator_id": "bounded_workcard"
    }
  ]
}

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    },
    {
      "id": "P-010",
      "name": "size-independent-init",
      "severity": "P0",
      "statement": "Initialization performs zero product-tree scans and creates exactly five init
records.",
      "validator_id": "zero_scan_init"
    },
    {
      "id": "P-011",
      "name": "explicit-external-bindings",
      "severity": "P1",
      "statement": "External standards and providers exist only in explicit digest-bound init
records.",
      "validator_id": "explicit_bindings"
    },
    {
      "id": "P-012",
      "name": "canonical-name-only",
      "severity": "P0",
      "statement": "The canonical identity is promin; aliases, redirects, and compatibility names are
rejected.",
      "validator_id": "canonical_name"
    },
    {
      "id": "P-013",
      "name": "safe-export",
      "severity": "P0",
      "statement": "Export is allowlisted, recursively bounded, symlink-safe, secret-aware, and
path-sanitized.",
      "validator_id": "safe_export"
    },
    {
      "id": "P-014",
      "name": "closed-acknowledgement",
      "severity": "P0",
      "statement": "Only a valid CLOSED transition with explicit close acknowledgement releases a
mutation slot.",
      "validator_id": "closed_ack"
    },
    {
      "id": "P-015",
      "name": "fail-closed-critical-fields",
      "severity": "P0",
      "statement": "Missing, unknown, stale, downgraded, or unbound critical fields reject before
mutation.",
      "validator_id": "critical_fields"
    },
    {
      "id": "P-016",
      "name": "preset-capability-ceiling",
      "severity": "P0",
      "statement": "A preset may configure provider needs and budgets but cannot grant action
authority.",
      "validator_id": "preset_no_authority"
    },
    {
      "id": "P-017",
      "name": "single-scan-inventory",
      "severity": "P1",
      "statement": "One requested inventory performs at most one full product-tree pass; all
projections derive from its stream.",
      "validator_id": "single_scan"
    },
    {
      "id": "P-018",
      "name": "idempotent-commands",
      "severity": "P0",
      "statement": "A duplicate idempotency key under the same activation and subject returns the
original result or a conflict.",
      "validator_id": "idempotent_commands"
    },
    {
      "id": "P-019",
      "name": "core-preset-separation",
      "severity": "P1",

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        "statement": "Core bundle identity excludes replaceable presets; Activation binds the selected
preset separately.",
        "validator_id": "core_preset_separation"
    },
    {
        "id": "P-020",
        "name": "acyclic-bootstrap-authority",
        "severity": "P0",
        "statement": "Init config defines root ceilings, never pre-activation runtime Grants.",
        "validator_id": "bootstrap_authority"
    },
    {
        "id": "P-021",
        "name": "normalization-collision-rejection",
        "severity": "P1",
        "statement": "Duplicate JSON keys and path or ID collisions after NFC/casefold normalization are
rejected.",
        "validator_id": "normalization_collision"
    },
    {
        "id": "P-022",
        "name": "provider-substitution-binding",
        "severity": "P1",
        "statement": "A provider identity or implementation-closure change invalidates the Activation
and every dependent continuation, evidence Artifact, projection, replay result, and current release
closure; it never silently changes authority.",
        "validator_id": "provider_substitution"
    },
    {
        "id": "P-023",
        "name": "exact-init-keyset",
        "severity": "P1",
        "statement": "Activation binds exactly project.json, standards.json, technologies.json, and
authority.json.",
        "validator_id": "exact_init_keyset"
    },
    {
        "id": "P-024",
        "name": "stale-projection-rejection",
        "severity": "P1",
        "statement": "Status, next, and WorkCard generation reject a projection whose source head or
activation differs.",
        "validator_id": "stale_projection"
    },
    {
        "id": "P-025",
        "name": "one-command-one-batch",
        "severity": "P0",
        "statement": "Every committed batch binds exactly one subject, validated command digest, command
intent digest, authorization digest, and idempotency key.",
        "validator_id": "one_command_one_batch"
    },
    {
        "id": "P-026",
        "name": "deterministic-retrieval",
        "severity": "P1",
        "statement": "Ranking ties are deterministic and acceptance-critical facts are added by typed
closure, not fuzzy ranking.",
        "validator_id": "deterministic_retrieval"
    },
    {
        "id": "P-027",
        "name": "strict-time-order",
        "severity": "P1",
        "statement": "Timestamps use canonical UTC seconds and issued, heartbeat, expiry, finish, and
decision order is semantically checked.",
        "validator_id": "strict_time"
    },
    {
        "id": "P-028",
        "name": "exact-standard-files",
        "severity": "P1",
        "statement": "Core and selected preset inputs reject symlinks and unexpected shadow files.",
        "validator_id": "exact_standard_files"
    },
    {
        "id": "P-029",
        "name": "root-authorization-is-bootstrap-only",

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    "severity": "P0",
    "statement": "Root command authorization signs one command intent and may issue the first
authority.manage Grant only; it cannot execute ordinary project commands.",
    "validator_id": "root_authorization_limit"
  },
  {
    "id": "P-030",
    "name": "configuration-exact-idempotency",
    "severity": "P1",
    "statement": "Repeated initialization succeeds only when the requested canonical configuration
digest exactly equals the existing Activation; otherwise it reports conflict.",
    "validator_id": "config_exact_idempotency"
  },
  {
    "id": "P-031",
    "name": "truthful-observability",
    "severity": "P1",
    "statement": "A command may report a scan, healthcheck, validation, or durability action only
when that action was actually executed and evidence-bound.",
    "validator_id": "truthful_observability"
  },
  {
    "id": "P-032",
    "name": "depth-aware-retrieval-budget",
    "severity": "P1",
    "statement": "Effective retrieval seed count reserves entity and relation capacity for typed
closure at the requested dependency depth; top_k is a ceiling, byte/entity/relation/fanout budgets all
remain effective, and no removed or unknown budget field may bypass a ceiling.",
    "validator_id": "depth_aware_retrieval"
  },
  {
    "id": "P-033",
    "name": "lease-only-for-mutation",
    "severity": "P1",
    "statement": "Mutation WorkCards require the current Lease; read-only WorkCards must not acquire
or carry a mutation Lease.",
    "validator_id": "lease_only_for_mutation"
  },
  {
    "id": "P-034",
    "name": "command-capability-scope",
    "severity": "P0",
    "statement": "Every command resolves to exactly one Core-owned capability; requested scope
retains every authorization constraint and explicitly contains every command effect target.",
    "validator_id": "command_capability_scope"
  },
  {
    "id": "P-035",
    "name": "grant-issuance-chain",
    "severity": "P0",
    "statement": "Every Grant proof binds claim_digest and is either the one bootstrap root
issuance, a verified team threshold, or an active authority.manage issuer Grant.",
    "validator_id": "grant_issuance_chain"
  },
  {
    "id": "P-036",
    "name": "verified-team-signatures",
    "severity": "P0",
    "statement": "Team-signature fields are never valid without threshold evaluation by the
signature adapter resolved from the exact technologies binding; process-local injected adapters are
explicitly non-reconstructable and cannot prove a cross-process CLI workflow.",
    "validator_id": "verified_team_signatures"
  },
  {
    "id": "P-037",
    "name": "provider-preflight-before-activation",
    "severity": "P1",
    "statement": "Every required provider identity and healthcheck succeeds before initialization is
committed; a declared provider is not evidence of availability.",
    "validator_id": "provider_preflight"
  },
  {
    "id": "P-038",
    "name": "canonical-state-machines",
    "severity": "P0",
    "statement": "Task and Lease transitions are accepted only by the policy-owned state machines;
generated lifecycle views cannot invent transitions.",

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        "validator_id": "state_machine_transition"
    },
    {
        "id": "P-039",
        "name": "decision-target-binding",
        "severity": "P0",
        "statement": "Every Decision names one typed target and the decision kind, requested scope,
candidate, and authority are consistent with that target.",
        "validator_id": "decision_target_binding"
    },
    {
        "id": "P-040",
        "name": "credit-requires-evidence",
        "severity": "P0",
        "statement": "Pass, fail, and blocked gate results cite at least one exact immutable Artifact
record reference; skipped results never receive pass credit.",
        "validator_id": "credit_requires_evidence"
    },
    {
        "id": "P-041",
        "name": "command-event-correspondence",
        "severity": "P0",
        "statement": "One command batch binds the exact authorization claim and contains exactly one
primary event equal to the validated command payload; auxiliary events are typed relations only and event
IDs are unique.",
        "validator_id": "command_event_correspondence"
    },
    {
        "id": "P-042",
        "name": "continuation-authorization-and-completeness",
        "severity": "P0",
        "statement": "A continuation is authenticated by a random per-Activation secret and binds the
exact subject, projection.read Grant claim, capability, requested scope, revocation state, normalized
query, deterministic cursor, Activation, HEAD, projection, ranking, complete budgets, depth,
implementation closure, issue time, and expiry; its complete page union emits every reachable entity and
relation exactly once without loss.",
        "validator_id": "continuation_completeness"
    },
    {
        "id": "P-043",
        "name": "candidate-snapshot-consistency",
        "severity": "P0",
        "statement": "The candidate recipe is applied in the single inventory pass; promin v1 permits
only provider-proven immutable VCS trees for credit, while best-effort observations are explicit and
receive no pass credit.",
        "validator_id": "candidate_snapshot_consistency"
    },
    {
        "id": "P-044",
        "name": "provider-adapter-dispatch",
        "severity": "P0",
        "statement": "Every technologies.json binding resolves to the adapter actually invoked and
carries a recomputed capability-specific implementation closure for the promin runtime, Python
interpreter, jsonschema or SQLite component where used, and adapter. Unsupported or drifted tuples reject,
and dependent evidence binds the exact implementation_closure_digest.",
        "validator_id": "provider_adapter_dispatch"
    },
    {
        "id": "P-045",
        "name": "current-release-closure",
        "severity": "P0",
        "statement": "A project-operational release Decision is immutable history; current product
release eligibility is derived and becomes false after a blocker, failed gate, stale or missing evidence,
implementation drift, or Activation drift. It is separate from the external StandardReleaseDecision for
promin distribution.",
        "validator_id": "current_release_closure"
    },
    {
        "id": "P-046",
        "name": "finding-disposition-evidence",
        "severity": "P0",
        "statement": "Every Finding cites exact immutable Artifact record references. Every resolve or
waive Decision cites evidence bound to the exact canonical Finding digest; waiver also requires an
exception-policy digest, future expiry, and separation of duties.",
        "validator_id": "finding_disposition_evidence"
    },
    {
        "id": "P-047",

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    "name": "candidate-delta-scope",
    "severity": "P0",
    "statement": "A diff Artifact binds base and new Candidate digests, the exact WorkCard digest,
and the complete changed-path set; every changed path is contained by both Task allowed_paths and the
authorizing Grant effect scope.",
    "validator_id": "candidate_delta_scope"
  },
  {
    "id": "P-048",
    "name": "verified-state-checkpoint",
    "severity": "P1",
    "statement": "Events remain the sole authority. A normal command appends O(batch), incrementally
updates changed projection rows/shards, and advances a verified bounded replay tail. Full derived-state
compaction occurs only after declared batch/byte thresholds or explicit recovery, publishes crash-safely
against exact HEAD, and missing, stale, or corrupt checkpoints require full verified fallback with equal
semantic digest.",
    "validator_id": "verified_state_checkpoint"
  },
  {
    "id": "P-049",
    "name": "profile-default-depth",
    "severity": "P1",
    "statement": "A profile default_dependency_depth applies only when depth is omitted; explicit
depth is accepted through the single Core dependency_depth_hard_max of 12 and remains subject to all
retrieval budgets.",
    "validator_id": "profile_default_depth"
  },
  {
    "id": "P-050",
    "name": "canonical-distribution-archive",
    "severity": "P0",
    "statement": "The sole package builder emits immutable candidate bytes and an external
StandardReleaseCandidateBinding. VERSION.json describes SemVer and package identity only; approval is
derived later and never changes the candidate ZIP.",

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    "validator_id": "canonical_distribution_archive"
  },
  {
    "id": "P-051",
    "name": "verified-inventory-provenance",
    "severity": "P0",
    "statement": "Public rebuild accepts only a verified InventoryResult or its persisted manifest;
caller provenance fields are reserved, data_class is forced to untrusted-source, and runtime injects the
exact path and digest.",
    "validator_id": "verified_inventory_provenance"
  },
  {
    "id": "P-052",
    "name": "supported-command-surface",
    "severity": "P1",
    "statement": "The vl preset advertises exactly init, doctor, status, next, validate, and
continue; no optional or administrative command catalogue may claim an absent executable path.",
    "validator_id": "supported_command_surface"
  },
  {
    "id": "P-053",
    "name": "profile-bound-scale",
    "severity": "P0",
    "statement": "Physical 100k evidence binds content-addressed raw inventory, query, continuation,
process-sample, phase-install, and operation-metric artifacts for at least 198999 explicitly recorded
Core-valid typed Relations and a representative mixed workload. The verifier recomputes every summary and
fails closed on truthful process/invoke status, commit p95/p99, bytes per changed record, checkpoint
count, semantic-ingestion time, search latency, incremental inventory memory amplification, absolute peak
RSS, database bytes, projection amplification, semantic inflation, continuation completeness, or
traversal-pass ceiling.",
    "validator_id": "profile_bound_scale"
  },
  {
    "id": "P-054",
    "name": "cross-platform-release-evidence",
    "severity": "P0",
    "statement": "The exact candidate ZIP must pass authenticated Windows and Linux platform
observations, clean-install, package, handle-relative path/race, locking, crash, replay, physical scale,
offline installed no-degradation, online compatibility, dependency/SBOM/RECORD/license closure, and typed
PDF gates. One StandardReleaseEvidenceManifest physically resolves exactly seven unique authority roles,
the exact non-authoritative eight-lane CPython 3.13/3.14 online/offline matrix, all four CPython 3.14
supplemental records, and every nested physical record by path, SHA-256, and byte count. All producer
attestations bind the same candidate and participate in one replay/SOD/chronology graph;

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matrix_authoritative, pass_credit, acceptance_pass, product_acceptance_pass, and public approval remain false.",

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    "validator_id": "cross_platform_release"
  },
  {
    "id": "P-055",
    "name": "standard-release-separation",
    "severity": "P0",
    "statement": "An external StandardReleaseDecision has explicit approve or reject outcome, binds
candidate and evidence manifest digests, follows the manifest-derived maximum evidence completion time,
stays within 300 seconds of verifier clock skew, and verifies decider, standard.distribute capability,
independently pinned trust root, Ed25519 provider, event-time key validity, nonce, signed claim digest,
and signature. The signed decision remains immutable history; current eligibility is derived;
product_acceptance_pass is false and product_public_approval is not_approved.",
    "validator_id": "standard_release_separation"
  },
  {
    "id": "P-056",
    "name": "required-gates-fail-closed",
    "severity": "P0",
    "statement": "No-degradation runs the complete required suite only through the clean installed
interpreter outside the install source, with isolated mode, cleared PYTHONPATH, import-origin assertion,
zero skipped tests, and full hash-bound JUnit/stdout/stderr bytes. Distribution acceptance executes every
required positive, mutation, concurrency, crash, replay, path, package, platform, Python-minor,
offline/online, scale, search, PDF, evidence-resolution, chronology, and signature gate; a skipped,
missing, stale, blocked, failed, unsigned, unresolved, source-shadowed, or unpinned gate receives no pass
credit.",
    "validator_id": "required_gates_fail_closed"
  },
  {
    "id": "P-057",
    "name": "research-draft-sanitation",
    "severity": "P0",
    "statement": "Research crosses into normative owners only through the non-persistent
MIGRATE_FACTS_NOT_FILES intake: every cited source is digest- and license-plan-bound, citation-required
claims cite allowed sources or remain blocked with a reason, and marketing, proof, distribution,
acceptance, or public-approval claims cannot become normative text.",
    "validator_id": "research_draft_sanitation"
  },
  {
    "id": "P-058",
    "name": "doctor-result-is-live-non-authority",
    "severity": "P0",
    "statement": "DoctorResult is an exact live-only derived health report with
report_authoritative=false, pass_credit=false, product_acceptance_pass=false,
product_public_approval=not_approved, zero product scans, and only its Core-owned operational metrics;
token, cost, usage, and goal/session/batch/workspace/per-run allocation are forbidden.",
    "validator_id": "doctor_result_non_authority"
  },
  {
    "id": "P-059",
    "name": "semantic-inheritance-is-structural",
    "severity": "P0",
    "statement": "The compiled projection preserves the complete SemanticModel owner: admission,
five independent granularities, promotion rules, exact entity purposes and admission rules, exact relation
domain/range/rules, and structured provider forbidden boundaries.",
    "validator_id": "semantic_inheritance"
  },
  {
    "id": "P-060",
    "name": "authority-owner-is-structural",
    "severity": "P0",
    "statement": "AuthorityModel is the single owner of the capability catalogue, delegation depth,
and exact separation-of-duty pairs; runtime policy may derive from that Activation-bound owner but may not
substitute hardcoded authority facts.",
    "validator_id": "authority_owner"
  },
  {
    "id": "P-061",
    "name": "gate-run-definition-is-independent",
    "severity": "P0",
    "statement": "An immutable GateRunDefinition independently owns the expected evidence class,
product-credit requirement, exact target scope, and candidate, policy, tool, provider, and input digests.
A run or gate command binds its exact definition digest before submitted Artifacts are evaluated.",
    "validator_id": "gate_run_definition_integrity"
  },
  {
    "id": "P-062",
```



```

    "name": "authority-runtime-policy-is-derived-live-only",
    "severity": "P0",
    "statement": "AuthorityRuntimePolicy is rebuilt from the installed AuthorityModel for every
mutation and replay, binds the exact owner digest, ordered capability IDs, structured separation-of-duty
rules and delegation ceiling, and is never persisted or imported as authority.",
    "validator_id": "authority_runtime_policy_projection"
  },
  {
    "id": "P-063",
    "name": "semantic-export-is-export-only",
    "severity": "P0",
    "statement": "SemanticExport is a strict bounded export-only semantic-state record binding the
exact Candidate, Activation, HEAD, Core, preset, implementation closure, providers, inputs, and output. It
is neither a package nor archive and cannot carry product acceptance, pass credit, public approval, or
release approval.",
    "validator_id": "semantic_export_integrity"
  },
  {
    "id": "P-064",
    "name": "operation-metrics-are-live-observations",
    "severity": "P0",
    "statement": "OperationMetrics is a strict live-only non-authoritative observation for one
command. It permits only Core-owned duration, changed-record, write, projection, checkpoint, and
physical-byte fields and forbids token, cost, usage, and goal/session/batch/workspace/per-run allocation.",
    "validator_id": "operation_metrics_non_authority"
  },
  {
    "id": "P-065",
    "name": "workcard-projection-is-v2-live-only",
    "severity": "P0",
    "statement": "RetrievalPage is the strict bounded live-only v2 retrieval result and exclusively
owns stream_cursor, next_stream_cursor, and retrieval continuation. WorkCardProjection is internal
selected-Task materialization and exposes no continuation or cursor fields.",
    "validator_id": "workcard_projection_v2"
  },
  {
    "id": "P-066",
    "name": "event-store-policy-is-derived-live-only",
    "severity": "P0",
    "statement": "EventStorePolicy is rebuilt from the installed AuthorityModel before
normalization, mutation, and replay. It binds exact command and batch bounds, envelope and commitment
fields, lease-bound commands and states, primary-event mapping, and UTC-second timestamps, with no
defaults or nullable critical fields.",
    "validator_id": "event_store_policy_projection"
  },
  {
    "id": "P-067",
    "name": "depends-on-graph-is-acyclic",
    "severity": "P0",
    "statement": "Every command, import, replay, and rebuild rejects a DEPENDS_ON cycle. Relation
domain and range validity alone cannot admit a cyclic dependency graph.",
    "validator_id": "dependency_graph_acyclicity"
  },
  {
    "id": "P-068",
    "name": "ready-frontier-is-current-derived-state",
    "severity": "P0",
    "statement": "ReadyFrontier is derived live-only from the current acyclic DEPENDS_ON graph. A
READY Task is returned only after every dependency is currently COMPLETED, every required GateResult is
current passing credit, and no current OPEN Finding blocks it; ordering and continuation are deterministic
and truncation is explicit.",
    "validator_id": "ready_frontier_projection"
  }
]
"record_type": {
  "const": "PolicySet"
},
"research_draft_sanitation": {
  "const": {
    "allowed_license_disposition": "allowed",
    "canonical_bytes_max": 1048576,
    "citation_required_rule": "source-refs-or-blocked-with-reason",
    "claim_digest_rule": "sha256(canonical complete claim excluding claim_digest and derived
normative_use_allowed)",
    "claim_kinds": [

```

```

        "research_question",
        "hypothesis",
        "assumption",
        "evidence_claim",
        "blocked_claim"
    ],
    "evidence_reference_rule": "citation_refs identify research sources; evidence_refs identify
independently resolved Artifact records; the sets are semantically distinct",
    "forbidden_normative_claim_classes": [
        "distribution",
        "marketing",
        "proof"
    ],
    "forbidden_requested_flag_mutations": [
        "acceptance_pass",
        "pass_credit",
        "product_acceptance_pass",
        "product_public_approval",
        "public_release_approved"
    ],
    "metadata_only_rule": "missing payload receipt is representable only by source_mode metadata-only
and every referencing claim remains blocked, non-authoritative, and normative_use_allowed=false",
    "migration_rule": "MIGRATE_FACTS_NOT_FILES",
    "normative_candidate_rule": "only a source-bound or verified evidence_claim classified fact with
eligible citations and resolved evidence_refs may become a sanitized normative candidate",
    "persistence": "transient-ingress-only",
    "result_flags": {
        "pass_credit": false,
        "product_acceptance_pass": false,
        "product_public_approval": "not_approved",
        "public_release_approved": false
    },
    "result_persistence": "derived-live-only",
    "result_record_type": "SanitizedResearchDraft",
    "source_binding": "source-id+trusted-source-locator+provenance-digest+artifact-id+artifact-record-d
igest+recomputed-content-digest+size+media-type+active-license-plan-digest",
    "source_license_rule": "allowed only when the active LicensesPlan binding review_state is reviewed
or source-verified",
    "source_payload_bytes_max": 16777216,
    "source_payload_total_bytes_max": 67108864,
    "verification_statuses": [
        "unverified",
        "source-bound",
        "verified",
        "blocked"
    ]
}
},
"schema_ref": {
    "const": "urn:promin:contracts:1#/$defs/PolicySet"
},
"semantic_export_contract": {
    "const": {
        "export_kind": "semantic-state",
        "input_digests_max": 128,
        "input_digests_rule": "sorted unique sha256(canonical exported record) values for the exact
replay-derived record set",
        "output_artifact_kind": "evidence",
        "output_artifact_record_digest_rule": "sha256(canonical finalized record with exact artifact and
authoritative commit_binding)",
        "output_bytes_max": 1073741824,
        "output_bytes_rule": "exact canonical SemanticStatePayload bytes stored in content-addressed
storage",
        "output_media_type": "application/vnd.promin.semantic-state+json",
        "output_record_type": "SemanticStatePayload",
        "output_records_max": 400000,
        "package_or_archive": false,
        "persistence": "export-only",
        "product_flags": {
            "pass_credit": false,
            "product_acceptance_pass": false,
            "product_public_approval": "not_approved",
            "public_release_approved": false
        },
        "record_order": "record-digest-ascending after exact digest deduplication",
        "required_bindings": [
            "candidate_digest",
            "activation_digest",

```

```

        "head_event_id",
        "head_sequence",
        "head_digest",
        "core_bundle_digest",
        "preset_digest",
        "implementation_closure_digest",
        "provider_binding_digest",
        "input_digests",
        "output_artifact_id",
        "output_artifact_record_digest",
        "output_digest",
        "output_size_bytes",
        "output_media_type"
    ]
}
},
"state_machines": {
    "const": {
        "capacity_release": {
            "closed_release": {
                "requires_field": "close_ack",
                "slot_reusable": true,
                "state": "CLOSED"
            },
            "terminal_reconciliation": {
                "fencing_token_binding": "exact-lease-fencing-token",
                "field": "capacity_reconciliation",
                "generation_binding": "exact-lease-generation",
                "manager_capability": "lease.manage",
                "required_fields": [
                    "reconciled_by",
                    "grant_id",
                    "grant_claim_digest",
                    "reconciled_at",
                    "generation",
                    "fencing_token"
                ],
                "slot_reusable_after_record": true,
                "states": [
                    "EXPIRED",
                    "REVOKED"
                ]
            }
        ],
        "unreconciled_slot_reusable": false
    },
    "lease": {
        "ACTIVE": [
            "CLOSING",
            "EXPIRED",
            "REVOKED"
        ],
        "CLOSED": [],
        "CLOSING": [
            "CLOSED",
            "EXPIRED",
            "REVOKED"
        ],
        "EXPIRED": [],
        "REVOKED": []
    },
    "task": {
        "BLOCKED": [
            "READY",
            "CANCELLED"
        ],
        "CANCELLED": [],
        "COMPLETED": [],
        "LEASED": [
            "RUNNING",
            "READY",
            "BLOCKED",
            "CANCELLED"
        ],
        "PLANNED": [
            "READY",
            "BLOCKED",
            "CANCELLED"
        ]
    }
}

```

```

    ],
    "READY": [
      "LEASED",
      "BLOCKED",
      "CANCELLED"
    ],
    "RUNNING": [
      "COMPLETED",
      "BLOCKED",
      "CANCELLED"
    ]
  }
},
"validator_registry_version": {
  "const": 1
},
"version": {
  "const": "1.0.0-alpha.3"
}
},

```

```

"required": [
  "record_type",
  "schema_ref",
  "document",
  "version",
  "function",
  "validator_registry_version",
  "optimization_gate",
  "state_machines",
  "policies",
  "derived_result_contracts",
  "gate_run_definition_contract",
  "research_draft_sanitation",
  "semantic_export_contract"
],
"type": "object"
}

```

Owner: *core/contracts.schema.json#/\$defs/PolicySet* | Validator: *urn:promin:contracts:1#/\$defs/PolicySet*

67. Preset

Type summary: type="object"; additionalProperties=false

Required: record_type, schema_ref, document, version, function, preset_id, default_profile, base_user_commands, required_provider_capabilities, optional_provider_capabilities, profiles, semantic_programming, forbidden_implicit_behavior

Property	Required	Structural rule
base_user_commands	yes	type="array"; const=["init", "doctor", "status", "next", "validate", "continue", "audit", "refresh", "context", "skills"]
default_profile	yes	\$ref="#/\$defs/Id"
document	yes	const="promin.preset"
forbidden_implicit_behavior	yes	type="array"; minItems=1; maxItems=16; uniqueItems=true
function	yes	type="string"; minLength=1
optional_provider_capabilities	yes	type="array"; minItems=0; maxItems=16; uniqueItems=true
preset_id	yes	\$ref="#/\$defs/Id"
profiles	yes	type="object"; additionalProperties=schema(keys:\$ref)
record_type	yes	const="Preset"
required_provider_capabilities	yes	type="array"; minItems=1; maxItems=16; uniqueItems=true
schema_ref	yes	const="urn:promin:contracts:1#/\$defs/Preset"
semantic_programming	yes	type="object"
version	yes	const="1.0.0-alpha.3"

```

{
  "additionalProperties": false,
  "properties": {
    "base_user_commands": {
      "const": [
        "init",

```

```

        "doctor",
        "status",
        "next",
        "validate",
        "continue",
        "audit",
        "refresh",
        "context",
        "skills"
    ],
    "type": "array"
},
"default_profile": {
    "$ref": "#/$defs/Id"
},
"document": {
    "const": "promin.preset"
},
"forbidden_implicit_behavior": {
    "items": {
        "$ref": "#/$defs/Id"
    },

```

```

        "maxItems": 16,
        "minItems": 1,
        "type": "array",
        "uniqueItems": true
    },
    "function": {
        "minLength": 1,
        "type": "string"
    },
    "optional_provider_capabilities": {
        "items": {
            "$ref": "#/$defs/Id"
        },
        "maxItems": 16,
        "minItems": 0,
        "type": "array",
        "uniqueItems": true
    },
    "preset_id": {
        "$ref": "#/$defs/Id"
    },
    "profiles": {
        "additionalProperties": {
            "$ref": "#/$defs/Profile"
        },
        "minProperties": 1,
        "type": "object"
    },
},

```

```

"record_type": {
    "const": "Preset"
},
"required_provider_capabilities": {
    "items": {
        "$ref": "#/$defs/Id"
    },
    "maxItems": 16,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
},
"schema_ref": {
    "const": "urn:promin:contracts:1#/$defs/Preset"
},
"semantic_programming": {
    "minProperties": 2,
    "type": "object"
},
"version": {
    "const": "1.0.0-alpha.3"
}
},
"required": [
    "record_type",
    "schema_ref",
    "document",
    "version",

```

```

    "function",
    "preset_id",
    "default_profile",
    "base_user_commands",
    "required_provider_capabilities",
    "optional_provider_capabilities",
    "profiles",
    "semantic_programming",
    "forbidden_implicit_behavior"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Preset | Validator: urn:promin:contracts:1#/\$defs/Preset

68. Profile

Type summary: type="object"; additionalProperties=false

Required: model_tier, max_context_bytes, max_entities, max_relations, max_fanout_per_entity, top_k, max_parallel_tasks, default_dependency_depth

Property	Required	Structural rule
default_dependency_depth	yes	type="integer"; minimum=1; maximum=12
max_context_bytes	yes	type="integer"; minimum=1024
max_entities	yes	type="integer"; minimum=1
max_fanout_per_entity	yes	type="integer"; minimum=1
max_parallel_tasks	yes	type="integer"; minimum=1
max_relations	yes	type="integer"; minimum=0
model_tier	yes	enum=["weak-local", "capable", "strong"]
top_k	yes	type="integer"; minimum=1

```

{
  "additionalProperties": false,
  "properties": {
    "default_dependency_depth": {
      "maximum": 12,
      "minimum": 1,
      "type": "integer"
    },
    "max_context_bytes": {
      "minimum": 1024,
      "type": "integer"
    },
    "max_entities": {
      "minimum": 1,
      "type": "integer"
    },
    "max_fanout_per_entity": {
      "minimum": 1,
      "type": "integer"
    },
    "max_parallel_tasks": {
      "minimum": 1,
      "type": "integer"
    },
    "max_relations": {
      "minimum": 0,
      "type": "integer"
    }
  },

```

```

    "model_tier": {
      "enum": [
        "weak-local",
        "capable",
        "strong"
      ]
    },
    "top_k": {
      "minimum": 1,
      "type": "integer"
    }
  },
  "required": [
    "model_tier",

```

```

    "max_context_bytes",
    "max_entities",
    "max_relations",
    "max_fanout_per_entity",
    "top_k",
    "max_parallel_tasks",
    "default_dependency_depth"
  ],
  "type": "object"
}

```

Owner: [core/contracts.schema.json#/\\$defs/Profile](#) | Validator: [urn:promin:contracts:1#/\\$defs/Profile](#)

69. ProjectInit

Type summary: type="object"; additionalProperties=false

Required: record_type, project_id, roots, candidate_recipe, preset_id, operating_profile, intent, resolved_profile, orchestration, telemetry

Property	Required	Structural rule
candidate_recipe	yes	\$ref="#/\$defs/CandidateRecipe"
intent	yes	type="object"; additionalProperties=false
operating_profile	yes	\$ref="#/\$defs/Id"
orchestration	yes	type="object"; additionalProperties=false
preset_id	yes	\$ref="#/\$defs/Id"
project_id	yes	\$ref="#/\$defs/Id"
record_type	yes	const="ProjectInit"
resolved_profile	yes	type="object"; additionalProperties=false
roots	yes	type="array"; minItems=1; maxItems=16
telemetry	yes	type="object"; additionalProperties=false

```

{
  "additionalProperties": false,
  "properties": {
    "candidate_recipe": {
      "$ref": "#/$defs/CandidateRecipe"
    },
    "intent": {
      "additionalProperties": false,
      "properties": {
        "autonomy": {
          "enum": [
            "ask",
            "safe-auto",
            "unsafe-auto"
          ]
        },
        "constraints": {
          "items": {
            "maxLength": 4096,
            "minLength": 1,
            "type": "string"
          },
          "maxItems": 256,
          "type": "array",
          "uniqueItems": true
        },
        "deliverables": {
          "items": {
            "maxLength": 4096,
            "minLength": 1,
            "type": "string"
          },
          "maxItems": 256,
          "type": "array",
          "uniqueItems": true
        },
        "goal": {
          "maxLength": 8192,
          "minLength": 1,
          "type": "string"
        }
      }
    },
    "resolved_profile": {
      "additionalProperties": false,
      "properties": {
        "autonomy": {
          "enum": [
            "ask",
            "safe-auto",
            "unsafe-auto"
          ]
        },
        "constraints": {
          "items": {
            "maxLength": 4096,
            "minLength": 1,
            "type": "string"
          },
          "maxItems": 256,
          "type": "array",
          "uniqueItems": true
        },
        "deliverables": {
          "items": {
            "maxLength": 4096,
            "minLength": 1,
            "type": "string"
          },
          "maxItems": 256,
          "type": "array",
          "uniqueItems": true
        },
        "goal": {
          "maxLength": 8192,
          "minLength": 1,
          "type": "string"
        }
      }
    },
    "roots": {
      "items": {
        "type": "string"
      },
      "minItems": 1,
      "maxItems": 16,
      "uniqueItems": true,
      "type": "array"
    }
  }
}

```

```

"non_goals": {
  "items": {
    "maxLength": 4096,
    "minLength": 1,
    "type": "string"
  },
  "maxItems": 256,
  "type": "array",
  "uniqueItems": true
},
"project_mode": {
  "enum": [
    "greenfield",
    "existing-code",
    "hybrid"

```

```

]
},
"references": {
  "items": {
    "maxLength": 4096,
    "minLength": 1,
    "type": "string"
  },
  "maxItems": 256,
  "type": "array",
  "uniqueItems": true
},
"reporting_language": {
  "enum": [
    "uk",
    "en"
  ]
},
"success_criteria": {
  "items": {
    "maxLength": 4096,
    "minLength": 1,
    "type": "string"
  },
  "maxItems": 256,
  "type": "array",
  "uniqueItems": true
},

```

```

"work_sources": {
  "items": {
    "maxLength": 4096,
    "minLength": 1,
    "type": "string"
  },
  "maxItems": 256,
  "type": "array",
  "uniqueItems": true
}
},
"required": [
  "project_mode",
  "goal",
  "success_criteria",
  "constraints",
  "non_goals",
  "deliverables",
  "references",
  "work_sources",
  "reporting_language",
  "autonomy"
],
"type": "object"
},
"operating_profile": {
  "$ref": "#/$defs/Id"
},

```

```

"orchestration": {
  "additionalProperties": false,
  "properties": {
    "enabled": {
      "const": true
    },
  },

```



```

    "enforcement": {
      "const": "required"
    },
    "host_source": {
      "enum": [
        "generic-agent-envelope",
        "explicit-promin",
        "inherited-compatible-host"
      ]
    },
    "model_routing": {
      "const": "cheapest-adequate-with-evidence-escalation"
    },
    "unregistered_work": {
      "const": "external-proposal-only"
    }
  },
  "required": [
    "enabled",
    "enforcement",
    "unregistered_work",
    "model_routing",

```

```

    "host_source"
  ],
  "type": "object"
},
"preset_id": {
  "$ref": "#/$defs/Id"
},
"project_id": {
  "$ref": "#/$defs/Id"
},
"record_type": {
  "const": "ProjectInit"
},
"resolved_profile": {
  "additionalProperties": false,
  "properties": {
    "authority_effect": {
      "const": "none"
    },
    "detected_technologies": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "confidence": {
            "maximum": 1,
            "minimum": 0,
            "type": "number"
          }
        }
      },

```

```

    "source_count_complete": {
      "type": "boolean"
    },
    "sources": {
      "items": {
        "maxLength": 4096,
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 64,
      "minItems": 0,
      "type": "array",
      "uniqueItems": true
    },
    "sources_truncated": {
      "type": "boolean"
    },
    "technology": {
      "$ref": "#/$defs/Id"
    },
    "total_source_count": {
      "minimum": 1,
      "type": "integer"
    }
  },
  "required": [
    "technology",
    "sources",

```

```

        "total_source_count",
        "sources_truncated",
        "source_count_complete",
        "confidence"
    ],
    "type": "object"
},
"maxItems": 128,
"type": "array"
},
"layers": {
    "items": {
        "$ref": "#/$defs/Id"
    },
    "maxItems": 32,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
},
"operation_profiles": {
    "items": {
        "additionalProperties": false,
        "properties": {
            "model_tier": {
                "enum": [
                    "tool-only",
                    "micro",
                    "standard",

```

```

                    "strong",
                    "critical-review"
                ]
            },
            "operation": {
                "$ref": "#/$defs/Id"
            },
            "reason": {
                "maxLength": 2048,
                "minLength": 1,
                "type": "string"
            }
        },
        "required": [
            "operation",
            "model_tier",
            "reason"
        ],
        "type": "object"
    },
    "maxItems": 64,
    "type": "array"
},
"profile_digest": {
    "$ref": "#/$defs/Digest"
},
"resolution": {
    "items": {

```

```

        "additionalProperties": false,
        "properties": {
            "confidence": {
                "maximum": 1,
                "minimum": 0,
                "type": "number"
            },
            "profile_id": {
                "$ref": "#/$defs/Id"
            },
            "reason": {
                "maxLength": 2048,
                "minLength": 1,
                "type": "string"
            }
        },
        "required": [
            "profile_id",
            "reason",
            "confidence"
        ],
        "type": "object"
    }
}

```

```

    },
    "maxItems": 64,
    "minItems": 1,
    "type": "array"
  },
  "revision": {
    "minimum": 1,
    "type": "integer"
  }
},
"required": [
  "revision",
  "layers",
  "detected_technologies",
  "operation_profiles",
  "resolution",
  "authority_effect",
  "profile_digest"
],
"type": "object"
},
"roots": {
  "items": {
    "$ref": "#/$defs/ProjectRoot"
  },
  "maxItems": 16,
  "minItems": 1,
  "type": "array"
},
"telemetry": {
  "additionalProperties": false,
  "properties": {
    "enabled": {
      "type": "boolean"
    }
  }
}

```

```

  },
  "export_enabled": {
    "const": false
  },
  "local_only": {
    "const": true
  },
  "max_log_bytes": {
    "maximum": 67108864,
    "minimum": 65536,
    "type": "integer"
  },
  "retention_records": {
    "maximum": 100000,
    "minimum": 100,
    "type": "integer"
  },
  "store_full_prompts": {
    "const": false
  },
  "store_secrets_or_pii": {
    "const": false
  },
  "store_source_bodies": {
    "const": false
  }
},
"required": [

```

```

  "enabled",
  "local_only",
  "export_enabled",
  "max_log_bytes",
  "retention_records",
  "store_full_prompts",
  "store_source_bodies",
  "store_secrets_or_pii"
],
"type": "object"
}
},
"required": [
  "record_type",
  "project_id",

```

```

    "roots",
    "candidate_recipe",
    "preset_id",
    "operating_profile",
    "intent",
    "resolved_profile",
    "orchestration",
    "telemetry"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ProjectInit | Validator: urn:promin:contracts:1#/\$defs/ProjectInit

70. ProjectRoot

Type summary: type="object"; additionalProperties=false

Required: path, kind

Property	Required	Structural rule
kind	yes	const="product"
path	yes	\$ref="#/\$defs/RelativePath"

```

{
  "additionalProperties": false,
  "properties": {
    "kind": {
      "const": "product"
    },
    "path": {
      "$ref": "#/$defs/RelativePath"
    }
  },
  "required": [
    "path",
    "kind"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ProjectRoot | Validator: urn:promin:contracts:1#/\$defs/ProjectRoot

71. ProjectionLimits

Type summary: type="object"; additionalProperties=false

Required: record_type, token_version, ranking_algorithm_id, traversal_algorithm_id, dependency_depth_hard_max, continuation_ttl_seconds_max, selected_profile_id, selected_profile_digest, persistent_entity_types, default_budget, hard_budget, default_depth, depth_min, depth_max, default_ttl_seconds, ttl_min_seconds, ttl_max_seconds, max_token_bytes, max_continuation_state_bytes, max_query_bytes, min_result_bytes, max_resume_binding_fields, max_resume_binding_key_chars, max_resume_binding_value_bytes, max_resume_binding_bytes, required_resume_binding_fields, policy_digest

Property	Required	Structural rule
continuation_ttl_seconds_max	yes	const=3600
default_budget	yes	\$ref="#/\$defs/Budget"
default_depth	yes	type="integer"; minimum=1; maximum=12
default_ttl_seconds	yes	const=900
dependency_depth_hard_max	yes	const=12
depth_max	yes	const=12
depth_min	yes	const=1
hard_budget	yes	\$ref="#/\$defs/Budget"
max_continuation_state_bytes	yes	const=16384
max_query_bytes	yes	const=4096
max_resume_binding_bytes	yes	const=4096
max_resume_binding_fields	yes	const=8
max_resume_binding_key_chars	yes	const=64
max_resume_binding_value_bytes	yes	const=512

Property	Required	Structural rule
max_token_bytes	yes	const=256
min_result_bytes	yes	const=1024
persistent_entity_types	yes	const=["Task", "Candidate", "Artifact", "Run", "Finding", "Decision", "GateResult", "Grant", "Lease"]
policy_digest	yes	\$ref="#/\$defs/Digest"
ranking_algorithm_id	yes	const="bm25-v1"
record_type	yes	const="ProjectionLimits"
required_resume_binding_fields	yes	const=["activation_digest", "capability_id", "grant_claim_digest", "grant_id", "implementation_closure_digest", "requested_scope_digest", "revocation_epoch", "subject_id"]
selected_profile_digest	yes	\$ref="#/\$defs/Digest"
selected_profile_id	yes	\$ref="#/\$defs/Id"
token_version	yes	const=2
traversal_algorithm_id	yes	const="typed-bfs-v2"
ttl_max_seconds	yes	const=3600
ttl_min_seconds	yes	const=1
<pre> { "additionalProperties": false, "properties": { "continuation_ttl_seconds_max": { "const": 3600 }, "default_budget": { "\$ref": "#/\$defs/Budget" }, "default_depth": { "maximum": 12, "minimum": 1, "type": "integer" }, "default_ttl_seconds": { "const": 900 }, "dependency_depth_hard_max": { "const": 12 }, "depth_max": { "const": 12 }, "depth_min": { "const": 1 }, "hard_budget": { "\$ref": "#/\$defs/Budget" }, "max_continuation_state_bytes": { "const": 16384 }, "max_query_bytes": { "const": 4096 }, "max_resume_binding_bytes": { "const": 4096 }, "max_resume_binding_fields": { "const": 8 }, "max_resume_binding_key_chars": { "const": 64 }, "max_resume_binding_value_bytes": { "const": 512 }, "max_token_bytes": { "const": 256 }, "min_result_bytes": { "const": 1024 }, }, </pre>		

```

"persistent_entity_types": {
  "const": [
    "Task",
    "Candidate",
    "Artifact",
    "Run",
    "Finding",
    "Decision",
    "GateResult",
    "Grant",
    "Lease"
  ]
},
"policy_digest": {
  "$ref": "#/$defs/Digest"
},
"ranking_algorithm_id": {
  "const": "bm25-v1"
},
"record_type": {
  "const": "ProjectionLimits"
},
"required_resume_binding_fields": {
  "const": [
    "activation_digest",
    "capability_id",
    "grant_claim_digest",
    "grant_id",
    "implementation_closure_digest",
    "requested_scope_digest",
    "revocation_epoch",
    "subject_id"
  ]
},
"selected_profile_digest": {
  "$ref": "#/$defs/Digest"
},
"selected_profile_id": {
  "$ref": "#/$defs/Id"
},
"token_version": {
  "const": 2
},
"traversal_algorithm_id": {
  "const": "typed-bfs-v2"
},
"ttl_max_seconds": {
  "const": 3600
},
"ttl_min_seconds": {
  "const": 1
}
},
"required": [
  "record_type",
  "token_version",
  "ranking_algorithm_id",
  "traversal_algorithm_id",
  "dependency_depth_hard_max",
  "continuation_ttl_seconds_max",
  "selected_profile_id",
  "selected_profile_digest",
  "persistent_entity_types",
  "default_budget",
  "hard_budget",
  "default_depth",
  "depth_min",
  "depth_max",
  "default_ttl_seconds",
  "ttl_min_seconds",
  "ttl_max_seconds",
  "max_token_bytes",
  "max_continuation_state_bytes",
  "max_query_bytes",
  "min_result_bytes",
  "max_resume_binding_fields",
  "max_resume_binding_key_chars",

```

```

    "max_resume_binding_value_bytes",
    "max_resume_binding_bytes",
    "required_resume_binding_fields",
    "policy_digest"
  ],
  "type": "object",
  "x-promin-ingress-operations": [
    "rebuild"
  ],
  "x-promin-persistence": "derived-live-only"
}

```

Owner: core/contracts.schema.json#/\$defs/ProjectionLimits | Validator: urn:promin:contracts:1#/\$defs/ProjectionLimits

72. ProminPlatformNoDegradationMatrix

Type summary: type="object"; additionalProperties=false

Required: record_type, standard_name, version, candidate_binding_digest, archive_sha256, archive_bytes, trust_configuration_sha256, lanes, lane_count, complete, semantic_validation_complete, unique_invocation_ids, unique_key_nonces, matrix_authoritative, pass_credit, acceptance_pass, product_acceptance_pass, product_public_approval, matrix_digest

Property	Required	Structural rule
acceptance_pass	yes	const=false
archive_bytes	yes	type="integer"; minimum=1
archive_sha256	yes	\$ref="#/\$defs/Digest"
candidate_binding_digest	yes	\$ref="#/\$defs/Digest"
complete	yes	const=true
lane_count	yes	const=8
lanes	yes	type="array"; minItems=8; maxItems=8
matrix_authoritative	yes	const=false
matrix_digest	yes	\$ref="#/\$defs/Digest"
pass_credit	yes	const=false
product_acceptance_pass	yes	const=false
product_public_approval	yes	const="not_approved"
record_type	yes	const="ProminPlatformNoDegradationMatrix"
semantic_validation_complete	yes	const=true
standard_name	yes	const="promin"
trust_configuration_sha256	yes	\$ref="#/\$defs/Digest"
unique_invocation_ids	yes	const=true
unique_key_nonces	yes	const=true
version	yes	const="1.0.0-alpha.3"

```

{
  "additionalProperties": false,
  "properties": {
    "acceptance_pass": {
      "const": false
    },
    "archive_bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "archive_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "complete": {
      "const": true
    },
    "lane_count": {
      "const": 8
    },
    "lanes": {

```

```

"items": {
  "additionalProperties": false,
  "properties": {
    "attestation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "evidence_role": {
      "enum": [
        "linux",
        "linux-no-degradation",
        "windows",
        "windows-no-degradation"
      ]
    },
    "install_mode": {
      "enum": [
        "offline-wheelhouse",
        "online-clean"
      ]
    },
    "invocation_id": {
      "$ref": "#/$defs/Id"
    },
    "key_id": {
      "$ref": "#/$defs/Id"
    },
    "lane_id": {
      "enum": [
        "linux-cp313-offline",
        "linux-cp313-online",
        "linux-cp314-offline",
        "linux-cp314-online",
        "windows-cp313-offline",
        "windows-cp313-online",
        "windows-cp314-offline",
        "windows-cp314-online"
      ]
    },
    "nonce": {
      "maxLength": 128,
      "minLength": 24,
      "type": "string"
    },
    "os": {
      "enum": [
        "linux",
        "windows"
      ]
    },
    "path": {
      "$ref": "#/$defs/RelativePath"
    },
    "python_abi": {
      "maxLength": 128,
      "minLength": 1,
      "type": "string"
    },
    "python_implementation": {
      "const": "cpython"
    },
    "python_version": {
      "maxLength": 128,
      "minLength": 1,
      "type": "string"
    },
    "record_type": {
      "enum": [
        "NoDegradationResult",
        "PlatformVerificationResult"
      ]
    },
    "result_digest": {
      "$ref": "#/$defs/Digest"
    },
    "semantic_valid": {
      "const": true
    }
  }
}

```



```

"sha256": {
  "$ref": "#/$defs/Digest"
},
"size_bytes": {
  "minimum": 1,
  "type": "integer"
},
"status": {
  "const": "pass"
},
"subject_id": {
  "$ref": "#/$defs/Id"

```

```

}
},
"required": [
  "lane_id",
  "path",
  "sha256",
  "size_bytes",
  "result_digest",
  "record_type",
  "status",
  "os",
  "python_version",
  "python_implementation",
  "python_abi",
  "install_mode",
  "evidence_role",
  "key_id",
  "subject_id",
  "attestation_digest",
  "invocation_id",
  "nonce",
  "semantic_valid"
],
"type": "object"
},
"maxItems": 8,
"minItems": 8,
"type": "array"

```

```

},
"matrix_authoritative": {
  "const": false
},
"matrix_digest": {
  "$ref": "#/$defs/Digest"
},
"pass_credit": {
  "const": false
},
"product_acceptance_pass": {
  "const": false
},
"product_public_approval": {
  "const": "not_approved"
},
"record_type": {
  "const": "ProminPlatformNoDegradationMatrix"
},
"semantic_validation_complete": {
  "const": true
},
"standard_name": {
  "const": "promin"
},
"trust_configuration_sha256": {
  "$ref": "#/$defs/Digest"
},

```

```

"unique_invocation_ids": {
  "const": true
},
"unique_key_nonces": {
  "const": true
},
"version": {
  "const": "1.0.0-alpha.3"
}

```

```

},
"required": [
  "record_type",
  "standard_name",
  "version",
  "candidate_binding_digest",
  "archive_sha256",
  "archive_bytes",
  "trust_configuration_sha256",
  "lanes",
  "lane_count",
  "complete",
  "semantic_validation_complete",
  "unique_invocation_ids",
  "unique_key_nonces",
  "matrix_authoritative",
  "pass_credit",
  "acceptance_pass",
  "product_acceptance_pass",

  "product_public_approval",
  "matrix_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ProminPlatformNoDegradationMatrix | Validator: urn:promin:contracts:1#/\$defs/ProminPlatformNoDegradationMatrix

73. ProviderBinding

Type summary: type="object"; additionalProperties=false; allOf[9]=keys:if,then,title | keys:if,then,title | keys:if,then,title | keys:if,then,title | keys:if,then,title | keys:if,then,title | keys:if,then,title

Required: capability_id, provider_id, version, invocation, purpose, required, healthcheck, license, identity, implementation_closure, dependency_receipt

Property	Required	Structural rule
capability_id	yes	enum=["control-runtime", "shape-validation", "content-identity", "local-serialization", "query-projection", "filesystem-inventory", "export-scan", "signature", "build-dependency"]
dependency_receipt	yes	\$ref="#/\$defs/ProviderDependencyReceipt"
healthcheck	yes	\$ref="#/\$defs/Healthcheck"
identity	yes	\$ref="#/\$defs/ProviderIdentity"
implementation_closure	yes	\$ref="#/\$defs/ImplementationClosure"
invocation	yes	\$ref="#/\$defs/Invocation"
license	yes	\$ref="#/\$defs/LicenseBinding"
provider_id	yes	\$ref="#/\$defs/Id"
purpose	yes	type="string"; minLength=1
required	yes	type="boolean"
version	yes	type="string"; minLength=1

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "if": {
        "properties": {
          "capability_id": {
            "const": "control-runtime"
          }
        }
      },
      "required": [
        "capability_id"
      ]
    },
    {
      "then": {
        "oneOf": [
          {
            "properties": {
              "identity": {
                "properties": {

```

```

        "kind": {
            "const": "file-digest"
        }
    },
    "required": [
        "kind"
    ]
},

```

```

    "implementation_closure": {
        "properties": {
            "adapters": {
                "items": {
                    "properties": {
                        "adapter_id": {
                            "const": "python-runtime-v1"
                        }
                    },
                    "required": [
                        "adapter_id"
                    ]
                }
            }
        },
        "required": [
            "adapters"
        ]
    },
    "invocation": {
        "properties": {
            "kind": {
                "const": "python-runtime"
            }
        },
        "required": [
            "kind"
        ]
    }
}

```

```

    }
}
],
"properties": {
    "implementation_closure": {
        "properties": {
            "adapters": {
                "items": {
                    "properties": {
                        "capability_id": {
                            "const": "control-runtime"
                        },
                        "protocol_id": {
                            "const": "promin.control-runtime.v1"
                        }
                    },
                    "required": [
                        "capability_id",
                        "protocol_id"
                    ]
                }
            }
        },
        "required": [
            "adapters"
        ]
    }
}

```

```

    },
    "title": "control-runtime binds its exact adapter protocol"
},
{
    "if": {
        "properties": {
            "capability_id": {
                "const": "shape-validation"
            }
        },
        "required": [
            "capability_id"
        ]
    }
}

```

```

    ]
  },
  "then": {
    "oneOf": [
      {
        "properties": {
          "identity": {
            "properties": {
              "kind": {
                "const": "file-digest"
              }
            }
          },
          "required": [
            "kind"
          ]
        }
      ]
    ]
  }
}

```

```

    },
    "implementation_closure": {
      "properties": {
        "adapters": {
          "items": {
            "properties": {
              "adapter_id": {
                "const": "python-runtime-v1"
              }
            },
            "required": [
              "adapter_id"
            ]
          }
        }
      },
      "required": [
        "adapters"
      ]
    },
    "invocation": {
      "properties": {
        "kind": {
          "const": "python-runtime"
        }
      },
      "required": [
        "kind"
      ]
    }
  }
}

```

```

    }
  }
},
"properties": {
  "implementation_closure": {
    "properties": {
      "adapters": {
        "items": {
          "properties": {
            "capability_id": {
              "const": "shape-validation"
            },
            "protocol_id": {
              "const": "promin.shape-validation.v1"
            }
          },
          "required": [
            "capability_id",
            "protocol_id"
          ]
        }
      }
    },
    "required": [
      "adapters"
    ]
  }
}

```

```

  }
},
"title": "shape-validation binds its exact adapter protocol"
},
{

```

```

    "if": {
      "properties": {
        "capability_id": {
          "const": "content-identity"
        }
      },
      "required": [
        "capability_id"
      ]
    },
    "then": {
      "oneOf": [
        {
          "properties": {
            "identity": {
              "properties": {
                "kind": {
                  "const": "file-digest"
                }
              },
              "required": [
                "kind"
              ]
            }
          },
          "implementation_closure": {
            "properties": {
              "adapters": {
                "items": {
                  "properties": {
                    "adapter_id": {
                      "const": "python-runtime-v1"
                    }
                  },
                  "required": [
                    "adapter_id"
                  ]
                }
              },
              "required": [
                "adapters"
              ]
            },
            "invocation": {
              "properties": {
                "kind": {
                  "const": "python-runtime"
                }
              },
              "required": [
                "kind"
              ]
            }
          }
        },
        {
          "properties": {
            "implementation_closure": {
              "properties": {
                "adapters": {
                  "items": {
                    "properties": {
                      "capability_id": {
                        "const": "content-identity"
                      },
                      "protocol_id": {
                        "const": "promin.content-identity.v1"
                      }
                    },
                    "required": [
                      "capability_id",
                      "protocol_id"
                    ]
                  }
                }
              },
              "required": [
                "adapters"
              ]
            }
          }
        }
      ]
    }
  ],
  "required": [
    "adapters"
  ]
}

```

```

    }
  }
},
"title": "content-identity binds its exact adapter protocol"
},
{
  "if": {
    "properties": {
      "capability_id": {
        "const": "local-serialization"
      }
    },
    "required": [
      "capability_id"
    ]
  },
  "then": {
    "oneOf": [
      {
        "properties": {
          "identity": {
            "properties": {
              "kind": {
                "const": "file-digest"
              }
            }
          },
          "required": [
            "kind"
          ]
        },
        "implementation_closure": {
          "properties": {
            "adapters": {
              "items": {
                "properties": {
                  "adapter_id": {
                    "const": "python-runtime-v1"
                  }
                },
                "required": [
                  "adapter_id"
                ]
              }
            },
            "required": [
              "adapters"
            ]
          },
          "invocation": {
            "properties": {
              "kind": {
                "const": "python-runtime"
              }
            },
            "required": [
              "kind"
            ]
          }
        }
      }
    ]
  },
  "properties": {
    "implementation_closure": {
      "properties": {
        "adapters": {
          "items": {
            "properties": {
              "capability_id": {
                "const": "local-serialization"
              },
              "protocol_id": {
                "const": "promin.local-serialization.v1"
              }
            },
            "required": [
              "capability_id",

```

```

        "protocol_id"
      ]
    }
  },
  "required": [
    "adapters"
  ]
},
{
  "title": "local-serialization binds its exact adapter protocol"
},
{
  "if": {
    "properties": {
      "capability_id": {
        "const": "query-projection"
      }
    },
    "required": [
      "capability_id"
    ]
  },
  "then": {
    "oneOf": [
      {
        "properties": {
          "identity": {
            "properties": {
              "kind": {
                "const": "file-digest"
              }
            }
          }
        },
        "required": [
          "kind"
        ]
      }
    ]
  },
  "implementation_closure": {
    "properties": {
      "adapters": {
        "items": {
          "properties": {
            "adapter_id": {
              "const": "python-runtime-v1"
            }
          },
          "required": [
            "adapter_id"
          ]
        }
      }
    },
    "required": [
      "adapters"
    ]
  },
  "invocation": {
    "properties": {
      "kind": {
        "const": "python-runtime"
      }
    }
  },
  "required": [
    "kind"
  ]
}
},
{
  "properties": {
    "implementation_closure": {
      "properties": {
        "adapters": {
          "items": {
            "properties": {
              "capability_id": {

```

```

        "const": "query-projection"
      },
      "protocol_id": {
        "const": "promin.query-projection.v1"
      }
    },
    "required": [
      "capability_id",
      "protocol_id"
    ]
  }
},
{
  "required": [
    "adapters"
  ]
},
{
  "title": "query-projection binds its exact adapter protocol"
},
{
  "if": {
    "properties": {
      "capability_id": {
        "const": "filesystem-inventory"
      }
    },
    "required": [
      "capability_id"
    ]
  },
  "then": {
    "oneOf": [
      {
        "properties": {
          "identity": {
            "properties": {
              "kind": {
                "const": "file-digest"
              }
            }
          }
        }
      }
    ]
  },
  "required": [
    "kind"
  ]
},
{
  "implementation_closure": {
    "properties": {
      "adapters": {
        "items": {
          "properties": {
            "adapter_id": {
              "const": "git-executable-v1"
            }
          }
        },
        "required": [
          "adapter_id"
        ]
      }
    }
  },
  "required": [
    "adapters"
  ]
},
{
  "invocation": {
    "properties": {
      "kind": {
        "const": "executable"
      }
    }
  },
  "required": [
    "kind"
  ]
}
}

```



```

    ],
    "properties": {
      "implementation_closure": {
        "properties": {
          "adapters": {
            "items": {
              "properties": {
                "capability_id": {
                  "const": "filesystem-inventory"
                },
                "protocol_id": {
                  "const": "promin.filesystem-inventory.v1"
                }
              },
              "required": [
                "capability_id",
                "protocol_id"
              ]
            }
          }
        }
      },
      "required": [
        "adapters"
      ]
    }
  },
  "title": "filesystem-inventory binds its exact adapter protocol"
},
{
  "if": {
    "properties": {
      "capability_id": {
        "const": "export-scan"
      }
    },
    "required": [
      "capability_id"
    ]
  },
  "then": {
    "oneOf": [
      {
        "properties": {
          "identity": {
            "properties": {
              "kind": {
                "const": "file-digest"
              }
            }
          },
          "required": [
            "kind"
          ]
        },
        "implementation_closure": {
          "properties": {
            "adapters": {
              "items": {
                "properties": {
                  "adapter_id": {
                    "const": "executable-export-scan-v1"
                  }
                },
                "required": [
                  "adapter_id"
                ]
              }
            }
          },
          "required": [
            "adapters"
          ]
        },
        "invocation": {
          "properties": {
            "kind": {

```

```

    "const": "executable"
  },
  "required": [
    "kind"
  ]
}
},
],
"properties": {
  "implementation_closure": {
    "properties": {
      "adapters": {
        "items": {
          "properties": {
            "capability_id": {
              "const": "export-scan"
            },
            "protocol_id": {
              "const": "promin.export-scan.v1"
            }
          },
          "required": [
            "capability_id",
            "protocol_id"
          ]
        }
      }
    },
    "required": [
      "adapters"
    ]
  },
  "title": "export-scan binds its exact adapter protocol"
},
{
  "if": {
    "properties": {
      "capability_id": {
        "const": "signature"
      }
    },
    "required": [
      "capability_id"
    ]
  },
  "then": {
    "oneOf": [
      {
        "properties": {
          "identity": {
            "properties": {
              "kind": {
                "const": "file-digest"
              }
            },
            "required": [
              "kind"
            ]
          },
          "implementation_closure": {
            "properties": {
              "adapters": {
                "items": {
                  "properties": {
                    "adapter_id": {
                      "const": "executable-signature-v1"
                    }
                  },
                  "required": [
                    "adapter_id"
                  ]
                }
              }
            }
          }
        }
      }
    ]
  }
},

```



```

    }
  },
  "required": [
    "adapters"
  ]
}
},
"if": {

```

```

  "properties": {
    "capability_id": {
      "const": "build-dependency"
    }
  },
  "required": [
    "capability_id"
  ]
},
"then": {
  "oneOf": [
    {
      "properties": {
        "identity": {
          "properties": {
            "kind": {
              "const": "file-digest"
            }
          },
          "required": [
            "kind"
          ]
        },
        "implementation_closure": {
          "properties": {
            "adapters": {
              "items": {

```

```

                "adapter_id": {
                  "const": "executable-build-dependency-v1"
                }
              },
              "required": [
                "adapter_id"
              ]
            }
          },
          "required": [
            "adapters"
          ]
        },
        "invocation": {
          "properties": {
            "kind": {
              "const": "executable"
            }
          },
          "required": [
            "kind"
          ]
        }
      ]
    }
  ],
  "properties": {

```

```

    "implementation_closure": {
      "properties": {
        "adapters": {
          "items": {
            "properties": {
              "capability_id": {
                "const": "build-dependency"
              },

```

```

        "protocol_id": {
          "const": "promin.build-dependency.v1"
        }
      },
      "required": [
        "capability_id",
        "protocol_id"
      ]
    }
  },
  "required": [
    "adapters"
  ]
}
},
"title": "build-dependency binds its exact adapter protocol"
}
],

```

```

"properties": {
  "capability_id": {
    "enum": [
      "control-runtime",
      "shape-validation",
      "content-identity",
      "local-serialization",
      "query-projection",
      "filesystem-inventory",
      "export-scan",
      "signature",
      "build-dependency"
    ]
  },
  "dependency_receipt": {
    "$ref": "#/$defs/ProviderDependencyReceipt"
  },
  "healthcheck": {
    "$ref": "#/$defs/Healthcheck"
  },
  "identity": {
    "$ref": "#/$defs/ProviderIdentity"
  },
  "implementation_closure": {
    "$ref": "#/$defs/ImplementationClosure"
  },
  "invocation": {
    "$ref": "#/$defs/Invocation"
  },

```

```

},
"license": {
  "$ref": "#/$defs/LicenseBinding"
},
"provider_id": {
  "$ref": "#/$defs/Id"
},
"purpose": {
  "minLength": 1,
  "type": "string"
},
"required": {
  "type": "boolean"
},
"version": {
  "minLength": 1,
  "type": "string"
}
},
"required": [
  "capability_id",
  "provider_id",
  "version",
  "invocation",
  "purpose",
  "required",
  "healthcheck",
  "license",

```

```

    "identity",
    "implementation_closure",
    "dependency_receipt"
  ],
  "type": "object"
}

```

Owner: [core/contracts.schema.json#/\\$defs/ProviderBinding](#) | Validator: [urn:promin:contracts:1#/\\$defs/ProviderBinding](#)

74. ProviderDependencyReceipt

Type summary: type="object"; additionalProperties=false

Required: complete, provider_owned_scope, os_substrate_excluded, components, aggregate_digest

Property	Required	Structural rule
aggregate_digest	yes	\$ref="#/\$defs/Digest"
complete	yes	const=true
components	yes	type="array"; minItems=1; maxItems=4096; uniqueItems=true
os_substrate_excluded	yes	const=true
provider_owned_scope	yes	const="configured-provider-root-and-runtime-components"

```

{
  "additionalProperties": false,
  "properties": {
    "aggregate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "complete": {
      "const": true
    },
    "components": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "component_id": {
            "$ref": "#/$defs/Id"
          },
          "component_kind": {
            "enum": [
              "native-runtime",
              "provider-file",
              "provider-tree",
              "python-distribution"
            ]
          },
          "digest": {
            "$ref": "#/$defs/Digest"
          },
          "size_bytes": {
            "minimum": 1,
            "type": "integer"
          },
          "source": {
            "maxLength": 4096,
            "minLength": 1,
            "type": "string"
          },
          "version": {
            "maxLength": 256,
            "minLength": 1,
            "type": "string"
          }
        },
        "required": [
          "component_id",
          "component_kind",
          "source",
          "version",
          "digest",
          "size_bytes"
        ],
        "type": "object"
      },
      "maxItems": 4096,
      "minItems": 1,

```

```

    "type": "array",
    "uniqueItems": true
  },
  "os_substrate_excluded": {
    "const": true
  },
  "provider_owned_scope": {
    "const": "configured-provider-root-and-runtime-components"
  }
},
"required": [
  "complete",
  "provider_owned_scope",
  "os_substrate_excluded",
  "components",
  "aggregate_digest"
],
"type": "object"
}

```

Owner: *core/contracts.schema.json#/\$defs/ProviderDependencyReceipt* | Validator: *urn:promin:contracts:1#/\$defs/ProviderDependencyReceipt*

75. ProviderIdentity

Type summary: type="object"; additionalProperties=false

Required: kind, digest, source

Property	Required	Structural rule
digest	yes	\$ref="#/\$defs/Digest"
kind	yes	enum=["file-digest", "module-file-digest", "distribution-tree-digest", "module-closure-digest", "platform-attestation"]
source	yes	type="string"; minLength=1

```

{
  "additionalProperties": false,
  "properties": {
    "digest": {
      "$ref": "#/$defs/Digest"
    },
    "kind": {
      "enum": [
        "file-digest",
        "module-file-digest",
        "distribution-tree-digest",
        "module-closure-digest",
        "platform-attestation"
      ]
    },
    "source": {
      "minLength": 1,
      "type": "string"
    }
  },
  "required": [
    "kind",
    "digest",
    "source"
  ],
  "type": "object"
}

```

Owner: *core/contracts.schema.json#/\$defs/ProviderIdentity* | Validator: *urn:promin:contracts:1#/\$defs/ProviderIdentity*

76. ProviderInvocationEvidence

Type summary: type="object"; additionalProperties=false; oneOf[52]=keys:properties keys:properties					
keys:properties keys:properties keys:properties keys:properties keys:properties keys:properties					
keys:properties keys:properties keys:properties keys:properties keys:properties keys:properties					
keys:properties keys:properties keys:properties keys:properties keys:properties keys:properties					
keys:properties keys:properties keys:properties keys:properties keys:properties keys:properties					
keys:properties keys:properties keys:properties keys:properties keys:properties keys:properties					
keys:properties keys:properties keys:properties keys:properties keys:properties keys:properties					
keys:properties keys:properties keys:properties keys:properties keys:properties keys:properties					
keys:properties keys:properties					

Required: capability_id, provider_id, invocation_kind, identity_kind, identity_digest, adapter_id, protocol_id, operation, operation_contract_digest, dependency_receipt_digest, implementation_closure_digest, invoked, started_at, completed_at, outcome, exit_code, invocation_request_digest, input_digest, output_digest, output_size_bytes, output_size_ceiling_bytes, stdout_capture_digest, stdout_capture_size_bytes, stdout_capture_truncated, stderr_capture_digest, stderr_capture_size_bytes, stderr_capture_truncated, invocation_receipt_digest, authoritative, pass_credit

Property	Required	Structural rule
adapter_id	yes	\$ref="#/\$defs/Id"
authoritative	yes	const=false
capability_id	yes	enum=["control-runtime", "shape-validation", "content-identity", "local-serialization", "query-projection", "filesystem-inventory", "export-scan", "signature", "build-dependency"]
completed_at	yes	\$ref="#/\$defs/Timestamp"
dependency_receipt_digest	yes	\$ref="#/\$defs/Digest"
exit_code	yes	oneOf[2]=type:integer type:null
identity_digest	yes	\$ref="#/\$defs/Digest"
identity_kind	yes	enum=["file-digest", "module-file-digest", "distribution-tree-digest", "module-closure-digest", "platform-attestation"]
implementation_closure_digest	yes	\$ref="#/\$defs/Digest"
input_digest	yes	\$ref="#/\$defs/Digest"
invocation_kind	yes	enum=["python-runtime", "python-module", "executable", "platform-service"]
invocation_receipt_digest	yes	\$ref="#/\$defs/Digest"
invocation_request_digest	yes	\$ref="#/\$defs/Digest"
invoked	yes	const=true
operation	yes	\$ref="#/\$defs/Id"
operation_contract_digest	yes	\$ref="#/\$defs/Digest"
outcome	yes	enum=["success", "failure", "blocked"]
output_digest	yes	\$ref="#/\$defs/Digest"
output_size_bytes	yes	type="integer"; minimum=0; maximum=1099511627776
output_size_ceiling_bytes	yes	type="integer"; minimum=0; maximum=1099511627776
pass_credit	yes	const=false
protocol_id	yes	\$ref="#/\$defs/Id"
provider_id	yes	\$ref="#/\$defs/Id"
started_at	yes	\$ref="#/\$defs/Timestamp"
stderr_capture_digest	yes	\$ref="#/\$defs/Digest"
stderr_capture_size_bytes	yes	type="integer"; minimum=0; maximum=1048576
stderr_capture_truncated	yes	type="boolean"
stdout_capture_digest	yes	\$ref="#/\$defs/Digest"
stdout_capture_size_bytes	yes	type="integer"; minimum=0; maximum=1048576
stdout_capture_truncated	yes	type="boolean"
{ "additionalProperties": false, "oneOf": [{ "properties": {		


```

    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "control-runtime"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "continue"
    },
    "operation_contract_digest": {
      "const": "a8a728c5c39a2c00083fc3b6463288e8b3797d871675d3d2e27f7feb1be7ff74"
    },
    "protocol_id": {
      "const": "promin.control-runtime.v1"
    }
  }
},

```

```

{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "control-runtime"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "doctor"
    },
    "operation_contract_digest": {
      "const": "dcf2e435cf7954edc17fd35fc7928951f9a607d2ee91343d1788c114d23cdead"
    },
    "protocol_id": {
      "const": "promin.control-runtime.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {

```

```

      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "control-runtime"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
    },
    "protocol_id": {
      "const": "promin.control-runtime.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"

```

```

    },
    "capability_id": {
        "const": "control-runtime"
    },
    "identity_kind": {
        "const": "file-digest"
    },
    "invocation_kind": {
        "const": "python-runtime"
    },
    "operation": {
        "const": "init"
    },
    "operation_contract_digest": {
        "const": "9c2e26f6706abd39bef62d2120fcd9785f4f1cf39cdae92c3e4fa8f20ff9dd12"
    },
    "protocol_id": {
        "const": "promin.control-runtime.v1"
    }
}
},
{
    "properties": {
        "adapter_id": {
            "const": "python-runtime-v1"
        },
        "capability_id": {
            "const": "control-runtime"
        },
        "identity_kind": {
            "const": "file-digest"
        },
        "invocation_kind": {
            "const": "python-runtime"
        },
        "operation": {
            "const": "next"
        },
        "operation_contract_digest": {
            "const": "b75143a53c7129f54c62fc5c0ebd97927a7732bb8fc6f91918d5858a9578fe5d"
        },
        "protocol_id": {
            "const": "promin.control-runtime.v1"
        }
    }
},
{
    "properties": {
        "adapter_id": {
            "const": "python-runtime-v1"
        },
        "capability_id": {
            "const": "control-runtime"
        },
        "identity_kind": {
            "const": "file-digest"
        },
        "invocation_kind": {
            "const": "python-runtime"
        },
        "operation": {
            "const": "provider-binding"
        },
        "operation_contract_digest": {
            "const": "135f8b5d81f1cec32ec18a36aledcb3455cd186266ed1a72a5b1ecb638102f6f"
        },
        "protocol_id": {
            "const": "promin.control-runtime.v1"
        }
    }
},
{
    "properties": {
        "adapter_id": {
            "const": "python-runtime-v1"
        },
        "capability_id": {

```

```

    "const": "control-runtime"
  },
  "identity_kind": {
    "const": "file-digest"
  },
  "invocation_kind": {
    "const": "python-runtime"
  },
  "operation": {
    "const": "status"
  },
  "operation_contract_digest": {
    "const": "0449eeb95febf5e15ffba4657cdb6cd606c0a12a148bd444384120faa5bc5eb1"
  },
  "protocol_id": {
    "const": "promin.control-runtime.v1"
  }
}
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "control-runtime"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "validate"
    },
    "operation_contract_digest": {
      "const": "6cab313c41c9cd37de52a75d0ba0676cfd9fb0a3fc53eb3f225521a02dcd8cf3"
    },
    "protocol_id": {
      "const": "promin.control-runtime.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "shape-validation"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "command"
    },
    "operation_contract_digest": {
      "const": "5ba4690e0a2adc6d06b74673ab39b0b714215a8543c188a7bed695b71a54c2bf"
    },
    "protocol_id": {
      "const": "promin.shape-validation.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "shape-validation"
    },

```

```

    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "continue"
    },
    "operation_contract_digest": {
      "const": "5ba4690e0a2adc6d06b74673ab39b0b714215a8543c188a7bed695b71a54c2bf"
    },
    "protocol_id": {
      "const": "promin.shape-validation.v1"
    }
  }
}

```

```

},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "shape-validation"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "export"
    },
    "operation_contract_digest": {
      "const": "5ba4690e0a2adc6d06b74673ab39b0b714215a8543c188a7bed695b71a54c2bf"
    },
    "protocol_id": {
      "const": "promin.shape-validation.v1"
    }
  }
},
{
  "properties": {

```

```

    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "shape-validation"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
    },
    "protocol_id": {
      "const": "promin.shape-validation.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },

```

```

    "capability_id": {
      "const": "shape-validation"
    },
    "identity_kind": {
      "const": "file-digest"
    }
  }
}

```

```

    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    },
    "operation": {
      "const": "import"
    },
    },
    "operation_contract_digest": {
      "const": "5ba4690e0a2adc6d06b74673ab39b0b714215a8543c188a7bed695b71a54c2bf"
    },
    },
    "protocol_id": {
      "const": "promin.shape-validation.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    },
    "capability_id": {
      "const": "shape-validation"
    },
    },
  },

```

```

    "identity_kind": {
      "const": "file-digest"
    },
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    },
    "operation": {
      "const": "provider-binding"
    },
    },
    "operation_contract_digest": {
      "const": "135f8b5d81f1cec32ec18a36a1edcb3455cd186266ed1a72a5b1ecb638102f6f"
    },
    },
    "protocol_id": {
      "const": "promin.shape-validation.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    },
    "capability_id": {
      "const": "shape-validation"
    },
    },
    "identity_kind": {
      "const": "file-digest"
    },
    },
  },

```

```

    "invocation_kind": {
      "const": "python-runtime"
    },
    },
    "operation": {
      "const": "rebuild"
    },
    },
    "operation_contract_digest": {
      "const": "5ba4690e0a2adc6d06b74673ab39b0b714215a8543c188a7bed695b71a54c2bf"
    },
    },
    "protocol_id": {
      "const": "promin.shape-validation.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    },
    "capability_id": {
      "const": "shape-validation"
    },
    },
    "identity_kind": {
      "const": "file-digest"
    },
    },
    "invocation_kind": {

```

```

    "const": "python-runtime"
  },
  "operation": {
    "const": "replay"
  },
  "operation_contract_digest": {
    "const": "5ba4690e0a2adc6d06b74673ab39b0b714215a8543c188a7bed695b71a54c2bf"
  },
  "protocol_id": {
    "const": "promin.shape-validation.v1"
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "content-identity"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "digest-bytes"
    }
  },
  "operation_contract_digest": {
    "const": "1898e33042a5a6c55d8c1687c07f53f3f8df020c5475a57e882ca54c9adcebfd"
  },
  "protocol_id": {
    "const": "promin.content-identity.v1"
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "content-identity"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "digest-file"
    },
    "operation_contract_digest": {
      "const": "3876865e4a21f0047d92daaca9b6d7177769130a97c781e90ce130ced513f60b"
    }
  },
  "protocol_id": {
    "const": "promin.content-identity.v1"
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "content-identity"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    }
  },

```

```

"operation": {
  "const": "digest-value"
},
"operation_contract_digest": {
  "const": "b0a54275c219f51585f519a4d9c1814478bf072dbeff050a92a80780eb492a39"
},
"protocol_id": {
  "const": "promin.content-identity.v1"
}

```

```

}
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "content-identity"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
    },
    "protocol_id": {
      "const": "promin.content-identity.v1"
    }
  }
},
{

```

```

"properties": {
  "adapter_id": {
    "const": "python-runtime-v1"
  },
  "capability_id": {
    "const": "content-identity"
  },
  "identity_kind": {
    "const": "file-digest"
  },
  "invocation_kind": {
    "const": "python-runtime"
  },
  "operation": {
    "const": "provider-binding"
  },
  "operation_contract_digest": {
    "const": "135f8b5d81f1cec32ec18a36a1edcb3455cd186266ed1a72a5b1ecb638102f6f"
  },
  "protocol_id": {
    "const": "promin.content-identity.v1"
  }
}
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"

```

```

},
"capability_id": {
  "const": "local-serialization"
},
"identity_kind": {
  "const": "file-digest"
},
"invocation_kind": {
  "const": "python-runtime"
},
"operation": {
  "const": "atomic-replace"

```

```

    },
    "operation_contract_digest": {
      "const": "afab188552442a9aa0a271a4021e25f96a34655eaa687bc4f82b934a04652706"
    },
    "protocol_id": {
      "const": "promin.local-serialization.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "local-serialization"
    }
  }
}

```

```

    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "directory-sync"
    },
    "operation_contract_digest": {
      "const": "83842a439d25cc9e85bd51afec2381576f2ff42390d364672b4ee0bd415b1b23"
    },
    "protocol_id": {
      "const": "promin.local-serialization.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "local-serialization"
    },
    "identity_kind": {
      "const": "file-digest"
    }
  }
}

```

```

    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "file-sync"
    },
    "operation_contract_digest": {
      "const": "3d757d7af8c2fcfb955b76db89a8917262aec19de8419305c6617a6f668cbefd"
    },
    "protocol_id": {
      "const": "promin.local-serialization.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "local-serialization"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    }
  }
}

```

```

    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {

```



```

        "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
    },
    "protocol_id": {
        "const": "promin.local-serialization.v1"
    }
},
{
    "properties": {
        "adapter_id": {
            "const": "python-runtime-v1"
        },
        "capability_id": {
            "const": "local-serialization"
        },
        "identity_kind": {
            "const": "file-digest"
        },
        "invocation_kind": {
            "const": "python-runtime"
        },
        "operation": {
            "const": "local-writer-lock"
        }
    }
},

```

```

    },
    "operation_contract_digest": {
        "const": "ad8ca42303233f79dfc5f72b097602947e2a3df3b24fa2b5d452e374505820c1"
    },
    "protocol_id": {
        "const": "promin.local-serialization.v1"
    }
},
{
    "properties": {
        "adapter_id": {
            "const": "python-runtime-v1"
        },
        "capability_id": {
            "const": "local-serialization"
        },
        "identity_kind": {
            "const": "file-digest"
        },
        "invocation_kind": {
            "const": "python-runtime"
        },
        "operation": {
            "const": "provider-binding"
        },
        "operation_contract_digest": {
            "const": "135f8b5d81f1cec32ec18a36a1edcb3455cd186266ed1a72a5b1ecb638102f6f"
        }
    }
},

```

```

    "protocol_id": {
        "const": "promin.local-serialization.v1"
    }
},
{
    "properties": {
        "adapter_id": {
            "const": "python-runtime-v1"
        },
        "capability_id": {
            "const": "local-serialization"
        },
        "identity_kind": {
            "const": "file-digest"
        },
        "invocation_kind": {
            "const": "python-runtime"
        },
        "operation": {
            "const": "recover"
        },
        "operation_contract_digest": {
            "const": "148d9b508acd481c22a9e74e9021bd7741bbbb4557bcaa0e863e8ab6b9481f98"
        }
    }
},

```

```

    "protocol_id": {
      "const": "promin.local-serialization.v1"
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "query-projection"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "continue"
      },
      "operation_contract_digest": {
        "const": "84f9a7260289ac24fea9192b7f97982539b86b489aba2f347c4d77a0b1121d1c"
      },
      "protocol_id": {
        "const": "promin.query-projection.v1"
      }
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "query-projection"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "healthcheck"
      },
      "operation_contract_digest": {
        "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
      },
      "protocol_id": {
        "const": "promin.query-projection.v1"
      }
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "query-projection"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "next"
      },
      "operation_contract_digest": {
        "const": "f9994e06188b1391ccdd2b1d592e9a1709db1874a5f437ac8bfebf4de349106"
      },
      "protocol_id": {
        "const": "promin.query-projection.v1"
      }
    }
  }
}

```

```

    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "query-projection"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "provider-binding"
      },
      "operation_contract_digest": {
        "const": "135f8b5d81f1cec32ec18a36aledcb3455cd186266ed1a72a5b1ecb638102f6f"
      },
      "protocol_id": {
        "const": "promin.query-projection.v1"
      }
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "query-projection"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "rebuild"
      },
      "operation_contract_digest": {
        "const": "825da07c8c0c880475d1e292bbc0ff059f10e3d47197d93da1cf1fc8c23d58e9"
      },
      "protocol_id": {
        "const": "promin.query-projection.v1"
      }
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "query-projection"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "status"
      },
      "operation_contract_digest": {
        "const": "6a48bc71d6f0876bde29d4267ebecf27c694e0652fd0f807aea49810a6d8b527"
      },
      "protocol_id": {
        "const": "promin.query-projection.v1"
      }
    }
  }
}

```

```

},
{
  "properties": {
    "adapter_id": {
      "const": "git-executable-v1"
    },
    "capability_id": {
      "const": "filesystem-inventory"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "053c4181d6d35bd4ca5a13b8a1f18c19571c21b3426644bfa6bda99066dce5b3"
    },
    "protocol_id": {
      "const": "promin.filesystem-inventory.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "git-executable-v1"
    },
    "capability_id": {
      "const": "filesystem-inventory"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "immutable-tree-object"
    },
    "operation_contract_digest": {
      "const": "7b9234db382b1728299f568503507fd04df4e1f19433c0e378c21378caad3241"
    },
    "protocol_id": {
      "const": "promin.filesystem-inventory.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "git-executable-v1"
    },
    "capability_id": {
      "const": "filesystem-inventory"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "immutable-tree-stream"
    },
    "operation_contract_digest": {
      "const": "a286c3c566c2af8bb74011affe25071137b9ec9452a472f0e4997c4dc1a1038e"
    },
    "protocol_id": {
      "const": "promin.filesystem-inventory.v1"
    }
  }
},
{

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```

"properties": {
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  },
  "capability_id": {
    "const": "filesystem-inventory"
  },
  "identity_kind": {
    "const": "file-digest"
  },
  "invocation_kind": {
    "const": "executable"
  },
  "operation": {
    "const": "provider-binding"
  },
  "operation_contract_digest": {
    "const": "561e1b0729e20cd9586ee796ce58ac2f2759bcdff6cd5e1c31524f89aa4cbd97"
  },
  "protocol_id": {
    "const": "promin.filesystem-inventory.v1"
  }
}
}

```

```

},
{
  "properties": {
    "adapter_id": {
      "const": "executable-export-scan-v1"
    },
    "capability_id": {
      "const": "export-scan"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "bounded-export-scan"
    },
    "operation_contract_digest": {
      "const": "7e0326ca387c256798bc2c37c1d1c3e0fa17440d775c9d0c7deafa7ada33d1c3"
    },
    "protocol_id": {
      "const": "promin.export-scan.v1"
    }
  }
},
{
  "properties": {

```

```

    "adapter_id": {
      "const": "executable-export-scan-v1"
    },
    "capability_id": {
      "const": "export-scan"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "116e5e8a519592f63585f0fae7b90f86863ba3369f0b006913a0e4f7d51eaf2e"
    },
    "protocol_id": {
      "const": "promin.export-scan.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {

```

```

    "const": "executable-export-scan-v1"
  },
  "capability_id": {
    "const": "export-scan"
  },
  "identity_kind": {
    "const": "file-digest"
  },
  "invocation_kind": {
    "const": "executable"
  },
  "operation": {
    "const": "provider-binding"
  },
  "operation_contract_digest": {
    "const": "561e1b0729e20cd9586ee796ce58ac2f2759bcdff6cd5e1c31524f89aa4cbd97"
  },
  "protocol_id": {
    "const": "promin.export-scan.v1"
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    }
  },
  "identity_kind": {
    "const": "file-digest"
  },
  "invocation_kind": {
    "const": "executable"
  },
  "operation": {
    "const": "healthcheck"
  },
  "operation_contract_digest": {
    "const": "0e2delaec47aec67eacbec7838d7f7f19f457bcff9c3fae57c9e88a72b6bb6be"
  },
  "protocol_id": {
    "const": "promin.signature.v1"
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-module-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "module-file-digest"
    }
  },
  "invocation_kind": {
    "const": "python-module"
  },
  "operation": {
    "const": "healthcheck"
  },
  "operation_contract_digest": {
    "const": "0e2delaec47aec67eacbec7838d7f7f19f457bcff9c3fae57c9e88a72b6bb6be"
  },
  "protocol_id": {
    "const": "promin.signature.v1"
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-signature-v1"
    }
  },

```

```

"capability_id": {
  "const": "signature"
},
"identity_kind": {
  "const": "file-digest"
},
"invocation_kind": {
  "const": "executable"
},

```

```

"operation": {
  "const": "provider-binding"
},
"operation_contract_digest": {
  "const": "340c605d4d232f122703a2d47141fd524232107e7f69a8968a53a4d4e3903a1d"
},
"protocol_id": {
  "const": "promin.signature.v1"
}
}
},
{
  "properties": {
    "adapter_id": {
      "const": "python-module-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "module-file-digest"
    },
    "invocation_kind": {
      "const": "python-module"
    },
    "operation": {
      "const": "provider-binding"
    },
  },

```

```

"operation_contract_digest": {
  "const": "340c605d4d232f122703a2d47141fd524232107e7f69a8968a53a4d4e3903a1d"
},
"protocol_id": {
  "const": "promin.signature.v1"
}
}
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "sign-digest"
    },
    "operation_contract_digest": {
      "const": "ba5393e6098a6e47715d7eab7df317b837c087766dd6dc82aa750da9fa9e3237"
    },
  },

```

```

"protocol_id": {
  "const": "promin.signature.v1"
}
}
},
{
  "properties": {
    "adapter_id": {
      "const": "python-module-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
  },

```

```

    },
    "identity_kind": {
      "const": "module-file-digest"
    },
    "invocation_kind": {
      "const": "python-module"
    },
    "operation": {
      "const": "sign-digest"
    },
    "operation_contract_digest": {
      "const": "ba5393e6098a6e47715d7eab7df317b837c087766dd6dc82aa750da9fa9e3237"
    },
    "protocol_id": {
      "const": "promin.signature.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "verify-digest"
    },
    "operation_contract_digest": {
      "const": "7985bee569d1fbd30f73edc3878027a7e2a388a86c02c7dad6894b0ef9d4e67b"
    },
    "protocol_id": {
      "const": "promin.signature.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-module-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "module-file-digest"
    },
    "invocation_kind": {
      "const": "python-module"
    },
    "operation": {
      "const": "verify-digest"
    },
    "operation_contract_digest": {
      "const": "7985bee569d1fbd30f73edc3878027a7e2a388a86c02c7dad6894b0ef9d4e67b"
    },
    "protocol_id": {
      "const": "promin.signature.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-build-dependency-v1"
    },
    "capability_id": {
      "const": "build-dependency"
    },
    "identity_kind": {

```



```

    "const": "file-digest"
  },
  "invocation_kind": {
    "const": "executable"
  },
  "operation": {
    "const": "healthcheck"
  },
  "operation_contract_digest": {
    "const": "116e5e8a519592f63585f0fae7b90f86863ba3369f0b006913a0e4f7d51eaf2e"
  },
  "protocol_id": {
    "const": "promin.build-dependency.v1"
  }
}
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-build-dependency-v1"
    },
    "capability_id": {
      "const": "build-dependency"
    }
  },
  "identity_kind": {
    "const": "file-digest"
  },
  "invocation_kind": {
    "const": "executable"
  },
  "operation": {
    "const": "import-build-facts"
  },
  "operation_contract_digest": {
    "const": "25063736ddf272a51989981c79fcbb8e9cf2decf040522d3e050b13f0e2457db"
  },
  "protocol_id": {
    "const": "promin.build-dependency.v1"
  }
}
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-build-dependency-v1"
    },
    "capability_id": {
      "const": "build-dependency"
    },
    "identity_kind": {
      "const": "file-digest"
    }
  },
  "invocation_kind": {
    "const": "executable"
  },
  "operation": {
    "const": "provider-binding"
  },
  "operation_contract_digest": {
    "const": "561e1b0729e20cd9586ee796ce58ac2f2759bcdff6cd5e1c31524f89aa4cbd97"
  },
  "protocol_id": {
    "const": "promin.build-dependency.v1"
  }
}
},
{
  "properties": {
    "adapter_id": {
      "$ref": "#/$defs/Id"
    },
    "authoritative": {
      "const": false
    },
    "capability_id": {
      "enum": [
        "control-runtime",

```

```

"shape-validation",
"content-identity",

"local-serialization",
"query-projection",
"filesystem-inventory",
"export-scan",
"signature",
"build-dependency"
]
},
"completed_at": {
  "$ref": "#/$defs/Timestamp"
},
"dependency_receipt_digest": {
  "$ref": "#/$defs/Digest"
},
"exit_code": {
  "oneOf": [
    {
      "type": "integer"
    },
    {
      "type": "null"
    }
  ]
},
"identity_digest": {
  "$ref": "#/$defs/Digest"
},
"identity_kind": {

```

```

  "enum": [
    "file-digest",
    "module-file-digest",
    "distribution-tree-digest",
    "module-closure-digest",
    "platform-attestation"
  ]
},
"implementation_closure_digest": {
  "$ref": "#/$defs/Digest"
},
"input_digest": {
  "$ref": "#/$defs/Digest"
},
"invocation_kind": {
  "enum": [
    "python-runtime",
    "python-module",
    "executable",
    "platform-service"
  ]
},
"invocation_receipt_digest": {
  "$ref": "#/$defs/Digest"
},
"invocation_request_digest": {
  "$ref": "#/$defs/Digest"
},

```

```

"invoked": {
  "const": true
},
"operation": {
  "$ref": "#/$defs/Id"
},
"operation_contract_digest": {
  "$ref": "#/$defs/Digest"
},
"outcome": {
  "enum": [
    "success",
    "failure",
    "blocked"
  ]
},
"output_digest": {
  "$ref": "#/$defs/Digest"
},

```

```

"output_size_bytes": {
  "maximum": 1099511627776,
  "minimum": 0,
  "type": "integer"
},
"output_size_ceiling_bytes": {
  "maximum": 1099511627776,
  "minimum": 0,
  "type": "integer"
},
"pass_credit": {
  "const": false
},
"protocol_id": {
  "$ref": "#/$defs/Id"
},
"provider_id": {
  "$ref": "#/$defs/Id"
},
"started_at": {
  "$ref": "#/$defs/Timestamp"
},
"stderr_capture_digest": {
  "$ref": "#/$defs/Digest"
},
"stderr_capture_size_bytes": {
  "maximum": 1048576,
  "minimum": 0,
  "type": "integer"
},
"stderr_capture_truncated": {
  "type": "boolean"
},
"stdout_capture_digest": {
  "$ref": "#/$defs/Digest"
},
"stdout_capture_size_bytes": {
  "maximum": 1048576,
  "minimum": 0,
  "type": "integer"
},
"stdout_capture_truncated": {
  "type": "boolean"
}
},
"required": [
  "capability_id",
  "provider_id",
  "invocation_kind",
  "identity_kind",
  "identity_digest",
  "adapter_id",
  "protocol_id",
  "operation",
  "operation_contract_digest",
  "dependency_receipt_digest",
  "implementation_closure_digest",
  "invoked",
  "started_at",
  "completed_at",
  "outcome",
  "exit_code",
  "invocation_request_digest",
  "input_digest",
  "output_digest",
  "output_size_bytes",
  "output_size_ceiling_bytes",
  "stdout_capture_digest",
  "stdout_capture_size_bytes",
  "stdout_capture_truncated",
  "stderr_capture_digest",
  "stderr_capture_size_bytes",
  "stderr_capture_truncated",
  "invocation_receipt_digest",
  "authoritative",
  "pass_credit"
],

```

```
"type": "object"
}
```

Owner: [core/contracts.schema.json#/\\$defs/ProviderInvocationEvidence](#) | Validator:
[urn:promin:contracts:1#/\\$defs/ProviderInvocationEvidence](#)

77. ReadyFrontier

Type summary: type="object"; additionalProperties=false; allOf[1]=keys:else,if,then

Required: record_type, activation_digest, head_digest, projection_digest, ordering, ready_tasks, evaluated_task_count, blocked_task_count, cycle_count, truncated, continuation, silent_truncation, projection_authoritative

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
blocked_task_count	yes	type="integer"; minimum=0
continuation	yes	oneOf[2]=type:null type:object
cycle_count	yes	type="integer"; const=0
evaluated_task_count	yes	type="integer"; minimum=0
head_digest	yes	oneOf[2]=#/\$defs/Digest type:null
ordering	yes	const=["created_at-ascending", "task_id-ascending"]
projection_authoritative	yes	const=false
projection_digest	yes	\$ref="#/\$defs/Digest"
ready_tasks	yes	type="array"; minItems=0; maxItems=12; uniqueItems=true
record_type	yes	const="ReadyFrontier"
silent_truncation	yes	const=false
truncated	yes	type="boolean"

```
{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "properties": {
          "continuation": {
            "type": "null"
          }
        }
      },
      "if": {
        "properties": {
          "truncated": {
            "const": true
          }
        }
      },
      "required": [
        "truncated"
      ]
    },
    {
      "then": {
        "properties": {
          "continuation": {
            "additionalProperties": false,
            "properties": {
              "activation_digest": {
                "$ref": "#/$defs/Digest"
              }
            }
          },
          "budget_digest": {
            "$ref": "#/$defs/Digest"
          },
          "cursor": {
            "minimum": 0,
            "type": "integer"
          },
          "depth": {
            "maximum": 12,
            "minimum": 1,
            "type": "integer"
          },
          "expiry": {
```

```

        "$ref": "#/$defs/Timestamp"
    },
    "head_digest": {
        "oneOf": [
            {
                "$ref": "#/$defs/Digest"
            },
            {
                "type": "null"
            }
        ]
    },
    "implementation_closure_digest": {
        "$ref": "#/$defs/Digest"
    },
    "projection_digest": {
        "$ref": "#/$defs/Digest"
    },
    "ranking": {
        "const": "bm25-v1"
    },
    "resume_binding_digest": {
        "$ref": "#/$defs/Digest"
    },
    "token": {
        "maxLength": 256,
        "minLength": 1,
        "pattern": "^promin-v2\\. [A-Za-z0-9_-]{22}\\. [0-9a-f]{64}\\. [A-Za-z0-9_-]{43}$",
        "type": "string"
    },
    "traversal": {
        "const": "typed-bfs-v2"
    },
    "version": {
        "const": 2
    }
},
"required": [
    "version",
    "traversal",
    "token",
    "cursor",
    "expiry",
    "activation_digest",
    "head_digest",
    "projection_digest",
    "implementation_closure_digest",
    "ranking",
    "depth",
    "budget_digest",
    "resume_binding_digest"
],
"type": "object"
}
}
}
},
"properties": {
    "activation_digest": {
        "$ref": "#/$defs/Digest"
    },
    "blocked_task_count": {
        "minimum": 0,
        "type": "integer"
    },
    "continuation": {
        "oneOf": [
            {
                "type": "null"
            }
        ]
    },
    {
        "additionalProperties": false,
        "properties": {
            "activation_digest": {
                "$ref": "#/$defs/Digest"
            }
        }
    }
}

```

```

    "budget_digest": {
      "$ref": "#/$defs/Digest"
    },
    "cursor": {
      "minimum": 0,
      "type": "integer"
    },
    "depth": {
      "maximum": 12,
      "minimum": 1,
      "type": "integer"
    },
    "expiry": {
      "$ref": "#/$defs/Timestamp"
    },
    "head_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {

```

```

          "type": "null"
        }
      ]
    },
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "projection_digest": {
      "$ref": "#/$defs/Digest"
    },
    "ranking": {
      "const": "bm25-v1"
    },
    "resume_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "token": {
      "maxLength": 256,
      "minLength": 1,
      "pattern": "^promin-v2\\. [A-Za-z0-9_-]{22}\\. [0-9a-f]{64}\\. [A-Za-z0-9_-]{43}$",
      "type": "string"
    },
    "traversal": {
      "const": "typed-bfs-v2"
    },
    "version": {
      "const": 2
    }
  }

```

```

},
"required": [
  "version",
  "traversal",
  "token",
  "cursor",
  "expiry",
  "activation_digest",
  "head_digest",
  "projection_digest",
  "implementation_closure_digest",
  "ranking",
  "depth",
  "budget_digest",
  "resume_binding_digest"
],
"type": "object"
}
]
},
"cycle_count": {
  "const": 0,
  "type": "integer"
},
"evaluated_task_count": {
  "minimum": 0,
  "type": "integer"
},
},

```

```

"head_digest": {
  "oneOf": [
    {
      "$ref": "#/$defs/Digest"
    },
    {
      "type": "null"
    }
  ]
},
"ordering": {
  "const": [
    "created_at-ascending",
    "task_id-ascending"
  ]
},
"projection_authoritative": {
  "const": false
},
"projection_digest": {
  "$ref": "#/$defs/Digest"
},
"ready_tasks": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "acceptance_predicate": {
        "maxLength": 256,

```

```

        "minLength": 1,
        "type": "string"
      },
      "candidate_digest": {
        "$ref": "#/$defs/Digest"
      },
      "created_at": {
        "$ref": "#/$defs/Timestamp"
      },
      "dependency_task_ids": {
        "items": {
          "$ref": "#/$defs/Id"
        },
        "maxItems": 32,
        "minItems": 0,
        "type": "array",
        "uniqueItems": true
      },
      "gate_result_digests": {
        "items": {
          "$ref": "#/$defs/Digest"
        },
        "maxItems": 64,
        "minItems": 0,
        "type": "array",
        "uniqueItems": true
      },
      "task_digest": {

```

```

        "$ref": "#/$defs/Digest"
      },
      "task_id": {
        "$ref": "#/$defs/Id"
      }
    },
    "required": [
      "task_id",
      "task_digest",
      "created_at",
      "candidate_digest",
      "acceptance_predicate",
      "dependency_task_ids",
      "gate_result_digests"
    ],
    "type": "object"
  },
  "maxItems": 12,
  "minItems": 0,
  "type": "array",
  "uniqueItems": true
},

```

```

    "record_type": {
      "const": "ReadyFrontier"
    },
    "silent_truncation": {
      "const": false
    },
  },
  "truncated": {
    "type": "boolean"
  }
},
"required": [
  "record_type",
  "activation_digest",
  "head_digest",
  "projection_digest",
  "ordering",
  "ready_tasks",
  "evaluated_task_count",
  "blocked_task_count",
  "cycle_count",
  "truncated",
  "continuation",
  "silent_truncation",
  "projection_authoritative"
],
"type": "object",
"x-promin-persistence": "derived-live-only"
}

```

Owner: core/contracts.schema.json#/\$defs/ReadyFrontier | Validator: urn:promin:contracts:1#/\$defs/ReadyFrontier

78. Relation

Type summary: type="object"; additionalProperties=false

Required: record_type, relation_id, kind, source_type, source_id, target_type, target_id, activation_digest, created_at

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
created_at	yes	\$ref="#/\$defs/Timestamp"
kind	yes	enum=["DEPENDS_ON", "READS", "PRODUCES", "VALIDATES", "CITES", "BLOCKS"]
record_type	yes	const="Relation"
relation_id	yes	\$ref="#/\$defs/Id"
source_id	yes	\$ref="#/\$defs/Id"
source_type	yes	\$ref="#/\$defs/Id"
target_id	yes	\$ref="#/\$defs/Id"
target_type	yes	\$ref="#/\$defs/Id"

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "created_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "kind": {
      "enum": [
        "DEPENDS_ON",
        "READS",
        "PRODUCES",
        "VALIDATES",
        "CITES",
        "BLOCKS"
      ]
    },
    "record_type": {
      "const": "Relation"
    },
    "relation_id": {
      "$ref": "#/$defs/Id"
    },
  },
}

```



```

    "source_id": {
      "$ref": "#/$defs/Id"
    },
    "source_type": {
      "$ref": "#/$defs/Id"
    },
    "target_id": {
      "$ref": "#/$defs/Id"
    },
    "target_type": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "record_type",
    "relation_id",
    "kind",
    "source_type",
    "source_id",
    "target_type",
    "target_id",
    "activation_digest",
    "created_at"
  ],
  "type": "object"
}

```

Owner: `core/contracts.schema.json#/$defs/Relation` | Validator: `urn:promin:contracts:1#/$defs/Relation`

79. RelationDef

Type summary: `type="object"; additionalProperties=false`

Required: `kind`, `source`, `target`, `rules`

Property	Required	Structural rule
<code>kind</code>	yes	<code>enum=["DEPENDS_ON", "READS", "PRODUCES", "VALIDATES", "CITES", "BLOCKS"]</code>
<code>rules</code>	yes	<code>type="array"; minItems=1; maxItems=8; uniqueItems=true</code>
<code>source</code>	yes	<code>type="array"; minItems=1; maxItems=4; uniqueItems=true</code>
<code>target</code>	yes	<code>type="array"; minItems=1; maxItems=4; uniqueItems=true</code>

```

{
  "additionalProperties": false,
  "properties": {
    "kind": {
      "enum": [
        "DEPENDS_ON",
        "READS",
        "PRODUCES",
        "VALIDATES",
        "CITES",
        "BLOCKS"
      ]
    },
    "rules": {
      "items": {
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 8,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "source": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 4,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "target": {
      "items": {

```

```

    "$ref": "#/$defs/Id"
  },
  "maxItems": 4,
  "minItems": 1,
  "type": "array",
  "uniqueItems": true
}
},
"required": [
  "kind",
  "source",
  "target",
  "rules"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/RelationDef | Validator: urn:promin:contracts:1#/\$defs/RelationDef

80. RelativePath

Type summary: type="string"; pattern="^(?!/)(?![A-Za-z:](?!.*\\))(?!.*(?:^/|\\.|\\.(?:/|\$))(?!.*//).+"; minLength=1; maxLength=4096

Required: -

```

{
  "maxLength": 4096,
  "minLength": 1,
  "pattern": "^(?!/)(?![A-Za-z:](?!.*\\))(?!.*(?:^/|\\.|\\.(?:/|$))(?!.*//).+",
  "type": "string"
}

```

Owner: core/contracts.schema.json#/\$defs/RelativePath | Validator: urn:promin:contracts:1#/\$defs/RelativePath

81. ReleaseEvidenceInvocation

Type summary: type="object"; additionalProperties=false

Required: invocation_id, operation, arguments_digest, started_at, completed_at, exit_code, platform_binding_digest

Property	Required	Structural rule
arguments_digest	yes	\$ref="#/\$defs/Digest"
completed_at	yes	\$ref="#/\$defs/Timestamp"
exit_code	yes	type="integer"
invocation_id	yes	\$ref="#/\$defs/Id"
operation	yes	\$ref="#/\$defs/Id"
platform_binding_digest	yes	\$ref="#/\$defs/Digest"
started_at	yes	\$ref="#/\$defs/Timestamp"

```

{
  "additionalProperties": false,
  "properties": {
    "arguments_digest": {
      "$ref": "#/$defs/Digest"
    },
    "completed_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "exit_code": {
      "type": "integer"
    },
    "invocation_id": {
      "$ref": "#/$defs/Id"
    },
    "operation": {
      "$ref": "#/$defs/Id"
    },
    "platform_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "started_at": {
      "$ref": "#/$defs/Timestamp"
    }
  },
  "required": [

```

```

    "invocation_id",
    "operation",

    "arguments_digest",
    "started_at",
    "completed_at",
    "exit_code",
    "platform_binding_digest"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ReleaseEvidenceInvocation | Validator:
urn:promin:contracts:1#/\$defs/ReleaseEvidenceInvocation

82. ReleaseEvidenceProducer

Type summary: type="object"; additionalProperties=false

Required: tool_path, tool_sha256, tool_version

Property	Required	Structural rule
tool_path	yes	enum=["tools/generate_human.py", "tools/promin_no_degradation.py", "tools/promin_package.py", "tools/promin_saturation.py", "tools/promin_saturation_audit.py", "tools/promin_validate.py"]
tool_sha256	yes	\$ref="#/\$defs/Digest"
tool_version	yes	\$ref="#/\$defs/SemVer"

```

{
  "additionalProperties": false,
  "properties": {
    "tool_path": {
      "enum": [
        "tools/generate_human.py",
        "tools/promin_no_degradation.py",
        "tools/promin_package.py",
        "tools/promin_saturation.py",
        "tools/promin_saturation_audit.py",
        "tools/promin_validate.py"
      ]
    },
    "tool_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "tool_version": {
      "$ref": "#/$defs/SemVer"
    }
  },
  "required": [
    "tool_path",
    "tool_sha256",
    "tool_version"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ReleaseEvidenceProducer | Validator:
urn:promin:contracts:1#/\$defs/ReleaseEvidenceProducer

83. ResearchDraftClaim

Type summary: type="object"; additionalProperties=false; allOf[1]=keys:else,if,then

Required: claim_id, claim_kind, claim_class, claim_digest, text, citation_required, citation_refs, evidence_refs, verification_status, normative_candidate, normative_use_allowed

Property	Required	Structural rule
blocked_reason	no	type="string"; minLength=1; maxLength=2048
citation_refs	yes	type="array"; maxItems=32; uniqueItems=true
citation_required	yes	type="boolean"
claim_class	yes	enum=["distribution", "fact", "marketing", "proof"]
claim_digest	yes	\$ref="#/\$defs/Digest"
claim_id	yes	\$ref="#/\$defs/Id"
claim_kind	yes	enum=["research_question", "hypothesis", "assumption", "evidence_claim", "blocked_claim"]

Property	Required	Structural rule
evidence_refs	yes	type="array"; maxItems=32; uniqueItems=true
normative_candidate	yes	type="boolean"
normative_use_allowed	yes	const=false
text	yes	type="string"; minLength=1; maxLength=8192
verification_status	yes	enum=["unverified", "source-bound", "verified", "blocked"]
<pre> { "additionalProperties": false, "allOf": [{ "else": { "not": { "required": ["blocked_reason"] } }, "if": { "anyOf": [{ "properties": { "claim_kind": { "const": "blocked_claim" } }, "required": ["claim_kind"] }, { "properties": { "verification_status": { "const": "blocked" } } }] }, "required": ["verification_status"] }], "then": { "required": ["blocked_reason"] } }, "properties": { "blocked_reason": { "maxLength": 2048, "minLength": 1, "type": "string" }, "citation_refs": { "items": { "\$ref": "#/\$defs/Id" }, "maxItems": 32, "type": "array", "uniqueItems": true }, "citation_required": { "type": "boolean" }, "claim_class": { "enum": ["distribution", "fact", "marketing", "proof"] }, "claim_digest": { </pre>		

```

    "$ref": "#/$defs/Digest"
  },
  "claim_id": {
    "$ref": "#/$defs/Id"
  },
  "claim_kind": {
    "enum": [
      "research_question",
      "hypothesis",
      "assumption",
      "evidence_claim",
      "blocked_claim"
    ]
  },
  "evidence_refs": {
    "items": {

```

```

      "additionalProperties": false,
      "properties": {
        "artifact_id": {
          "$ref": "#/$defs/Id"
        },
        "artifact_record_digest": {
          "$ref": "#/$defs/Digest"
        }
      },
      "required": [
        "artifact_id",
        "artifact_record_digest"
      ],
      "type": "object"
    },
    "maxItems": 32,
    "type": "array",
    "uniqueItems": true
  },
  "normative_candidate": {
    "type": "boolean"
  },
  "normative_use_allowed": {
    "const": false
  },
  "text": {
    "maxLength": 8192,
    "minLength": 1,

```

```

    "type": "string"
  },
  "verification_status": {
    "enum": [
      "unverified",
      "source-bound",
      "verified",
      "blocked"
    ]
  },
  "required": [
    "claim_id",
    "claim_kind",
    "claim_class",
    "claim_digest",
    "text",
    "citation_required",
    "citation_refs",
    "evidence_refs",
    "verification_status",
    "normative_candidate",
    "normative_use_allowed"
  ],
  "type": "object"
}

```

Owner: [core/contracts.schema.json#/\\$defs/ResearchDraftClaim](#) | Validator: [urn:promin:contracts:1#/\\$defs/ResearchDraftClaim](#)

84. ResearchDraftIntake

Type summary: type="object"; additionalProperties=false

Required: record_type, migration_rule, license_plan_digest, sources, claims, requested_flag_mutations

Property	Required	Structural rule
claims	yes	type="array"; minItems=1; maxItems=256
license_plan_digest	yes	\$ref="#/\$defs/Digest"
migration_rule	yes	const="MIGRATE_FACTS_NOT_FILES"
record_type	yes	const="ResearchDraftIntake"
requested_flag_mutations	yes	type="array"; const=[]
sources	yes	type="array"; minItems=1; maxItems=128

```
{
  "additionalProperties": false,
  "properties": {
    "claims": {
      "items": {
        "$ref": "#/$defs/ResearchDraftClaim"
      },
      "maxItems": 256,
      "minItems": 1,
      "type": "array"
    },
    "license_plan_digest": {
      "$ref": "#/$defs/Digest"
    },
    "migration_rule": {
      "const": "MIGRATE_FACTS_NOT_FILES"
    },
    "record_type": {
      "const": "ResearchDraftIntake"
    },
    "requested_flag_mutations": {
      "const": [],
      "type": "array"
    },
    "sources": {
      "items": {
        "$ref": "#/$defs/ResearchSourceIntake"
      },
      "maxItems": 128,
      "minItems": 1,
      "type": "array"
    }
  },
  "required": [
    "record_type",
    "migration_rule",
    "license_plan_digest",
    "sources",
    "claims",
    "requested_flag_mutations"
  ],
  "type": "object",
  "x-promin-ingress-operations": [
    "transient"
  ],
  "x-promin-max-canonical-bytes": 1048576,
  "x-promin-persistence": "transient-ingress-only"
}
```

Owner: [core/contracts.schema.json#/\\$defs/ResearchDraftIntake](#) | Validator: [urn:promin:contracts:1#/\\$defs/ResearchDraftIntake](#)

85. ResearchSourceIntake

Type summary: type="object"; additionalProperties=false; allOf[1]=keys:else,if,then

Required: source_id, source_mode, source_locator, source_provenance_digest, media_type, license_provider_id, license_expression, license_plan_digest

Property	Required	Structural rule
artifact_id	no	\$ref="#/\$defs/Id"
artifact_record_digest	no	\$ref="#/\$defs/Digest"
content_digest	no	\$ref="#/\$defs/Digest"
license_expression	yes	type="string"; minLength=1; maxLength=512
license_plan_digest	yes	\$ref="#/\$defs/Digest"

Property	Required	Structural rule
license_provider_id	yes	\$ref="#/\$defs/Id"
media_type	yes	type="string"; minLength=1; maxLength=256
missing_reason	no	type="string"; minLength=1; maxLength=2048
size_bytes	no	type="integer"; minimum=0
source_id	yes	\$ref="#/\$defs/Id"
source_locator	yes	type="string"; minLength=1; maxLength=2048
source_mode	yes	enum=["captured-payload", "metadata-only"]
source_provenance_digest	yes	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "anyOf": [
            {
              "required": [
                "artifact_id"
              ]
            },
            {
              "required": [
                "artifact_record_digest"
              ]
            },
            {
              "required": [
                "content_digest"
              ]
            },
            {
              "required": [
                "size_bytes"
              ]
            }
          ]
        }
      }
    }
  ],
  "required": [
    "missing_reason"
  ]
},
{
  "if": {
    "properties": {
      "source_mode": {
        "const": "captured-payload"
      }
    }
  },
  "required": [
    "source_mode"
  ]
},
{
  "then": {
    "not": {
      "required": [
        "missing_reason"
      ]
    },
    "required": [
      "artifact_id",
      "artifact_record_digest",
      "content_digest",
      "size_bytes"
    ]
  }
}
]
},
{
  "properties": {
    "artifact_id": {
      "$ref": "#/$defs/Id"
    },
    "artifact_record_digest": {

```

```

    "$ref": "#/$defs/Digest"
  },
  "content_digest": {
    "$ref": "#/$defs/Digest"
  },
  "license_expression": {
    "maxLength": 512,
    "minLength": 1,
    "type": "string"
  },
  "license_plan_digest": {
    "$ref": "#/$defs/Digest"
  },
  "license_provider_id": {
    "$ref": "#/$defs/Id"
  },
  "media_type": {
    "maxLength": 256,
    "minLength": 1,
    "type": "string"
  },
  "missing_reason": {
    "maxLength": 2048,
    "minLength": 1,
    "type": "string"
  },
  "size_bytes": {
    "minimum": 0,
    "type": "integer"
  },
  "source_id": {
    "$ref": "#/$defs/Id"
  },
  "source_locator": {
    "maxLength": 2048,
    "minLength": 1,
    "type": "string"
  },
  "source_mode": {
    "enum": [
      "captured-payload",
      "metadata-only"
    ]
  },
  "source_provenance_digest": {
    "$ref": "#/$defs/Digest"
  }
},
"required": [
  "source_id",
  "source_mode",
  "source_locator",
  "source_provenance_digest",
  "media_type",
  "license_provider_id",
  "license_expression",
  "license_plan_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ResearchSourceIntake | Validator: urn:promin:contracts:1#/\$defs/ResearchSourceIntake

86. RetrievalPage

Type summary: type="object"; additionalProperties=false

Required: record_type, query, activation_digest, depth, ranking, budget, head_digest, projection_digest, entities, relations, evidence, truncated, continuation, continuation_version, stream_cursor, next_stream_cursor, effective_top_k, selected_seed_count, refinement_required, refinement_hints, unselected_matches_traversable, selected_closure_complete, silent_truncation, projection_authoritative

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
budget	yes	\$ref="#/\$defs/Budget"
continuation	yes	oneOf[2]=type:null type:object

Property	Required	Structural rule
continuation_version	yes	const=2
depth	yes	type="integer"; minimum=1; maximum=12
effective_top_k	yes	type="integer"; minimum=1; maximum=12
entities	yes	type="array"; maxItems=32
evidence	yes	type="array"; maxItems=32
head_digest	yes	oneOf[2]=#/\$defs/Digest type:null
next_stream_cursor	yes	type="integer"; minimum=0
projection_authoritative	yes	const=false
projection_digest	yes	\$ref="#/\$defs/Digest"
query	yes	type="string"; minLength=1; maxLength=4096
ranking	yes	const="bm25-v1"
record_type	yes	const="RetrievalPage"
refinement_hints	yes	type="array"; maxItems=8; uniqueItems=true
refinement_required	yes	type="boolean"
relations	yes	type="array"; maxItems=48
selected_closure_complete	yes	type="boolean"
selected_seed_count	yes	type="integer"; minimum=0
silent_truncation	yes	const=false
stream_cursor	yes	type="integer"; minimum=0
truncated	yes	type="boolean"
unselected_matches_traversable	yes	const=false

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "budget": {
      "$ref": "#/$defs/Budget"
    },
    "continuation": {
      "oneOf": [
        {
          "type": "null"
        },
        {
          "additionalProperties": false,
          "properties": {
            "activation_digest": {
              "$ref": "#/$defs/Digest"
            },
            "budget_digest": {
              "$ref": "#/$defs/Digest"
            },
            "cursor": {
              "minimum": 0,
              "type": "integer"
            },
            "depth": {
              "maximum": 12,
              "minimum": 1,
              "type": "integer"
            },
            "expiry": {
              "$ref": "#/$defs/Timestamp"
            },
            "head_digest": {
              "oneOf": [
                {
                  "$ref": "#/$defs/Digest"
                },
                {
                  "type": "null"
                }
              ]
            }
          }
        }
      ]
    }
  }
}

```

```

    ],
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "projection_digest": {
      "$ref": "#/$defs/Digest"
    },
    "ranking": {
      "const": "bm25-v1"
    },
    "resume_binding_digest": {
      "$ref": "#/$defs/Digest"
    }
  },

```

```

    },
    "token": {
      "maxLength": 256,
      "minLength": 1,
      "pattern": "^promin-v2\\. [A-Za-z0-9_-]{22}\\.[0-9a-f]{64}\\.[A-Za-z0-9_-]{43}$",
      "type": "string"
    },
    "traversal": {
      "const": "typed-bfs-v2"
    },
    "version": {
      "const": 2
    }
  },
  "required": [
    "version",
    "traversal",
    "token",
    "cursor",
    "expiry",
    "activation_digest",
    "head_digest",
    "projection_digest",
    "implementation_closure_digest",
    "ranking",
    "depth",
    "budget_digest",
    "resume_binding_digest"
  ]
}

```

```

    ],
    "type": "object"
  }
]
},
"continuation_version": {
  "const": 2
},
"depth": {
  "maximum": 12,
  "minimum": 1,
  "type": "integer"
},
"effective_top_k": {
  "maximum": 12,
  "minimum": 1,
  "type": "integer"
},
"entities": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "data_class": {
        "enum": [
          "operational-record",
          "untrusted-source"
        ]
      }
    }
  }
},

```

```

"entity_type": {
  "enum": [
    "Task",
    "Candidate",
    "Artifact",
    "Run",
    "Finding",
    "Decision",

```

```

        "GateResult",
        "Grant",
        "Lease"
    ]
},
"id": {
    "$ref": "#/$defs/Id"
},
"payload": {
    "type": "object"
},
"source_digest": {
    "$ref": "#/$defs/Digest"
}
},
"required": [
    "id",
    "entity_type",
    "data_class",
    "payload",

```

```

        "source_digest"
    ],
    "type": "object"
},
"maxItems": 32,
"type": "array"
},
"evidence": {
    "items": {
        "$ref": "#/$defs/Artifact"
    },
    "maxItems": 32,
    "type": "array"
},
"head_digest": {
    "oneOf": [
        {
            "$ref": "#/$defs/Digest"
        },
        {
            "type": "null"
        }
    ]
}
},
"next_stream_cursor": {
    "minimum": 0,
    "type": "integer"
},

```

```

"projection_authoritative": {
    "const": false
},
"projection_digest": {
    "$ref": "#/$defs/Digest"
},
"query": {
    "maxLength": 4096,
    "minLength": 1,
    "type": "string"
},
"ranking": {
    "const": "bm25-v1"
},
"record_type": {
    "const": "RetrievalPage"
},
"refinement_hints": {
    "items": {
        "maxLength": 512,
        "minLength": 1,
        "type": "string"
    },
    "maxItems": 8,
    "type": "array",
    "uniqueItems": true
},
"refinement_required": {

```

```

    "type": "boolean"
  },
  "relations": {
    "items": {
      "$ref": "#/$defs/Relation"
    },
    "maxItems": 48,
    "type": "array"
  },
  "selected_closure_complete": {
    "type": "boolean"
  },
  "selected_seed_count": {
    "minimum": 0,
    "type": "integer"
  },
  "silent_truncation": {
    "const": false
  },
  "stream_cursor": {
    "minimum": 0,
    "type": "integer"
  },
  "truncated": {
    "type": "boolean"
  },
  "unselected_matches_traversable": {
    "const": false
  }
},
"required": [
  "record_type",
  "query",
  "activation_digest",
  "depth",
  "ranking",
  "budget",
  "head_digest",
  "projection_digest",
  "entities",
  "relations",
  "evidence",
  "truncated",
  "continuation",
  "continuation_version",
  "stream_cursor",
  "next_stream_cursor",
  "effective_top_k",
  "selected_seed_count",
  "refinement_required",
  "refinement_hints",
  "unselected_matches_traversable",
  "selected_closure_complete",
  "silent_truncation",
  "projection_authoritative"
],
"type": "object",
"x-promin-persistence": "derived-live-only"
}

```

Owner: [core/contracts.schema.json#/\\$defs/RetrievalPage](#) | Validator: [urn:promin:contracts:1#/\\$defs/RetrievalPage](#)

87. RootAuthority

Type summary: type="object"; additionalProperties=false

Required: subject_id, capability_ceiling, scope

Property	Required	Structural rule
capability_ceiling	yes	type="array"; minItems=1; maxItems=32; uniqueItems=true
scope	yes	type="array"; minItems=1; maxItems=32
subject_id	yes	\$ref="#/\$defs/Id"

```

{
  "additionalProperties": false,
  "properties": {
    "capability_ceiling": {

```

```

    "items": {
      "$ref": "#/$defs/Id"
    },
    "maxItems": 32,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "scope": {
    "items": {
      "$ref": "#/$defs/ScopeSelector"
    },
    "maxItems": 32,
    "minItems": 1,
    "type": "array"
  },
  "subject_id": {
    "$ref": "#/$defs/Id"
  }
},
"required": [
  "subject_id",
  "capability_ceiling",
  "scope"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/RootAuthority | Validator: urn:promin:contracts:1#/\$defs/RootAuthority

88. RootAuthorization

Type summary: type="object"; additionalProperties=false

Required: kind, subject_id, proofs

Property	Required	Structural rule
kind	yes	const="root"
proofs	yes	type="array"; minItems=1; maxItems=32; uniqueItems=true
subject_id	yes	\$ref="#/\$defs/Id"

```

{
  "additionalProperties": false,
  "properties": {
    "kind": {
      "const": "root"
    },
    "proofs": {
      "items": {
        "$ref": "#/$defs/RootCommandProof"
      },
      "maxItems": 32,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "kind",
    "subject_id",
    "proofs"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/RootAuthorization | Validator: urn:promin:contracts:1#/\$defs/RootAuthorization

89. RootCommandProof

Type summary: oneOf[2]=#/\$defs/LocalRootCommandProof | #/\$defs/SignatureRootCommandProof

Required: -

```

{
  "oneOf": [

```

```

    {
      "$ref": "#/$defs/LocalRootCommandProof"
    },
    {
      "$ref": "#/$defs/SignatureRootCommandProof"
    }
  ]
}

```

Owner: *core/contracts.schema.json#/\$defs/RootCommandProof* | Validator: *urn:promin:contracts:1#/\$defs/RootCommandProof*

90. Run

Type summary: type="object"; additionalProperties=false

Required: record_type, run_id, run_kind, task_id, candidate_digest, policy_digest, tool_digest, input_digests, status, started_at, finished_at, activation_digest, definition_digest, implementation_closure_digest, provider_binding_digest

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
candidate_digest	yes	\$ref="#/\$defs/Digest"
definition_digest	yes	\$ref="#/\$defs/Digest"
finished_at	yes	\$ref="#/\$defs/Timestamp"
implementation_closure_digest	yes	\$ref="#/\$defs/Digest"
input_digests	yes	type="array"; minItems=0; maxItems=128; uniqueItems=true
policy_digest	yes	\$ref="#/\$defs/Digest"
provider_binding_digest	yes	\$ref="#/\$defs/Digest"
record_type	yes	const="Run"
run_id	yes	\$ref="#/\$defs/Id"
run_kind	yes	enum=["execution", "validation"]
started_at	yes	\$ref="#/\$defs/Timestamp"
status	yes	enum=["pass", "fail", "blocked", "skipped", "error"]
task_id	yes	\$ref="#/\$defs/Id"
tool_digest	yes	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "definition_digest": {
      "$ref": "#/$defs/Digest"
    },
    "finished_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "input_digests": {
      "items": {
        "$ref": "#/$defs/Digest"
      },
      "maxItems": 128,
      "minItems": 0,
      "type": "array",
      "uniqueItems": true
    },
    "policy_digest": {
      "$ref": "#/$defs/Digest"
    },
    "provider_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "record_type": {
      "const": "Run"
    }
  }
}

```

```

    },
    "run_id": {
      "$ref": "#/$defs/Id"
    },
    },
    "run_kind": {
      "enum": [
        "execution",
        "validation"
      ]
    },
    },
    "started_at": {
      "$ref": "#/$defs/Timestamp"
    },
    },
    "status": {
      "enum": [
        "pass",
        "fail",
        "blocked",
        "skipped",
        "error"
      ]
    }
  ]
},
"task_id": {
  "$ref": "#/$defs/Id"
},
"tool_digest": {
  "$ref": "#/$defs/Digest"
}
},
"required": [
  "record_type",
  "run_id",
  "run_kind",
  "task_id",
  "candidate_digest",
  "policy_digest",
  "tool_digest",
  "input_digests",
  "status",
  "started_at",
  "finished_at",
  "activation_digest",
  "definition_digest",
  "implementation_closure_digest",
  "provider_binding_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Run | Validator: urn:promin:contracts:1#/\$defs/Run

91. SanitizedResearchDraft

Type summary: type="object"; additionalProperties=false

Required: record_type, migration_rule, license_plan_digest, sources, normative_candidates, excluded_claims, requested_flag_mutations, pass_credit, product_acceptance_pass, product_public_approval, public_release_approved

Property	Required	Structural rule
excluded_claims	yes	type="array"; minItems=0; maxItems=256; uniqueItems=true
license_plan_digest	yes	\$ref="#/\$defs/Digest"
migration_rule	yes	const="MIGRATE_FACTS_NOT_FILES"
normative_candidates	yes	type="array"; minItems=0; maxItems=256; uniqueItems=true
pass_credit	yes	const=false
product_acceptance_pass	yes	const=false
product_public_approval	yes	const="not_approved"
public_release_approved	yes	const=false
record_type	yes	const="SanitizedResearchDraft"
requested_flag_mutations	yes	type="array"; const=[]
sources	yes	type="array"; minItems=1; maxItems=128; uniqueItems=true
<pre> { "additionalProperties": false, </pre>		

```

"properties": {
  "excluded_claims": {
    "items": {
      "additionalProperties": false,
      "allOf": [
        {
          "else": {
            "not": {
              "required": [
                "blocked_reason"
              ]
            }
          },
          "if": {
            "anyOf": [
              {
                "properties": {
                  "claim_kind": {
                    "const": "blocked_claim"
                  }
                },
                "required": [
                  "claim_kind"
                ]
              },
              {
            }
          }
        }
      ]
    }
  },
  "then": {
    "required": [
      "blocked_reason"
    ]
  }
}

```

```

    "properties": {
      "verification_status": {
        "const": "blocked"
      }
    },
    "required": [
      "verification_status"
    ]
  }
},
"then": {
  "required": [
    "blocked_reason"
  ]
}
},
"properties": {
  "blocked_reason": {
    "maxLength": 2048,
    "minLength": 1,
    "type": "string"
  },
  "citation_refs": {
    "items": {
      "$ref": "#/$defs/Id"
    }
  },

```

```

    "maxItems": 32,
    "type": "array",
    "uniqueItems": true
  },
  "citation_required": {
    "type": "boolean"
  },
  "claim_class": {
    "enum": [
      "distribution",
      "fact",
      "marketing",
      "proof"
    ]
  },
  "claim_digest": {
    "$ref": "#/$defs/Digest"
  },
  "claim_id": {
    "$ref": "#/$defs/Id"
  },
  "claim_kind": {
    "enum": [
      "research_question",

```



```
"hypothesis",
"assumption",
"evidence_claim",
"blocked_claim"
```

```
]
},
"evidence_refs": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "artifact_id": {
        "$ref": "#/$defs/Id"
      },
      "artifact_record_digest": {
        "$ref": "#/$defs/Digest"
      }
    },
    "required": [
      "artifact_id",
      "artifact_record_digest"
    ],
    "type": "object"
  },
  "maxItems": 32,
  "type": "array",
  "uniqueItems": true
},
"normative_candidate": {
  "type": "boolean"
},
"normative_use_allowed": {
  "const": false
```

```
},
"reason": {
  "maxLength": 2048,
  "minLength": 1,
  "type": "string"
},
"text": {
  "maxLength": 8192,
  "minLength": 1,
  "type": "string"
},
"verification_status": {
  "enum": [
    "unverified",
    "source-bound",
    "verified",
    "blocked"
  ]
},
"required": [
  "claim_id",
  "claim_kind",
  "claim_class",
  "claim_digest",
  "text",
  "citation_required",
  "citation_refs",
```

```
"evidence_refs",
"verification_status",
"normative_candidate",
"normative_use_allowed",
"reason"
],
"type": "object"
},
"maxItems": 256,
"minItems": 0,
"type": "array",
"uniqueItems": true
},
"license_plan_digest": {
  "$ref": "#/$defs/Digest"
},
"migration_rule": {
```

```

    "const": "MIGRATE_FACTS_NOT_FILES"
  },
  "normative_candidates": {
    "items": {
      "additionalProperties": false,
      "properties": {
        "citation_refs": {
          "items": {
            "$ref": "#/$defs/Id"
          },
          "maxItems": 32,

```

```

          "minItems": 1,
          "type": "array",
          "uniqueItems": true
        },
        "citation_required": {
          "const": true
        },
        "claim_class": {
          "const": "fact"
        },
        "claim_digest": {
          "$ref": "#/$defs/Digest"
        },
        "claim_id": {
          "$ref": "#/$defs/Id"
        },
        "claim_kind": {
          "const": "evidence_claim"
        },
        "evidence_refs": {
          "items": {
            "additionalProperties": false,
            "properties": {
              "artifact_id": {
                "$ref": "#/$defs/Id"
              },
              "artifact_record_digest": {
                "$ref": "#/$defs/Digest"

```

```

            }
          },
          "required": [
            "artifact_id",
            "artifact_record_digest"
          ],
          "type": "object"
        },
        "maxItems": 32,
        "minItems": 1,
        "type": "array",
        "uniqueItems": true
      },
      "normative_candidate": {
        "const": true
      },
      "normative_use_allowed": {
        "const": true
      },
      "source_bindings": {
        "items": {
          "additionalProperties": false,
          "properties": {
            "artifact_id": {
              "$ref": "#/$defs/Id"
            },
            "artifact_record_digest": {
              "$ref": "#/$defs/Digest"

```

```

            },
            "content_digest": {
              "$ref": "#/$defs/Digest"
            },
            "license_expression": {
              "maxLength": 512,
              "minLength": 1,
              "type": "string"
            },
            "license_plan_digest": {

```

```

    "$ref": "#/$defs/Digest"
  },
  "license_provider_id": {
    "$ref": "#/$defs/Id"
  },
  "media_type": {
    "maxLength": 256,
    "minLength": 1,
    "type": "string"
  },
  "size_bytes": {
    "minimum": 0,
    "type": "integer"
  },
  "source_id": {
    "$ref": "#/$defs/Id"
  },
  "source_locator": {

```

```

    "maxLength": 2048,
    "minLength": 1,
    "type": "string"
  },
  "source_mode": {
    "enum": [
      "captured-payload",
      "metadata-only"
    ]
  },
  "source_provenance_digest": {
    "$ref": "#/$defs/Digest"
  }
},
"required": [
  "source_id",
  "source_mode",
  "source_locator",
  "source_provenance_digest",
  "artifact_id",
  "artifact_record_digest",
  "content_digest",
  "size_bytes",
  "media_type",
  "license_provider_id",
  "license_expression",
  "license_plan_digest"
],

```

```

    "type": "object"
  },
  "maxItems": 32,
  "minItems": 1,
  "type": "array",
  "uniqueItems": true
},
"text": {
  "maxLength": 8192,
  "minLength": 1,
  "type": "string"
},
"verification_status": {
  "enum": [
    "source-bound",
    "verified"
  ]
}
},
"required": [
  "claim_id",
  "claim_kind",
  "claim_class",
  "claim_digest",
  "text",
  "citation_required",
  "citation_refs",
  "evidence_refs",

```

```

  "verification_status",
  "normative_candidate",
  "normative_use_allowed",

```

```

        "source_bindings"
      ],
      "type": "object"
    },
    "maxItems": 256,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
  },
  "pass_credit": {
    "const": false
  },
  "product_acceptance_pass": {
    "const": false
  },
  "product_public_approval": {
    "const": "not_approved"
  },
  "public_release_approved": {
    "const": false
  },
  "record_type": {
    "const": "SanitizedResearchDraft"
  },
  "requested_flag_mutations": {

```

```

    "const": [],
    "type": "array"
  },
  "sources": {
    "items": {
      "additionalProperties": false,
      "allOf": [
        {
          "else": {
            "not": {
              "anyOf": [
                {
                  "required": [
                    "artifact_id"
                  ]
                },
                {
                  "required": [
                    "artifact_record_digest"
                  ]
                },
                {
                  "required": [
                    "content_digest"
                  ]
                }
              ],
            },
            {
              "required": [

```

```

                "size_bytes"
              ]
            }
          ],
          "required": [
            "missing_reason"
          ]
        },
        "if": {
          "properties": {
            "source_mode": {
              "const": "captured-payload"
            }
          },
          "required": [
            "source_mode"
          ]
        },
        "then": {
          "not": {
            "required": [
              "missing_reason"
            ]
          }
        }
      ],

```

```

        "required": [
            "artifact_id",
            "artifact_record_digest",

            "content_digest",
            "size_bytes"
        ]
    }
},
"properties": {
    "artifact_id": {
        "$ref": "#/$defs/Id"
    },
    "artifact_record_digest": {
        "$ref": "#/$defs/Digest"
    },
    "content_digest": {
        "$ref": "#/$defs/Digest"
    },
    "license_expression": {
        "maxLength": 512,
        "minLength": 1,
        "type": "string"
    },
    "license_plan_digest": {
        "$ref": "#/$defs/Digest"
    },
    "license_provider_id": {
        "$ref": "#/$defs/Id"
    },
    "media_type": {
        "maxLength": 256,
        "minLength": 1,
        "type": "string"
    },
    "missing_reason": {
        "maxLength": 2048,
        "minLength": 1,
        "type": "string"
    },
    "size_bytes": {
        "minimum": 0,
        "type": "integer"
    },
    "source_id": {
        "$ref": "#/$defs/Id"
    },
    "source_locator": {
        "maxLength": 2048,
        "minLength": 1,
        "type": "string"
    },
    "source_mode": {
        "enum": [
            "captured-payload",
            "metadata-only"
        ]
    },
    "source_provenance_digest": {
        "$ref": "#/$defs/Digest"
    }
},
"required": [
    "source_id",
    "source_mode",
    "source_locator",
    "source_provenance_digest",
    "media_type",
    "license_provider_id",
    "license_expression",
    "license_plan_digest"
],
"type": "object"
},
"maxItems": 128,
"minItems": 1,
"type": "array",

```

```

    "uniqueItems": true
  },
  "required": [
    "record_type",
    "migration_rule",
    "license_plan_digest",
    "sources",
    "normative_candidates",
    "excluded_claims",
    "requested_flag_mutations",
    "pass_credit",
    "product_acceptance_pass",
    "product_public_approval",
    "public_release_approved"
  ],
  "type": "object",
  "x-promin-persistence": "derived-live-only"
}

```

Owner: `core/contracts.schema.json#/$defs/SanitizedResearchDraft` | Validator: `urn:promin:contracts:1#/$defs/SanitizedResearchDraft`

92. SaturationAudit

Type summary: `type="object"; additionalProperties=false`

Required: `record_type`, `status`, `candidate_binding_digest`, `zero_new_iterations`, `new_findings`, `pass_credit`, `artifact_binding`, `families`, `collection`, `iterations`, `consecutive_full_zero_new`, `requirements`, `predeclared_zero_new`, `acceptance_pass`, `product_acceptance_pass`, `producer`, `invocation`, `raw_artifact_manifest_digest`, `producer_attestation`, `result_digest`

Property	Required	Structural rule
<code>acceptance_pass</code>	yes	<code>const=false</code>
<code>artifact_binding</code>	yes	<code>type="object"</code>
<code>candidate_binding_digest</code>	yes	<code>\$ref="#/\$defs/Digest"</code>
<code>collection</code>	yes	<code>type="object"; additionalProperties=false</code>
<code>consecutive_full_zero_new</code>	yes	<code>const=3</code>
<code>families</code>	yes	<code>type="array"; minItems=1; uniqueItems=true</code>
<code>invocation</code>	yes	<code>\$ref="#/\$defs/ReleaseEvidenceInvocation"</code>
<code>iterations</code>	yes	<code>type="array"; minItems=3; maxItems=18</code>
<code>new_findings</code>	yes	<code>const=0</code>
<code>pass_credit</code>	yes	<code>const=false</code>
<code>predeclared_zero_new</code>	yes	<code>const=false</code>
<code>producer</code>	yes	<code>\$ref="#/\$defs/ReleaseEvidenceProducer"</code>
<code>producer_attestation</code>	yes	<code>\$ref="#/\$defs/EvidenceProducerAttestation"</code>
<code>product_acceptance_pass</code>	yes	<code>const=false</code>
<code>raw_artifact_manifest_digest</code>	yes	<code>\$ref="#/\$defs/Digest"</code>
<code>record_type</code>	yes	<code>const="SaturationAudit"</code>
<code>requirements</code>	yes	<code>type="object"; additionalProperties=false</code>
<code>result_digest</code>	yes	<code>\$ref="#/\$defs/Digest"</code>
<code>status</code>	yes	<code>const="pass"</code>
<code>zero_new_iterations</code>	yes	<code>const=3</code>

```

{
  "additionalProperties": false,
  "properties": {
    "acceptance_pass": {
      "const": false
    },
    "artifact_binding": {
      "type": "object"
    },
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "collection": {

```

```

    "additionalProperties": false,
    "properties": {
      "catalogue_digest": {},
      "families_discovered": {},
      "production_probes_discovered": {}
    },
    "required": [
      "catalogue_digest",
      "families_discovered",
      "production_probes_discovered"
    ],
    "type": "object"
  },
  "consecutive_full_zero_new": {
    "const": 3
  }
}

```

```

  },
  "families": {
    "items": {
      "type": "string"
    },
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "invocation": {
    "$ref": "#/$defs/ReleaseEvidenceInvocation"
  },
  "iterations": {
    "items": {
      "additionalProperties": false,
      "properties": {
        "finding_fingerprints": {},
        "focused_and_integration_tests": {
          "additionalProperties": false,
          "properties": {
            "cache_provider_disabled": {},
            "elapsed_ms": {},
            "exit_code": {},
            "log_bytes": {},
            "log_path": {},
            "log_sha256": {},
            "marker_expression": {},
            "output_digest": {}
          }
        }
      }
    }
  }
}

```

```

    "pytest_only_sentinel_executed": {},
    "pytest_only_sentinel_token_digest": {},
    "runner": {},
    "sentinel_bytes": {},
    "sentinel_path": {},
    "sentinel_sha256": {}
  },
  "required": [
    "exit_code",
    "elapsed_ms",
    "output_digest",
    "runner",
    "marker_expression",
    "cache_provider_disabled",
    "pytest_only_sentinel_executed",
    "pytest_only_sentinel_token_digest",
    "log_path",
    "log_sha256",
    "log_bytes",
    "sentinel_path",
    "sentinel_sha256",
    "sentinel_bytes"
  ],
  "type": "object"
},
"full_system_corpus_rerun": {},
"iteration": {},
"mutation_families_run": {},

```

```

"mutation_seed": {},
"new_finding_classes": {},
"physical_runtime_saturation": {
  "additionalProperties": false,
  "properties": {
    "artifact_binding_digest": {},
  }
}

```

```

"current_release_regression": {},
"elapsed_ms": {},
"execution_route": {},
"exit_code": {},
"files": {},
"forced_continuation_chains": {},
"forced_union_matches": {},
"fresh_control_state": {
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "core_bundle_digest": {
      "$ref": "#/$defs/Digest"
    },
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "init_record_count": {
      "const": 5
    }
  },

```

```

    "physical_corpus_recipe": {
      "additionalProperties": false,
      "properties": {
        "bytes": {
          "minimum": 1,
          "type": "integer"
        },
        "file_count": {
          "const": 100000
        },
        "sha256": {
          "$ref": "#/$defs/Digest"
        }
      },
      "required": [
        "sha256",
        "bytes",
        "file_count"
      ],
      "type": "object"
    },
    "preset_digest": {
      "$ref": "#/$defs/Digest"
    },
    "product_tree_reused": {
      "type": "boolean"
    },
    "product_tree_scans": {

```

```

      "const": 0
    },
    "semantic_state_reused": {
      "const": false
    },
    "status": {
      "const": "created"
    }
  },
  "required": [
    "status",
    "product_tree_scans",
    "init_record_count",
    "activation_digest",
    "implementation_closure_digest",
    "core_bundle_digest",
    "preset_digest",
    "semantic_state_reused",
    "product_tree_reused",
    "physical_corpus_recipe"
  ],
  "type": "object"
},
"immutable_snapshot_creditable": {},
"invocation_id": {},
"log_bytes": {},
"log_path": {},
"log_sha256": {},

```



```

    "metrics": {},
    "output_digest": {},
    "performance_profile": {},
    "platform_binding_digest": {},
    "pytest_marker": {},
    "pytest_selection_command": {},
    "pytest_selection_command_executed": {},
    "queries": {},
    "record_bytes": {
        "minimum": 1,
        "type": "integer"
    },
    "record_path": {},
    "record_sha256": {},
    "result_digest": {}
},
"required": [
    "exit_code",
    "elapsed_ms",
    "output_digest",
    "log_path",
    "log_sha256",
    "log_bytes",
    "files",
    "queries",
    "performance_profile",
    "pytest_marker",
    "pytest_selection_command",
    "pytest_selection_command_executed",
    "execution_route",
    "fresh_control_state",
    "metrics",
    "artifact_binding_digest",
    "immutable_snapshot_creditable",
    "platform_binding_digest",
    "record_path",
    "record_sha256",
    "record_bytes",
    "result_digest",
    "invocation_id",
    "forced_continuation_chains",
    "forced_union_matches",
    "current_release_regression"
],
"type": "object"
},
"silent_truncations": {},
"status": {},
"streak_reset_reason": {},
"unexpected_accepts": {},
"unexpected_crashes": {},
"unexpected_rejects": {},
"zero_new_streak_after": {},
"zero_new_streak_before": {}
},
"required": [
    "finding_fingerprints",
    "focused_and_integration_tests",
    "full_system_corpus_rerun",
    "iteration",
    "mutation_families_run",
    "mutation_seed",
    "new_finding_classes",
    "physical_runtime_saturation",
    "silent_truncations",
    "status",
    "streak_reset_reason",
    "unexpected_accepts",
    "unexpected_crashes",
    "unexpected_rejects",
    "zero_new_streak_after",
    "zero_new_streak_before"
],
"type": "object"
},
"maxItems": 18,
"minItems": 3,
"type": "array"

```

```

},
"new_findings": {
  "const": 0
},
"pass_credit": {
  "const": false
},
"predeclared_zero_new": {
  "const": false
},
"producer": {
  "$ref": "#/$defs/ReleaseEvidenceProducer"
},
"producer_attestation": {
  "$ref": "#/$defs/EvidenceProducerAttestation"
},
"product_acceptance_pass": {
  "const": false
},
"raw_artifact_manifest_digest": {
  "$ref": "#/$defs/Digest"
},
"record_type": {
  "const": "SaturationAudit"
},
"requirements": {
  "additionalProperties": false,
  "properties": {
    "actual_runtime_queries": {},
    "configured_iteration_limit": {},
    "core_valid_relations": {},
    "depths": {},
    "focused_pytest_marker": {},
    "fresh_control_state_per_iteration": {
      "const": true
    },
    "init_product_tree_scans": {
      "const": 0
    },
    "maximum_iterations": {},
    "minimum_consecutive_full_zero_new": {},
    "missing_scale_workspace_fails": {},
    "mixed_query_classes": {},
    "performance_profile": {},
    "physical_files": {},
    "physical_pytest_command": {},
    "physical_pytest_marker": {},
    "profile_bound_fail_closed": {},
    "raw_file_proxy_ratio": {},
    "semantic_corpus_relations": {},
    "semantic_corpus_tasks": {},
    "semantic_state_reused": {
      "const": false
    },
    "synthetic_task_ratio": {}
  },
  "required": [
    "minimum_consecutive_full_zero_new",
    "maximum_iterations",
    "configured_iteration_limit",
    "physical_files",
    "core_valid_relations",
    "actual_runtime_queries",
    "depths",
    "raw_file_proxy_ratio",
    "synthetic_task_ratio",
    "semantic_corpus_tasks",
    "semantic_corpus_relations",
    "mixed_query_classes",
    "performance_profile",
    "profile_bound_fail_closed",
    "focused_pytest_marker",
    "physical_pytest_marker",
    "physical_pytest_command",
    "missing_scale_workspace_fails",
    "fresh_control_state_per_iteration",
    "semantic_state_reused",

```

```

        "init_product_tree_scans"
      ],
      "type": "object"
    },
    "result_digest": {
      "$ref": "#/$defs/Digest"
    },
    "status": {
      "const": "pass"
    },
    "zero_new_iterations": {
      "const": 3
    }
  }
},
"required": [
  "record_type",
  "status",
  "candidate_binding_digest",
  "zero_new_iterations",
  "new_findings",
  "pass_credit",
  "artifact_binding",
  "families",
  "collection",
  "iterations",
  "consecutive_full_zero_new",
  "requirements",
  "predeclared_zero_new",
  "acceptance_pass",
  "product_acceptance_pass",
  "producer",
  "invocation",
  "raw_artifact_manifest_digest",
  "producer_attestation",
  "result_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/SaturationAudit | Validator: urn:promin:contracts:1#/\$defs/SaturationAudit

93. SaturationEvidence

Type summary: type="object"; additionalProperties=false

Required: record_type, status, candidate_binding_digest, artifact_binding, artifact_binding_unchanged, workspace_initialization, runtime_binding, physical_files, core_valid_relations, runtime_queries, silent_truncations, selected_closure_union_completeness, memory_amplification_at_most_32, core_valid_relations_exact_198999, broad_query_refinement_required, high_cardinality_terms_verified, content_search_verified, miss_behavior_verified, hostile_proxy_content_verified, exact_artifact_search_verified, mixed_query_classes_complete, continuation_token_bytes_at_most_256, continuation_state_bytes_at_most_16384, continuation_token_overhead_at_most_10_percent, pass_credit, physical, inventory, projection, query_authorization, search, resources, performance, contract_predicates, raw_artifact_manifest, current_release_regression, claim_scope, acceptance_pass, product_acceptance_pass, public_release_approved, producer, invocation, raw_artifact_manifest_digest, producer_attestation, result_digest

Property	Required	Structural rule
acceptance_pass	yes	const=false
artifact_binding	yes	type="object"
artifact_binding_unchanged	yes	const=true
broad_query_refinement_required	yes	type="boolean"
candidate_binding_digest	yes	\$ref="#/\$defs/Digest"
claim_scope	yes	type="string"; minLength=1
content_search_verified	yes	type="boolean"
continuation_state_bytes_at_most_16384	yes	type="boolean"
continuation_token_bytes_at_most_256	yes	type="boolean"
continuation_token_overhead_at_most_10_percent	yes	type="boolean"
contract_predicates	yes	type="object"; additionalProperties=false

Property	Required	Structural rule
core_valid_relations	yes	const=198999
core_valid_relations_exact_19899	yes	const=true
current_release_regression	yes	type="object"; additionalProperties=false
exact_artifact_search_verified	yes	type="boolean"
high_cardinality_terms_verified	yes	type="boolean"
hostile_proxy_content_verified	yes	type="boolean"
inventory	yes	type="object"; additionalProperties=false
invocation	yes	\$ref="#/\$defs/ReleaseEvidenceInvocation"
memory_amplification_at_most_32	yes	type="boolean"
miss_behavior_verified	yes	type="boolean"
mixed_query_classes_complete	yes	type="boolean"
pass_credit	yes	const=false
performance	yes	type="object"; additionalProperties=false
physical	yes	type="object"; additionalProperties=false
physical_files	yes	const=100000
producer	yes	\$ref="#/\$defs/ReleaseEvidenceProducer"
producer_attestation	yes	\$ref="#/\$defs/EvidenceProducerAttestation"
product_acceptance_pass	yes	const=false
projection	yes	type="object"; additionalProperties=false
public_release_approved	yes	const=false
query_authorization	yes	type="object"; additionalProperties=false
raw_artifact_manifest	yes	type="object"; additionalProperties=false
raw_artifact_manifest_digest	yes	\$ref="#/\$defs/Digest"
record_type	yes	const="SaturationEvidence"
resources	yes	type="object"; additionalProperties=false
result_digest	yes	\$ref="#/\$defs/Digest"
runtime_binding	yes	type="object"; additionalProperties=false
runtime_queries	yes	const=600
search	yes	type="object"; additionalProperties=false
selected_closure_union_completeness	yes	type="number"; minimum=0.0; maximum=1.0
silent_truncations	yes	const=0
status	yes	enum=["pass", "fail"]
workspace_initialization	yes	type="object"; additionalProperties=false
<pre> { "additionalProperties": false, "properties": { "acceptance_pass": { "const": false }, "artifact_binding": { "type": "object" }, "artifact_binding_unchanged": { "const": true }, "broad_query_refinement_required": { "type": "boolean" }, "candidate_binding_digest": { "\$ref": "#/\$defs/Digest" }, "claim_scope": { "minLength": 1, "type": "string" } } } </pre>		

```

},
"content_search_verified": {
  "type": "boolean"
},
"continuation_state_bytes_at_most_16384": {
  "type": "boolean"
},
"continuation_token_bytes_at_most_256": {
  "type": "boolean"
},
"continuation_token_overhead_at_most_10_percent": {
  "type": "boolean"
},
"contract_predicates": {
  "additionalProperties": false,
  "properties": {
    "broad_query_refinement_required": {},
    "content_search_verified": {},
    "continuation_state_bytes_at_most_16384": {},
    "continuation_token_bytes_at_most_256": {},
    "continuation_token_overhead_at_most_10_percent": {},
    "continuation_union_complete": {},
    "core_valid_relations_exact": {
      "const": true
    },
    "exact_artifact_binding_unchanged": {},
    "exact_artifact_search_verified": {},
    "high_cardinality_terms_verified": {},
    "hostile_proxy_content_verified": {},
    "inventory_incremental_memory_amplification_at_most_32": {},
    "inventory_passes_exact": {},
    "miss_behavior_verified": {},
    "mixed_query_classes_complete": {},
    "physical_relation_artifact_coverage_complete": {},
    "raw_file_proxy_ratio_exact": {},
    "rebuild_digest_equal": {},
    "rebuild_product_passes_zero": {},
    "runtime_depths_1_through_12": {},
    "runtime_queries_exact": {
      "const": true
    },
    "runtime_query_budget_bounded": {},
    "selected_closure_union_complete": {},
    "semantic_commit_count_exact": {},
    "silent_truncations_zero": {},
    "synthetic_task_ratio_zero": {}
  },
  "required": [
    "broad_query_refinement_required",
    "content_search_verified",
    "continuation_state_bytes_at_most_16384",
    "continuation_token_bytes_at_most_256",
    "continuation_token_overhead_at_most_10_percent",
    "continuation_union_complete",
    "core_valid_relations_exact",
    "exact_artifact_binding_unchanged",
    "exact_artifact_search_verified",
    "high_cardinality_terms_verified",
    "hostile_proxy_content_verified",
    "inventory_incremental_memory_amplification_at_most_32",
    "inventory_passes_exact",
    "miss_behavior_verified",
    "mixed_query_classes_complete",
    "physical_relation_artifact_coverage_complete",
    "raw_file_proxy_ratio_exact",
    "rebuild_digest_equal",
    "rebuild_product_passes_zero",
    "runtime_depths_1_through_12",
    "runtime_queries_exact",
    "runtime_query_budget_bounded",
    "selected_closure_union_complete",
    "semantic_commit_count_exact",
    "silent_truncations_zero",
    "synthetic_task_ratio_zero"
  ],
  "type": "object"
},

```

```

"core_valid_relations": {
  "const": 198999
},
"core_valid_relations_exact_198999": {
  "const": true
},
"current_release_regression": {
  "additionalProperties": false,
  "properties": {
    "after": {},
    "before": {},
    "eligibility_remained_false": {},
    "historical_decision_preserved": {}
  },

```

```

    "required": [
      "before",
      "after",
      "eligibility_remained_false",
      "historical_decision_preserved"
    ],
    "type": "object"
  },
  "exact_artifact_search_verified": {
    "type": "boolean"
  },
  "high_cardinality_terms_verified": {
    "type": "boolean"
  },
  "hostile_proxy_content_verified": {
    "type": "boolean"
  },
  "inventory": {
    "additionalProperties": false,
    "properties": {
      "candidate_digest": {},
      "elapsed_ms": {},
      "entries": {},
      "passes": {},
      "snapshot": {},
      "stream_bytes": {}
    },
    "required": [

```

```

      "candidate_digest",
      "elapsed_ms",
      "entries",
      "passes",
      "snapshot",
      "stream_bytes"
    ],
    "type": "object"
  },
  "invocation": {
    "$ref": "#/$defs/ReleaseEvidenceInvocation"
  },
  "memory_amplification_at_most_32": {
    "type": "boolean"
  },
  "miss_behavior_verified": {
    "type": "boolean"
  },
  "mixed_query_classes_complete": {
    "type": "boolean"
  },
  "pass_credit": {
    "const": false
  },
  "performance": {
    "additionalProperties": false,
    "properties": {
      "all_within_profile": {},

```

```

    "compatible_platforms": {
      "const": [
        "linux",
        "windows"
      ]
    },
    "observed": {},

```

```

"platform_binding": {
  "additionalProperties": false,
  "properties": {
    "binding_digest": {
      "pattern": "^sha256:[0-9a-f]{64}$",
      "type": "string"
    },
    "machine": {
      "minLength": 1,
      "type": "string"
    },
    "profile_key": {
      "minLength": 1,
      "type": "string"
    },
    "python_executable_sha256": {
      "pattern": "^sha256:[0-9a-f]{64}$",
      "type": "string"
    },
    "python_implementation": {
      "const": "CPython"
    }
  },
  "python_version": {
    "pattern": "^[0-9]+\\.?[0-9]+\\.?[0-9]+$",
    "type": "string"
  },
  "release": {
    "minLength": 1,
    "type": "string"
  },
  "sqlite_version": {
    "pattern": "^[0-9]+\\.?[0-9]+\\.?[0-9]+$",
    "type": "string"
  },
  "system": {
    "enum": [
      "darwin",
      "linux",
      "windows"
    ]
  }
},
"required": [
  "binding_digest",
  "system",
  "release",
  "machine",
  "python_implementation",
  "python_version",
  "python_executable_sha256",
  "sqlite_version",
  "profile_key"
],
"type": "object"
},
"predicates": {},
"profile_id": {
  "const": "portable-local-v1"
},
"requires_same_runner_no_degradation": {
  "const": true
},
"thresholds": {
  "additionalProperties": false,
  "properties": {
    "commit_bytes_per_changed_record_max": {
      "const": 32768
    },
    "commit_p95_ms_max": {
      "const": 500
    },
    "commit_p99_ms_max": {
      "const": 1000
    },
    "database_bytes_max": {
      "const": 402653184
    }
  }
},

```

```

        "p50_ms_max": {
          "const": 100
        },
        "p95_ms_max": {
          "const": 150
        },
        "p99_ms_max": {
          "const": 500
        },
        "peak_rss_bytes_max": {
          "const": 805306368
        },
        "projection_amplification_max": {
          "const": 20
        },
        "runtime_checkpoint_count_max": {
          "const": 24
        },
        "semantic_inflation_max": {
          "const": 3.25
        },
        "semantic_ingestion_seconds_max": {
          "const": 600
        }
      },
      "required": [
        "p50_ms_max",
        "p95_ms_max",
        "p99_ms_max",
        "peak_rss_bytes_max",
        "database_bytes_max",
        "projection_amplification_max",
        "semantic_inflation_max",
        "commit_p95_ms_max",
        "commit_p99_ms_max",
        "commit_bytes_per_changed_record_max",
        "runtime_checkpoint_count_max",
        "semantic_ingestion_seconds_max"
      ],
      "type": "object"
    },
    "required": [
      "all_within_profile",
      "compatible_platforms",
      "observed",
      "platform_binding",
      "predicates",
      "profile_id",
      "requires_same_runner_no_degradation",
      "thresholds"
    ],
    "type": "object"
  },
  "physical": {
    "additionalProperties": false,
    "properties": {
      "explicit_semantic_corpus": {
        "additionalProperties": false,
        "properties": {
          "conflicting_exact_id_text": {},
          "continuation_query_ids": {},
          "depths": {},
          "generation": {},
          "harness_generated": {
            "const": true
          }
        },
        "high_fanout": {},
        "physical_relation_fixture": {
          "additionalProperties": false,
          "properties": {
            "artifact_target_count": {},
            "artifact_target_coverage": {},
            "relation_count": {},
            "relation_kind": {},
            "relations_per_atomic_batch_max": {},
            "target_type": {},
            "task_count": {}
          }
        }
      }
    }
  }
}

```



```

    },
    "required": [
      "task_count",
      "relation_count",
      "relation_kind",
      "target_type",
      "artifact_target_count",
      "artifact_target_coverage",
      "relations_per_atomic_batch_max"
    ],
    "type": "object"
  },
  "physical_relation_fixture_reused": {
    "const": false
  },
  "product_acceptance_credit": {
    "const": false
  },
  "query_ids": {},
  "record_type": {
    "const": "SaturationSemanticCorpus"
  },
  "relation_count": {},
  "reused": {
    "const": false
  },
  "search_fixture": {
    "additionalProperties": false,
    "properties": {
      "depths": {},
      "high_fanout": {},
      "relation_count": {},
      "task_count": {}
    }
  },

```

```

    "required": [
      "task_count",
      "relation_count",
      "depths",
      "high_fanout"
    ],
    "type": "object"
  },
  "search_fixture_reused": {
    "const": false
  },
  "task_count": {}
},
"required": [
  "record_type",
  "generation",
  "harness_generated",
  "product_acceptance_credit",
  "task_count",
  "relation_count",
  "depths",
  "high_fanout",
  "conflicting_exact_id_text",
  "query_ids",
  "continuation_query_ids",
  "reused",
  "search_fixture",
  "physical_relation_fixture",

```

```

  "search_fixture_reused",
  "physical_relation_fixture_reused"
],
"type": "object"
},
"files": {},
"generation_elapsed_ms": {},
"inventory_relations": {},
"raw_file_proxies": {},
"raw_file_proxy_ratio": {},
"raw_files": {},
"relations": {},
"reused_product": {},
"semantic_proxies": {},
"synthetic_task_count": {},

```

```

    "synthetic_task_ratio": {},
    "synthetic_tasks": {},
    "vcs_commit": {},
    "vcs_provider_version": {},
    "vcs_tree_digest": {},
    "vcs_tree_files": {}
  },

```

```

    "required": [
      "explicit_semantic_corpus",
      "files",
      "generation_elapsed_ms",
      "inventory_relations",
      "raw_file_proxies",

```

```

      "raw_file_proxy_ratio",
      "raw_files",
      "relations",
      "reused_product",
      "semantic_proxies",
      "synthetic_task_count",
      "synthetic_task_ratio",
      "synthetic_tasks",
      "vcs_commit",
      "vcs_provider_version",
      "vcs_tree_digest",
      "vcs_tree_files"
    ],

```

```

    "type": "object"
  },

```

```

  "physical_files": {
    "const": 100000
  },

```

```

  "producer": {
    "$ref": "#/$defs/ReleaseEvidenceProducer"
  },

```

```

  "producer_attestation": {
    "$ref": "#/$defs/EvidenceProducerAttestation"
  },

```

```

  "product_acceptance_pass": {
    "const": false
  },

```

```

  "projection": {

```

```

    "additionalProperties": false,
    "properties": {
      "database_bytes": {},
      "elapsed_ms": {},
      "entity_count": {},
      "equal_semantic_digest": {},
      "implementation_closure_digest": {
        "$ref": "#/$defs/Digest"
      },
    },
    "initial_inventory_passes": {},
    "initial_product_passes": {},
    "inventory_integrity": {},
    "inventory_projection_amplification": {},
    "projection_amplification": {},
    "rebuild_inventory_passes": {},
    "rebuild_product_passes": {},
    "relation_count": {},
    "semantic_digest": {},
    "semantic_inflation": {}
  },

```

```

    "required": [
      "database_bytes",
      "elapsed_ms",
      "entity_count",
      "equal_semantic_digest",
      "implementation_closure_digest",
      "initial_inventory_passes",
      "initial_product_passes",

```

```

      "inventory_integrity",
      "inventory_projection_amplification",
      "projection_amplification",
      "rebuild_inventory_passes",
      "rebuild_product_passes",
      "relation_count",
      "semantic_digest",
      "semantic_inflation"
    ]
  }
}

```

```

    ],
    "type": "object"
  },
  "public_release_approved": {
    "const": false
  },
  "query_authorization": {
    "additionalProperties": false,
    "properties": {
      "activation_digest": {
        "$ref": "#/$defs/Digest"
      },
      "capability_id": {
        "const": "projection.read"
      },
      "claim_digest": {
        "$ref": "#/$defs/Digest"
      },
      "expires_at": {
        "$ref": "#/$defs/Timestamp"
      }
    }
  },

```

```

  },
  "grant_id": {
    "$ref": "#/$defs/Id"
  },
  "scope": {
    "items": {
      "type": "object"
    },
    "maxItems": 128,
    "minItems": 1,
    "type": "array"
  },
  "subject_id": {
    "$ref": "#/$defs/Id"
  }
},
"required": [
  "activation_digest",
  "capability_id",
  "claim_digest",
  "expires_at",
  "grant_id",
  "scope",
  "subject_id"
],
"type": "object"
},
"raw_artifact_manifest": {

```

```

  "additionalProperties": false,
  "properties": {
    "artifact_count": {
      "const": 6
    },
    "artifacts": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "bytes": {
            "maximum": 268435456,
            "minimum": 1,
            "type": "integer"
          },
          "media_type": {
            "enum": [
              "application/json",
              "application/x-ndjson"
            ]
          },
          "path": {
            "pattern": "^raw/",
            "type": "string"
          },
          "records": {
            "maximum": 100000,
            "minimum": 1,
            "type": "integer"
          }
        }
      }
    }
  }
}

```

```

    },
    "role": {
      "enum": [
        "inventory-stream",
        "query-results",
        "process-samples",
        "continuation-state-manifest",
        "phase-log",
        "operation-metrics"
      ]
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "role",
    "path",
    "media_type",
    "sha256",
    "bytes",
    "records"
  ],
  "type": "object"
},
"maxItems": 6,
"minItems": 6,
"type": "array"

```

```

},
"evidence_class": {
  "const": "harness_generated"
},
"inventory_identity_digest": {
  "$ref": "#/$defs/Digest"
},
"inventory_stream_digest": {
  "$ref": "#/$defs/Digest"
},
"manifest_digest": {
  "$ref": "#/$defs/Digest"
},
"path_scope": {
  "const": "saturation-result-directory"
},
"product_acceptance_credit": {
  "const": false
},
"record_type": {
  "const": "SaturationRawArtifactManifest"
}
},
"required": [
  "record_type",
  "path_scope",
  "evidence_class",
  "product_acceptance_credit",

```

```

  "artifacts",
  "artifact_count",
  "inventory_stream_digest",
  "inventory_identity_digest",
  "manifest_digest"
],
"type": "object"
},
"raw_artifact_manifest_digest": {
  "$ref": "#/$defs/Digest"
},
"record_type": {
  "const": "SaturationEvidence"
},
"resources": {
  "additionalProperties": false,
  "properties": {
    "inventory_absolute_rss_amplification": {},
    "inventory_incremental_memory_amplification": {},
    "inventory_pipeline_incremental_peak_bytes": {},
    "inventory_pipeline_peak_rss_bytes": {},
    "inventory_stage_rss": {},

```

```

    "memory_amplification_metric": {},
    "peak_rss_bytes": {},
    "projection_stage_rss": {}
  },
  "required": [
    "inventory_absolute_rss_amplification",
    "inventory_incremental_memory_amplification",
    "inventory_pipeline_incremental_peak_bytes",
    "inventory_pipeline_peak_rss_bytes",
    "inventory_stage_rss",
    "memory_amplification_metric",
    "peak_rss_bytes",
    "projection_stage_rss"
  ],
  "type": "object"
},
"result_digest": {
  "$ref": "#/$defs/Digest"
},
"runtime_binding": {
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "platform_binding_digest": {
      "pattern": "^sha256:[0-9a-f]{64}$",
      "type": "string"
    }
  }
},
"required": [
    "activation_digest",
    "implementation_closure_digest",
    "platform_binding_digest"
  ],
  "type": "object"
},
"runtime_queries": {
  "const": 600
},
"search": {
  "additionalProperties": false,
  "properties": {
    "actual_runtime_queries": {},
    "broad_query_refinement_required": {},
    "content_search_verified": {},
    "continuation_state": {},
    "continuation_token_overhead_at_most_10_percent": {},
    "continuation_union_complete": {},
    "continuation_union_completeness": {},
    "continuations_checked": {},
    "depth_counts": {},
    "depth_max": {},
    "depth_min": {},
    "exact_artifact_search_verified": {},
    "explicit_truncations": {},
    "forced_continuation_chains": {},
    "forced_depths": {},
    "forced_query_budget": {},
    "forced_union_matches": {},
    "high_cardinality_terms_verified": {},
    "hostile_proxy_content_verified": {},
    "maximum_continuation_token_bytes": {},
    "miss_behavior_verified": {},
    "mixed_query_classes_complete": {},
    "p50_ms": {},
    "p95_ms": {},
    "p99_ms": {},
    "pages_observed": {},
    "promin_service_continuation_calls": {},
    "promin_service_search_calls": {},
    "query_class_depths": {},
    "query_class_latency_ms": {},
    "query_mix": {},

```

```

    "result_digest": {},
    "runtime_ingress": {},
    "runtime_query_budget": {},
    "selected_closure_chains": {},
    "selected_closure_union_completeness": {},
    "silent_truncations": {}
  },

```

```

  "required": [
    "actual_runtime_queries",
    "broad_query_refinement_required",
    "content_search_verified",
    "continuation_state",
    "continuation_token_overhead_at_most_10_percent",

```

```

    "continuation_union_complete",
    "continuation_union_completeness",
    "continuations_checked",
    "depth_counts",
    "depth_max",
    "depth_min",
    "exact_artifact_search_verified",
    "explicit_truncations",
    "forced_continuation_chains",
    "forced_depths",
    "forced_query_budget",
    "forced_union_matches",
    "high_cardinality_terms_verified",
    "hostile_proxy_content_verified",
    "maximum_continuation_token_bytes",
    "miss_behavior_verified",
    "mixed_query_classes_complete",
    "p50_ms",
    "p95_ms",
    "p99_ms",
    "pages_observed",
    "promin_service_continuation_calls",
    "promin_service_search_calls",
    "query_class_depths",
    "query_class_latency_ms",
    "query_mix",
    "result_digest",
    "runtime_ingress",

```

```

    "runtime_query_budget",
    "selected_closure_chains",
    "selected_closure_union_completeness",
    "silent_truncations"
  ],

```

```

  "type": "object"
},

```

```

"selected_closure_union_completeness": {
  "maximum": 1.0,
  "minimum": 0.0,
  "type": "number"
},

```

```

"silent_truncations": {
  "const": 0
},

```

```

"status": {
  "enum": [
    "pass",
    "fail"
  ]
},

```

```

"workspace_initialization": {
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "implementation_closure_digest": {

```

```

      "$ref": "#/$defs/Digest"
    },

```

```

    "init_record_count": {
      "const": 5
    },

```

```

    "init_records": {
      "const": [
        "activation.json",

```

```

        "authority.json",
        "project.json",
        "standards.json",
        "technologies.json"
    ]
},
"product_tree_scans": {
    "const": 0
},
"project_id": {
    "const": "promin-physical-saturation"
},
"record_type": {
    "const": "SaturationWorkspaceInitialization"
},
"snapshot_provider_id": {
    "const": "git-filesystem-inventory"
},
"snapshot_provider_version": {
    "minLength": 1,

```

```

        "type": "string"
    },
    "status": {
        "enum": [
            "created",
            "reused"
        ]
    }
},
"required": [
    "activation_digest",
    "implementation_closure_digest",
    "init_record_count",
    "init_records",
    "product_tree_scans",
    "project_id",
    "record_type",
    "snapshot_provider_id",
    "snapshot_provider_version",
    "status"
],
"type": "object"
}
},

```

```

"required": [
    "record_type",
    "status",
    "candidate_binding_digest",

```

```

"artifact_binding",
"artifact_binding_unchanged",
"workspace_initialization",
"runtime_binding",
"physical_files",
"core_valid_relations",
"runtime_queries",
"silent_truncations",
"selected_closure_union_completeness",
"memory_amplification_at_most_32",
"core_valid_relations_exact_198999",
"broad_query_refinement_required",
"high_cardinality_terms_verified",
"content_search_verified",
"miss_behavior_verified",
"hostile_proxy_content_verified",
"exact_artifact_search_verified",
"mixed_query_classes_complete",
"continuation_token_bytes_at_most_256",
"continuation_state_bytes_at_most_16384",
"continuation_token_overhead_at_most_10_percent",
"pass_credit",
"physical",
"inventory",
"projection",
"query_authorization",
"search",
"resources",

```

Owner: core/contracts.schema.json#/\$defs/SaturationEvidence | Validator: urn:promin:contracts:1#/\$defs/SaturationEvidence

Owner: core/contracts.schema.json#/\$defs/ScopeSelector | Validator: urn:promin:contracts:1#/\$defs/ScopeSelector

Owner: core/contracts.schema.json#/\$defs/SemVer | Validator: urn:promin:contracts:1#/\$defs/SemVer

Required: record_type, export_kind, package_or_archive, candidate_digest, activation_digest, head_event_id, head_sequence, head_digest, core_bundle_digest, preset_digest, implementation_closure_digest, provider_binding_digest, input_digests, output_artifact_id, output_artifact_record_digest, output_digest, output_size_bytes, output_media_type, pass_credit, product_acceptance_pass, product_public_approval, public_release_approved

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
candidate_digest	yes	\$ref="#/\$defs/Digest"
core_bundle_digest	yes	\$ref="#/\$defs/Digest"
export_kind	yes	const="semantic-state"
head_digest	yes	oneOf[2]=#/\$defs/Digest type:null
head_event_id	yes	oneOf[2]=#/\$defs/Id type:null
head_sequence	yes	type="integer"; minimum=0
implementation_closure_digest	yes	\$ref="#/\$defs/Digest"
input_digests	yes	type="array"; minItems=0; maxItems=128; uniqueItems=true
output_artifact_id	yes	\$ref="#/\$defs/Id"
output_artifact_record_digest	yes	\$ref="#/\$defs/Digest"
output_digest	yes	\$ref="#/\$defs/Digest"
output_media_type	yes	const="application/vnd.promin.semantic-state+json"
output_size_bytes	yes	type="integer"; minimum=1; maximum=1073741824
package_or_archive	yes	const=false
pass_credit	yes	const=false
preset_digest	yes	\$ref="#/\$defs/Digest"
product_acceptance_pass	yes	const=false
product_public_approval	yes	const="not_approved"
provider_binding_digest	yes	\$ref="#/\$defs/Digest"
public_release_approved	yes	const=false
record_type	yes	const="SemanticExport"

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "core_bundle_digest": {
      "$ref": "#/$defs/Digest"
    },
    "export_kind": {
      "const": "semantic-state"
    },
    "head_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    },
    "head_event_id": {
      "oneOf": [
        {
          "$ref": "#/$defs/Id"
        },
        {
          "type": "null"
        }
      ]
    },
    "head_sequence": {
      "type": "integer"; minimum=0
    },
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "input_digests": {
      "type": "array"; minItems=0; maxItems=128; uniqueItems=true
    },
    "output_artifact_id": {
      "$ref": "#/$defs/Id"
    },
    "output_artifact_record_digest": {
      "$ref": "#/$defs/Digest"
    },
    "output_digest": {
      "$ref": "#/$defs/Digest"
    },
    "output_media_type": {
      "const": "application/vnd.promin.semantic-state+json"
    },
    "output_size_bytes": {
      "type": "integer"; minimum=1; maximum=1073741824
    },
    "package_or_archive": {
      "const": false
    },
    "pass_credit": {
      "const": false
    },
    "preset_digest": {
      "$ref": "#/$defs/Digest"
    },
    "product_acceptance_pass": {
      "const": false
    },
    "product_public_approval": {
      "const": "not_approved"
    },
    "provider_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "public_release_approved": {
      "const": false
    },
    "record_type": {
      "const": "SemanticExport"
    }
  }
}

```

```

"head_sequence": {
  "minimum": 0,
  "type": "integer"
},
"implementation_closure_digest": {
  "$ref": "#/$defs/Digest"
},
"input_digests": {
  "items": {
    "$ref": "#/$defs/Digest"
  },
  "maxItems": 128,
  "minItems": 0,
  "type": "array",
  "uniqueItems": true
},
"output_artifact_id": {
  "$ref": "#/$defs/Id"
},
"output_artifact_record_digest": {
  "$ref": "#/$defs/Digest"

```

```

},
"output_digest": {
  "$ref": "#/$defs/Digest"
},
"output_media_type": {
  "const": "application/vnd.promin.semantic-state+json"
},
"output_size_bytes": {
  "maximum": 1073741824,
  "minimum": 1,
  "type": "integer"
},
"package_or_archive": {
  "const": false
},
"pass_credit": {
  "const": false
},
"preset_digest": {
  "$ref": "#/$defs/Digest"
},
"product_acceptance_pass": {
  "const": false
},
"product_public_approval": {
  "const": "not_approved"
},
"provider_binding_digest": {

```

```

  "$ref": "#/$defs/Digest"
},
"public_release_approved": {
  "const": false
},
"record_type": {
  "const": "SemanticExport"
}
},
"required": [
  "record_type",
  "export_kind",
  "package_or_archive",
  "candidate_digest",
  "activation_digest",
  "head_event_id",
  "head_sequence",
  "head_digest",
  "core_bundle_digest",
  "preset_digest",
  "implementation_closure_digest",
  "provider_binding_digest",
  "input_digests",
  "output_artifact_id",
  "output_artifact_record_digest",
  "output_digest",
  "output_size_bytes",
  "output_media_type",

```

```

    "pass_credit",
    "product_acceptance_pass",
    "product_public_approval",
    "public_release_approved"
  ],
  "type": "object",
  "x-promin-ingress-operations": [
    "export"
  ],
  "x-promin-persistence": "export-only"
}

```

Owner: core/contracts.schema.json#/\$defs/SemanticExport | Validator: urn:promin:contracts:1#/\$defs/SemanticExport

97. SemanticModel

Type summary: type="object"; additionalProperties=false

Required: record_type, schema_ref, document, version, function, persistent_entities, relations, derived_projection_kinds, value_objects, design_rules, technology_capabilities, admission

Property	Required	Structural rule
admission	yes	const=schema(keys:derived_by_default,entity,relation)
derived_projection_kinds	yes	const=[16 items]
design_rules	yes	const=schema(keys:artifact_subtypes,candidate_consistency,granularities_are_independent,implementation_closure,inventory_projection,promotion_test,provider_invocation_evidence,semantic_objects_default)
document	yes	const="promin.semantic-model"
function	yes	const="Own the minimal persistent vocabulary, relation grammar, derived-projection boundary, and provider capability contracts."
persistent_entities	yes	const=[9 items]; maxItems=9
record_type	yes	const="SemanticModel"
relations	yes	const=[6 items]; maxItems=6
schema_ref	yes	const="urn:promin:contracts:1#/\$defs/SemanticModel"
technology_capabilities	yes	const=[9 items]
value_objects	yes	const=schema(keys:CandidateDelta,Command,Comparator,ContinuationToken,CurrentReleaseStatus,ImplementationClosure,Operation,PlanProposal,Policy,Predicate,ProviderInvocationEvidence,Query,StandardReleaseDecision,StateCheckpoint,WorkCard)
version	yes	const="1.0.0-alpha.3"

```

{
  "additionalProperties": false,
  "properties": {
    "admission": {
      "const": {
        "derived_by_default": "Each raw inventory file yields exactly one derived Artifact proxy. Raw-file proxies never create a persistent Task or relation; persistence still requires an independent invariant.",
        "entity": "EntityUtility=QueriesEnabled+RulesEnabled+AutomationEnabled-StorageCost-MaintenanceCost-NavigationNoise > 0",
        "relation": "RelationUtility=QueriesEnabled+RulesEnabled+SchedulingGain-WriteCost-FanoutNoise-Ambiguity > 0"
      }
    },
    "derived_projection_kinds": {
      "const": [
        "InventoryEntry",
        "SearchDocument",
        "InventoryProjectionRow",
        "ProjectionLimits",
        "RetrievalPage",
        "ReadyFrontier",
        "WorkCardProjection",
        "WorkCard",
        "NextResult",
        "ContinueResult",
        "NavigationView",
        "StateCheckpoint",
        "CurrentReleaseStatus",
        "DoctorResult",
        "OperationMetrics",
        "Report"
      ]
    }
  }
}

```

```

    ],
    },
    "design_rules": {
        "const": {
            "artifact_subtypes": {
                "candidate-delta": "An Artifact with kind diff binds one base Candidate, one new Candidate, one
WorkCard, and the complete normalized changed-path set.",
                "evidence": "An evidence Artifact binds exact Candidate, policy, tool, inputs, and every invoked
provider adapter; Finding disposition evidence additionally binds the canonical Finding digest."
            },
            "candidate_consistency": {
                "creditable": "Only an immutable-vcs-tree provider may produce a creditable Candidate in promin
v1.",
                "observational": "A best-effort filesystem observation is explicit, non-creditable, and cannot
receive product or release pass credit.",
                "recipe_binding": "Candidate identity binds the exact candidate_recipe digest, including
include, exclude, path identity, collision, and symlink rules."
            },
            "granularities_are_independent": [
                "semantic",
                "public",
                "compilation",
                "runtime-dispatch",
                "documentation"
            ],
            "implementation_closure": {
                "activation_binding": "Activation binds the aggregate digest of every capability-specific
implementation closure.",
                "binding_shape": "Each TechnologiesInit binding carries runtime, interpreter, zero or more
capability components, exactly one matching adapter, and a recomputed closure_digest.",
                "required_components": {
                    "control-runtime": [
                        "promin-runtime",
                        "python-interpreter"
                    ],
                    "query-projection": [
                        "promin-runtime",
                        "python-interpreter",
                        "sqlite"
                    ],
                    "shape-validation": [
                        "promin-runtime",
                        "python-interpreter",
                        "jsonschema"
                    ]
                },
                "stale_rule": "A runtime, interpreter, component, adapter, or closure digest change invalidates
the Activation and every dependent evidence Artifact, projection, continuation, and current release
closure."
            },
            "inventory_projection": {
                "current_inventory_persistence": "derived-rebuild-only",
                "current_inventory_record_type": "CurrentInventory",
                "data_class": "untrusted-source",
                "derived_entity_type": "Artifact",
                "input": "Only a verified InventoryResult or its persisted manifest may enter the public rebuild
path.",
                "input_manifest_persistence": "transient-ingress-only",
                "input_manifest_record_type": "InventoryInputManifest",
                "jsonl_row_fields": [
                    "path",
                    "digest",
                    "size",
                    "search_text"
                ],
                "one_file_rule": "Each verified raw file produces exactly one InventoryProjectionRow and one
deterministic derived Artifact proxy.",
                "prohibited_derivations": [
                    "Task",
                    "READS",
                    "PRODUCES"
                ],
                "reserved_provenance": [
                    "data_class",
                    "inventory_path",
                    "inventory_digest"
                ],
                "runtime_owned_fields": [

```

```

        "path",
        "digest",
        "size",
        "inventory_path",
        "inventory_digest"
    ],
    "search_text_rule": "exact NFC-normalized canonical path, UTF-8 bounded by Conformance
scale_contracts.inventory.search_text_bytes_max",
    "stream_artifact_kind": "report",
    "stream_artifact_record_digest_rule": "sha256(canonical finalized record with exact artifact and
authoritative commit_binding)",
    "stream_media_type": "application/x-ndjson"
},
"promotion_test": [
    "independent-invariant",
    "independent-policy",
    "replaceability",
    "measurable-algorithm",
    "failure-boundary",
    "reuse",
    "domain-significance"
],
"provider_invocation_evidence": {
    "diagnostic_capture_bytes_max": 1048576,
    "diagnostic_capture_fields": [
        "stdout_capture_digest",
        "stdout_capture_size_bytes",
        "stdout_capture_truncated",
        "stderr_capture_digest",
        "stderr_capture_size_bytes",
        "stderr_capture_truncated"
    ],
    "full_output_bytes_hard_max": 1099511627776,
    "full_output_fields": [
        "output_digest",
        "output_size_bytes",
        "output_size_ceiling_bytes"
    ],
    "full_output_rule": "output_digest and output_size_bytes cover the complete provider operation
output stream; they are never derived from the bounded diagnostic captures",
    "immutable_tree_stream_rule": "filesystem-inventory immutable-tree-stream digests and counts the
complete raw archive stream while stdout/stderr captures remain bounded diagnostics",
    "invocation_receipt_digest_rule": "sha256(canonical ProviderInvocationEvidence without
invocation_receipt_digest)",
    "invocation_request_digest_rule": "sha256(canonical pre-invocation request receipt binding exact
argv or module call, configured cwd, stdin digest, selected output ceiling, environment allowlist,
provider identity, dependency receipt, and implementation closure)",
    "selected_ceiling_rule": "output_size_ceiling_bytes is the exact operation or
configured-workspace ceiling selected before invocation, does not exceed full_output_bytes_hard_max, and
rejects output_size_bytes above it"
},
"semantic_objects_default": [
    "private",
    "final",
    "immutable-or-stateless",
    "no-runtime-dispatch",
    "no-separate-build-root",
    "bounded-context-local"
]
}
},
"document": {
    "const": "promin.semantic-model"
},
"function": {
    "const": "Own the minimal persistent vocabulary, relation grammar, derived-projection boundary, and
provider capability contracts."
},
"persistent_entities": {
    "const": [
        {
            "admission_rule": "independent-scheduling-and-acceptance-boundary",
            "kind": "Task",
            "purpose": "Represent one schedulable unit with one acceptance predicate and bounded mutation
scope."
        },
        {
            "admission_rule": "independent-candidate-identity-and-consistency-boundary",

```

```

    "kind": "Candidate",
    "purpose": "Identify either one provider-proven immutable product snapshot or one explicitly
non-creditable observation, independently from control state and bound to the exact candidate recipe."
  },
  {
    "admission_rule": "independent-content-identity-provenance-and-retention-boundary",
    "kind": "Artifact",

    "purpose": "Identify immutable product, evidence, log, report, or diff bytes with provenance and
retention metadata; raw-file inventory produces only derived Artifact proxies, while a diff may carry the
canonical Candidate-delta change manifest without creating another persistent entity."
  },
  {
    "admission_rule": "independent-execution-attempt-and-evidence-boundary",
    "kind": "Run",
    "purpose": "Record one execution or validation attempt against exact candidate, policy, tool,
and input digests."
  },
  {
    "admission_rule": "independent-defect-lifecycle-and-blocking-boundary",
    "kind": "Finding",
    "purpose": "Record an observed defect or policy violation with explicit effective-blocking
semantics."
  },
  {
    "admission_rule": "independent-human-judgement-and-target-binding-boundary",
    "kind": "Decision",
    "purpose": "Record a project-operational acceptance, rejection, resolution, waiver, promotion,
or release judgement. It is distinct from the external StandardReleaseDecision for the standard
distribution."
  },
  {
    "admission_rule": "immutable-task-owned-gate-definition-plus-current-run-and-finalized-artifact-r
ecord-bindings",
    "kind": "GateResult",
    "purpose": "Record one gate outcome against the exact immutable Task-owned GateRunDefinition,
current Run, and finalized evidence Artifact records; skipped, blocked, failed, stale, or unresolved
evidence never receives pass credit."
  },
  {
    "admission_rule": "independent-capability-claim-and-revocation-boundary",
    "kind": "Grant",
    "purpose": "Prove one subject capability within one activation, time interval, nonce, and scope."
  },
  {
    "admission_rule": "independent-mutation-ownership-and-fencing-boundary",
    "kind": "Lease",
    "purpose": "Coordinate temporary mutation ownership through generation, fencing, expiry,
heartbeat, and close acknowledgement."
  },
  ],
  "maxItems": 9
},
"record_type": {
  "const": "SemanticModel"
},
"relations": {
  "const": [
    {
      "kind": "DEPENDS_ON",
      "rules": [
        "acyclic-for-ready-frontier",
        "accepted-current-outcome-required"
      ],
      "source": [
        "Task"
      ],
      "target": [
        "Task"
      ]
    },
    {
      "kind": "READS",
      "rules": [
        "read-does-not-imply-authority"
      ],
      "source": [
        "Task",

```

```

    "Run"
  ],
  "target": [
    "Candidate",
    "Artifact"
  ]
},
{
  "kind": "PRODUCES",
  "rules": [
    "content-or-record-identity-required"
  ],
  "source": [
    "Task",
    "Run"
  ],
  "target": [
    "Candidate",
    "Artifact",
    "Finding",
    "Decision"
  ]
},
{

```

```

  "kind": "VALIDATES",
  "rules": [
    "candidate-policy-tool-input-binding-required"
  ],
  "source": [
    "Run"
  ],
  "target": [
    "Candidate",
    "Artifact",
    "Task"
  ]
},
{
  "kind": "CITES",
  "rules": [
    "selector-and-content-resolution-required"
  ],
  "source": [
    "Finding",
    "Decision"
  ],
  "target": [
    "Artifact",
    "Run",
    "Finding"
  ]
},

```

```

{
  "kind": "BLOCKS",
  "rules": [
    "effective-only-while-open-and-blocking"
  ],
  "source": [
    "Finding"
  ],
  "target": [
    "Task",
    "Candidate",
    "Decision"
  ]
}
],
"maxItems": 6
},
"schema_ref": {
  "const": "urn:promin:contracts:1#/$defs/SemanticModel"
},
"technology_capabilities": {
  "const": [
    {
      "adapter_protocol": {
        "dependency_receipt_rule": {
          "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",

```

```

"complete": true,
"component_fields": [
  "component_id",
  "component_kind",
  "source",
  "version",
  "digest",
  "size_bytes"
],
"component_order": "component-id-ascending",
"os_substrate_excluded": true,
"provider_owned_scope": "configured-provider-root-and-runtime-components",
"unresolved_dependencies": "reject"
},
"operation_contracts": {
  "continue": {
    "request": "validated-continuation-token",
    "response": "WorkCardProjection"
  },
  "doctor": {
    "request": "installed-activation",
    "response": "DoctorResult"
  },
  "healthcheck": {
    "request": "configured-healthcheck",
    "response": "bounded-process-observation"
  },
  "init": {
    "request": "explicit-init-plans",
    "response": "InitResult"
  },
  "next": {
    "request": "authorized-bounded-query",
    "response": "WorkCardProjection"
  },
  "provider-binding": {
    "request": "configured-provider-binding",
    "response": "content-addressed-receipt"
  },
  "status": {
    "request": "installed-activation",
    "response": "StatusResult"
  },
  "validate": {
    "request": "installed-activation",
    "response": "ValidationResult"
  }
},
"protocol_id": "promin.control-runtime.v1",
"receipt_persistence": "content-addressed-receipt",
"supported_bindings": [
  {
    "adapter_id": "python-runtime-v1",
    "identity_kind": "file-digest",
    "invocation_kind": "python-runtime"
  }
],
"supported_commands": [
  "continue",
  "doctor",
  "healthcheck",
  "init",
  "next",
  "provider-binding",
  "status",
  "validate"
],
"unknown_binding": "reject"
},
"forbidden_boundary": {
  "effect": "reject",
  "forbids": [
    "authority-invention",
    "policy-invention",
    "provider-binding-invention"
  ]
},
"normative_authority": false

```



```

    },
    "id": "control-runtime",
    "task": "Parse explicit configuration, canonicalize JSON, and run local CLI orchestration."
  },
  {
    "adapter_protocol": {
      "dependency_receipt_rule": {
        "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
        "complete": true,
        "component_fields": [
          "component_id",
          "component_kind",
          "source",
          "version",
          "digest",
          "size_bytes"
        ],
        "component_order": "component-id-ascending",
        "os_substrate_excluded": true,
        "provider_owned_scope": "configured-provider-root-and-runtime-components",
        "unresolved_dependencies": "reject"
      },
      "operation_contracts": {
        "command": {
          "request": "strict-json-record",
          "response": "Draft-2020-12-plus-semantic-result"
        },
        "continue": {
          "request": "strict-json-record",
          "response": "Draft-2020-12-plus-semantic-result"
        },
        "export": {
          "request": "strict-json-record",
          "response": "Draft-2020-12-plus-semantic-result"
        },
        "healthcheck": {
          "request": "configured-healthcheck",
          "response": "bounded-process-observation"
        },
        "import": {
          "request": "strict-json-record",
          "response": "Draft-2020-12-plus-semantic-result"
        },
        "provider-binding": {
          "request": "configured-provider-binding",
          "response": "content-addressed-receipt"
        },
        "rebuild": {
          "request": "strict-json-record",
          "response": "Draft-2020-12-plus-semantic-result"
        },
        "replay": {
          "request": "strict-json-record",
          "response": "Draft-2020-12-plus-semantic-result"
        }
      },
      "protocol_id": "promin.shape-validation.v1",
      "receipt_persistence": "content-addressed-receipt",
      "supported_bindings": [
        {
          "adapter_id": "python-runtime-v1",
          "identity_kind": "file-digest",
          "invocation_kind": "python-runtime"
        }
      ],
      "supported_commands": [
        "command",
        "continue",
        "export",
        "healthcheck",
        "import",
        "provider-binding",
        "rebuild",
        "replay"
      ],
      "unknown_binding": "reject"
    }
  },

```

```

    "forbidden_boundary": {
      "effect": "reject",
      "forbids": [
        "authority-decision",
        "cross-record-semantic-decision"
      ],
      "normative_authority": false
    },
    "id": "shape-validation",
    "task": "Validate the single Draft 2020-12 contract bundle with enforced formats and local
references."
  },
  {
    "adapter_protocol": {
      "dependency_receipt_rule": {
        "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
        "complete": true,
        "component_fields": [
          "component_id",
          "component_kind",
          "source",
          "version",
          "digest",
          "size_bytes"
        ],
        "component_order": "component-id-ascending",
        "os_substrate_excluded": true,
        "provider_owned_scope": "configured-provider-root-and-runtime-components",
        "unresolved_dependencies": "reject"
      },
      "operation_contracts": {
        "digest-bytes": {
          "request": "bounded-bytes",
          "response": "lowercase-sha256"
        },
        "digest-file": {
          "request": "no-follow-regular-file",
          "response": "lowercase-sha256"
        },
        "digest-value": {
          "request": "canonical-json-value",
          "response": "lowercase-sha256"
        },
        "healthcheck": {
          "request": "configured-healthcheck",
          "response": "bounded-process-observation"
        },
        "provider_binding": {
          "request": "configured-provider-binding",
          "response": "content-addressed-receipt"
        }
      },
      "protocol_id": "promin.content-identity.v1",
      "receipt_persistence": "content-addressed-receipt",
      "supported_bindings": [
        {
          "adapter_id": "python-runtime-v1",
          "identity_kind": "file-digest",
          "invocation_kind": "python-runtime"
        }
      ],
      "supported_commands": [
        "digest-bytes",
        "digest-file",
        "digest-value",
        "healthcheck",
        "provider-binding"
      ],
      "unknown_binding": "reject"
    },
    "forbidden_boundary": {
      "effect": "reject",
      "forbids": [
        "digest-as-action-authority"
      ],
      "normative_authority": false
    },
  },

```

```

    "id": "content-identity",
    "task": "Compute content digests and content-addressed object identities."
  },
  {
    "adapter_protocol": {
      "dependency_receipt_rule": {
        "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
        "complete": true,
        "component_fields": [
          "component_id",
          "component_kind",
          "source",
          "version",
          "digest",
          "size_bytes"
        ],
        "component_order": "component-id-ascending",
        "os_substrate_excluded": true,
        "provider_owned_scope": "configured-provider-root-and-runtime-components",
        "unresolved_dependencies": "reject"
      },
      "operation_contracts": {
        "atomic_replace": {
          "request": "same-directory-staged-bytes",

          "response": "durable-replacement"
        },
        "directory_sync": {
          "request": "owning-directory",
          "response": "durability-observation"
        },
        "file_sync": {
          "request": "open-regular-file",
          "response": "durability-observation"
        },
        "healthcheck": {
          "request": "configured-healthcheck",
          "response": "bounded-process-observation"
        },
        "local_writer_lock": {
          "request": "activation-bound-lock",
          "response": "exclusive-lease"
        },
        "provider_binding": {
          "request": "configured-provider-binding",
          "response": "content-addressed-receipt"
        },
        "recover": {
          "request": "durable-journal-state",
          "response": "verified-head"
        }
      },
      "protocol_id": "promin.local-serialization.v1",

      "receipt_persistence": "content-addressed-receipt",
      "supported_bindings": [
        {
          "adapter_id": "python-runtime-v1",
          "identity_kind": "file-digest",
          "invocation_kind": "python-runtime"
        }
      ],
      "supported_commands": [
        "atomic_replace",
        "directory_sync",
        "file_sync",
        "healthcheck",
        "local_writer_lock",
        "provider_binding",
        "recover"
      ],
      "unknown_binding": "reject"
    },
    "forbidden_boundary": {
      "effect": "reject",
      "forbids": [
        "distributed-consensus-claim"
      ],
      "normative_authority": false
    }
  },

```

```

    "id": "local-serialization",
    "task": "Provide one local-writer lock, atomic replace, file sync, directory sync, and recovery
primitives."

```

```

},
{
  "adapter_protocol": {
    "dependency_receipt_rule": {
      "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
      "complete": true,
      "component_fields": [
        "component_id",
        "component_kind",
        "source",
        "version",
        "digest",
        "size_bytes"
      ],
      "component_order": "component-id-ascending",
      "os_substrate_excluded": true,
      "provider_owned_scope": "configured-provider-root-and-runtime-components",
      "unresolved_dependencies": "reject"
    },
    "operation_contracts": {
      "continue": {
        "request": "authenticated-ready-frontier-continuation",
        "response": "ContinueResult"
      },
      "healthcheck": {
        "request": "configured-healthcheck",
        "response": "bounded-process-observation"
      }
    }
  },

```

```

    "next": {
      "request": "authorized-bounded-ready-query",
      "response": "NextResult"
    },
    "provider_binding": {
      "request": "configured-provider-binding",
      "response": "content-addressed-receipt"
    },
    "rebuild": {
      "request": "verified-event-and-inventory-input",
      "response": "current-projection-state"
    },
    "status": {
      "request": "projection-file",
      "response": "current-projection-state"
    }
  },
  "protocol_id": "promin.query-projection.v1",
  "receipt_persistence": "content-addressed-receipt",
  "supported_bindings": [
    {
      "adapter_id": "python-runtime-v1",
      "identity_kind": "file-digest",
      "invocation_kind": "python-runtime"
    }
  ],
  "supported_commands": [
    "continue",

```

```

    "healthcheck",
    "next",
    "provider-binding",
    "rebuild",
    "status"
  ],
  "unknown_binding": "reject"
},
"forbidden_boundary": {
  "effect": "reject",
  "forbids": [
    "projection-as-normative-authority"
  ],
  "normative_authority": false
},
"id": "query-projection",
"task": "Build disposable status and bounded typed-retrieval projections; derive ReadyFrontier,
materialize one exact WorkCardProjection, and return Core-owned NextResult or ContinueResult envelopes for

```

the six supported CLI commands."

```
    },
    {
      "adapter_protocol": {
        "dependency_receipt_rule": {
          "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
          "complete": true,
          "component_fields": [
            "component_id",
            "component_kind",
            "source",
            "version",
            "digest",
            "size_bytes"
          ],
          "component_order": "component-id-ascending",
          "os_substrate_excluded": true,
          "provider_owned_scope": "configured-provider-root-and-runtime-components",
          "unresolved_dependencies": "reject"
        },
        "operation_contracts": {
          "healthcheck": {
            "request": "configured-bounded-git-healthcheck",
            "response": "exit-zero-captured-stdout-stderr"
          },
          "immutable-tree-object": {
            "request": "argv:rev-parse --verify HEAD^{tree};cwd:configured-root;stdin:none",
            "response": "exit-zero;stdout:one-lowercase-hex-object-id-plus-LF;stderr:captured"
          },
          "immutable-tree-stream": {
            "request": "argv:archive --format=tar <verified-tree-object>;cwd:configured-root;stdin:none",
            "response": "exit-zero;stdout:raw-git-tar-stream;stderr:captured"
          },
          "provider-binding": {
            "request": "configured-executable-file-digest",
            "response": "content-addressed-receipt"
          }
        },
        "protocol_id": "promin.filesystem-inventory.v1",
        "receipt_persistence": "content-addressed-receipt",
        "supported_bindings": [
          {
            "adapter_id": "git-executable-v1",
            "identity_kind": "file-digest",
            "invocation_kind": "executable"
          }
        ],
        "supported_commands": [
          "healthcheck",
          "immutable-tree-object",
          "immutable-tree-stream",
          "provider-binding"
        ],
        "unknown_binding": "reject"
      },
      "forbidden_boundary": {
        "effect": "reject",
        "forbids": [
          "implicit-initialization-scan"
        ],
        "normative_authority": false
      },
      "id": "filesystem-inventory",
      "task": "Apply the exact candidate recipe in one product pass, use no-follow reads, and publish either a provider-proven immutable VCS tree stream or an explicit non-creditable observation."
    },
    {
      "adapter_protocol": {
        "dependency_receipt_rule": {
          "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
          "complete": true,
          "component_fields": [
            "component_id",
            "component_kind",
            "source",
            "version",
```

```

        "digest",
        "size_bytes"
    ],
    "component_order": "component-id-ascending",
    "os_substrate_excluded": true,
    "provider_owned_scope": "configured-provider-root-and-runtime-components",
    "unresolved_dependencies": "reject"
},
"operation_contracts": {
    "bounded-export-scan": {
        "request": "canonical-json-stdin:{operation,root,allowlist_digest,max_files,max_bytes}",
        "response": "exit-zero;canonical-json-stdout:{status,files_scanned,bytes_scanned,violations,
output_digest};stderr:captured"
    },
    "healthcheck": {
        "request": "configured-bounded-healthcheck",
        "response": "exit-zero-captured-stdout-stderr"
    },
    "provider-binding": {
        "request": "configured-executable-file-digest",
        "response": "content-addressed-receipt"
    }
}

```

```

},
"protocol_id": "promin.export-scan.v1",
"receipt_persistence": "content-addressed-receipt",
"supported_bindings": [
    {
        "adapter_id": "executable-export-scan-v1",
        "identity_kind": "file-digest",
        "invocation_kind": "executable"
    }
],
"supported_commands": [
    "bounded-export-scan",
    "healthcheck",
    "provider-binding"
],
"unknown_binding": "reject"
},
"forbidden_boundary": {
    "effect": "reject",
    "forbids": [
        "package-as-authorization-boundary"
    ],
    "normative_authority": false
},
"id": "export-scan",
"task": "Perform bounded recursive export inspection, allowlisting, secret classification, and
symlink rejection."
},
{

```

```

"adapter_protocol": {
    "dependency_receipt_rule": {
        "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
        "complete": true,
        "component_fields": [
            "component_id",
            "component_kind",
            "source",
            "version",
            "digest",
            "size_bytes"
        ],
        "component_order": "component-id-ascending",
        "os_substrate_excluded": true,
        "provider_owned_scope": "configured-provider-root-and-runtime-components",
        "unresolved_dependencies": "reject"
    },
    "operation_contracts": {
        "healthcheck": {
            "request": "configured-bounded-signature-healthcheck",
            "response": "exit-zero-captured-stdout-stderr"
        },
        "provider-binding": {
            "request": "configured-module-or-executable-digest",
            "response": "content-addressed-receipt"
        }
    },
}

```

```

"sign-digest": {
  "request": "domain-separated-canonical-claim-digest",

  "response": "configured-adapter-signature"
},
"verify-digest": {
  "request": "domain-separated-canonical-claim-digest-plus-signature",
  "response": "strict-boolean-verification"
}
},
"protocol_id": "promin.signature.v1",
"receipt_persistence": "content-addressed-receipt",
"supported_bindings": [
  {
    "adapter_id": "executable-signature-v1",
    "identity_kind": "file-digest",
    "invocation_kind": "executable"
  },
  {
    "adapter_id": "python-module-signature-v1",
    "identity_kind": "module-file-digest",
    "invocation_kind": "python-module"
  }
],
"supported_commands": [
  "healthcheck",
  "provider-binding",
  "sign-digest",
  "verify-digest"
],
"unknown_binding": "reject"
},
"forbidden_boundary": {
  "effect": "reject",
  "forbids": [
    "core-signer-implementation-lock-in"
  ],
  "normative_authority": false
},
"id": "signature",
"task": "Dispatch the exact configured signature adapter to sign or verify activation, Grant,
and Decision digests for team trust mode, and evidence the invoked adapter identity."
},
{
  "adapter_protocol": {
    "dependency_receipt_rule": {
      "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
      "complete": true,
      "component_fields": [
        "component_id",
        "component_kind",
        "source",
        "version",
        "digest",
        "size_bytes"
      ],
      "component_order": "component-id-ascending",
      "os_substrate_excluded": true,
      "provider_owned_scope": "configured-provider-root-and-runtime-components",
      "unresolved_dependencies": "reject"
    },
    "operation_contracts": {
      "healthcheck": {
        "request": "configured-bounded-healthcheck",
        "response": "exit-zero-captured-stdout-stderr"
      },
      "import-build-facts": {
        "request": "canonical-json-stdin:{operation,project_root,input_digests,max_files,max_bytes}"
        ,
        "response": "exit-zero;canonical-json-stdout:{status,facts,input_digests,output_digest};std
err:captured"
      },
      "provider-binding": {
        "request": "configured-executable-file-digest",
        "response": "content-addressed-receipt"
      }
    },
    "protocol_id": "promin.build-dependency.v1",

```

```

    "receipt_persistence": "content-addressed-receipt",
    "supported_bindings": [
      {
        "adapter_id": "executable-build-dependency-v1",
        "identity_kind": "file-digest",
        "invocation_kind": "executable"
      }
    ],
    "supported_commands": [
      "healthcheck",
      "import-build-facts",
      "provider-binding"
    ],
    "unknown_binding": "reject"
  },
  "forbidden_boundary": {
    "effect": "reject",
    "forbids": [
      "semantic-model-redefinition"
    ],
    "normative_authority": false
  },
  "id": "build-dependency",
  "task": "Import project-specific build and dependency facts as derived projections."
}
]
},
"value_objects": {
  "const": {
    "CandidateDelta": "A diff Artifact payload binding base and new Candidate digests, WorkCard digest, and the complete normalized changed-path set.",
    "Command": "A state-changing request validated before commit.",
    "Comparator": "A pure ordering or equality rule.",
    "ContinuationToken": "An authenticated bounded runtime value signed with the random per-Activation secret. It binds subject, exact projection.read Grant claim, requested scope, revocation state, normalized query, cursor, Activation, HEAD, projection, ranking, complete budgets, depth, implementation closure, issue time, and expiry; it never grants authority by itself.",
    "CurrentReleaseStatus": "A derived evaluation of an immutable historical release Decision against the current closure, blockers, evidence, providers, and Activation.",
    "ImplementationClosure": "A canonical capability-specific binding of the promin runtime, Python interpreter, jsonschema or SQLite component where used, exactly one adapter, and the recomputed aggregate closure digest.",
    "Operation": "One completed semantic step with explicit input, output, failure, and evidence semantics.",
    "PlanProposal": "A non-authoritative bounded proposal that derives candidate Tasks and relations from project intent, repository facts, backlog, and references; it cannot mutate operational state or grant authority.",
    "Policy": "A machine-addressable invariant owned by policy-set.json.",
    "Predicate": "A pure decision rule.",
    "ProviderInvocationEvidence": "The exact capability, provider, invocation kind, identity kind and digest, and adapter ID actually used by one operation.",
    "Query": "A read-only request that cannot mutate authoritative state.",
    "StandardReleaseDecision": "An external human-reviewable distribution record bound to the exact promin ZIP, Core, preset, tools, implementation closure, tests, platform results, saturation, PDFs, and evidence manifest. It never grants product acceptance.",
    "StateCheckpoint": "A disposable self-digested state or journal index bound to an exact authoritative HEAD; it cannot grant authority or acceptance.",
    "WorkCard": "A bounded generated context package for one Task that separately binds the projection.read query Grant claim and the task.execute holder Grant; neither binding implies acceptance."
  }
},
"version": {
  "const": "1.0.0-alpha.3"
}
},
"required": [
  "record_type",
  "schema_ref",
  "document",
  "version",
  "function",
  "persistent_entities",
  "relations",
  "derived_projection_kinds",
  "value_objects",
  "design_rules",
  "technology_capabilities",
  "admission"
]

```



```

    ],
    "type": "object"
  }

```

Owner: core/contracts.schema.json#/\$defs/SemanticModel | Validator: urn:promin:contracts:1#/\$defs/SemanticModel

98. SemanticStatePayload

Type summary: type="object"; additionalProperties=false

Required: record_type, candidate_digest, activation_digest, head_event_id, head_sequence, head_digest, records

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
candidate_digest	yes	\$ref="#/\$defs/Digest"
head_digest	yes	oneOf[2]=#/\$defs/Digest type:null
head_event_id	yes	oneOf[2]=#/\$defs/Id type:null
head_sequence	yes	type="integer"; minimum=0
record_type	yes	const="SemanticStatePayload"
records	yes	type="array"; minItems=0; maxItems=400000

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "head_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    },
    "head_event_id": {
      "oneOf": [
        {
          "$ref": "#/$defs/Id"
        },
        {
          "type": "null"
        }
      ]
    },
    "head_sequence": {
      "minimum": 0,
      "type": "integer"
    },
    "record_type": {
      "const": "SemanticStatePayload"
    },
    "records": {
      "items": {
        "oneOf": [
          {
            "$ref": "#/$defs/Activation"
          },
          {
            "$ref": "#/$defs/Task"
          },
          {
            "$ref": "#/$defs/Candidate"
          },
          {
            "$ref": "#/$defs/Artifact"
          },
          {
            "$ref": "#/$defs/Run"
          }
        ]
      }
    }
  }
}

```

```

    {
      "$ref": "#/$defs/Finding"
    },
    {
      "$ref": "#/$defs/Decision"
    },
    {
      "$ref": "#/$defs/Grant"
    },
    {
      "$ref": "#/$defs/Lease"
    },
    {
      "$ref": "#/$defs/GateResult"
    },
    {
      "$ref": "#/$defs/GrantRevocation"
    },
    {
      "$ref": "#/$defs/Relation"
    }
  ]
},
"maxItems": 400000,
"minItems": 0,
"type": "array"
}
},
"required": [
  "record_type",
  "candidate_digest",
  "activation_digest",
  "head_event_id",
  "head_sequence",
  "head_digest",
  "records"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/SemanticStatePayload | Validator: urn:promin:contracts:1#/\$defs/SemanticStatePayload

99. SignatureActivationProof

Type summary: type="object"; additionalProperties=false

Required: kind, key_id, algorithm, signed_digest, signature

Property	Required	Structural rule
algorithm	yes	type="string"; minLength=1
key_id	yes	\$ref="#/\$defs/Id"
kind	yes	const="signature"
signature	yes	type="string"; minLength=1
signed_digest	yes	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "properties": {
    "algorithm": {
      "minLength": 1,
      "type": "string"
    },
    "key_id": {
      "$ref": "#/$defs/Id"
    },
    "kind": {
      "const": "signature"
    },
    "signature": {
      "minLength": 1,
      "type": "string"
    },
    "signed_digest": {
      "$ref": "#/$defs/Digest"
    }
  }
},

```

```

"required": [
  "kind",
  "key_id",
  "algorithm",
  "signed_digest",
  "signature"
],

"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/SignatureActivationProof | Validator: urn:promin:contracts:1#/\$defs/SignatureActivationProof

100. SignatureRootCommandProof

Type summary: type="object"; additionalProperties=false

Required: kind, key_id, algorithm, signed_intent_digest, signature

Property	Required	Structural rule
algorithm	yes	type="string"; minLength=1
key_id	yes	\$ref="#/\$defs/Id"
kind	yes	const="signature-command"
signature	yes	type="string"; minLength=1
signed_intent_digest	yes	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "properties": {
    "algorithm": {
      "minLength": 1,
      "type": "string"
    },
    "key_id": {
      "$ref": "#/$defs/Id"
    },
    "kind": {
      "const": "signature-command"
    },
    "signature": {
      "minLength": 1,
      "type": "string"
    },
    "signed_intent_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "kind",
    "key_id",
    "algorithm",
    "signed_intent_digest",
    "signature"
  ],

  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/SignatureRootCommandProof | Validator: urn:promin:contracts:1#/\$defs/SignatureRootCommandProof

101. SignatureTrustProof

Type summary: type="object"; additionalProperties=false

Required: kind, key_id, algorithm, signed_digest, signature

Property	Required	Structural rule
algorithm	yes	type="string"; minLength=1; maxLength=128
key_id	yes	\$ref="#/\$defs/Id"
kind	yes	const="signature"
signature	yes	type="string"; minLength=1; maxLength=8192
signed_digest	yes	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "properties": {
    "algorithm": {
      "maxLength": 128,
      "minLength": 1,
      "type": "string"
    },
    "key_id": {
      "$ref": "#/$defs/Id"
    },
    "kind": {
      "const": "signature"
    },
    "signature": {
      "maxLength": 8192,
      "minLength": 1,
      "type": "string"
    },
    "signed_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "kind",
    "key_id",
    "algorithm",
    "signed_digest",
    "signature"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/SignatureTrustProof | Validator: urn:promin:contracts:1#/\$defs/SignatureTrustProof

102. StandardBinding

Type summary: type="object"; additionalProperties=false

Required: standard_id, version, source_uri, source_digest, purpose

Property	Required	Structural rule
purpose	yes	type="string"; minLength=1
source_digest	yes	\$ref="#/\$defs/Digest"
source_uri	yes	type="string"; minLength=1
standard_id	yes	\$ref="#/\$defs/Id"
version	yes	type="string"; minLength=1

```

{
  "additionalProperties": false,
  "properties": {
    "purpose": {
      "minLength": 1,
      "type": "string"
    },
    "source_digest": {
      "$ref": "#/$defs/Digest"
    },
    "source_uri": {
      "minLength": 1,
      "type": "string"
    },
    "standard_id": {
      "$ref": "#/$defs/Id"
    },
    "version": {
      "minLength": 1,
      "type": "string"
    }
  },
  "required": [
    "standard_id",
    "version",
    "source_uri",
    "source_digest",
    "purpose"
  ]
}

```

```

    },
    "type": "object"
  }

```

Owner: core/contracts.schema.json#/\$defs/StandardBinding | Validator: urn:promin:contracts:1#/\$defs/StandardBinding

103. StandardManifest

Type summary: type="object"; additionalProperties=false

Required: record_type, schema_ref, canonical_name, aliases, canonical_name_only, version, function, core_components, bundle_digest, core_artifact_count, schema_definition_count, selected_preset_count, persistent_entity_kind_count, relation_kind_count, technology_capability_count, action_capability_count, policy_count, acceptance_predicate_count, mutation_family_count

Property	Required	Structural rule
acceptance_predicate_count	yes	type="integer"; const=102
action_capability_count	yes	type="integer"; const=16
aliases	yes	type="array"; maxItems=0
bundle_digest	yes	\$ref="#/\$defs/Digest"
canonical_name	yes	const="promin"
canonical_name_only	yes	const=true
core_artifact_count	yes	type="integer"; const=6
core_components	yes	type="array"; minItems=5; maxItems=5
function	yes	type="string"; minLength=1
mutation_family_count	yes	type="integer"; const=77
persistent_entity_kind_count	yes	type="integer"; const=9
policy_count	yes	type="integer"; const=68
record_type	yes	const="StandardManifest"
relation_kind_count	yes	type="integer"; const=6
schema_definition_count	yes	type="integer"; const=127
schema_ref	yes	const="urn:promin:contracts:1#/\$defs/StandardManifest"
selected_preset_count	yes	type="integer"; const=1
technology_capability_count	yes	type="integer"; const=9
version	yes	const="1.0.0-alpha.3"

```

{
  "additionalProperties": false,
  "properties": {
    "acceptance_predicate_count": {
      "const": 102,
      "type": "integer"
    },
    "action_capability_count": {
      "const": 16,
      "type": "integer"
    },
    "aliases": {
      "maxItems": 0,
      "type": "array"
    },
    "bundle_digest": {
      "$ref": "#/$defs/Digest"
    },
    "canonical_name": {
      "const": "promin"
    },
    "canonical_name_only": {
      "const": true
    },
    "core_artifact_count": {
      "const": 6,
      "type": "integer"
    },
    "core_components": {
      "items": {
        "$ref": "#/$defs/CoreComponent"
      }
    }
  }
}

```

```

    },
    "maxItems": 5,
    "minItems": 5,
    "type": "array"
  },
  "function": {
    "minLength": 1,
    "type": "string"
  },
  "mutation_family_count": {
    "const": 77,
    "type": "integer"
  },
  "persistent_entity_kind_count": {
    "const": 9,
    "type": "integer"
  },
  "policy_count": {
    "const": 68,
    "type": "integer"
  },
  "record_type": {
    "const": "StandardManifest"
  },
  "relation_kind_count": {
    "const": 6,
    "type": "integer"
  },
  "schema_definition_count": {
    "const": 127,
    "type": "integer"
  },
  "schema_ref": {
    "const": "urn:promin:contracts:1#/$defs/StandardManifest"
  },
  "selected_preset_count": {
    "const": 1,
    "type": "integer"
  },
  "technology_capability_count": {
    "const": 9,
    "type": "integer"
  },
  "version": {
    "const": "1.0.0-alpha.3"
  }
},
"required": [
  "record_type",
  "schema_ref",
  "canonical_name",
  "aliases",
  "canonical_name_only",
  "version",
  "function",
  "core_components",
  "bundle_digest",
  "core_artifact_count",
  "schema_definition_count",
  "selected_preset_count",
  "persistent_entity_kind_count",
  "relation_kind_count",
  "technology_capability_count",
  "action_capability_count",
  "policy_count",
  "acceptance_predicate_count",
  "mutation_family_count"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/StandardManifest | Validator: urn:promin:contracts:1#/\$defs/StandardManifest

104. StandardReleaseCandidateBinding

Type summary: type="object"; additionalProperties=false

Required: record_type, standard_name, version, archive_sha256, archive_bytes, archive_member_manifest_digest, package_manifest_digest, checksums_digest, core_bundle_digest, preset_digest, package_tool_digest, validator_digest, test_manifest_digest, portable_implementation_closure_digest, evidence_tool_digests, candidate_binding_digest

Property	Required	Structural rule
archive_bytes	yes	type="integer"; minimum=1
archive_member_manifest_digest	yes	\$ref="#/\$defs/Digest"
archive_sha256	yes	\$ref="#/\$defs/Digest"
candidate_binding_digest	yes	\$ref="#/\$defs/Digest"
checksums_digest	yes	\$ref="#/\$defs/Digest"
core_bundle_digest	yes	\$ref="#/\$defs/Digest"
evidence_tool_digests	yes	type="object"; additionalProperties=false
package_manifest_digest	yes	\$ref="#/\$defs/Digest"
package_tool_digest	yes	\$ref="#/\$defs/Digest"
portable_implementation_closure_digest	yes	\$ref="#/\$defs/Digest"
preset_digest	yes	\$ref="#/\$defs/Digest"
record_type	yes	const="StandardReleaseCandidateBinding"
standard_name	yes	const="promin"
test_manifest_digest	yes	\$ref="#/\$defs/Digest"
validator_digest	yes	\$ref="#/\$defs/Digest"
version	yes	\$ref="#/\$defs/SemVer"

```

{
  "additionalProperties": false,
  "properties": {
    "archive_bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "archive_member_manifest_digest": {
      "$ref": "#/$defs/Digest"
    },
    "archive_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "checksums_digest": {
      "$ref": "#/$defs/Digest"
    },
    "core_bundle_digest": {
      "$ref": "#/$defs/Digest"
    },
    "evidence_tool_digests": {
      "additionalProperties": false,
      "properties": {
        "tools/generate_human.py": {
          "$ref": "#/$defs/Digest"
        },
        "tools/promin_no_degradation.py": {
          "$ref": "#/$defs/Digest"
        },
        "tools/promin_package.py": {
          "$ref": "#/$defs/Digest"
        },
        "tools/promin_saturation.py": {
          "$ref": "#/$defs/Digest"
        },
        "tools/promin_saturation_audit.py": {
          "$ref": "#/$defs/Digest"
        },
        "tools/promin_validate.py": {
          "$ref": "#/$defs/Digest"
        }
      }
    },
    "required": [

```

```

        "tools/generate_human.py",
        "tools/promin_no_degradation.py",
        "tools/promin_package.py",
        "tools/promin_saturation.py",
        "tools/promin_saturation_audit.py",
        "tools/promin_validate.py"
    ],
    "type": "object"
},
"package_manifest_digest": {
    "$ref": "#/$defs/Digest"
},
"package_tool_digest": {
    "$ref": "#/$defs/Digest"
},
"portable_implementation_closure_digest": {
    "$ref": "#/$defs/Digest"
},
"preset_digest": {
    "$ref": "#/$defs/Digest"
},
"record_type": {
    "const": "StandardReleaseCandidateBinding"
},
"standard_name": {
    "const": "promin"
},
"test_manifest_digest": {
    "$ref": "#/$defs/Digest"
},
"validator_digest": {
    "$ref": "#/$defs/Digest"
},
"version": {
    "$ref": "#/$defs/SemVer"
}
},
"required": [
    "record_type",
    "standard_name",
    "version",
    "archive_sha256",
    "archive_bytes",
    "archive_member_manifest_digest",
    "package_manifest_digest",
    "checksums_digest",
    "core_bundle_digest",
    "preset_digest",
    "package_tool_digest",
    "validator_digest",
    "test_manifest_digest",
    "portable_implementation_closure_digest",
    "evidence_tool_digests",
    "candidate_binding_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/StandardReleaseCandidateBinding | Validator: urn:promin:contracts:1#/\$defs/StandardReleaseCandidateBinding

105. StandardReleaseDecision

Type summary: type="object"; additionalProperties=false

Required: record_type, decision_id, standard_name, version, candidate_binding_digest, evidence_manifest_digest, outcome, decider_id, release_capability, trust_root_id, signature_provider_id, key_id, nonce, decided_at, signed_claim_digest, signature

Property	Required	Structural rule
candidate_binding_digest	yes	\$ref="#/\$defs/Digest"
decided_at	yes	\$ref="#/\$defs/Timestamp"
decider_id	yes	\$ref="#/\$defs/Id"
decision_id	yes	\$ref="#/\$defs/Id"
evidence_manifest_digest	yes	\$ref="#/\$defs/Digest"

Property	Required	Structural rule
key_id	yes	\$ref="#/\$defs/Id"
nonce	yes	type="string"; minLength=24; maxLength=128
outcome	yes	enum=["approve", "reject"]
record_type	yes	const="StandardReleaseDecision"
release_capability	yes	const="standard.distribute"
signature	yes	type="string"; minLength=80; maxLength=128
signature_provider_id	yes	const="cryptography-ed25519-v1"
signed_claim_digest	yes	\$ref="#/\$defs/Digest"
standard_name	yes	const="promin"
trust_root_id	yes	\$ref="#/\$defs/Id"
version	yes	\$ref="#/\$defs/SemVer"

```

{
  "additionalProperties": false,
  "properties": {
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "decided_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "decider_id": {
      "$ref": "#/$defs/Id"
    },
    "decision_id": {
      "$ref": "#/$defs/Id"
    },
    "evidence_manifest_digest": {
      "$ref": "#/$defs/Digest"
    },
    "key_id": {
      "$ref": "#/$defs/Id"
    },
    "nonce": {
      "maxLength": 128,
      "minLength": 24,
      "type": "string"
    },
    "outcome": {
      "enum": [
        "approve",
        "reject"
      ]
    },
    "record_type": {
      "const": "StandardReleaseDecision"
    },
    "release_capability": {
      "const": "standard.distribute"
    },
    "signature": {
      "maxLength": 128,
      "minLength": 80,
      "type": "string"
    },
    "signature_provider_id": {
      "const": "cryptography-ed25519-v1"
    },
    "signed_claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "standard_name": {
      "const": "promin"
    },
    "trust_root_id": {
      "$ref": "#/$defs/Id"
    },
    "version": {
      "$ref": "#/$defs/SemVer"
    }
  }
}

```

```

},
"required": [
  "record_type",
  "decision_id",
  "standard_name",
  "version",
  "candidate_binding_digest",
  "evidence_manifest_digest",
  "outcome",
  "decider_id",
  "release_capability",
  "trust_root_id",
  "signature_provider_id",
  "key_id",
  "nonce",
  "decided_at",
  "signed_claim_digest",
  "signature"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/StandardReleaseDecision | Validator: urn:promin:contracts:1#/\$defs/StandardReleaseDecision

106. StandardReleaseEvidenceEntry

Type summary: type="object"; additionalProperties=false

Required: evidence_id, evidence_role, path, sha256, size_bytes, record_type, status, candidate_binding_digest, predicates

Property	Required	Structural rule
candidate_binding_digest	yes	\$ref="#/\$defs/Digest"
evidence_id	yes	\$ref="#/\$defs/Id"
evidence_role	yes	enum=["linux", "windows", "physical-scale", "saturation-audit", "human-documents", "linux-no-degradation", "windows-no-degradation"]
path	yes	\$ref="#/\$defs/RelativePath"
predicates	yes	type="array"; minItems=1; maxItems=64
record_type	yes	\$ref="#/\$defs/Id"
sha256	yes	\$ref="#/\$defs/Digest"
size_bytes	yes	type="integer"; minimum=1
status	yes	const="pass"

```

{
  "additionalProperties": false,
  "properties": {
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "evidence_id": {
      "$ref": "#/$defs/Id"
    },
    "evidence_role": {
      "enum": [
        "linux",
        "windows",
        "physical-scale",
        "saturation-audit",
        "human-documents",
        "linux-no-degradation",
        "windows-no-degradation"
      ]
    },
    "path": {
      "$ref": "#/$defs/RelativePath"
    },
    "predicates": {
      "items": {
        "$ref": "#/$defs/StandardReleaseEvidencePredicate"
      },
      "maxItems": 64,
      "minItems": 1,
      "type": "array"
    }
  }
}

```

```

    },
    "record_type": {
      "$ref": "#/$defs/Id"
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    },
    "size_bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "status": {
      "const": "pass"
    }
  },
  "required": [
    "evidence_id",
    "evidence_role",
    "path",
    "sha256",
    "size_bytes",
    "record_type",
    "status",
    "candidate_binding_digest",
    "predicates"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/StandardReleaseEvidenceEntry | Validator: urn:promin:contracts:1#/\$defs/StandardReleaseEvidenceEntry

107. StandardReleaseEvidenceManifest

Type summary: type="object"; additionalProperties=false

Required: record_type, standard_name, version, candidate_binding_digest, max_evidence_completed_at, entries, matrix_aggregate, supplemental_lanes, evidence_manifest_digest

Property	Required	Structural rule
candidate_binding_digest	yes	\$ref="#/\$defs/Digest"
entries	yes	type="array"; minItems=7; maxItems=7
evidence_manifest_digest	yes	\$ref="#/\$defs/Digest"
matrix_aggregate	yes	\$ref="#/\$defs/StandardReleaseMatrixBinding"
max_evidence_completed_at	yes	\$ref="#/\$defs/Timestamp"
record_type	yes	const="StandardReleaseEvidenceManifest"
standard_name	yes	const="promin"
supplemental_lanes	yes	type="array"; minItems=4; maxItems=4
version	yes	\$ref="#/\$defs/SemVer"

```

{
  "additionalProperties": false,
  "properties": {
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "entries": {
      "items": {
        "$ref": "#/$defs/StandardReleaseEvidenceEntry"
      },
      "maxItems": 7,
      "minItems": 7,
      "type": "array"
    },
    "evidence_manifest_digest": {
      "$ref": "#/$defs/Digest"
    },
    "matrix_aggregate": {
      "$ref": "#/$defs/StandardReleaseMatrixBinding"
    },
    "max_evidence_completed_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "record_type": {

```

```

    "const": "StandardReleaseEvidenceManifest"
  },
  "standard_name": {
    "const": "promin"
  },
  "supplemental_lanes": {
    "items": {
      "$ref": "#/$defs/StandardReleaseSupplementalLane"
    },
    "maxItems": 4,
    "minItems": 4,
    "type": "array"
  },
  "version": {
    "$ref": "#/$defs/SemVer"
  }
},
"required": [
  "record_type",
  "standard_name",
  "version",
  "candidate_binding_digest",
  "max_evidence_completed_at",
  "entries",
  "matrix_aggregate",
  "supplemental_lanes",
  "evidence_manifest_digest"
],
"type": "object"
}

```

Owner: *core/contracts.schema.json#/\$defs/StandardReleaseEvidenceManifest* | Validator: *urn:promin:contracts:1#/\$defs/StandardReleaseEvidenceManifest*

108. StandardReleaseEvidencePredicate

Type summary: type="object"; additionalProperties=false

Required: pointer, equals

Property	Required	Structural rule
equals	yes	composed structural definition
pointer	yes	type="string"; pattern="^/"; maxLength=1024

```

{
  "additionalProperties": false,
  "properties": {
    "equals": {},
    "pointer": {
      "maxLength": 1024,
      "pattern": "^/",
      "type": "string"
    }
  },
  "required": [
    "pointer",
    "equals"
  ],
  "type": "object"
}

```

Owner: *core/contracts.schema.json#/\$defs/StandardReleaseEvidencePredicate* | Validator: *urn:promin:contracts:1#/\$defs/StandardReleaseEvidencePredicate*

109. StandardReleaseMatrixBinding

Type summary: type="object"; additionalProperties=false

Required: path, sha256, size_bytes, record_type, matrix_digest, lane_count, matrix_authoritative, pass_credit, acceptance_pass, product_acceptance_pass, product_public_approval

Property	Required	Structural rule
acceptance_pass	yes	const=false
lane_count	yes	const=8
matrix_authoritative	yes	const=false
matrix_digest	yes	\$ref="#/\$defs/Digest"

Property	Required	Structural rule
pass_credit	yes	const=false
path	yes	const="matrix-current/platform-no-degradation-matrix.json"
product_acceptance_pass	yes	const=false
product_public_approval	yes	const="not_approved"
record_type	yes	const="ProminPlatformNoDegradationMatrix"
sha256	yes	\$ref="#/\$defs/Digest"
size_bytes	yes	type="integer"; minimum=1

```
{
  "additionalProperties": false,
  "properties": {
    "acceptance_pass": {
      "const": false
    },
    "lane_count": {
      "const": 8
    },
    "matrix_authoritative": {
      "const": false
    },
    "matrix_digest": {
      "$ref": "#/$defs/Digest"
    },
    "pass_credit": {
      "const": false
    },
    "path": {
      "const": "matrix-current/platform-no-degradation-matrix.json"
    },
    "product_acceptance_pass": {
      "const": false
    },
    "product_public_approval": {
      "const": "not_approved"
    },
    "record_type": {
      "const": "ProminPlatformNoDegradationMatrix"
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    },
    "size_bytes": {
      "minimum": 1,
      "type": "integer"
    }
  },
  "required": [
    "path",
    "sha256",
    "size_bytes",
    "record_type",
    "matrix_digest",
    "lane_count",
    "matrix_authoritative",
    "pass_credit",
    "acceptance_pass",
    "product_acceptance_pass",
    "product_public_approval"
  ],
  "type": "object",
  "x-promin-persistence": "derived-live-only"
}
```

Owner: core/contracts.schema.json#/\$defs/StandardReleaseMatrixBinding | Validator:
urn:promin:contracts:1#/\$defs/StandardReleaseMatrixBinding

110. StandardReleaseSupplementalLane

Type summary: type="object"; additionalProperties=false

Required: lane_id, path, sha256, size_bytes, record_type, status, candidate_binding_digest, result_digest, evidence_role, producer_attestation_digest

Property	Required	Structural rule
candidate_binding_digest	yes	\$ref="#/\$defs/Digest"
evidence_role	yes	enum=["linux", "linux-no-degradation", "windows", "windows-no-degradation"]
lane_id	yes	enum=["linux-cp314-offline", "linux-cp314-online", "windows-cp314-offline", "windows-cp314-online"]
path	yes	\$ref="#/\$defs/RelativePath"
producer_attestation_digest	yes	\$ref="#/\$defs/Digest"
record_type	yes	enum=["NoDegradationResult", "PlatformVerificationResult"]
result_digest	yes	\$ref="#/\$defs/Digest"
sha256	yes	\$ref="#/\$defs/Digest"
size_bytes	yes	type="integer"; minimum=1
status	yes	const="pass"

```

{
  "additionalProperties": false,
  "properties": {
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "evidence_role": {
      "enum": [
        "linux",
        "linux-no-degradation",
        "windows",
        "windows-no-degradation"
      ]
    },
    "lane_id": {
      "enum": [
        "linux-cp314-offline",
        "linux-cp314-online",
        "windows-cp314-offline",
        "windows-cp314-online"
      ]
    },
    "path": {
      "$ref": "#/$defs/RelativePath"
    },
    "producer_attestation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "record_type": {
      "enum": [
        "NoDegradationResult",
        "PlatformVerificationResult"
      ]
    },
    "result_digest": {
      "$ref": "#/$defs/Digest"
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    },
    "size_bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "status": {
      "const": "pass"
    }
  },
  "required": [
    "lane_id",
    "path",
    "sha256",
    "size_bytes",
    "record_type",
    "status",
    "candidate_binding_digest",
    "result_digest",
    "evidence_role",
    "producer_attestation_digest"
  ]
}

```

```

    },
    "type": "object"
  }

```

Owner: core/contracts.schema.json#/\$defs/StandardReleaseSupplementalLane | Validator: urn:promin:contracts:1#/\$defs/StandardReleaseSupplementalLane

111. StandardReleaseTrustConfiguration

Type summary: type="object"; additionalProperties=false; allOf[7]=keys:properties | keys:properties | keys:properties | keys:properties | keys:properties | keys:properties | keys:properties

Required: record_type, trust_root_id, signature_provider_id, algorithm, keys

Property	Required	Structural rule
algorithm	yes	const="Ed25519"
keys	yes	type="array"; minItems=1; maxItems=32; uniqueItems=true
record_type	yes	const="StandardReleaseTrustConfiguration"
signature_provider_id	yes	const="cryptography-ed25519-v1"
trust_root_id	yes	\$ref="#/\$defs/Id"

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "properties": {
        "keys": {
          "contains": {
            "properties": {
              "evidence_roles": {
                "contains": {
                  "const": "human-documents"
                }
              }
            },
            "required": [
              "evidence_roles"
            ]
          },
          "maxContains": 1,
          "minContains": 1
        }
      }
    },
    {
      "properties": {
        "keys": {
          "contains": {
            "properties": {
              "evidence_roles": {
                "contains": {
                  "const": "linux"
                }
              }
            },
            "required": [
              "evidence_roles"
            ]
          },
          "maxContains": 1,
          "minContains": 1
        }
      }
    },
    {
      "properties": {
        "keys": {
          "contains": {
            "properties": {
              "evidence_roles": {
                "contains": {
                  "const": "linux-no-degradation"
                }
              }
            }
          }
        }
      }
    }
  ],
  "properties": {
    "keys": {
      "contains": {
        "properties": {
          "evidence_roles": {
            "contains": {
              "const": "linux-no-degradation"
            }
          }
        }
      }
    }
  }
}

```

```

        "required": [
            "evidence_roles"
        ]
    },
    "maxContains": 1,
    "minContains": 1
}
},
{
    "properties": {
        "keys": {
            "contains": {
                "properties": {
                    "evidence_roles": {
                        "contains": {
                            "const": "physical-scale"
                        }
                    }
                },
                "required": [
                    "evidence_roles"
                ]
            },
            "maxContains": 1,
            "minContains": 1
        }
    }
},
{

```

```

        "properties": {
            "keys": {
                "contains": {
                    "properties": {
                        "evidence_roles": {
                            "contains": {
                                "const": "saturation-audit"
                            }
                        }
                    },
                    "required": [
                        "evidence_roles"
                    ]
                },
                "maxContains": 1,
                "minContains": 1
            }
        },
        {
            "properties": {
                "keys": {
                    "contains": {
                        "properties": {
                            "evidence_roles": {
                                "contains": {
                                    "const": "windows"
                                }
                            }
                        }
                    }
                }
            }
        },
        {

```

```

            }
        },
        "required": [
            "evidence_roles"
        ]
    },
    "maxContains": 1,
    "minContains": 1
}
},
{
    "properties": {
        "keys": {
            "contains": {
                "properties": {
                    "evidence_roles": {
                        "contains": {
                            "const": "windows-no-degradation"
                        }
                    }
                }
            }
        }
    }
}

```



```

    }
  }
},
"required": [
  "evidence_roles"
]
},
"maxContains": 1,
"minContains": 1

```

```

    }
  }
},
"properties": {
  "algorithm": {
    "const": "Ed25519"
  },
  "keys": {
    "items": {
      "additionalProperties": false,
      "allOf": [
        {
          "if": {
            "properties": {
              "capabilities": {
                "const": [
                  "evidence.produce"
                ]
              }
            }
          },
          "then": {
            "properties": {
              "evidence_roles": {
                "maxItems": 1,
                "minItems": 1
              }
            }
          }
        }
      ]
    }
  }
},

```

```

    "platforms": {
      "minItems": 1
    }
  }
},
{
  "if": {
    "properties": {
      "capabilities": {
        "const": [
          "standard.distribute"
        ]
      }
    }
  },
  "then": {
    "properties": {
      "evidence_roles": {
        "maxItems": 0
      },
      "platforms": {
        "maxItems": 0
      }
    }
  }
},
{
  "if": {
    "properties": {
      "evidence_roles": {
        "const": [
          "human-documents"
        ]
      }
    }
  },
  "then": {
    "properties": {
      "platforms": {

```

```

        "const": [
            "windows"
        ]
    }
}
},
{
    "if": {
        "properties": {
            "evidence_roles": {
                "const": [
                    "linux"
                ]
            }
        }
    }
}

```

```

},
"then": {
    "properties": {
        "platforms": {
            "const": [
                "linux"
            ]
        }
    }
}
},
{
    "if": {
        "properties": {
            "evidence_roles": {
                "const": [
                    "linux-no-degradation"
                ]
            }
        }
    },
    "then": {
        "properties": {
            "platforms": {
                "const": [
                    "linux"
                ]
            }
        }
    }
}

```

```

    }
}
},
{
    "if": {
        "properties": {
            "evidence_roles": {
                "const": [
                    "physical-scale"
                ]
            }
        }
    },
    "then": {
        "properties": {
            "platforms": {
                "const": [
                    "windows"
                ]
            }
        }
    }
}
},
{
    "if": {
        "properties": {
            "evidence_roles": {
                "const": [
                    "saturation-audit"
                ]
            }
        }
    }
}
},

```

```

        "then": {
          "properties": {
            "platforms": {
              "const": [
                "windows"
              ]
            }
          }
        },
      },
    {
      "if": {
        "properties": {
          "evidence_roles": {
            "const": [
              "windows"
            ]
          }
        }
      },
    },
    "then": {
      "properties": {
        "platforms": {
          "const": [
            "windows"
          ]
        }
      }
    },
  },
  {
    "if": {
      "properties": {
        "evidence_roles": {
          "const": [
            "windows-no-degradation"
          ]
        }
      }
    },
    "then": {
      "properties": {
        "platforms": {
          "const": [
            "windows"
          ]
        }
      }
    }
  }
},
],

```

```

"properties": {
  "capabilities": {
    "items": {
      "enum": [
        "evidence.produce",
        "standard.distribute"
      ]
    },
    "maxItems": 1,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "evidence_roles": {
    "items": {
      "enum": [
        "human-documents",
        "linux",
        "linux-no-degradation",
        "physical-scale",
        "saturation-audit",
        "windows",
        "windows-no-degradation"
      ]
    },
    "maxItems": 1,

```

```

        "type": "array",
        "uniqueItems": true
    },
    "key_id": {
        "$ref": "#/$defs/Id"
    },
    "not_after": {
        "$ref": "#/$defs/Timestamp"
    },
    "not_before": {
        "$ref": "#/$defs/Timestamp"
    },
    "platforms": {
        "items": {
            "enum": [
                "cross-platform",
                "darwin",
                "linux",
                "windows"
            ]
        },
        "maxItems": 4,
        "type": "array",
        "uniqueItems": true
    },
    "public_key": {
        "maxLength": 44,
        "minLength": 44,
        "pattern": "^[A-Za-z0-9+/{43}=$",
        "type": "string"
    },
    "revoked": {
        "type": "boolean"
    },
    "subject_id": {
        "$ref": "#/$defs/Id"
    }
},
"required": [
    "key_id",
    "subject_id",
    "capabilities",
    "evidence_roles",
    "platforms",
    "public_key",
    "not_before",
    "not_after",
    "revoked"
],
"type": "object"
},
"maxItems": 32,
"minItems": 1,
"type": "array",
"uniqueItems": true
},
"record_type": {
    "const": "StandardReleaseTrustConfiguration"
},
"signature_provider_id": {
    "const": "cryptography-ed25519-v1"
},
"trust_root_id": {
    "$ref": "#/$defs/Id"
}
},
"required": [
    "record_type",
    "trust_root_id",
    "signature_provider_id",
    "algorithm",
    "keys"
],
"type": "object"
}

```

112. StandardsInit

Type summary: type="object"; additionalProperties=false

Required: record_type, bindings

Property	Required	Structural rule
bindings	yes	type="array"; minItems=0; maxItems=64
record_type	yes	const="StandardsInit"

```
{
  "additionalProperties": false,
  "properties": {
    "bindings": {
      "items": {
        "$ref": "#/$defs/StandardBinding"
      },
      "maxItems": 64,
      "minItems": 0,
      "type": "array"
    },
    "record_type": {
      "const": "StandardsInit"
    }
  },
  "required": [
    "record_type",
    "bindings"
  ],
  "type": "object"
}
```

Owner: core/contracts.schema.json#/\$defs/StandardsInit | Validator: urn:promin:contracts:1#/\$defs/StandardsInit

113. StateBindingUpdate

Type summary: type="object"; additionalProperties=false; oneOf[2]=keys:properties,required | keys:not,properties

Required: leaf_type, leaf_id, operation

Property	Required	Structural rule
leaf_id	yes	\$ref="#/\$defs/Id"
leaf_type	yes	enum=["Activation", "Task", "Candidate", "Artifact", "Run", "Finding", "Decision", "GateResult", "Grant", "Lease", "Relation"]
operation	yes	enum=["delete", "set"]
value_digest	no	\$ref="#/\$defs/Digest"

```
{
  "additionalProperties": false,
  "oneOf": [
    {
      "properties": {
        "operation": {
          "const": "set"
        }
      },
      "required": [
        "value_digest"
      ]
    },
    {
      "not": {
        "required": [
          "value_digest"
        ]
      },
      "properties": {
        "operation": {
          "const": "delete"
        }
      }
    }
  ],
  "type": "object"
}
```

```

"properties": {
  "leaf_id": {
    "$ref": "#/$defs/Id"
  },
  "leaf_type": {
    "enum": [
      "Activation",
      "Task",
      "Candidate",
      "Artifact",
      "Run",
      "Finding",
      "Decision",
      "GateResult",
      "Grant",
      "Lease",
      "Relation"
    ]
  },
  "operation": {
    "enum": [
      "delete",
      "set"
    ]
  },
  "value_digest": {
    "$ref": "#/$defs/Digest"
  }
},
"required": [
  "leaf_type",
  "leaf_id",
  "operation"
],
"type": "object"
}

```

Owner: *core/contracts.schema.json#/\$defs/StateBindingUpdate* | Validator: *urn:promin:contracts:1#/\$defs/StateBindingUpdate*

114. StateMachines

Type summary: `const=`schema(keys:capacity_release,lease,task)

Required: -

```

{
  "const": {
    "capacity_release": {
      "closed_release": {
        "requires_field": "close_ack",
        "slot_reusable": true,
        "state": "CLOSED"
      },
      "terminal_reconciliation": {
        "fencing_token_binding": "exact-lease-fencing-token",
        "field": "capacity_reconciliation",
        "generation_binding": "exact-lease-generation",
        "manager_capability": "lease.manage",
        "required_fields": [
          "reconciled_by",
          "grant_id",
          "grant_claim_digest",
          "reconciled_at",
          "generation",
          "fencing_token"
        ],
        "slot_reusable_after_record": true,
        "states": [
          "EXPIRED",
          "REVOKED"
        ]
      },
      "unreconciled_slot_reusable": false
    },
    "lease": {
      "ACTIVE": [
        "CLOSING",
        "EXPIRED",

```

```

    "REVOKED"
  ],
  "CLOSED": [],
  "CLOSING": [
    "CLOSED",
    "EXPIRED",
    "REVOKED"
  ],
  "EXPIRED": [],
  "REVOKED": []
},
"task": {
  "BLOCKED": [
    "READY",
    "CANCELLED"
  ],
  "CANCELLED": [],
  "COMPLETED": [],
  "LEASED": [
    "RUNNING",
    "READY",
    "BLOCKED",
    "CANCELLED"
  ],
  "PLANNED": [
    "READY",
    "BLOCKED",
    "CANCELLED"
  ],
  "READY": [
    "LEASED",
    "BLOCKED",
    "CANCELLED"
  ],
  "RUNNING": [
    "COMPLETED",
    "BLOCKED",
    "CANCELLED"
  ]
}
}
}

```

Owner: core/contracts.schema.json#/\$defs/StateMachines | Validator: urn:promin:contracts:1#/\$defs/StateMachines

115. StructuralBudgets

Type summary: type="object"; additionalProperties=false

Required: core_normative_artifacts, selected_presets, authority_bearing_init_records, persistent_entity_kinds_max, relation_kinds_max, base_user_commands_max

Property	Required	Structural rule
authority_bearing_init_records	yes	type="integer"; minimum=1
base_user_commands_max	yes	type="integer"; minimum=1
core_normative_artifacts	yes	type="integer"; minimum=1
persistent_entity_kinds_max	yes	type="integer"; minimum=1
relation_kinds_max	yes	type="integer"; minimum=1
selected_presets	yes	type="integer"; minimum=1

```

{
  "additionalProperties": false,
  "properties": {
    "authority_bearing_init_records": {
      "minimum": 1,
      "type": "integer"
    },
    "base_user_commands_max": {
      "minimum": 1,
      "type": "integer"
    },
    "core_normative_artifacts": {
      "minimum": 1,
      "type": "integer"
    },
    "persistent_entity_kinds_max": {

```

```

    "minimum": 1,
    "type": "integer"
  },
  "relation_kinds_max": {
    "minimum": 1,
    "type": "integer"
  },
  "selected_presets": {
    "minimum": 1,
    "type": "integer"
  }
}
},
"required": [
  "core_normative_artifacts",
  "selected_presets",
  "authority_bearing_init_records",
  "persistent_entity_kinds_max",
  "relation_kinds_max",
  "base_user_commands_max"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/StructuralBudgets | Validator: urn:promin:contracts:1#/\$defs/StructuralBudgets

116. SubjectInit

Type summary: type="object"; additionalProperties=false

Required: subject_id, kind, display_name

Property	Required	Structural rule
display_name	yes	type="string"; minLength=1
kind	yes	enum=["human", "service", "agent"]
subject_id	yes	\$ref="#/\$defs/Id"

```

{
  "additionalProperties": false,
  "properties": {
    "display_name": {
      "minLength": 1,
      "type": "string"
    },
    "kind": {
      "enum": [
        "human",
        "service",
        "agent"
      ]
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "subject_id",
    "kind",
    "display_name"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/SubjectInit | Validator: urn:promin:contracts:1#/\$defs/SubjectInit

117. Task

Type summary: type="object"; additionalProperties=false

Required: record_type, task_id, state, required_capability, acceptance_predicate, allowed_paths, activation_digest, candidate_digest, created_at, gate_run_definitions

Property	Required	Structural rule
acceptance_predicate	yes	type="string"; minLength=1
activation_digest	yes	\$ref="#/\$defs/Digest"
allowed_paths	yes	type="array"; minItems=0; maxItems=64; uniqueItems=true

Property	Required	Structural rule
candidate_digest	yes	\$ref="#/\$defs/Digest"
created_at	yes	\$ref="#/\$defs/Timestamp"
gate_run_definitions	yes	type="array"; minItems=1; maxItems=64
operation_profile_id	no	\$ref="#/\$defs/Id"
orchestration_required	no	const=true
recommended_model_tier	no	enum=["tool-only", "micro", "standard", "strong", "critical-review"]
record_type	yes	const="Task"
required_capability	yes	\$ref="#/\$defs/Id"
state	yes	\$ref="#/\$defs/TaskState"
task_id	yes	\$ref="#/\$defs/Id"
<pre> { "additionalProperties": false, "properties": { "acceptance_predicate": { "minLength": 1, "type": "string" }, "activation_digest": { "\$ref": "#/\$defs/Digest" }, "allowed_paths": { "items": { "\$ref": "#/\$defs/RelativePath" }, "maxItems": 64, "minItems": 0, "type": "array", "uniqueItems": true }, "candidate_digest": { "\$ref": "#/\$defs/Digest" }, "created_at": { "\$ref": "#/\$defs/Timestamp" }, "gate_run_definitions": { "items": { "additionalProperties": false, "properties": { "definition": { "\$ref": "#/\$defs/GateRunDefinition" }, "definition_digest": { "\$ref": "#/\$defs/Digest" } }, "required": ["definition_digest", "definition"], "type": "object" }, "maxItems": 64, "minItems": 1, "type": "array" }, "operation_profile_id": { "\$ref": "#/\$defs/Id" }, "orchestration_required": { "const": true }, "recommended_model_tier": { "enum": ["tool-only", "micro", "standard", "strong", "critical-review"] } } } </pre>		

```

    },
    "record_type": {
      "const": "Task"
    },
    "required_capability": {
      "$ref": "#/$defs/Id"
    },
    "state": {
      "$ref": "#/$defs/TaskState"
    },
    "task_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "record_type",
    "task_id",
    "state",
    "required_capability",
    "acceptance_predicate",
    "allowed_paths",
    "activation_digest",
    "candidate_digest",
    "created_at",
    "gate_run_definitions"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Task | Validator: urn:promin:contracts:1#/\$defs/Task

118. TaskState

Type summary: enum=["BLOCKED", "CANCELLED", "COMPLETED", "LEASED", "PLANNED", "READY", "RUNNING"]

Required: -

```

{
  "enum": [
    "BLOCKED",
    "CANCELLED",
    "COMPLETED",
    "LEASED",
    "PLANNED",
    "READY",
    "RUNNING"
  ]
}

```

Owner: core/contracts.schema.json#/\$defs/TaskState | Validator: urn:promin:contracts:1#/\$defs/TaskState

119. TaskStateMachine

Type summary: const=schema(keys:BLOCKED,CANCELLED,COMPLETED,LEASED,PLANNED,READY,RUNNING)

Required: -

```

{
  "const": {
    "BLOCKED": [
      "READY",
      "CANCELLED"
    ],
    "CANCELLED": [],
    "COMPLETED": [],
    "LEASED": [
      "RUNNING",
      "READY",
      "BLOCKED",
      "CANCELLED"
    ],
    "PLANNED": [
      "READY",
      "BLOCKED",
      "CANCELLED"
    ],
    "READY": [
      "LEASED",
      "BLOCKED",
      "CANCELLED"
    ]
  }
}

```

```

    ],
    "RUNNING": [
      "COMPLETED",
      "BLOCKED",
      "CANCELLED"
    ]
  }
}

```

Owner: core/contracts.schema.json#/\$defs/TaskStateMachine | Validator: urn:promin:contracts:1#/\$defs/TaskStateMachine

120. TaskTransition

Type summary: type="object"; additionalProperties=false

Required: task_id, from_state, to_state, reason

Property	Required	Structural rule
from_state	yes	\$ref="#/\$defs/TaskState"
reason	yes	type="string"; minLength=1
task_id	yes	\$ref="#/\$defs/Id"
to_state	yes	\$ref="#/\$defs/TaskState"

```

{
  "additionalProperties": false,
  "properties": {
    "from_state": {
      "$ref": "#/$defs/TaskState"
    },
    "reason": {
      "minLength": 1,
      "type": "string"
    },
    "task_id": {
      "$ref": "#/$defs/Id"
    },
    "to_state": {
      "$ref": "#/$defs/TaskState"
    }
  },
  "required": [
    "task_id",
    "from_state",
    "to_state",
    "reason"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/TaskTransition | Validator: urn:promin:contracts:1#/\$defs/TaskTransition

121. TeamPolicy

Type summary: type="object"; additionalProperties=false

Required: threshold, key_ids, signature_provider_capability

Property	Required	Structural rule
key_ids	yes	type="array"; minItems=1; maxItems=32; uniqueItems=true
signature_provider_capability	yes	const="signature"
threshold	yes	type="integer"; minimum=1

```

{
  "additionalProperties": false,
  "properties": {
    "key_ids": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 32,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "signature_provider_capability": {
      "const": "signature"
    },
    "threshold": {
      "minimum": 1,
      "type": "integer"
    }
  },
  "required": [
    "key_ids",
    "signature_provider_capability",
    "threshold"
  ],
  "type": "object"
}

```

```

    },
    "threshold": {
      "minimum": 1,
      "type": "integer"
    }
  },
  "required": [
    "threshold",
    "key_ids",
    "signature_provider_capability"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/TeamPolicy | Validator: urn:promin:contracts:1#/\$defs/TeamPolicy

122. TechnologiesInit

Type summary: type="object"; additionalProperties=false

Required: record_type, bindings

Property	Required	Structural rule
bindings	yes	type="array"; minItems=1; maxItems=32
record_type	yes	const="TechnologiesInit"

```

{
  "additionalProperties": false,
  "properties": {
    "bindings": {
      "items": {
        "$ref": "#/$defs/ProviderBinding"
      },
      "maxItems": 32,
      "minItems": 1,
      "type": "array"
    },
    "record_type": {
      "const": "TechnologiesInit"
    }
  },
  "required": [
    "record_type",
    "bindings"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/TechnologiesInit | Validator: urn:promin:contracts:1#/\$defs/TechnologiesInit

123. TechnologyCapabilityDef

Type summary: type="object"; additionalProperties=false

Required: id, task, forbidden_boundary, adapter_protocol

Property	Required	Structural rule
adapter_protocol	yes	type="object"; additionalProperties=false
forbidden_boundary	yes	type="object"; additionalProperties=false
id	yes	enum=["control-runtime", "shape-validation", "content-identity", "local-serialization", "query-projection", "filesystem-inventory", "export-scan", "signature", "build-dependency"]
task	yes	type="string"; minLength=1

```

{
  "additionalProperties": false,
  "properties": {
    "adapter_protocol": {
      "additionalProperties": false,
      "properties": {
        "dependency_receipt_rule": {
          "const": {
            "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
            "complete": true,
            "component_fields": [
              "component_id",
              "component_kind",

```

```

        "source",
        "version",
        "digest",
        "size_bytes"
    ],
    "component_order": "component-id-ascending",
    "os_substrate_excluded": true,
    "provider_owned_scope": "configured-provider-root-and-runtime-components",
    "unresolved_dependencies": "reject"
}
},
"operation_contracts": {
    "additionalProperties": {
        "additionalProperties": false,
        "properties": {

```

```

            "request": {
                "maxLength": 1024,
                "minLength": 1,
                "type": "string"
            },
            "response": {
                "maxLength": 1024,
                "minLength": 1,
                "type": "string"
            }
        },
        "required": [
            "request",
            "response"
        ],
        "type": "object"
    },
    "maxProperties": 32,
    "minProperties": 1,
    "propertyNames": {
        "$ref": "#/$defs/Id"
    },
    "type": "object"
},
"protocol_id": {
    "$ref": "#/$defs/Id"
},
"receipt_persistence": {

```

```

    "const": "content-addressed-receipt"
},
"supported_bindings": {
    "items": {
        "additionalProperties": false,
        "properties": {
            "adapter_id": {
                "$ref": "#/$defs/Id"
            },
            "identity_kind": {
                "enum": [
                    "file-digest",
                    "module-file-digest",
                    "distribution-tree-digest",
                    "module-closure-digest",
                    "platform-attestation"
                ]
            },
            "invocation_kind": {
                "enum": [
                    "python-runtime",
                    "python-module",
                    "executable",
                    "platform-service"
                ]
            }
        }
    },
    "required": [

```

```

        "adapter_id",
        "identity_kind",
        "invocation_kind"
    ],
    "type": "object"
},

```

```

    "maxItems": 8,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "supported_commands": {
    "items": {
      "$ref": "#/$defs/Id"
    },
    "maxItems": 32,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "unknown_binding": {
    "const": "reject"
  }
},
"required": [
  "dependency_receipt_rule",
  "operation_contracts",
  "protocol_id",

```

```

  "receipt_persistence",
  "supported_bindings",
  "supported_commands",
  "unknown_binding"
],
"type": "object"
},
"forbidden_boundary": {
  "additionalProperties": false,
  "properties": {
    "effect": {
      "const": "reject"
    },
  },
  "forbids": {
    "items": {
      "maxLength": 128,
      "minLength": 1,
      "type": "string"
    },
    "maxItems": 16,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "normative_authority": {
    "const": false
  }
},

```

```

"required": [
  "effect",
  "forbids",
  "normative_authority"
],
"type": "object"
},
"id": {
  "enum": [
    "control-runtime",
    "shape-validation",
    "content-identity",
    "local-serialization",
    "query-projection",
    "filesystem-inventory",
    "export-scan",
    "signature",
    "build-dependency"
  ]
},
"task": {
  "minLength": 1,
  "type": "string"
}
},
"required": [
  "id",
  "task",

```

```

    "forbidden_boundary",
    "adapter_protocol"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/TechnologyCapabilityDef | Validator: urn:promin:contracts:1#/\$defs/TechnologyCapabilityDef

124. Timestamp

Type summary: type="string"; format="date-time"; pattern="^\d{4}-\d{2}-\d{2}T\d{2}:\d{2}:\d{2}Z\$"

Required: -

```

{
  "format": "date-time",
  "pattern": "^\d{4}-\d{2}-\d{2}T\d{2}:\d{2}:\d{2}Z$",
  "type": "string"
}

```

Owner: core/contracts.schema.json#/\$defs/Timestamp | Validator: urn:promin:contracts:1#/\$defs/Timestamp

125. TrustKey

Type summary: type="object"; additionalProperties=false

Required: key_id, subject_id, algorithm, public_key, fingerprint

Property	Required	Structural rule
algorithm	yes	type="string"; minLength=1
fingerprint	yes	\$ref="#/\$defs/Digest"
key_id	yes	\$ref="#/\$defs/Id"
public_key	yes	type="string"; minLength=1
subject_id	yes	\$ref="#/\$defs/Id"

```

{
  "additionalProperties": false,
  "properties": {
    "algorithm": {
      "minLength": 1,
      "type": "string"
    },
    "fingerprint": {
      "$ref": "#/$defs/Digest"
    },
    "key_id": {
      "$ref": "#/$defs/Id"
    },
    "public_key": {
      "minLength": 1,
      "type": "string"
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "key_id",
    "subject_id",
    "algorithm",
    "public_key",
    "fingerprint"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/TrustKey | Validator: urn:promin:contracts:1#/\$defs/TrustKey

126. WorkCard

Type summary: type="object"; additionalProperties=false; allOf[2]=keys:else,if,then | keys:else,if,then,title

Required: record_type, task_id, operation_mode, operation, acceptance_predicate, allowed_paths, holder_grant_id, candidate_digest, activation_digest, context_digest, stop_conditions, budget, truncated, query_grant_id, query_grant_claim_digest

Property	Required	Structural rule
acceptance_predicate	yes	type="string"; minLength=1
activation_digest	yes	\$ref="#/\$defs/Digest"
allowed_paths	yes	type="array"; minItems=0; maxItems=64; uniqueItems=true
budget	yes	\$ref="#/\$defs/Budget"
candidate_digest	yes	\$ref="#/\$defs/Digest"
context_digest	yes	\$ref="#/\$defs/Digest"
continuation_query	no	type="string"; minLength=1
fencing_token	no	type="integer"; minimum=1
holder_grant_id	yes	\$ref="#/\$defs/Id"
lease_generation	no	type="integer"; minimum=1
lease_id	no	\$ref="#/\$defs/Id"
operation	yes	type="string"; minLength=1
operation_mode	yes	enum=["read", "mutate"]
operation_profile_id	no	\$ref="#/\$defs/Id"
orchestration_required	no	const=true
query_grant_claim_digest	yes	\$ref="#/\$defs/Digest"
query_grant_id	yes	\$ref="#/\$defs/Id"
recommended_model_tier	no	enum=["tool-only", "micro", "standard", "strong", "critical-review"]
record_type	yes	const="WorkCard"
stop_conditions	yes	type="array"; minItems=1; maxItems=16; uniqueItems=true
task_id	yes	\$ref="#/\$defs/Id"
truncated	yes	type="boolean"

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "required": [
            "continuation_query"
          ]
        }
      },
      "if": {
        "properties": {
          "truncated": {
            "const": true
          }
        }
      },
      "then": {
        "required": [
          "continuation_query"
        ]
      }
    },
    {
      "else": {
        "not": {
          "anyOf": [
            {
              "required": [
                "lease_id"
              ]
            },
            {
              "required": [
                "lease_generation"
              ]
            },
            {
              "required": [
                "fencing_token"
              ]
            }
          ]
        }
      }
    }
  ]
}

```



```

        ]
      }
    ]
  }
},
"if": {
  "properties": {
    "operation_mode": {
      "const": "mutate"
    }
  },
  "required": [
    "operation_mode"
  ]
},
"then": {
  "required": [
    "lease_id",
    "lease_generation",
    "fencing_token"
  ]
},
"title": "require current Lease generation and fence for mutation"
},
],
"properties": {
  "acceptance_predicate": {
    "minLength": 1,
    "type": "string"
  },
  "activation_digest": {
    "$ref": "#/$defs/Digest"
  },
  "allowed_paths": {
    "items": {
      "$ref": "#/$defs/RelativePath"
    },
    "maxItems": 64,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
  },
  "budget": {
    "$ref": "#/$defs/Budget"
  },
  "candidate_digest": {
    "$ref": "#/$defs/Digest"
  },
  "context_digest": {
    "$ref": "#/$defs/Digest"
  },
  "continuation_query": {
    "minLength": 1,
    "type": "string"
  },
  "fencing_token": {
    "minimum": 1,
    "type": "integer"
  },
  "holder_grant_id": {
    "$ref": "#/$defs/Id"
  },
  "lease_generation": {
    "minimum": 1,
    "type": "integer"
  },
  "lease_id": {
    "$ref": "#/$defs/Id"
  },
  "operation": {
    "minLength": 1,
    "type": "string"
  },
  "operation_mode": {
    "enum": [
      "read",
      "mutate"
    ]
  }
}

```

```

    ],
    "operation_profile_id": {
      "$ref": "#/$defs/Id"
    },
    "orchestration_required": {
      "const": true
    },
    "query_grant_claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "query_grant_id": {
      "$ref": "#/$defs/Id"
    },
    "recommended_model_tier": {
      "enum": [
        "tool-only",
        "micro",
        "standard",
        "strong",
        "critical-review"
      ]
    }
  ]
}

```

```

},
"record_type": {
  "const": "WorkCard"
},
"stop_conditions": {
  "items": {
    "minLength": 1,
    "type": "string"
  },
  "maxItems": 16,
  "minItems": 1,
  "type": "array",
  "uniqueItems": true
},
"task_id": {
  "$ref": "#/$defs/Id"
},
"truncated": {
  "type": "boolean"
}
},
"required": [
  "record_type",
  "task_id",
  "operation_mode",
  "operation",
  "acceptance_predicate",
  "allowed_paths",

```

```

  "holder_grant_id",
  "candidate_digest",
  "activation_digest",
  "context_digest",
  "stop_conditions",
  "budget",
  "truncated",
  "query_grant_id",
  "query_grant_claim_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/WorkCard | Validator: urn:promin:contracts:1#/\$defs/WorkCard

127. WorkCardProjection

Type summary: type="object"; additionalProperties=false

Required: record_type, activation_digest, head_digest, projection_digest, task_id, task_digest, task, budget, projection_authoritative

Property	Required	Structural rule
activation_digest	yes	\$ref="#/\$defs/Digest"
budget	yes	\$ref="#/\$defs/Budget"
head_digest	yes	oneOf[2]=#/\$defs/Digest type:null

Property	Required	Structural rule
projection_authoritative	yes	const=false
projection_digest	yes	\$ref="#/\$defs/Digest"
record_type	yes	const="WorkCardProjection"
task	yes	type="object"; additionalProperties=false
task_digest	yes	\$ref="#/\$defs/Digest"
task_id	yes	\$ref="#/\$defs/Id"
<pre> { "additionalProperties": false, "properties": { "activation_digest": { "\$ref": "#/\$defs/Digest" }, "budget": { "\$ref": "#/\$defs/Budget" }, "head_digest": { "oneOf": [{ "\$ref": "#/\$defs/Digest" }, { "type": "null" }] }, "projection_authoritative": { "const": false }, "projection_digest": { "\$ref": "#/\$defs/Digest" }, "record_type": { "const": "WorkCardProjection" } }, "task": { "additionalProperties": false, "properties": { "data_class": { "enum": ["operational-record", "untrusted-source"] }, "entity_type": { "enum": ["Task", "Candidate", "Artifact", "Run", "Finding", "Decision", "GateResult", "Grant", "Lease"] }, "id": { "\$ref": "#/\$defs/Id" }, "payload": { "type": "object" } }, "source_digest": { "\$ref": "#/\$defs/Digest" } }, "required": ["id", "entity_type", "data_class", "payload", "source_digest"] } </pre>		

```

    ],
    "type": "object"
  },
  "task_digest": {
    "$ref": "#/$defs/Digest"
  },
  "task_id": {
    "$ref": "#/$defs/Id"
  }
},
"required": [
  "record_type",
  "activation_digest",
  "head_digest",
  "projection_digest",
  "task_id",
  "task_digest",
  "task",
  "budget",
  "projection_authoritative"
],
"type": "object",
"x-promin-persistence": "derived-live-only"
}

```

Owner: core/contracts.schema.json#/\$defs/WorkCardProjection | Validator: urn:promin:contracts:1#/\$defs/WorkCardProjection

Policy registry - 68 policies

ID	Severity	Name	Statement	Validator ID	Owner
P-001	P0	no-false-success	Gate pass credit is true exactly when normalized status is pass and every required gate passes.	no_false_success	core/policy-set.json
P-002	P0	activation-binding	Every authoritative record binds the exact active Core, preset, init, and profile configuration.	activation_binding	core/policy-set.json
P-003	P0	grant-validity	Subject, capability, conjunctive scope, interval, nonce, exact claim digest, proof chain, revocation state, and Activation all match.	grant_validity	core/policy-set.json
P-004	P0	lease-fence	Every mutation uses the current active Lease generation and monotonic fencing token.	lease_fence	core/policy-set.json
P-005	P0	candidate-evidence-binding	Every evidence Artifact requires an explicit diagnostic, gate, or product purpose and binds Activation, candidate, policy, provider/tool, and input digests inside the immutable event payload.	candidate_evidence_binding	core/policy-set.json
P-006	P0	atomic-event-commit	One command produces one immutable batch under lock, compare-head, sync, atomic replace, and recovery.	atomic_event_commit	core/policy-set.json
P-007	P0	projection-is-derived	Projection deletion and deterministic replay cannot change normative meaning.	projection_derived	core/policy-set.json
P-008	P1	minimal-graph	Each verified raw file creates exactly one non-authoritative derived Artifact proxy. Raw inventory creates no Task, READS, or PRODUCES record; Tasks and typed relations require explicit semantic content.	minimal_graph	core/policy-set.json
P-009	P0	bounded-workcard	Every WorkCard stays within both Core hard ceilings and its selected profile byte, entity, relation, fanout, and top-k budgets; every truncation is explicit and losslessly resumable.	bounded_workcard	core/policy-set.json
P-010	P0	size-independent-init	Initialization performs zero product-tree scans and creates exactly five init records.	zero_scan_init	core/policy-set.json
P-011	P1	explicit-external-bindings	External standards and providers exist only in explicit digest-bound init records.	explicit_bindings	core/policy-set.json
P-012	P0	canonical-name-only	The canonical identity is promin; aliases, redirects, and compatibility names are rejected.	canonical_name	core/policy-set.json
P-013	P0	safe-export	Export is allowlisted, recursively bounded, symlink-safe, secret-aware, and path-sanitized.	safe_export	core/policy-set.json
P-014	P0	closed-acknowledgment	Only a valid CLOSED transition with explicit close acknowledgement releases a mutation slot.	closed_ack	core/policy-set.json
P-015	P0	fail-closed-critical-fields	Missing, unknown, stale, downgraded, or unbound critical fields reject before mutation.	critical_fields	core/policy-set.json
P-016	P0	preset-capability-ceiling	A preset may configure provider needs and budgets but cannot grant action authority.	preset_no_authority	core/policy-set.json
P-017	P1	single-scan-inventory	One requested inventory performs at most one full product-tree pass; all projections derive from its stream.	single_scan	core/policy-set.json
P-018	P0	idempotent-commands	A duplicate idempotency key under the same activation and subject returns the original result or a conflict.	idempotent_commands	core/policy-set.json
P-019	P1	core-preset-separation	Core bundle identity excludes replaceable presets; Activation binds the selected preset separately.	core_preset_separation	core/policy-set.json
P-020	P0	acyclic-bootstrap-authority	Init config defines root ceilings, never pre-activation runtime Grants.	bootstrap_authority	core/policy-set.json
P-021	P1	normalization-collision-rejection	Duplicate JSON keys and path or ID collisions after NFC/casefold normalization are rejected.	normalization_collision	core/policy-set.json

ID	Severity	Name	Statement	Validator ID	Owner
P-022	P1	provider-substitution-binding	A provider identity or implementation-closure change invalidates the Activation and every dependent continuation, evidence Artifact, projection, replay result, and current release closure; it never silently changes authority.	provider_substitution	core/policy-set.json
P-023	P1	exact-init-keyset	Activation binds exactly project.json, standards.json, technologies.json, and authority.json.	exact_init_keyset	core/policy-set.json
P-024	P1	stale-projection-rejection	Status, next, and WorkCard generation reject a projection whose source head or activation differs.	stale_projection	core/policy-set.json
P-025	P0	one-command-one-batch	Every committed batch binds exactly one subject, validated command digest, command intent digest, authorization digest, and idempotency key.	one_command_one_batch	core/policy-set.json
P-026	P1	deterministic-retrieval	Ranking ties are deterministic and acceptance-critical facts are added by typed closure, not fuzzy ranking.	deterministic_retrieval	core/policy-set.json
P-027	P1	strict-time-order	Timestamps use canonical UTC seconds and issued, heartbeat, expiry, finish, and decision order is semantically checked.	strict_time	core/policy-set.json
P-028	P1	exact-standard-files	Core and selected preset inputs reject symlinks and unexpected shadow files.	exact_standard_files	core/policy-set.json
P-029	P0	root-authorization-is-bootstrap-only	Root command authorization signs one command intent and may issue the first authority.manage Grant only; it cannot execute ordinary project commands.	root_authorization_limit	core/policy-set.json
P-030	P1	configuration-exact-idempotency	Repeated initialization succeeds only when the requested canonical configuration digest exactly equals the existing Activation; otherwise it reports conflict.	config_exact_idempotency	core/policy-set.json
P-031	P1	truthful-observability	A command may report a scan, healthcheck, validation, or durability action only when that action was actually executed and evidence-bound.	truthful_observability	core/policy-set.json
P-032	P1	depth-aware-retrieval-budget	Effective retrieval seed count reserves entity and relation capacity for typed closure at the requested dependency depth; top_k is a ceiling, byte/entity/relation/fanout budgets all remain effective, and no removed or unknown budget field may bypass a ceiling.	depth_aware_retrieval	core/policy-set.json
P-033	P1	lease-only-for-mutation	Mutation WorkCards require the current Lease; read-only WorkCards must not acquire or carry a mutation Lease.	lease_only_for_mutation	core/policy-set.json
P-034	P0	command-capability-scope	Every command resolves to exactly one Core-owned capability; requested scope retains every authorization constraint and explicitly contains every command effect target.	command_capability_scope	core/policy-set.json
P-035	P0	grant-issuance-chain	Every Grant proof binds claim_digest and is either the one bootstrap root issuance, a verified team threshold, or an active authority.manage issuer Grant.	grant_issuance_chain	core/policy-set.json
P-036	P0	verified-team-signatures	Team-signature fields are never valid without threshold evaluation by the signature adapter resolved from the exact technologies binding; process-local injected adapters are explicitly non-reconstructable and cannot prove a cross-process CLI workflow.	verified_team_signatures	core/policy-set.json
P-037	P1	provider-preflight-before-activation	Every required provider identity and healthcheck succeeds before initialization is committed; a declared provider is not evidence of availability.	provider_preflight	core/policy-set.json
P-038	P0	canonical-state-machines	Task and Lease transitions are accepted only by the policy-owned state machines; generated lifecycle views cannot invent transitions.	state_machine_transition	core/policy-set.json

ID	Severity	Name	Statement	Validator ID	Owner
P-039	P0	decision-target-binding	Every Decision names one typed target and the decision kind, requested scope, candidate, and authority are consistent with that target.	decision_target_binding	core/policy-set.json
P-040	P0	credit-requires-evidence	Pass, fail, and blocked gate results cite at least one exact immutable Artifact record reference; skipped results never receive pass credit.	credit_requires_evidence	core/policy-set.json
P-041	P0	command-event-correspondence	One command batch binds the exact authorization claim and contains exactly one primary event equal to the validated command payload; auxiliary events are typed relations only and event IDs are unique.	command_event_correspondence	core/policy-set.json
P-042	P0	continuation-authorization-and-completeness	A continuation is authenticated by a random per-Activation secret and binds the exact subject, projection.read Grant claim, capability, requested scope, revocation state, normalized query, deterministic cursor, Activation, HEAD, projection, ranking, complete budgets, depth, implementation closure, issue time, and expiry; its complete page union emits every reachable entity and relation exactly once without loss.	continuation_completeness	core/policy-set.json
P-043	P0	candidate-snapshot-consistency	The candidate recipe is applied in the single inventory pass; promin v1 permits only provider-proven immutable VCS trees for credit, while best-effort observations are explicit and receive no pass credit.	candidate_snapshot_consistency	core/policy-set.json
P-044	P0	provider-adapter-dispatch	Every technologies.json binding resolves to the adapter actually invoked and carries a recomputed capability-specific implementation closure for the promin runtime, Python interpreter, jsonschema or SQLite component where used, and adapter. Unsupported or drifted tuples reject, and dependent evidence binds the exact implementation_closure_digest.	provider_adapter_dispatch	core/policy-set.json
P-045	P0	current-release-closure	A project-operational release Decision is immutable history; current product release eligibility is derived and becomes false after a blocker, failed gate, stale or missing evidence, implementation drift, or Activation drift. It is separate from the external StandardReleaseDecision for promin distribution.	current_release_closure	core/policy-set.json
P-046	P0	finding-disposition-evidence	Every Finding cites exact immutable Artifact record references. Every resolve or waive Decision cites evidence bound to the exact canonical Finding digest; waiver also requires an exception-policy digest, future expiry, and separation of duties.	finding_disposition_evidence	core/policy-set.json
P-047	P0	candidate-delta-scope	A diff Artifact binds base and new Candidate digests, the exact WorkCard digest, and the complete changed-path set; every changed path is contained by both Task allowed_paths and the authorizing Grant effect scope.	candidate_delta_scope	core/policy-set.json
P-048	P1	verified-state-checkpoint	Events remain the sole authority. A normal command appends O(batch), incrementally updates changed projection rows/shards, and advances a verified bounded replay tail. Full derived-state compaction occurs only after declared batch/byte thresholds or explicit recovery, publishes crash-safely against exact HEAD, and missing, stale, or corrupt checkpoints require full verified fallback with equal semantic digest.	verified_state_checkpoint	core/policy-set.json
P-049	P1	profile-default-depth	A profile default_dependency_depth applies only when depth is omitted; explicit depth is accepted through the single Core dependency_depth_hard_max of 12 and remains subject to all retrieval budgets.	profile_default_depth	core/policy-set.json

ID	Severity	Name	Statement	Validator ID	Owner
P-050	P0	canonical-distribution-archive	The sole package builder emits immutable candidate bytes and an external StandardReleaseCandidateBinding. VERSION.json describes SemVer and package identity only; approval is derived later and never changes the candidate ZIP.	canonical_distribution_archive	core/policy-set.json
P-051	P0	verified-inventory-provenance	Public rebuild accepts only a verified InventoryResult or its persisted manifest; caller provenance fields are reserved, data_class is forced to untrusted-source, and runtime injects the exact path and digest.	verified_inventory_provenance	core/policy-set.json
P-052	P1	supported-command-surface	The v1 preset advertises exactly init, doctor, status, next, validate, and continue; no optional or administrative command catalogue may claim an absent executable path.	supported_command_surface	core/policy-set.json
P-053	P0	profile-bound-scale	Physical 100k evidence binds content-addressed raw inventory, query, continuation, process-sample, phase-log, and operation-metric artifacts for at least 198999 explicitly recorded Core-valid typed Relations and a representative mixed workload. The verifier recomputes every summary and fails closed on truthful process/invocation status, commit p95/p99, bytes per changed record, checkpoint count, semantic-ingestion time, search latency, incremental inventory memory amplification, absolute peak RSS, database bytes, projection amplification, semantic inflation, continuation completeness, or traversal-pass ceiling.	profile_bound_scale	core/policy-set.json
P-054	P0	cross-platform-release-evidence	The exact candidate ZIP must pass authenticated Windows and Linux platform observations, clean-install, package, handle-relative path/race, locking, crash, replay, physical scale, offline installed no-degradation, online compatibility, dependency/SBOM/RECORD/license closure, and typed PDF gates. One StandardReleaseEvidenceManifest physically resolves exactly seven unique authority roles, the exact non-authoritative eight-lane CPython 3.13/3.14 online/offline matrix, all four CPython 3.14 supplemental records, and every nested physical record by path, SHA-256, and byte count. All producer attestations bind the same candidate and participate in one replay/SOD/chronology graph; matrix_authoritative, pass_credit, acceptance_pass, product_acceptance_pass, and public approval remain false.	cross_platform_release	core/policy-set.json
P-055	P0	standard-release-separation	An external StandardReleaseDecision has explicit approve or reject outcome, binds candidate and evidence manifest digests, follows the manifest-derived maximum evidence completion time, stays within 300 seconds of verifier clock skew, and verifies decider, standard.distribute capability, independently pinned trust root, Ed25519 provider, event-time key validity, nonce, signed claim digest, and signature. The signed decision remains immutable history; current eligibility is derived; product_acceptance_pass is false and product_public_approval is not_approved.	standard_release_separation	core/policy-set.json

ID	Severity	Name	Statement	Validator ID	Owner
P-056	P0	required-gates-fail-closed	No-degradation runs the complete required suite only through the clean installed interpreter outside the install source, with isolated mode, cleared PYTHONPATH, import-origin assertion, zero skipped tests, and full hash-bound JUnit/stdout/stderr bytes. Distribution acceptance executes every required positive, mutation, concurrency, crash, replay, path, package, platform, Python-minor, offline/online, scale, search, PDF, evidence-resolution, chronology, and signature gate; a skipped, missing, stale, blocked, failed, unsigned, unresolved, source-shadowed, or unpinned gate receives no pass credit.	required_gates_fail_closed	core/policy-set.json
P-057	P0	research-draft-sanitation	Research crosses into normative owners only through the non-persistent MIGRATE_FACTS_NOT_FILES intake: every cited source is digest- and license-plan-bound, citation-required claims cite allowed sources or remain blocked with a reason, and marketing, proof, distribution, acceptance, or public-approval claims cannot become normative text.	research_draft_sanitation	core/policy-set.json
P-058	P0	doctor-result-is-live-non-authority	DoctorResult is an exact live-only derived health report with report_authoritative=false, pass_credit=false, product_acceptance_pass=false, product_public_approval=not_approved, zero product scans, and only its Core-owned operational metrics; token, cost, usage, and goal/session/batch/workspace/per-run allocation are forbidden.	doctor_result_non_authority	core/policy-set.json
P-059	P0	semantic-inheritance-is-structural	The compiled projection preserves the complete SemanticModel owner: admission, five independent granularities, promotion rules, exact entity purposes and admission rules, exact relation domain/range/rules, and structured provider forbidden boundaries.	semantic_inheritance	core/policy-set.json
P-060	P0	authority-owner-is-structural	AuthorityModel is the single owner of the capability catalogue, delegation depth, and exact separation-of-duty pairs; runtime policy may derive from that Activation-bound owner but may not substitute hardcoded authority facts.	authority_owner	core/policy-set.json
P-061	P0	gate-run-definition-is-independent	An immutable GateRunDefinition independently owns the expected evidence class, product-credit requirement, exact target scope, and candidate, policy, tool, provider, and input digests. A run or gate command binds its exact definition digest before submitted Artifacts are evaluated.	gate_run_definition_integrity	core/policy-set.json
P-062	P0	authority-runtime-policy-is-derived-live-only	AuthorityRuntimePolicy is rebuilt from the installed AuthorityModel for every mutation and replay, binds the exact owner digest, ordered capability IDs, structured separation-of-duty rules and delegation ceiling, and is never persisted or imported as authority.	authority_runtime_policy_projection	core/policy-set.json
P-063	P0	semantic-export-is-export-only	SemanticExport is a strict bounded export-only semantic-state record binding the exact Candidate, Activation, HEAD, Core, preset, implementation closure, providers, inputs, and output. It is neither a package nor archive and cannot carry product acceptance, pass credit, public approval, or release approval.	semantic_export_integrity	core/policy-set.json
P-064	P0	operation-metrics-is-live-observations	OperationMetrics is a strict live-only non-authoritative observation for one command. It permits only Core-owned duration, changed-record, write, projection, checkpoint, and physical-byte fields and forbids token, cost, usage, and goal/session/batch/workspace/per-run allocation.	operation_metrics_non_authority	core/policy-set.json

ID	Severity	Name	Statement	Validator ID	Owner
P-065	P0	workcard-projection-is-v2-live-only	RetrievalPage is the strict bounded live-only v2 retrieval result and exclusively owns stream_cursor, next_stream_cursor, and retrieval continuation. WorkCardProjection is internal selected-Task materialization and exposes no continuation or cursor fields.	workcard_projection_v2	core/policy-set.json
P-066	P0	event-store-policy-is-derived-live-only	EventStorePolicy is rebuilt from the installed AuthorityModel before normalization, mutation, and replay. It binds exact command and batch bounds, envelope and commitment fields, lease-bound commands and states, primary-event mapping, and UTC-second timestamps, with no defaults or nullable critical fields.	event_store_policy_projection	core/policy-set.json
P-067	P0	depends-on-graph-is-acyclic	Every command, import, replay, and rebuild rejects a DEPENDS_ON cycle. Relation domain and range validity alone cannot admit a cyclic dependency graph.	dependency_graph_acyclicity	core/policy-set.json
P-068	P0	ready-frontier-is-current-derived-state	ReadyFrontier is derived live-only from the current acyclic DEPENDS_ON graph. A READY Task is returned only after every dependency is currently COMPLETED, every required GateResult is current passing credit, and no current OPEN Finding blocks it; ordering and continuation are deterministic and truncation is explicit.	ready_frontier_projection	core/policy-set.json

Owner: core/policy-set.json | Validator: validator_registry_version=1

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P-029 | root_authorization_limit | core/policy-set.json

P-030 | config_exact_idempotency | core/policy-set.json

P-031 | truthful_observability | core/policy-set.json

P-032 | depth_aware_retrieval | core/policy-set.json

P-033 | lease_only_for_mutation | core/policy-set.json

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P-035 | grant_issuance_chain | core/policy-set.json

P-036 | verified_team_signatures | core/policy-set.json

P-037 | provider_preflight | core/policy-set.json

P-038 | state_machine_transition | core/policy-set.json

P-039 | decision_target_binding | core/policy-set.json

P-040 | credit_requires_evidence | core/policy-set.json

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P-057 | research_draft_sanitation | core/policy-set.json

P-058 | doctor_result_non_authority | core/policy-set.json

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P-061 | gate_run_definition_integrity | core/policy-set.json

P-062 | authority_runtime_policy_projection | core/policy-set.json

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P-066 | event_store_policy_projection | core/policy-set.json

P-067 | dependency_graph_acyclicity | core/policy-set.json

P-068 | readv_frontier_projection | core/policy-set.json

Owner: core/policy-set.json | Validator: policies[].validator_id

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#	Predicate ID	Validator / predicate ID	Owner
1	canonical-name-only	canonical-name-only	core/conformance.json
2	exact-six-core-artifacts	exact-six-core-artifacts	core/conformance.json
3	preset-outside-core-digest	preset-outside-core-digest	core/conformance.json
4	exact-one-selected-preset	exact-one-selected-preset	core/conformance.json
5	exact-five-init-records	exact-five-init-records	core/conformance.json
6	zero-product-scan-during-init	zero-product-scan-during-init	core/conformance.json
7	single-pass-inventory	single-pass-inventory	core/conformance.json
8	no-false-success	no-false-success	core/conformance.json
9	strict-critical-fields	strict-critical-fields	core/conformance.json
10	activation-digest-binding	activation-digest-binding	core/conformance.json
11	root-ceiling-without-bootstrap-grant-cycle	root-ceiling-without-bootstrap-grant-cycle	core/conformance.json
12	grant-scope-expiry-nonce-proof	grant-scope-expiry-nonce-proof	core/conformance.json
13	lease-generation-fence-and-close-ack	lease-generation-fence-and-close-ack	core/conformance.json
14	candidate-evidence-provider-binding	candidate-evidence-provider-binding	core/conformance.json
15	one-command-one-atomic-batch	one-command-one-atomic-batch	core/conformance.json
16	deterministic-replay	deterministic-replay	core/conformance.json
17	projection-disposable-and-current	projection-disposable-and-current	core/conformance.json
18	bounded-workcard-through-depth-twelve	bounded-workcard-through-depth-twelve	core/conformance.json
19	typed-reference-domain-range-closure	typed-reference-domain-range-closure	core/conformance.json
20	dependency-graph-acyclic-and-current-ready-frontier	dependency-graph-acyclic-and-current-ready-frontier	core/conformance.json
21	safe-recursive-export	safe-recursive-export	core/conformance.json
22	idempotent-state-commands	idempotent-state-commands	core/conformance.json
23	normalization-collision-rejection	normalization-collision-rejection	core/conformance.json
24	provider-substitution-invalidates-derived-state	provider-substitution-invalidates-derived-state	core/conformance.json
25	rollback-and-recutover-equivalence	rollback-and-recutover-equivalence	core/conformance.json
26	zero-forbidden-name-occurrences	zero-forbidden-name-occurrences	core/conformance.json
27	zero-unowned-normative-facts	zero-unowned-normative-facts	core/conformance.json
28	zero-core-provider-lock-in	zero-core-provider-lock-in	core/conformance.json
29	zero-compiled-cache-files	zero-compiled-cache-files	core/conformance.json
30	physical-100k-one-artifact-proxy-per-file	physical-100k-one-artifact-proxy-per-file	core/conformance.json
31	external-standard-release-decision-exact-binding	external-standard-release-decision-exact-binding	core/conformance.json
32	root-bootstrap-command-limited	root-bootstrap-command-limited	core/conformance.json
33	deterministic-activation-identity	deterministic-activation-identity	core/conformance.json
34	configuration-exact-idempotency	configuration-exact-idempotency	core/conformance.json
35	truthful-operation-reporting	truthful-operation-reporting	core/conformance.json
36	depth-aware-seed-budget	depth-aware-seed-budget	core/conformance.json
37	lossless-continuation-in-reference-100k-scenario	lossless-continuation-in-reference-100k-scenario	core/conformance.json
38	read-workcard-without-lease	read-workcard-without-lease	core/conformance.json
39	mutation-workcard-requires-current-lease	mutation-workcard-requires-current-lease	core/conformance.json
40	command-capability-and-scope-resolution	command-capability-and-scope-resolution	core/conformance.json
41	grant-issuance-chain-closure	grant-issuance-chain-closure	core/conformance.json
42	team-signatures-cryptographically-verified	team-signatures-cryptographically-verified	core/conformance.json
43	required-provider-preflight-before-commit	required-provider-preflight-before-commit	core/conformance.json

#	Predicate ID	Validator / predicate ID	Owner
44	policy-owned-task-and-lease-state-machines	policy-owned-task-and-lease-state-machines	core/conformance.json
45	decision-kind-target-scope-binding	decision-kind-target-scope-binding	core/conformance.json
46	non-skipped-credit-result-has-evidence	non-skipped-credit-result-has-evidence	core/conformance.json
47	exact-command-primary-event-correspondence	exact-command-primary-event-correspondence	core/conformance.json
48	conjunctive-scope-containment	conjunctive-scope-containment	core/conformance.json
49	authorization-binds-exact-grant-claim	authorization-binds-exact-grant-claim	core/conformance.json
50	command-effect-target-in-requested-scope	command-effect-target-in-requested-scope	core/conformance.json
51	lease-holder-has-task-execute-grant	lease-holder-has-task-execute-grant	core/conformance.json
52	decision-grant-matches-command-authorization	decision-grant-matches-command-authorization	core/conformance.json
53	artifact-kind-resolves-correct-capability	artifact-kind-resolves-correct-capability	core/conformance.json
54	continuation-v2-complete-page-union	continuation-v2-complete-page-union	core/conformance.json
55	candidate-recipe-applied-once	candidate-recipe-applied-once	core/conformance.json
56	creditable-candidate-immutable-vcs-tree	creditable-candidate-immutable-vcs-tree	core/conformance.json
57	selected-provider-adapter-evidence	selected-provider-adapter-evidence	core/conformance.json
58	team-signed-cli-end-to-end	team-signed-cli-end-to-end	core/conformance.json
59	current-release-closure-revalidation	current-release-closure-revalidation	core/conformance.json
60	finding-bound-resolution-and-waiver-evidence	finding-bound-resolution-and-waiver-evidence	core/conformance.json
61	candidate-delta-within-task-and-grant-scope	candidate-delta-within-task-and-grant-scope	core/conformance.json
62	verified-head-checkpoint-delta-replay	verified-head-checkpoint-delta-replay	core/conformance.json
63	profile-default-with-core-depth-twelve	profile-default-with-core-depth-twelve	core/conformance.json
64	canonical-exact-zip-distribution	canonical-exact-zip-distribution	core/conformance.json
65	current-interpreter-and-clean-install-modes	current-interpreter-and-clean-install-modes	core/conformance.json
66	required-no-degradation-tests-fail-closed	required-no-degradation-tests-fail-closed	core/conformance.json
67	continuation-secret-subject-grant-binding	continuation-secret-subject-grant-binding	core/conformance.json
68	implementation-closure-bound-and-current	implementation-closure-bound-and-current	core/conformance.json
69	verified-inventory-result-provenance	verified-inventory-result-provenance	core/conformance.json
70	exact-six-base-command-surface	exact-six-base-command-surface	core/conformance.json
71	zero-dead-evidence-budget-fields	zero-dead-evidence-budget-fields	core/conformance.json
72	profile-bound-physical-100k-performance	profile-bound-physical-100k-performance	core/conformance.json
73	windows-linux-exact-zip-matrix	windows-linux-exact-zip-matrix	core/conformance.json
74	fresh-semantic-control-state-per-saturation-iteration	fresh-semantic-control-state-per-saturation-iteration	core/conformance.json
75	three-zero-new-saturation-iterations	three-zero-new-saturation-iterations	core/conformance.json
76	standard-distribution-separate-from-product-acceptance	standard-distribution-separate-from-product-acceptance	core/conformance.json
77	candidate-evidence-decision-acyclic-chain	candidate-evidence-decision-acyclic-chain	core/conformance.json
78	evidence-manifest-physical-resolution	evidence-manifest-physical-resolution	core/conformance.json
79	evidence-manifest-exact-eight-lane-matrix-resolution	evidence-manifest-exact-eight-lane-matrix-resolution	core/conformance.json
80	evidence-manifest-four-cp314-supplemental-records	evidence-manifest-four-cp314-supplemental-records	core/conformance.json
81	evidence-manifest-path-digest-size-closure	evidence-manifest-path-digest-size-closure	core/conformance.json
82	evidence-manifest-nested-and-supplemental-chronology	evidence-manifest-nested-and-supplemental-chronology	core/conformance.json
83	seven-distinct-role-producer-scopes	seven-distinct-role-producer-scopes	core/conformance.json
84	signed-standard-decision-authority-proof	signed-standard-decision-authority-proof	core/conformance.json
85	semver-from-candidate-binding	semver-from-candidate-binding	core/conformance.json

#	Predicate ID	Validator / predicate ID	Owner
86	typed-human-document-verification	typed-human-document-verification	core/conformance.json
87	broad-query-bounded-seed-refinement	broad-query-bounded-seed-refinement	core/conformance.json
88	continuation-token-bytes-at-most-256	continuation-token-bytes-at-most-256	core/conformance.json
89	inventory-stream-memory-amplification-at-most-32	inventory-stream-memory-amplification-at-most-32	core/conformance.json
90	explicit-scale-marker-selection	explicit-scale-marker-selection	core/conformance.json
91	authenticated-role-platform-evidence-attestations	authenticated-role-platform-evidence-attestations	core/conformance.json
92	installed-platform-observation-closure	installed-platform-observation-closure	core/conformance.json
93	handle-relative-evidence-root-safety	handle-relative-evidence-root-safety	core/conformance.json
94	raw-scale-summary-recomputation	raw-scale-summary-recomputation	core/conformance.json
95	bounded-incremental-commit-and-compaction	bounded-incremental-commit-and-compaction	core/conformance.json
96	offline-installed-no-degradation	offline-installed-no-degradation	core/conformance.json
97	offline-release-online-compatibility	offline-release-online-compatibility	core/conformance.json
98	decision-after-evidence-with-bounded-skew	decision-after-evidence-with-bounded-skew	core/conformance.json
99	single-derived-platform-closure-owner	single-derived-platform-closure-owner	core/conformance.json
100	truthful-process-invocation-status	truthful-process-invocation-status	core/conformance.json
101	product-public-approval-separation	product-public-approval-separation	core/conformance.json
102	historical-decision-current-eligibility-separation	historical-decision-current-eligibility-separation	core/conformance.json

Standard distribution success never grants product acceptance. Live product credit separately requires a creditable immutable-vcs-tree Candidate, all P0/P1 findings closed with exact target-bound evidence, current project release closure eligibility, rollback and recutover equivalence, and an authorized human project Decision. `product_acceptance_pass` remains false for promin v1 distribution evidence.

Owner: `core/conformance.json` | Validator: `required_acceptance`

Mutation catalog - 77 families

#	Mutation family	Mutation probe ID	Owner
1	status-credit-contradiction	status-credit-contradiction	core/conformance.json
2	missing-critical-field	missing-critical-field	core/conformance.json
3	unknown-critical-field	unknown-critical-field	core/conformance.json
4	activation-drift	activation-drift	core/conformance.json
5	grant-replay	grant-replay	core/conformance.json
6	scope-escalation	scope-escalation	core/conformance.json
7	expired-grant	expired-grant	core/conformance.json
8	revoked-grant	revoked-grant	core/conformance.json
9	stale-fence	stale-fence	core/conformance.json
10	candidate-drift	candidate-drift	core/conformance.json
11	provider-substitution	provider-substitution	core/conformance.json
12	event-fork	event-fork	core/conformance.json
13	batch-command-mismatch	batch-command-mismatch	core/conformance.json
14	duplicate-idempotency-key	duplicate-idempotency-key	core/conformance.json
15	projection-corruption	projection-corruption	core/conformance.json
16	projection-staleness	projection-staleness	core/conformance.json
17	reference-break	reference-break	core/conformance.json
18	relation-domain-break	relation-domain-break	core/conformance.json
19	context-flood	context-flood	core/conformance.json
20	symlink-core-input	symlink-core-input	core/conformance.json
21	shadow-core-file	shadow-core-file	core/conformance.json
22	symlink-product-entry	symlink-product-entry	core/conformance.json
23	secret-export	secret-export	core/conformance.json
24	archive-expansion	archive-expansion	core/conformance.json
25	preset-authority-escalation	preset-authority-escalation	core/conformance.json
26	normalization-key-collision	normalization-key-collision	core/conformance.json
27	path-casefold-collision	path-casefold-collision	core/conformance.json
28	forbidden-name-reintroduction	forbidden-name-reintroduction	core/conformance.json
29	reinit-configuration-conflict	reinit-configuration-conflict	core/conformance.json
30	clock-order-violation	clock-order-violation	core/conformance.json
31	capability-confusion	capability-confusion	core/conformance.json
32	command-scope-escalation	command-scope-escalation	core/conformance.json
33	grant-issuer-chain-break	grant-issuer-chain-break	core/conformance.json
34	unverified-signature-acceptance	unverified-signature-acceptance	core/conformance.json
35	provider-declared-but-unavailable	provider-declared-but-unavailable	core/conformance.json
36	illegal-task-transition	illegal-task-transition	core/conformance.json
37	illegal-lease-transition	illegal-lease-transition	core/conformance.json
38	decision-target-confusion	decision-target-confusion	core/conformance.json
39	credit-without-evidence	credit-without-evidence	core/conformance.json
40	command-primary-event-mismatch	command-primary-event-mismatch	core/conformance.json
41	duplicate-event-id	duplicate-event-id	core/conformance.json
42	scope-selector-drop-broadening	scope-selector-drop-broadening	core/conformance.json
43	authorization-grant-claim-substitution	authorization-grant-claim-substitution	core/conformance.json
44	command-effect-outside-requested-scope	command-effect-outside-requested-scope	core/conformance.json
45	lease-holder-grant-mismatch	lease-holder-grant-mismatch	core/conformance.json
46	decision-grant-provenance-mismatch	decision-grant-provenance-mismatch	core/conformance.json

#	Mutation family	Mutation probe ID	Owner
47	artifact-kind-capability-confusion	artifact-kind-capability-confusion	core/conformance.json
48	continuation-frontier-loss	continuation-frontier-loss	core/conformance.json
49	continuation-duplicate-emission	continuation-duplicate-emission	core/conformance.json
50	candidate-recipe-bypass	candidate-recipe-bypass	core/conformance.json
51	observational-candidate-pass-credit	observational-candidate-pass-credit	core/conformance.json
52	snapshot-provider-substitution	snapshot-provider-substitution	core/conformance.json
53	unsupported-provider-adapter	unsupported-provider-adapter	core/conformance.json
54	stale-release-closure-credit	stale-release-closure-credit	core/conformance.json
55	finding-evidence-target-mismatch	finding-evidence-target-mismatch	core/conformance.json
56	candidate-delta-scope-escape	candidate-delta-scope-escape	core/conformance.json
57	checkpoint-head-mismatch	checkpoint-head-mismatch	core/conformance.json
58	checkpoint-corruption-credit	checkpoint-corruption-credit	core/conformance.json
59	normal-command-full-replay	normal-command-full-replay	core/conformance.json
60	profile-default-depth-bypass	profile-default-depth-bypass	core/conformance.json
61	continuation-public-material-mismatch	continuation-public-material-mismatch	core/conformance.json
62	inventory-provenance-override	inventory-provenance-override	core/conformance.json
63	implementation-closure-drift	implementation-closure-drift	core/conformance.json
64	noncanonical-archive-layout	noncanonical-archive-layout	core/conformance.json
65	inventory-graph-inflation	inventory-graph-inflation	core/conformance.json
66	workcard-budget-bypass	workcard-budget-bypass	core/conformance.json
67	fabricated-evidence-digest	fabricated-evidence-digest	core/conformance.json
68	unsigned-standard-decision	unsigned-standard-decision	core/conformance.json
69	unconfigured-standard-decision-key	unconfigured-standard-decision-key	core/conformance.json
70	standard-decision-version-replay	standard-decision-version-replay	core/conformance.json
71	candidate-decision-cycle	candidate-decision-cycle	core/conformance.json
72	broad-query-corpus-pagination	broad-query-corpus-pagination	core/conformance.json
73	continuation-token-context-inflation	continuation-token-context-inflation	core/conformance.json
74	inventory-stream-materialization	inventory-stream-materialization	core/conformance.json
75	pdf-header-only-credit	pdf-header-only-credit	core/conformance.json
76	scale-environment-skip-credit	scale-environment-skip-credit	core/conformance.json
77	content-search-omission	content-search-omission	core/conformance.json

Owner: core/conformance.json | Validator: mutation_families

State machines

Task

State	Allowed next state
BLOCKED	READY, CANCELLED
CANCELLED	-
COMPLETED	-
LEASED	RUNNING, READY, BLOCKED, CANCELLED
PLANNED	READY, BLOCKED, CANCELLED
READY	LEASED, BLOCKED, CANCELLED
RUNNING	COMPLETED, BLOCKED, CANCELLED

Owner: core/policy-set.json | Validator: state_machines.task + state_machine_transition

Lease

State	Allowed next state
ACTIVE	CLOSING, EXPIRED, REVOKED
CLOSED	-
CLOSING	CLOSED, EXPIRED, REVOKED
EXPIRED	-
REVOKED	-

Owner: *core/policy-set.json* | Validator: *state_machines.lease + state_machine_transition*

```
{'closed_release': {'requires_field': 'close_ack', 'slot_reusable': True, 'state': 'CLOSED'}, '
terminal_reconciliation': {'fencing_token_binding': 'exact-lease-fencing-token', 'field': '
capacity_reconciliation', 'generation_binding': 'exact-lease-generation', 'manager_capability': '
lease.manage', 'required_fields': ['reconciled_by', 'grant_id', 'grant_claim_digest', 'reconciled_at', '
generation', 'fencing_token'], 'slot_reusable_after_record': True, 'states': ['EXPIRED', 'REVOKED']}, '
unreconciled_slot_reusable': False}
```

Owner: *core/policy-set.json* | Validator: *state_machines.capacity_release*

Capabilities and command authorization

Capability	Purpose	Owner	Validator
standard.activate	Activate one digest-bound repository configuration.	core/authority-model.json	command_capability_scope
authority.manage	Issue or revoke runtime Grants within a configured root ceiling.	core/authority-model.json	command_capability_scope
task.plan	Create, split, order, or block Tasks.	core/authority-model.json	command_capability_scope
task.execute	Execute one Task inside its WorkCard and valid Lease.	core/authority-model.json	command_capability_scope
lease.manage	Acquire, heartbeat, close, expire, or revoke Leases.	core/authority-model.json	command_capability_scope
evidence.publish	Publish candidate-bound evidence Artifacts.	core/authority-model.json	command_capability_scope
validation.evaluate	Evaluate gates without promotion or release authority.	core/authority-model.json	command_capability_scope
finding.record	Record Findings.	core/authority-model.json	command_capability_scope
finding.resolve	Resolve one Finding only with evidence bound to that Finding's canonical digest.	core/authority-model.json	command_capability_scope
finding.waive	Waive one Finding only through an expiring exception-policy Decision, Finding-bound evidence, and separation of duties.	core/authority-model.json	command_capability_scope
candidate.promote	Promote a validated Candidate.	core/authority-model.json	command_capability_scope
release.decide	Record an immutable historical release judgement bound to an exact acceptance-closure and provider-binding digest; current eligibility remains derived.	core/authority-model.json	command_capability_scope
export.create	Create a sanitized external package.	core/authority-model.json	command_capability_scope
migration.manage	Execute cutover, rollback, and deterministic recutover.	core/authority-model.json	command_capability_scope
projection.rebuild	Rebuild or verify non-authoritative projections.	core/authority-model.json	command_capability_scope
projection.read	Read one bounded current projection or resume its continuation under the exact current Grant claim; it grants no mutation, validation, promotion, product acceptance, or standard distribution authority.	core/authority-model.json	command_capability_scope

Command to capability resolution

Command kind	Capability resolution	Owner	Validator
task.record	task.plan	core/authority-model.json	command_capability_scope
task.transition	task.plan	core/authority-model.json	command_capability_scope
candidate.record	task.execute	core/authority-model.json	command_capability_scope
artifact.record	{ "capability_by_value": { "diff": "task.execute", "evidence": "evidence.publish", "log": "task.execute", "product": "task.execute", "report": "projection.rebuild", "payload_field": "artifact_kind" }	core/authority-model.json	command_capability_scope
run.record	{ "capability_by_value": { "execution": "task.execute", "validation": "validation.evaluate", "payload_field": "run_kind" }	core/authority-model.json	command_capability_scope
gate.record	validation.evaluate	core/authority-model.json	command_capability_scope
finding.record	finding.record	core/authority-model.json	command_capability_scope
decision.record	{ "capability_by_value": { "accept": "validation.evaluate", "promote": "candidate.promote", "reject": "validation.evaluate", "release": "release.decide", "resolve": "finding.resolve", "revoke": "authority.manage", "waive": "finding.waive", "payload_field": "decision_kind" }	core/authority-model.json	command_capability_scope
grant.issue	authority.manage	core/authority-model.json	command_capability_scope
grant.revoke	authority.manage	core/authority-model.json	command_capability_scope
lease.record	lease.manage	core/authority-model.json	command_capability_scope

Command effect scope

Command kind	Mode	Target rule	Owner	Validator
task.record	payload-id	{ "id_field": "task_id", "kind": "task" }	core/authority-model.json	command_capability_scope
task.transition	payload-id	{ "id_field": "task_id", "kind": "task" }	core/authority-model.json	command_capability_scope
candidate.record	payload-id	{ "id_field": "candidate_id", "kind": "candidate" }	core/authority-model.json	command_capability_scope
artifact.record	payload-id	{ "id_field": "artifact_id", "kind": "artifact" }	core/authority-model.json	command_capability_scope
run.record	payload-multi-id	{ "targets": [{ "id_field": "task_id", "kind": "task" }, { "id_field": "candidate_digest", "kind": "candidate" }] }	core/authority-model.json	command_capability_scope
gate.record	payload-id	{ "id_field": "candidate_digest", "kind": "candidate" }	core/authority-model.json	command_capability_scope
finding.record	payload-id	{ "id_field": "finding_id", "kind": "finding" }	core/authority-model.json	command_capability_scope
decision.record	payload-dynamic-id-or-referenced-grant-scope	{ "grant_target_type": "Grant", "id_field": "target_id", "kind_field": "target_type", "kind_map": { "Candidate": "candidate", "Finding": "finding", "Task": "task" } }	core/authority-model.json	command_capability_scope
grant.issue	payload-scope	{ "scope_field": "scope" }	core/authority-model.json	command_capability_scope
grant.revoke	referenced-grant-scope	{ "grant_id_field": "grant_id" }	core/authority-model.json	command_capability_scope
lease.record	payload-id	{ "id_field": "task_id", "kind": "task" }	core/authority-model.json	command_capability_scope

Lease-bound mutation claim

A lease-bound command is admissible only when authorization proves the exact command capability, holder_authorization independently proves task.execute, both exact Grant claims bind the same subject, Activation and conjunctive effect scope, the referenced Lease is current and ACTIVE at issued_at, and task, generation, fence, candidate, context, and WorkCard digests match exactly.

Claim component	Canonical value	Owner	Validator
command_authorization_source	CommandRequest.authorization.grant_id + grant_claim_digest	core/authority-model.js on	lease_fence + command_capability_scope
grant_separation_rule	The command authorization Grant and holder Grant are resolved and validated independently; they may be the same record only when the required command capability is task.execute.	core/authority-model.js on	lease_fence + command_capability_scope
holder_grant_capability	task.execute	core/authority-model.js on	lease_fence + command_capability_scope
holder_grant_field	holder_authorization.grant_id	core/authority-model.js on	lease_fence + command_capability_scope
lease_bound_command_kinds	["artifact.record", "run.record", "gate.record", "finding.record"]	core/authority-model.js on	lease_fence + command_capability_scope
lease_bound_task_transition_states	["LEASED", "RUNNING", "COMPLETED"]	core/authority-model.js on	lease_fence + command_capability_scope
lease_required_state	ACTIVE	core/authority-model.js on	lease_fence + command_capability_scope
required_command_fields	["workcard_task_id", "holder_authorization", "lease_id", "lease_generation", "fencing_token", "workcard_digest", "context_digest"]	core/authority-model.js on	lease_fence + command_capability_scope
required_workcard_fields	["holder_grant_id", "lease_id", "lease_generation", "fencing_token", "context_digest"]	core/authority-model.js on	lease_fence + command_capability_scope
stale_or_unresolved	reject	core/authority-model.js on	lease_fence + command_capability_scope
Binding equality		Owner	Validator
CommandRequest.workcard_task_id == WorkCard.task_id == Lease.task_id		core/authority-model.json	lease_fence + command_capability_scope
CommandRequest.subject_id == Lease.holder_subject_id		core/authority-model.json	lease_fence + command_capability_scope
CommandRequest.holder_authorization.grant_id == WorkCard.holder_grant_id == Lease.holder_grant_id		core/authority-model.json	lease_fence + command_capability_scope
CommandRequest.lease_id == WorkCard.lease_id == Lease.lease_id		core/authority-model.json	lease_fence + command_capability_scope
CommandRequest.lease_generation == WorkCard.lease_generation == Lease.generation		core/authority-model.json	lease_fence + command_capability_scope
CommandRequest.fencing_token == WorkCard.fencing_token == Lease.fencing_token		core/authority-model.json	lease_fence + command_capability_scope
CommandRequest.context_digest == WorkCard.context_digest		core/authority-model.json	lease_fence + command_capability_scope
CommandRequest.workcard_digest == sha256(canonical WorkCard)		core/authority-model.json	lease_fence + command_capability_scope
<pre>{ "binding_equalities": ["CommandRequest.workcard_task_id == WorkCard.task_id == Lease.task_id",</pre>			

```

"CommandRequest.subject_id == Lease.holder_subject_id",
"CommandRequest.holder_authorization.grant_id == WorkCard.holder_grant_id == Lease.holder_grant_id",
"CommandRequest.lease_id == WorkCard.lease_id == Lease.lease_id",
"CommandRequest.lease_generation == WorkCard.lease_generation == Lease.generation",
"CommandRequest.fencing_token == WorkCard.fencing_token == Lease.fencing_token",
"CommandRequest.context_digest == WorkCard.context_digest",
"CommandRequest.workcard_digest == sha256(canonical WorkCard)"
],
"claim_rule": "A lease-bound command is admissible only when authorization proves the exact command
capability, holder_authorization independently proves task.execute, both exact Grant claims bind the same
subject, Activation and conjunctive effect scope, the referenced Lease is current and ACTIVE at issued_at,
and task, generation, fence, candidate, context, and WorkCard digests match exactly.",
"command_authorization_source": "CommandRequest.authorization.grant_id + grant_claim_digest",
"grant_separation_rule": "The command authorization Grant and holder Grant are resolved and validated
independently; they may be the same record only when the required command capability is task.execute.",
"holder_grant_capability": "task.execute",
"holder_grant_field": "holder_authorization.grant_id",
"lease_bound_command_kinds": [
  "artifact.record",
  "run.record",
  "gate.record",
  "finding.record"
],
"lease_bound_task_transition_states": [
  "LEASED",
  "RUNNING",
  "COMPLETED"
],
"lease_required_state": "ACTIVE",

"required_command_fields": [
  "workcard_task_id",
  "holder_authorization",
  "lease_id",
  "lease_generation",
  "fencing_token",
  "workcard_digest",
  "context_digest"
],
"required_workcard_fields": [
  "holder_grant_id",
  "lease_id",
  "lease_generation",
  "fencing_token",
  "context_digest"
],
"stale_or_unresolved": "reject"
}

```

Owner: core/authority-model.json; core/policy-set.json | Validator: command_mutation_claim_rule + lease_fence + command_capability_scope

Replay-complete Evidence and CAS

Evidence field	Rule	Schema
Artifact.required	["record_type", "artifact_id", "artifact_kind", "digest", "media_type", "size_bytes", "retention_class", "created_at"]	Artifact
EvidenceBinding.required	["activation_digest", "implementation_closure_digest", "candidate_digest", "policy_digest", "tool_digest", "input_digests"]	EvidenceBinding
outcome	["pass", "fail", "blocked", "skipped", "error"]	Artifact
evidence_class	["product-execution", "validator", "harness-generated", "migration"]	Artifact
product_credit_eligible	true only for fresh resolved passing product-execution evidence	Artifact.allOf
GateResult.evidence_artifacts	exact artifact_id + artifact_record_digest for pass, fail, or blocked	GateResult.oneOf

Owner: core/contracts.schema.json; core/semantic-model.json; core/policy-set.json | Validator: \$defs/Artifact + \$defs/EvidenceBinding + \$defs/GateResult + candidate_evidence_binding + credit_requires_evidence

Separation of duties

ID	Rule	Constraint	Owner	Validator
SOD-001	lease-is-not-acceptance	{ "forbid_lease_derived_capabilities": ["validation.evaluate", "finding.resolve", "finding.waive", "candidate.promote", "release.decide"], "scope_kind": "lease" }	core/authority-model.json	SOD-001
SOD-002	executor-not-release-decider	{ "forbid_same_subject_for_same_candidate": ["task.execute", "release.decide"], "scope_kind": "candidate" }	core/authority-model.json	SOD-002
SOD-003	validator-not-self-promoter	{ "forbid_same_subject_for_same_candidate": ["validation.evaluate", "candidate.promote"], "scope_kind": "candidate" }	core/authority-model.json	SOD-003
SOD-004	finder-not-sole-waiver	{ "forbid_same_subject_for_same_finding": ["finding.record", "finding.waive"], "scope_kind": "finding" }	core/authority-model.json	SOD-004
SOD-005	evidence-producer-not-distribution-decider	{ "forbid_same_key_capabilities": ["evidence.produce", "standard.distribute"], "forbid_same_subject_for_same_candidate": ["evidence.produce", "standard.distribute"], "scope_kind": "candidate-and-key" }	core/authority-model.json	SOD-005

Owner: core/authority-model.json | Validator: capabilities + command_capability_rules + command_effect_scope_rules + separation_of_duties

Trust, bootstrap, and role presets

authority.json configures root ceilings and Activation only. Root command proofs bind one command intent and may issue only the first authority.manage Grant. Every Grant proof binds claim_digest; later Grants cite an active authority.manage issuer Grant or team signatures verified by the exact signature adapter selected in technologies.json.

Trust mode	Definition	Owner	Validator
local-owner	{ "boundary": "single-machine owner-controlled root authority", "signer_required": false }	core/authority-model.json	grant_validity
team-signed	{ "boundary": "multiple actors or remote transport", "minimum_threshold": 1, "provider_capability": "signature", "provider_resolution": "Resolve the exact technologies.json signature binding for init and every ActivationGuard, service, CLI, replay, and mutation path. A process-local injected adapter is explicitly non-reconstructable and valid only while bound to the same project, Activation, provider, and identity; cross-process CLI workflows require a reconstructable configured adapter.", "signer_required": true, "unsupported_binding": "reject-before-workflow" }	core/authority-model.json	grant_validity

Role preset	Capabilities	Authority status	Owner
auditor	projection.read, finding.record, finding.resolve, export.create	non-authoritative	core/authority-model.json
root-owner	standard.activate, authority.manage, task.plan, lease.manage, finding.resolve, finding.waive, candidate.promote, release.decide, export.create, migration.manage, projection.rebuild, projection.read	non-authoritative	core/authority-model.json
validator	projection.read, validation.evaluate, evidence.publish, finding.record	non-authoritative	core/authority-model.json
worker	projection.read, task.execute, evidence.publish	non-authoritative	core/authority-model.json

Owner: core/authority-model.json | Validator: bootstrap_rule + trust_modes + role_presets_non_authoritative

Semantic vocabulary

Entity	Purpose		Owner	Validator
Task	Represent one schedulable unit with one acceptance predicate and bounded mutation scope.		core/semantic-model.json	admission.entity
Candidate	Identify either one provider-proven immutable product snapshot or one explicitly non-creditable observation, independently from control state and bound to the exact candidate recipe.		core/semantic-model.json	admission.entity
Artifact	Identify immutable product, evidence, log, report, or diff bytes with provenance and retention metadata; raw-file inventory produces only derived Artifact proxies, while a diff may carry the canonical Candidate-delta change manifest without creating another persistent entity.		core/semantic-model.json	admission.entity
Run	Record one execution or validation attempt against exact candidate, policy, tool, and input digests.		core/semantic-model.json	admission.entity
Finding	Record an observed defect or policy violation with explicit effective-blocking semantics.		core/semantic-model.json	admission.entity
Decision	Record a project-operational acceptance, rejection, resolution, waiver, promotion, or release judgement. It is distinct from the external StandardReleaseDecision for the standard distribution.		core/semantic-model.json	admission.entity
GateResult	Record one gate outcome against the exact immutable Task-owned GateRunDefinition, current Run, and finalized evidence Artifact records; skipped, blocked, failed, stale, or unresolved evidence never receives pass credit.		core/semantic-model.json	admission.entity
Grant	Prove one subject capability within one activation, time interval, nonce, and scope.		core/semantic-model.json	admission.entity
Lease	Coordinate temporary mutation ownership through generation, fencing, expiry, heartbeat, and close acknowledgement.		core/semantic-model.json	admission.entity
Relation	Source	Target	Rules	Validator
DEPENDS_ON	Task	Task	acyclic-for-ready-frontier, accepted-current-outcome-required	admission.relation
READS	Task, Run	Candidate, Artifact	read-does-not-imply-authority	admission.relation
PRODUCES	Task, Run	Candidate, Artifact, Finding, Decision	content-or-record-identity-required	admission.relation
VALIDATES	Run	Candidate, Artifact, Task	candidate-policy-tool-input-binding-required	admission.relation
CITES	Finding, Decision	Artifact, Run, Finding	selector-and-content-resolution-required	admission.relation
BLOCKS	Finding	Task, Candidate, Decision	effective-only-while-open-and-blocking	admission.relation
Value object		Meaning	Owner	Validator
CandidateDelta		A diff Artifact payload binding base and new Candidate digests, WorkCard digest, and the complete normalized changed-path set.	core/semantic-model.json	design_rules
Command		A state-changing request validated before commit.	core/semantic-model.json	design_rules
Comparator		A pure ordering or equality rule.	core/semantic-model.json	design_rules
ContinuationToken		An authenticated bounded runtime value signed with the random per-Activation secret. It binds subject, exact projection.read Grant claim, requested scope, revocation state, normalized query, cursor, Activation, HEAD, projection, ranking, complete budgets, depth, implementation closure, issue time, and expiry; it never grants authority by itself.	core/semantic-model.json	design_rules
CurrentReleaseStatus		A derived evaluation of an immutable historical release Decision against the current closure, blockers, evidence, providers, and Activation.	core/semantic-model.json	design_rules

Value object	Meaning	Owner	Validator
ImplementationClosure	A canonical capability-specific binding of the promin runtime, Python interpreter, jsonschema or SQLite component where used, exactly one adapter, and the recomputed aggregate closure digest.	core/semantic-model.json	design_rules
Operation	One completed semantic step with explicit input, output, failure, and evidence semantics.	core/semantic-model.json	design_rules
PlanProposal	A non-authoritative bounded proposal that derives candidate Tasks and relations from project intent, repository facts, backlog, and references; it cannot mutate operational state or grant authority.	core/semantic-model.json	design_rules
Policy	A machine-addressable invariant owned by policy-set.json.	core/semantic-model.json	design_rules
Predicate	A pure decision rule.	core/semantic-model.json	design_rules
ProviderInvocationEvidence	The exact capability, provider, invocation kind, identity kind and digest, and adapter ID actually used by one operation.	core/semantic-model.json	design_rules
Query	A read-only request that cannot mutate authoritative state.	core/semantic-model.json	design_rules
StandardReleaseDecision	An external human-reviewable distribution record bound to the exact promin ZIP, Core, preset, tools, implementation closure, tests, platform results, saturation, PDFs, and evidence manifest. It never grants product acceptance.	core/semantic-model.json	design_rules
StateCheckpoint	A disposable self-digested state or journal index bound to an exact authoritative HEAD; it cannot grant authority or acceptance.	core/semantic-model.json	design_rules
WorkCard	A bounded generated context package for one Task that separately binds the projection.read query Grant claim and the task.execute holder Grant; neither binding implies acceptance.	core/semantic-model.json	design_rules

Derived projection	Authority	Owner	Validator
InventoryEntry	derived and non-authoritative	core/semantic-model.json	derived_projection_kinds
SearchDocument	derived and non-authoritative	core/semantic-model.json	derived_projection_kinds
InventoryProjectionRow	derived and non-authoritative	core/semantic-model.json	derived_projection_kinds
ProjectionLimits	derived and non-authoritative	core/semantic-model.json	derived_projection_kinds
RetrievalPage	derived and non-authoritative	core/semantic-model.json	derived_projection_kinds
ReadyFrontier	derived and non-authoritative	core/semantic-model.json	derived_projection_kinds
WorkCardProjection	derived and non-authoritative	core/semantic-model.json	derived_projection_kinds
WorkCard	derived and non-authoritative	core/semantic-model.json	derived_projection_kinds
NextResult	derived and non-authoritative	core/semantic-model.json	derived_projection_kinds
ContinueResult	derived and non-authoritative	core/semantic-model.json	derived_projection_kinds
NavigationView	derived and non-authoritative	core/semantic-model.json	derived_projection_kinds
StateCheckpoint	derived and non-authoritative	core/semantic-model.json	derived_projection_kinds
CurrentReleaseStatus	derived and non-authoritative	core/semantic-model.json	derived_projection_kinds
DoctorResult	derived and non-authoritative	core/semantic-model.json	derived_projection_kinds
OperationMetrics	derived and non-authoritative	core/semantic-model.json	derived_projection_kinds
Report	derived and non-authoritative	core/semantic-model.json	derived_projection_kinds

Owner: core/semantic-model.json | Validator: admission + design_rules + persistent_entities + relations + value_objects + derived_projection_kinds

Technology boundaries

Capability	Task	Forbidden boundary	Owner	Validator
control-runtime	Parse explicit configuration, canonicalize JSON, and run local CLI orchestration.	{"effect": "reject", "forbids": ["authority-invention", "policy-invention", "provider-binding-invention"], "normative_authority": false}	core/semantic-model.json	technology_capabilities
shape-validation	Validate the single Draft 2020-12 contract bundle with enforced formats and local references.	{"effect": "reject", "forbids": ["authority-decision", "cross-record-semantic-decision"], "normative_authority": false}	core/semantic-model.json	technology_capabilities
content-identity	Compute content digests and content-addressed object identities.	{"effect": "reject", "forbids": ["digest-as-action-authority"], "normative_authority": false}	core/semantic-model.json	technology_capabilities
local-serialization	Provide one local-writer lock, atomic replace, file sync, directory sync, and recovery primitives.	{"effect": "reject", "forbids": ["distributed-consensus-claim"], "normative_authority": false}	core/semantic-model.json	technology_capabilities
query-projection	Build disposable status and bounded typed-retrieval projections; derive ReadyFrontier, materialize one exact WorkCardProjection, and return Core-owned NextResult or ContinueResult envelopes for the six supported CLI commands.	{"effect": "reject", "forbids": ["projection-as-normative-authority"], "normative_authority": false}	core/semantic-model.json	technology_capabilities
filesystem-inventory	Apply the exact candidate recipe in one product pass, use no-follow reads, and publish either a provider-proven immutable VCS tree stream or an explicit non-creditable observation.	{"effect": "reject", "forbids": ["implicit-initialization-scan"], "normative_authority": false}	core/semantic-model.json	technology_capabilities
export-scan	Perform bounded recursive export inspection, allowlisting, secret classification, and symlink rejection.	{"effect": "reject", "forbids": ["package-as-authorization-boundary"], "normative_authority": false}	core/semantic-model.json	technology_capabilities
signature	Dispatch the exact configured signature adapter to sign or verify activation, Grant, and Decision digests for team trust mode, and evidence the invoked adapter identity.	{"effect": "reject", "forbids": ["core-signer-implementation-lock-in"], "normative_authority": false}	core/semantic-model.json	technology_capabilities
build-dependency	Import project-specific build and dependency facts as derived projections.	{"effect": "reject", "forbids": ["semantic-model-redefinition"], "normative_authority": false}	core/semantic-model.json	technology_capabilities

Owner: core/semantic-model.json | Validator: technology_capabilities

Conformance budgets and scale

The physical lane uses 100,000 raw files and exactly one derived Artifact proxy per file. Raw inventory creates no Task or Relation semantics. An explicit authorized harness corpus brings the projection to at least 198,999 Core-valid typed Relations. At least 600 actual runtime queries mix exact identities, content-only and high-cardinality terms, broad bounded refinement, misses, hostile identity text, and forced continuation across depths 1-12. This bounded physical proof is not a full 1000 by 1000 execution and must not be replaced by one.

Structural budget	Value	Owner	Validator
authority_bearing_init_records	5	core/conformance.json	authority_bearing_init_records
base_user_commands_max	10	core/conformance.json	base_user_commands_max
core_normative_artifacts	6	core/conformance.json	core_normative_artifacts
persistent_entity_kinds_max	9	core/conformance.json	persistent_entity_kinds_max
relation_kinds_max	6	core/conformance.json	relation_kinds_max

Structural budget	Value	Owner	Validator
resolved_plan_bytes_max	8192	core/conformance.json	resolved_plan_bytes_max
selected_presets	1	core/conformance.json	selected_presets
resolved_plan_source_samples_max	64	core/conformance.json	resolved_plan_source_samples_max
WorkCard ceiling	Value	Owner	Validator
max_bytes	16384	core/conformance.json	bounded-workcard-through-depth-twelve
max_entities	32	core/conformance.json	bounded-workcard-through-depth-twelve
max_fanout_per_entity	8	core/conformance.json	bounded-workcard-through-depth-twelve
max_relations	48	core/conformance.json	bounded-workcard-through-depth-twelve
top_k	12	core/conformance.json	bounded-workcard-through-depth-twelve
Scale contract	Definition	Owner	Validator
event_commit	{"complexity": "O(batch)", "head_lookup": "O(1)"}	core/conformance.json	event_commit
init	{"complexity": "O(core+plan)", "project_tree_passes": 0}	core/conformance.json	init
inventory	{"complexity": "O(files+bytes)", "creditable_consistency_modes": ["immutable-vcs-tree"], "derived_artifact_proxies_per_raw_file": 1, "manifest_verified_jsonl": true, "memory_amplification_max": 32, "observational_mode_pass_credit": false, "project_tree_passes_max": 1, "search_text_bytes_max": 4096, "synthetic_relations_per_raw_file": 0, "synthetic_tasks_per_raw_file": 0}	core/conformance.json	inventory
normal_one_shot_replay	{"checkpoint_authoritative": false, "complexity": "O(delta-since-verified-checkpoint)", "full_replay_allowed": "doctor, validate, recovery, or verified fallback only"}	core/conformance.json	normal_one_shot_replay
projection	{"public_rebuild_ingress": "verified InventoryResult or persisted verified manifest only", "rebuild_product_tree_passes": 0, "reserved_provenance_rejected": true}	core/conformance.json	projection
replay	{"complexity": "O(events)"}	core/conformance.json	replay
saturation	{"fresh_control_state_per_iteration": true, "init_product_tree_scans": 0, "semantic_state_reused": false}	core/conformance.json	saturation
workcard	{"broad_query_behavior": "deterministic bounded top-k seeds, refinement_required=true, no corpus pagination", "complexity": "O(top_k+capped_typed_closure)", "continuation_state_bytes_max": 16384, "continuation_token_bytes_max": 256, "corpus_independent_output_bound": true, "seed_budget": "effective_top_k=min(profile.top_k,max(2,floor(max_entities/(1+observed_branching*requested_depth))))", "selected_closure_union_completeness": 1}	core/conformance.json	workcard
Reference benchmark	Value	Owner	Validator
core_valid_relation_count	198999	core/conformance.json	reference-benchmark
depths_tested	[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]	core/conformance.json	reference-benchmark
file_count	100000	core/conformance.json	reference-benchmark
forced_continuation	True	core/conformance.json	reference-benchmark
inventory_passes	1	core/conformance.json	reference-benchmark

Reference benchmark	Value	Owner	Validator
performance_profiles	{ "portable-local-v1": { "commit_bytes_per_changed_record_max": 32768, "commit_p95_ms_max": 500, "commit_p99_ms_max": 1000, "compatible_platforms": ["linux", "windows"], "database_bytes_max": 402653184, "p50_ms_max": 100, "p95_ms_max": 150, "p99_ms_max": 500, "peak_rss_bytes_max": 805306368, "projection_amplification_max": 20, "requires_same_runner_no_degradation": true, "runtime_checkpoint_count_max": 24, "semantic_inflation_max": 3.25, "semantic_ingestion_seconds_max": 600 } }	core/conformance.json	reference-benchmark
physical_relation_corpus	{ "artifact_target_coverage": 1, "expected_counts_source": "fixture-manifest", "generation": "explicit-authorized-command-events", "relation_count": 198971, "relation_kind": "READS", "separate_from_raw_inventory": true, "task_count": 1567, "total_core_valid_relations": 198999 }	core/conformance.json	reference-benchmark
query_count	600	core/conformance.json	reference-benchmark
raw_file_proxy_ratio	1	core/conformance.json	reference-benchmark
rebuild_product_passes	0	core/conformance.json	reference-benchmark
reference_targets_are_environment_specific	True	core/conformance.json	reference-benchmark
semantic_corpus	{ "depths": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12], "expected_counts_source": "fixture-manifest", "relation_count": 28, "relation_kind": "DEPENDS_ON", "separate_from_raw_inventory": true, "task_count": 32 }	core/conformance.json	reference-benchmark
silent_truncations_max	0	core/conformance.json	reference-benchmark
synthetic_task_ratio	0	core/conformance.json	reference-benchmark
Owner: core/conformance.json Validator: structural_budgets + workcard_hard_ceiling + scale_contracts + reference_benchmarks			

Preset and profiles

Select Semantic Programming plus Morok/Tower decomposition, retrieval, and parallelism budgets without granting action authority.

Profile	Budgets	Owner	Validator
morok-local	{ "default_dependency_depth": 2, "max_context_bytes": 8192, "max_entities": 16, "max_fanout_per_entity": 4, "max_parallel_tasks": 1, "max_relations": 24, "model_tier": "weak-local", "top_k": 8 }	presets/semantic-morok-tower.json	profiles

Profile	Budgets	Owner	Validator
tower-capable	{"default_dependency_depth": 4, "max_context_bytes": 12288, "max_entities": 24, "max_fanout_per_entity": 6, "max_parallel_tasks": 3, "max_relations": 36, "model_tier": "capable", "top_k": 10}	presets/semantic-morok-tower.json	profiles
tower-strong	{"default_dependency_depth": 6, "max_context_bytes": 16384, "max_entities": 32, "max_fanout_per_entity": 8, "max_parallel_tasks": 6, "max_relations": 48, "model_tier": "strong", "top_k": 12}	presets/semantic-morok-tower.json	profiles
Preset field	Value	Owner	Validator
base_user_commands	["init", "doctor", "status", "next", "validate", "continue", "audit", "refresh", "context", "skills"]	presets/semantic-morok-tower.json	base_user_commands
required_provider_capabilities	["control-runtime", "shape-validation", "content-identity", "local-serialization", "query-projection"]	presets/semantic-morok-tower.json	required_provider_capabilities
forbidden_implicit_behavior	["full-scan-on-init", "provider-discovery-as-authority", "implicit-action-grant", "unbounded-context-expansion", "search-ranking-as-acceptance-proof"]	presets/semantic-morok-tower.json	forbidden_implicit_behavior
semantic_programming	{"default_visibility": "private", "object_promotion_criteria": ["independent-invariant", "independent-policy", "replaceability", "measurable-algorithm", "failure-boundary", "reuse", "domain-significance"], "pipeline": ["validate-input", "prepare-change", "execute-operation", "record-result", "publish-evidence"]}	presets/semantic-morok-tower.json	semantic_programming
additional_provider_capabilities	["filesystem-inventory", "export-scan", "signature", "build-dependency"]	presets/semantic-morok-tower.json	additional_provider_capabilities
Owner: presets/semantic-morok-tower.json Validator: Preset schema + preset field identifiers			

Top-level schema ingress

#	Accepted root definition	Owner	Validator
1	#\$defs/StandardManifest	core/contracts.schema.json	#\$defs/StandardManifest
2	#\$defs/SemanticModel	core/contracts.schema.json	#\$defs/SemanticModel
3	#\$defs/AuthorityModel	core/contracts.schema.json	#\$defs/AuthorityModel
4	#\$defs/PolicySet	core/contracts.schema.json	#\$defs/PolicySet
5	#\$defs/Conformance	core/contracts.schema.json	#\$defs/Conformance
6	#\$defs/Preset	core/contracts.schema.json	#\$defs/Preset
7	#\$defs/ProjectInit	core/contracts.schema.json	#\$defs/ProjectInit
8	#\$defs/StandardsInit	core/contracts.schema.json	#\$defs/StandardsInit
9	#\$defs/TechnologiesInit	core/contracts.schema.json	#\$defs/TechnologiesInit
10	#\$defs/AuthorityInit	core/contracts.schema.json	#\$defs/AuthorityInit
11	#\$defs/Activation	core/contracts.schema.json	#\$defs/Activation
12	#\$defs/Grant	core/contracts.schema.json	#\$defs/Grant
13	#\$defs/Lease	core/contracts.schema.json	#\$defs/Lease
14	#\$defs/Task	core/contracts.schema.json	#\$defs/Task
15	#\$defs/Candidate	core/contracts.schema.json	#\$defs/Candidate
16	#\$defs/Artifact	core/contracts.schema.json	#\$defs/Artifact
17	#\$defs/Run	core/contracts.schema.json	#\$defs/Run
18	#\$defs/Finding	core/contracts.schema.json	#\$defs/Finding
19	#\$defs/Decision	core/contracts.schema.json	#\$defs/Decision
20	#\$defs/GateResult	core/contracts.schema.json	#\$defs/GateResult
21	#\$defs/Relation	core/contracts.schema.json	#\$defs/Relation
22	#\$defs/WorkCard	core/contracts.schema.json	#\$defs/WorkCard
23	#\$defs/CommandRequest	core/contracts.schema.json	#\$defs/CommandRequest
24	#\$defs/CommandResult	core/contracts.schema.json	#\$defs/CommandResult
25	#\$defs/EventBatch	core/contracts.schema.json	#\$defs/EventBatch

Owner: core/contracts.schema.json | Validator: oneOf + Draft 2020-12